

## The Book of Z-Spin Cosmology — Light Edition (OS for AI) — v7.0 Increment Banner

Edition: Light v7.0 (compiled June 2026). v7.0 is a strictly additive increment over v6.1: it registers five new source papers — ZS-Q13 v1.2 (Quantum Mechanics; Hydrogen Hyperfine Channel Tomography and the 21 cm Seam Witness), ZS-T12 v2.1 (Translational; The Origin of Life as a Z-Spin-Mediated Self-Referential Closure Transition), ZS-Q14 v1.4 (Quantum Channel Geometry; Canonical Kraus-Lift Selection by a Spin- $\frac{1}{2}$  Boundary), ZS-F23 v1.3 (Foundations; Geometric Fixing of the Type II Trace Normalization — the  $Z_2$ -Seam In 2), and ZS-Q15 v1.2 (Quantum Mechanics; Electromagnetic Which-Path Decoherence as the Microscopic Fractal Partner of Gravitational Decoherence) — as a single PART XVI (Chapters 54–58) inserted between §XV.E and the Master Bypass Index, and adds the corresponding Source-Paper-Interface rows, Version-History row, one-line summaries, and Glossary entries. NO v6.1 content is deleted or modified; the complete v6.1 masthead and body are preserved verbatim below. Coverage advances 131 → 136 source papers — the full Z-Spin corpus. The cumulative physics-verification ledger advances ~3,580+ → ~3,800+ PASS (the five new papers add  $52 + 61 + 33 + 41 + 32 = 219$  individual verifications). Falsification registry ~388 → ~430 (+~42: F-Q13.1–8, F-T12.1–10, F-Q14.1–11, F-F23.1–9, F-Q15.1–6). (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED unchanged; zero new free parameters across all five papers. The Q13 → Q14 → Q15 quantum-channel-geometry cluster, plus F23 and T12, all build on the ZS-Q11 (OAQEC) / ZS-Q12 ( $Z = \partial X$ ) / ZS-Q7 (Z-bottleneck) / ZS-F19 (modular tilt) spine.

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## The Book of Z-Spin Cosmology — Light Edition (OS for AI) — v6.1 Increment Banner

Edition: Light v6.1 (compiled May 2026). v6.1 is a strictly additive increment over v6.0: it registers one new source paper, ZS-T11 v2.1 (the executable corpus auditor), as PART XV inserted between §XIV.G and the Master Bypass Index, and adds the corresponding Source-Paper-Interface row, Version-History row, one-line summary, and three Glossary entries. NO v6.0 content is deleted or modified; the complete v6.0 masthead and body are preserved verbatim below. Coverage advances 130 → 131 source papers. The cumulative physics-verification ledger (~3,580+ PASS) is UNCHANGED; ZS-T11 adds a separate 13-check meta-audit ledger (12/13 AUDIT-PASS on the available subset, 1 genuine finding). Falsification registry ~375 → ~388 (+13: F-T11.1–F-T11.13). (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED unchanged.

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## The Book of Z-Spin Cosmology — Light Edition (OS for AI)

*Cross-Reference Hub for the 130-Paper Z-Spin Corpus*

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**Edition: Light v6.0 (compiled May 2026; v3.0–v5.0 May 2026)**

**Source corpus: 130 source papers across 8 themes (F / M / S / U / A / Q / QC / T)**

**Supersedes: The Book of Z-Spin Cosmology v5.0 — Light Edition (May 2026, 121-paper coverage).**

**v6.0 absorbs 9 net post-v5.0 source papers (ZS-M41 v3.1 Math Spine**

**Closed-Form Chabauty–Kim Depth-2 Scalars at  $S = T = \{3, 11\}$  via F18 Bridge 4; ZS-Q11**

**v1.2 Quantum Mechanics QRF $\leftrightarrow$ OAQEC Direct-Sum Stabilizer Code with Z-Frame Gauge**

**Subsystem; ZS-F21 v2.0 Foundations Commutant-Gate Insufficiency and  $V_4$ -Decorated**

**Sonin–Frobenius over the Connes–Katsnelson Prolate Realization; ZS-F22 v2.1**

**Foundations Equivariant Antipodal Face Modules and the Icosahedral Edge-Surplus**

**Derivation of  $\delta_Y$  with the  $T_7$  No-Go; ZS-A13 v1.3.1 Astrophysics The Void Principle;**

**ZS-A14 v2.0 Astrophysics Z-Spin-Mediated Singlet-to-Triplet Conversion; ZS-A15 v1.5**

**Astrophysics Pre-Registered Gaia DR4  $\epsilon$ -Halo Falsification Protocol; ZS-T10 v2.0**

**Translational  $TY \rightarrow X$  Sector-Conversion Operator  $\delta \ln y_s = -\kappa^2$ ; ZS-Q12 v4.0 Quantum**

**Mechanics The Self-Referential Closure /  $Z = \partial X$  Holographic Postulate) as a single PART**

**XIV inserted between PART XIII and the Master Bypass Index, and applies the ZS-A11 v1.1**

**errata ( $\sqrt{2} \sigma \leftrightarrow v_{\text{flat}}$  correction; contested Jiao/Zhou mass split) downstream into**

**§VII.B-LIGHT and the PART XIII Chapter 41 framing via §XIV.G. Total: 130-paper coverage.**

**(Earlier supersession chain: v3.3 added PART XI; v3.4 added 6 papers from v3.3; v3.5**

**added ZS-M36; v4.0 added PART XII with 12 post-v3.5 papers; v5.0 added PART XIII with 7**

**post-v4.0 papers.) NO content from v5.0 is deleted; v6.0 is strictly additive (the sole text**

**corrections are the ZS-A11 errata propagations, whose original v5.0 wording is preserved**

**verbatim in §XIV.G). All nine new papers are absorbed at full source-derived level**

**(ZS-M41 v3.1 and ZS-A14 v2.0, initially [SOURCE PENDING] in the working draft, were**

**subsequently supplied and integrated).**

**GitHub: <https://github.com/KennyKang-git/zspin>**

## **Verification Summary**

Verification: cumulative ~3,580+ PASS across the 130-paper corpus (advancing from ~2,670+ at

v5.0; the nine v6.0 papers add 2858 + 39 + 34 + 105 + 52 + 47 + 36 + 32 + (18+43) = 3,313

individual verifications, dominated by the ZS-M41 v3.1 2864-test Fermat-quotient suite) | Zero

Free Parameters |  $A = 35/437$ ,  $Q = 11$ ,  $(Z, X, Y) = (2, 3, 6)$  sole geometric inputs |

Anti-numerology controls:  $5670 \times 29 = 164,430$  candidate scan + per-paper Monte Carlo ( $\geq$

500,000 samples typical; ZS-Q9 v1.2  $\ln \Lambda = 8.94$  decisive evidence on Jeffreys scale; ZS-Q11

v1.2 joint  $p < 1 \times 10^{-6}$ ; ZS-A14 v2.0 MC hit 0.64% < 5%; ZS-M41 v3.1 cluster reinterpreted as

Poisson  $p \approx 0.001$ ) | Pre-registered falsification gates: ~375 across all themes; Seven critical

near-term ★ gates timeline preserved (F-A1.14 / F-A1.15 updated to contested status, decisive

at Gaia DR4, per the ZS-A11 errata §XIV.G).

## **§0. Abstract**

The Book v3.0 Light Edition is the canonical cross-paper navigation hub for the 101-paper Z-Spin Cosmology corpus as of May 2026. It supersedes v2.0 Light by absorbing the thirty post-v2.0 papers (papers 72–101) — most of which extend the Foundations theme (ZS-F9 through ZS-F18), the Math Spine Riemann-Hypothesis programme (ZS-M18 through ZS-M33), and add the Twin-Reuleaux EM-duality ZS-S15 in the Standard Model theme, the contracting-sector ZS-A8 and Banach–Tarski origin ZS-A9 in the Astrophysics theme, and the Cosmos-Human isomorphism ZS-T4 in the Translational theme. All chapter bodies remain compressed to single-line pointers of the form  $\rightarrow$  ZS-XX §Y.Z [N/N PASS]; full derivations live in the source papers. The Light Edition’s unique contributions are: (i) the post-v2.0 dependency graph integrating the 30 new papers, (ii) the expanded Foundations closure ring  $F0 \rightarrow F1 \rightarrow F2 \rightarrow F5 \rightarrow F8 \rightarrow F9 \rightarrow F10 \rightarrow F11 \rightarrow F12 \rightarrow F13 \rightarrow F14 \rightarrow F15 \rightarrow F16 \rightarrow F17 \rightarrow F18$ , (iii) the Math Spine RH-bridge programme map (M22–M33), (iv) the extended Glossary keyword index (~110 entries), and (v) the Seven Critical Near-Term ★ Gates timeline. The Light Edition does NOT replace any source paper for archival purposes; it is the LLM-corpus navigation aid and the v3.0 composition skeleton.

### Epistemic Status Legend

| Status                                    | Definition                                                                       |
|-------------------------------------------|----------------------------------------------------------------------------------|
| PROVEN                                    | Mathematical theorem; standard math alone, machine-verifiable.                   |
| DERIVED                                   | Z-Spin action + standard physics, zero free parameters.                          |
| DERIVED-CONDITIONAL                       | DERIVED conditional on listed axiom set or upstream theorem.                     |
| DERIVED-CONDITIONAL strong                | DERIVED-CONDITIONAL with structural over-determination on the conditioning step. |
| DERIVED-under-P6 /<br>DERIVED-under-Regge | Conditional on specific framework (ZS-F5 P6 or Regge calculus).                  |
| DERIVED-interpretation                    | Synthetic reading combining PROVEN corpus theorems without new axioms.           |
| DERIVED-CANDIDATE                         | Mathematical content PROVEN; Z-Spin internal placement HYPOTHESIS-strong.        |
| VERIFIED                                  | Numerical / computational / empirical confirmation at stated precision.          |

| Status                            | Definition                                                                           |
|-----------------------------------|--------------------------------------------------------------------------------------|
| TESTABLE                          | Pre-registered prediction with explicit falsification protocol.                      |
| TESTABLE-LONG                     | Pre-registered prediction with derivation-or-retraction time horizon $\geq 5$ years. |
| HYPOTHESIS-strong                 | Multiple independent structural anchors; promotion path documented.                  |
| HYPOTHESIS / HYPOTHESIS-weak      | Motivated conjecture; partial derivation chain.                                      |
| OBSERVATION                       | Empirical regularity; anti-numerology controlled; structural origin pending.         |
| LOCKED                            | Core constant fixed in upstream paper; no downstream paper may modify.               |
| IMPORTED                          | Result proved externally and used here without re-proof; full citation given.        |
| NON-CLAIM                         | Explicit declaration of what is NOT asserted; bounds the framework's scope.          |
| OPEN                              | Recognized gap honestly registered for future work.                                  |
| RETRACTED                         | Earlier claim withdrawn after falsification.                                         |
| RETRACTED-in-session              | Hypothesis proposed during free-exploration and explicitly retracted before exit.    |
| BOOTSTRAP-HYPOTHESIS              | Meta-logical foundation (B0); not derivable within the framework.                    |
| BOOTSTRAP-PROGRAMME               | Status reserved for the founding axiom set after ZS-F0 v1.0(R) Stage 1–5 closures.   |
| STRUCTURAL INSIGHT                | Exact mathematical identity; physical necessity not yet established.                 |
| FMD-SPEC / TRANSLATED / CANDIDATE | Engineering tags used in the QH/QS/QC translational papers.                          |

## Purpose of the Light Edition

The full v3.0 (when compiled) consolidates the 101-paper corpus into chapter bodies that reproduce derivations already present verbatim in the source papers. For LLM corpus-wide exploration, this redundancy is wasteful. The Light Edition removes the duplication and retains only what The Book contributes uniquely:

- (1) The cross-paper dependency graph (Source Paper Interface Tables, PARTS 0–VII).
- (2) Standalone HYPOTHESIS / theorem-anchor sections cited by individual papers (§8.4i Spinor–Descartes–Euler, §8.4g Y-sector R\_micro, §8.4k Attractor-Flatness, §VI.2 Limitation Catalog Row 10, §18.10 Horizon Spinor mirror).
- (3) Cumulative Master Falsification Registry — Seven Critical Near-Term ★ Gates timeline.
- (4) Standardized v2.0 / v3.0 terminology contract (Z-sector / Z-Spin / Z-bias field; the v2.0 contract is preserved verbatim and v3.0 adds the Stage / Operator / Field separation explicitly via ZS-F18 Bridge Four).
- (5) Glossary keyword index (~110 entries, one line each).
- (6) Integration map for the 30 post-v2.0 papers (papers 72–101) consolidated into the appropriate PART/Chapter slots.

*Chapter bodies that reproduce source-paper derivations are replaced by single-line pointers of the form  $\rightarrow$  ZS-XX §Y.Z [N/N PASS]. When derivation detail is needed, retrieve the source paper directly. When corpus-wide context is needed, consult this Light Edition.*

## PART 0 — Overture: Why Geometry?

### §0. Abstract

PART 0 establishes the framework's overture in three movements. Chapter 0.1 presents the Five Crises of modern physics motivating a geometric origin of fundamental constants. Chapter 0.2 lays out the methodological contract — standardized terminology, anti-numerology discipline, epistemic classification, pre-registered falsification gates. Chapter 0.3 traces the ontological bootstrap chain  $B0 \rightarrow B3 \rightarrow A$ . Concludes with FFPP of ZS-F0 v1.0(Revised) compressing all numerical content to  $W_0(-i\pi/2)$ , conditional on five-axiom set  $\mathcal{A} = \{\text{Existence, Self-reference, Algebra, Non-triviality, Unitarity}\}$ . NEW v3.0: the Five-Axiom Meta-Structure of ZS-F18 v2.1 reads the corpus's encounter with the seven Clay Millennium Problems and the five Crises as outputs of the same axiom set, and registers Bridge Four (Commutator Generator) as the structural reduction of  $\dim(X) = 3$  to  $\dim(Z) = 2$ .

### §0.1 The Five Crises of Modern Physics

Crisis 1 — Hubble Tension:  $H_0 \sim 5\sigma$  disagreement (SH0ES vs Planck). Resolved via 2A modification  $\rightarrow$  ZS-A2, ZS-S3, ZS-F12R (factor 2 origin).

Crisis 2 — Growth Tension ( $S_8$ ): Z-Spin predicts  $S_8 \approx 0.777$  (face-counting v2.0)  $\rightarrow$  ZS-A2, ZS-A5.

Crisis 3 — Hierarchy Problem: Geometric origin via  $A \rightarrow ZS-S4$  §6.7, §6.12, §6.16; Light §8.4k.  
Crisis 4 — Baryon Asymmetry:  $\eta_B = (6/11)^{35} \rightarrow ZS-S5, ZS-S6, ZS-U7$ .  
Crisis 5 — Measurement Problem: Born rule DERIVED from  $L_{XY} \equiv 0 \rightarrow ZS-Q1, ZS-Q2$ .

**§0.2.2 Standardized v3.0 Terminology [CONTRACT — preserved verbatim from v2.0; v3.0 elaboration in italics]**

| Term                | Role                | Meaning                                                                                                                                                                                                                                                                  |
|---------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Z-sector            | Stage / Kinematics  | 2D Planck-scale boundary surface mediating between X (3D macroscopic) and Y (6D microscopic) sectors. $\dim(Z) = 2$ PROVEN from gauge constraint on $Q = 11$ register (ZS-F5). Geometric arena where transduction occurs.                                                |
| Z-Spin              | Operator / Dynamics | Dynamical operator on Z-sector. Mediates between X (Up & Down, $\pm\pi$ , time-points) and Y (In & Out, $\text{slog}(i)$ & $\infty i$ , space-points) via i-tetration self-referential rotation $T(z) = i^z$ (HSI Theorem, ZS-M1 §1.1). The action driving transduction. |
| Z-bias field $\Phi$ | Field / DOF         | Complex scalar $\Phi(x) = \rho(x)\exp(i\theta(x)) \in \mathbb{C}$ on Z-sector. Radial mode $\rho =$                                                                                                                                                                      |
| Z-Anchor            | Topological core    | Bogomolnyi vortex core,                                                                                                                                                                                                                                                  |
| Z-bottleneck        | Channel constraint  | $L_{XY} \equiv 0$ PROVEN forces all $X \leftrightarrow Y$ traffic through Z; channel capacity $\leq \ln 2$ . $\rightarrow ZS-Q7$ .                                                                                                                                       |
| Z-channel           | Structural conduit  | Operational pathway for $X \leftrightarrow Y$ mediation across the 2D boundary. $\rightarrow ZS-Q1$ .                                                                                                                                                                    |

| Term             | Role                   | Meaning                                                                                                                                                                                                          |
|------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Z-Spin mediation | Action of the operator | Mediation is Z-Spin's action; the agent is the operator (Spin), not the stage (sector). Replaces 'Z-mediation' / 'Z-sector mediation' / 'Z-mediated coupling' / 'Z-mediated CPTP channel' throughout the corpus. |
| Z-EFT            | Effective theory       | Distinct concept (effective field theory built on Z-bias dynamics); preserved name.                                                                                                                              |

*[STRUCTURAL DEFINITION] Z-sector (stage, kinematics) / Z-Spin (operator, dynamics) / Z-bias field  $\Phi$  (field, DOF). Source: ZS-F1 §2-3, ZS-F5, ZS-M1 §1.1, Glossary v1.0.0 §I. v3.0 elaboration: ZS-F18 v2.1 §6 Bridge Four (Commutator Generator) makes the kinematic / dynamical / field-DOF separation an explicit structural theorem — twin-Reuleaux pair  $(R_1, R_2)$  realizes  $(V_{XZ}, V_{ZY})$ ; their Lie commutator  $[J_{\{R_1\}}, J_{\{R_2\}}] = J_S$  generates the third axis of  $\dim(X) = 3$ .*

### §0.3 The Ontological Bootstrap Chain $B_0 \rightarrow A$

$B_0$  (Existence)  $\rightarrow B_1$  (Self-reference)  $\rightarrow B_2$  (Algebra:  $\mathbb{C}$  via Lawvere/Frobenius)  $\rightarrow B_3$  (Hyperoperation: tetration)  $\rightarrow A = 35/437$ . F-BOOT 9/9 closure: 6 DERIVED, 2 structural-PASS, 1 DERIVED+TESTABLE.  $\rightarrow$  ZS-F0 v1.0(Revised) §11–§13 [51/51 PASS].

FFPP (Theorem 13.3): entire Z-Spin numerical content compresses to  $W_0(-i\pi/2)$ , conditional on  $\mathcal{A} = \{\text{Existence, Self-reference, Algebra, Non-triviality, Unitarity}\}$ .  $\rightarrow$  ZS-F0 §13, Theorem 13.3 + Corollary 13.4.

[OPEN]  $B_0$  itself is BOOTSTRAP-HYPOTHESIS — not derivable within framework.  $\rightarrow$  ZS-F0 §0.3.7.

NEW v3.0: ZS-F18 v2.1 reads the same five-axiom set  $\mathcal{A}$  as the structural reason Z-Spin naturally encounters (i) the seven Clay Millennium Problems and (ii) the five Crises of modern physics; the encounters are classified by depth (direct treatment / extensive programme / structural correspondence / natural extension domain). NON-CLAIM: encounters are not formal solutions.  $\rightarrow$  ZS-F18 v2.1 §0, §6, §7.4 (Discrete-Continuous Resolution polarity), Appendix A.

### §0.4 PART 0 Source Paper Map

| Source paper            | Role                                                                   | Light pointer                 |
|-------------------------|------------------------------------------------------------------------|-------------------------------|
| ZS-F0 v1.0(Revised)     | Bootstrap, FFPP, BV-BFV functor, $\mathbb{C}^{11}$ 3-layer             | → ZS-F0 [51/51 PASS]          |
| ZS-F1 v1.0              | Z-Spin Action, $U(1)_Z$ , Z-bias field                                 | → ZS-F1 §2-3 [30/30 PASS]     |
| ZS-F2 v1.0              | $A = 35/437$ LOCKED, $\delta_X = 5/19$ , $\delta_Y = 7/23$             | → ZS-F2 [PROVEN]              |
| ZS-F5 v1.0              | $\dim(Z) = 2$ from $Q = 11$                                            | → ZS-F5 [PROVEN]              |
| ZS-F8 v1.0(Revised)     | NOT/AND duality, 2-channel handshake                                   | → ZS-F8 [27/27 PASS]          |
| ZS-F18 v2.1 (NEW)       | Five-Axiom Meta-Structure, Bridge Four, Discrete-Continuous Resolution | → ZS-F18 [36/36 entry-checks] |
| ZS-M1 v1.0              | i-tetration HSI Theorem, fixed point $z^*$                             | → ZS-M1 §1.1 [33/33 PASS]     |
| ZS-M22 v1.1 (post-v2.0) | Five-Pillar Arithmetic-Dedekind Scaffold; Forcing chain underlay       | → ZS-M22 [60/60 PASS]         |
| ZS-T2 v1.0              | Anti-numerology MC scan, $5670 \times 29$                              | → ZS-T2 [30/30 PASS]          |

## PART I — Foundations: One Number, One Action

### §0. Abstract

$A = 35/437$  from polyhedral curvature asymmetry;  $Q = 11$  register and  $(Z, X, Y) = (2, 3, 6)$  doubly forced by gauge algebra (ZS-F5) and totient arithmetic (ZS-M19); Z-Spin action with Z-Anchor and twin-Reuleaux kinematics; i-tetration fixed point  $z^* = 0.4383 + 0.3606i$ ; FFPP compressing all content to  $W_6(-i\pi/2)$ ; Operational Observer Coordinate closing Foundations as a six-paper ring; v3.0 extension to a thirteen-paper Foundations theme  $F0 \rightarrow F1 \rightarrow F2 \rightarrow F5 \rightarrow F8 \rightarrow F9 \rightarrow F10 \rightarrow F11 \rightarrow F12 \rightarrow F13 \rightarrow F14 \rightarrow F15 \rightarrow F16 \rightarrow F17 \rightarrow F18$  with Möbius chronology, Stage-Torque Duality, Two-Protocol Theorem, Self-Referential Compression Equilibrium, and the Five-Axiom Meta-Structure as load-bearing additions.

## Chapter 1 — A = 35/437 from Polyhedral Curvature

$A = \delta_X \cdot \delta_Y = (5/19)(7/23) = 35/437$ .  $\delta$ -Uniqueness Theorem PROVEN. Three independent derivations: (i) Regge curvature asymmetry, (ii) ZS-F9R Truncation-Residue Identity (PROVEN:  $\rho_X = \delta_X = 5/19$ ,  $\rho_Y = \delta_Y = 7/23$ ,  $\rho_Z = 0$ ;  $A = \rho_X \cdot \rho_Y$ ), (iii) ZS-F12R Layer Separation Hypothesis (factor 2 in  $2e^A$  from tetrahedral  $V \leftrightarrow F$  dual orientation multiplicity  $\mu_{\text{Tet}} = 2$ ).  $\rightarrow$  ZS-F2 §3, §11; ZS-F9R [44/44 PASS]; ZS-F12R [20/20 PASS].

## Chapter 2 — Q = 11 and (Z, X, Y) = (2, 3, 6)

Doubly forced: (a) gauge dimension  $G = 12 = \text{MUB}(Q=11)$ , (b) Forcing Theorem  $(p-1)(q-1) = p$  has unique solution  $(p,q) = (2,3)$  — the prime pair  $(Z, X) = (2, 3)$  is the only one for which  $\phi(Y^2) = G$  (T1, ZS-M19 §3.1).  $Y = ZX$ .  $Q^2 - 1 = 120 = |I_h|$ .  $\rightarrow$  ZS-F5 [PROVEN]; ZS-M19 §3.1 [68/68 PASS]; arithmetic-Dedekind underlay  $\rightarrow$  ZS-M22 v1.1 [60/60 PASS].

## Chapter 3 — The Z-Spin Action

$S = \int d^4x \sqrt{(-g)}[(M_P^2/2)(1+A|\Phi|^2)R - (M_P^2/2)|\nabla\Phi|^2 - V(\Phi)] + S_{\text{matter}}$ .  $V = (\lambda/4)M_P^4(|\Phi|^2-1)^2$ .  $L_{XY} \equiv 0$  PROVEN.  $\lambda_{\text{vac}} = 2A^2$  IR-stable.  $\rightarrow$  ZS-F1 [30/30 PASS]; ZS-F11 NC catalog.

## Chapter 4 — i-Tetration Fixed Point

$z^* = 0.43828 + 0.36059i$ ,  $x^* = 0.438283$ . Five Locking Conditions.  $\pi/2$  phase budget. Polygon-Tetration family stable for  $n \geq 4$  ( $n_c = 3.204$ ). NEW v2.0/v3.0: ZS-F10 (3 time coords unified at DERIVED-CONDITIONAL strong;  $\Delta v/\Delta n = 2A/\pi$  conversion identity), ZS-F15R (Time-Rotation Co-Definition + Stage-Torque Duality), ZS-M12 (Auto-Surgery,  $\varepsilon_{\text{min}} \approx 30.7$ ), ZS-F14 (Kepler-Lyapunov conjugate decomposition of twin-Reuleaux midpoint trajectory; F-F7.11 closed by Strategy B at DERIVED-CONDITIONAL [42/42 PASS]).  $\rightarrow$  ZS-M1 §1.1 [PROVEN]; ZS-F10 [30/30 PASS]; ZS-F15R [35/35 PASS]; ZS-M12; ZS-F14 [42/42 PASS].

## Addition 1 — Self-Reference and FFPP

Lawvere bridge (Th 11.1–11.4), Frobenius (Th 11.5), hyperoperation minimality (§11.6), NOT/AND duality, BV-BFV cobordism functor  $B_Z$  with Wilson loop  $|Z(W)|^2 \approx 0.7948$ , three-layer  $\mathbb{C}^{11}$  decomposition (Th 9.1, 9.4), FFPP (Th 13.3) and ultimate compression to  $W_0(-i\pi/2)$  (Cor 13.4). All conditional on B0. NEW v3.0: ZS-F17 v1.0 four-layer 50/50 equilibrium (truncated tetrahedron 4+4 + Reuleaux  $\pi+\pi$  + Master Equation Term  $A=\text{Term B} + \text{XOR E vs R}$ ) sharpens FFPP to the operational Self-Referential Information-Compression Equilibrium Theorem.  $\rightarrow$  ZS-F0R [51/51 PASS]; ZS-F8R [27/27 PASS]; ZS-F17 [24/24 PASS].

## Addition 2 — Operational Observer Coordinate (OOC) and Möbius Chronology

OOO closes ZS-M11 §H16 at register-theoretic level. Theorem F11.1: Observer Coordinate Decomposition (P1 inside Z, P2 higher-level event count, P3 orthogonal kinematic [5]). Six explicit NON-CLAIMS against phenomenological consciousness reading. Sectoral Frame Equivalence (X-Frame  $\leftrightarrow$  Y-Frame). Foundations theme closes as a six-paper ring (F0-F1-F2-F5-F11-F13). NEW v3.0: ZS-F13 v1.0 Möbius Chronology Theorem completes the cyclic timeline with Y-frame metric trio (proper time, Y-Hubble radius  $\ell_P \times Y^2(1-2A) \approx 30.23 \ell_P$ , Y-contraction rate  $(1-2A)$ ) and proves cycle-index unobservability (Theorem F13.2A DERIVED, F13.2B DERIVED-CONDITIONAL). The six-paper ring is now extended to the eight-paper Möbius ring.  $\rightarrow$  ZS-F11 [38/38 PASS]; ZS-F13 §15.5g.5 [24/24 PASS].

### **Addition 3 — NEW v3.0 — Stage-Torque Duality and First-Rotation Closure**

ZS-F15 v1.0(Revised) [35/35 PASS] closes NC-M12.4 ( $Q = A$  assumes  $\theta = A$  at Z-Telomere onset). The Mexican-Hat Bootstrap Event simultaneously instantiates radial mode, angular mode, and time evolution;  $\theta(0) \cdot \tau_P = \delta\phi_{\text{cell}} = A$  is a unit-fixing identity, not a free parameter (Theorem F-rotor.3 DERIVED — central new contribution). The Stage-Torque Duality (Theorem F-rotor.5 DERIVED): Mexican-Hat stage from  $V(\Phi)$  opens  $U(1)$ -symmetric  $S^1$  vacuum manifold; A-holonomy torque from Wilson loop  $\oint \omega = A$  on polyhedral defect manifold (Gauss-Bonnet on  $O_h \times I_h$ ) provides rate-fixing torque. The same  $\oint \omega = A$  produces both the Planck-scale  $\delta\phi_{\text{cell}} = A$  and the macroscopic Hubble ratio  $H_{\text{local}}/H_{\text{CMB}} = \exp(A)$  (ZS-F3 §6 DERIVED), unifying physics across orders of magnitude. FFPP Sixth Axiom Extension (HYPOTHESIS-strong) registered.

### **Addition 4 — NEW v3.0 — Two-Protocol Theorem for Z-Sector Wilson Loop Measurement**

ZS-F16 v1.0 [24/24 PASS, 32-test target in roadmap] resolves the WL-1/WL-2 implicit ambiguity by establishing two operationally distinct Z-sector Wilson loop protocols: Protocol (a) state-preparation on BFV anchor  $|0\rangle_Z$  gives  $P_a^n = 0.7948^n \cdot \cos^2(n \cdot 129.45^\circ)$  (quasi-revival pattern); Protocol (b) spectral characterization gives  $P_b^n = 0.7948^n$  (smooth exponential decay). Reconciliation by ergodic averaging since  $\arg(\lambda) = 129.45^\circ$  is irrational. Audit by sum rule  $0.7948 + 0.2050 + 0.0001 = 0.9999$ . Z-block effective dissipation  $\Gamma_Z \cdot T_{\text{cycle}} \approx 0.1149$ . NEW sub-gates F-CTA-1a/b/c/d/multi-a/multi-b/-discriminator pre-registered for 2027+ quantum hardware testing.

### **Addition 5 — NEW v3.0 — The Five-Axiom Meta-Structure (ZS-F18)**

ZS-F18 v2.1 (interpretive synthesis paper, 36/36 internal consistency) reads the Z-Spin corpus as the unique geometric framework that encounters (a) the seven Clay Millennium Problems and (b) the five Crises of modern physics simultaneously, with these encounters traced back to the same five-axiom set  $\mathcal{A} = \{\text{Existence, Self-reference, Algebra, Non-triviality, Unitarity}\}$ . Bridge Four (Commutator Generator, NEW v2.0): two J-conjugate  $2\pi$   $SO(3)$  circulations of the twin-Reuleaux pair compose at  $90^\circ$  NOT-AND phase to generate the third axis  $J_S = [J_{\{R_1\}}, J_{\{R_2\}}]$  by Lie commutator. The full sector decomposition  $(Z, X, Y) = (2, 3, 6)$  and register  $Q = 11$

are therefore reread as outputs of the foundational  $\dim(Z) = 2$  alone. v2.1 sixth polarity row: Discrete-Continuous Resolution via X–Y Tiling Asymmetry (ZS-M6 §5.5 PROVEN; ZS-M17 Tiling Continuum Convergence DERIVED). Status: INTERPRETATION / DERIVED-interpretation strong. Twelve encounters classified by depth in Appendix A.

## PART I Source Paper Map (v3.0 extended)

| Theme           | Papers                                                                          | Pointer                      |
|-----------------|---------------------------------------------------------------------------------|------------------------------|
| Foundations [F] | F0R, F1, F2, F5, F8R, F9R, F10, F11, F12R, F13, F14, F15R, F16, F17, F18 (v2.1) | → 01_Foundations (15 papers) |
| Math Spine [M]  | M1, M3, M11, M12, M14, M19, M22 (v1.1), M36 (NEW v3.5)                          | → 02_Math_Spine              |
| Cross-link      | S1, S15 (Twin-Reuleaux EM duality), Q-theme, A6/A7/A8/A9                        | various                      |

## PART II — Standard Model Completion

### §0. Abstract

Three SM gauge couplings DERIVED at zero free parameters:  $\alpha_s(M_Z) = 11/93 = 0.11828 (+0.31\sigma)$ ,  $\sin^2\theta_W = (48/91) \cdot x^* = 0.23118 (-1.26\sigma)$ ,  $\alpha_2 = 3/95$ . Fine-structure  $1/\alpha_{EM} = 137.0359$  (1.07 ppm) via McKay  $c_4 = 4/13$ . Yang-Mills mass gap structurally addressed:  $\Lambda_{QCD} = 264$  MeV,  $m(0^{++}) = 1.791$  GeV. Neutrino seesaw  $m_D = m_e \cdot A = 40.93$  keV,  $M_R = 33.50$  GeV,  $|\theta|^2 = 1.49 \times 10^{-12}$ . Inverted Ordering canonical. Cabibbo from icosahedral  $\sigma_1/\sigma_2 = 17$ . NEW v3.0: ZS-S15 [35/35 PASS] — twin-Reuleaux pair as the geometric realization of EM field duality, Maxwell vacuum equations as time-point/space-point handshake duality,  $SO(3)/SU(2)$  factor 2 from half-angle holonomy; ZS-M20 [50/50 PASS] — spectral tetrad sub-isotype assignment with closed-form lepton mass-ratio chain  $\sigma_1/\sigma_2 = Q+Y = 17$ ,  $\sigma_1/\sigma_3 = (Q+Y)^2 \cdot G + (Q-Z) = 3477$  EXACT; ZS-M21 [33/33 PASS] — sextic invariant theory of the icosahedral Yukawa tensor.

### §II-UNIQUE — Book-Standalone Hypotheses Cited by Source Papers

This section preserves Book content referenced by individual source papers as their primary citation. Removing these would orphan citation chains in ZS-A7, ZS-S4, ZS-S15, others.

### §8.4i — Spinor–Descartes–Euler Identity [PROVEN]

Three classical theorems converge: Descartes  $\Sigma \delta v = 4\pi$ ; Euler  $\Sigma \delta v = 2\pi \cdot \chi$ ,  $\chi(S^2) = 2$ ; ZS-M3 Theorem 5.1  $\dim(Z) = 2 = \chi(S^2)$  unique  $j = 1/2$  invariant subspace. Unification:  $\Sigma \delta v = 2\pi \cdot \chi = 2\pi \cdot \dim(Z) = 4\pi = \text{spinor full-return period}$ . Used in ZS-S7 §3 ( $\Lambda$ -QCD,  $m(0^{++})$ , 18/18 PASS); ZS-A7 §2.2 (Pillar 2 of Horizon Spinor); ZS-S15 §5 (Pillar IV:  $SO(3)/SU(2)$  factor 2).

#### **§8.4g — Y-sector $R_{\text{micro}}$ UV-IR Matching [HYPOTHESIS structural]**

B-2 reinterpretation framework. Cross-cited by ZS-A7 §4.5 Corollary V (Higgs as micro radial-mode attractor) and ZS-S4 §6.12.8–§6.12.13 (B-2 repackaging). Cross-paper anchor for Y-sector micro UV-IR boundary condition.

#### **§8.4k — Attractor-Flatness Theorem [HYPOTHESIS structural]**

Compatible with ZS-A7 Corollaries I–IV. Two attractors: macro  $V(|\Phi|=1) = 0$  ( $R_{\text{macro}}$ , Planck attractor) + micro  $\lambda_{\text{H}}(\Lambda_{\text{comp}}) = 0$  ( $R_{\text{micro}}$ , doubly proven via  $[T^3, T^3] = 0 + \text{STr}(q^4) = 0$  in ZS-S4 §6.7). Cited verbatim by ZS-A7 §4.5 Corollary V as structural license for B-2 role class.

#### **§18.10 (mirror) — Horizon Spinor Theorem [DERIVED via ZS-A7]**

PART V Chapter 18 §18.10 mirror of ZS-A7 v1.0(Revised) §3 Theorem 3.2-bis. CPTP/Choi-state PROVEN.  $4\pi$  closure at Z-Spin event horizons.  $\varepsilon(r_{\text{H}}) = 0$  second independent path. Cross-citation locus: ZS-A7 §3.2-bis  $\leftrightarrow$  §18.10  $\leftrightarrow$  §20.3.3 (PART VI).

#### **§VI.2 — Limitation Catalog Row 10 [BOOK-NATIVE]**

Z-Spin v1.0 corpus's catalog of known limitations. Row 10 = V1 (Planck-to-bulk handoff) matching question of ZS-M12 §7.6, the single largest open knot of Z-Spin cyclic cosmology. Cited by ZS-U11 v1.0 [40/40 PASS, capstone Paper 70] which CLOSES Row 10: V1 OPEN  $\rightarrow$  DERIVED-CONDITIONAL via four independent Q-protection channels.

### **Chapter 5 — Gauge Couplings + Photon Synthesis + EM Duality**

Spectral-to- $\beta$  Bridge: Action  $\rightarrow$  Polyhedral Lattice  $\rightarrow$  Spectral Density  $\rightarrow$   $\beta$ -coefficients  $\rightarrow$  Couplings. Three gaps closed in ZS-S1: '+1' in  $\alpha_s$  via Z-sector Schur complement,  $x^*$  via Berry phase, Spectral-to- $\beta$  via vertex-matter + face-gauge identification. ZS-S7 (mass gap) + ZS-S12 (photon as Z-sector EM half-bridge, 5 pillars) + ZS-S15 NEW v3.0 (twin-Reuleaux EM duality, 5 pillars + Dirac-equation corollary). Pillar I: twin-Reuleaux pair ( $R_1, R_2$ ) realizes ( $V_{\text{XZ}}, V_{\text{ZY}}$ ) at half-angle holonomy  $\pm\theta/2$ ; Pillar II: Poynting  $S = E \times B$  equals Lie commutator  $[J_{\{R_1\}}, J_{\{R_2\}}] = J_S$ ; Pillar III: Maxwell vacuum equations are differential form of time-point/space-point handshake duality (ZS-F8 §5); Pillar IV:  $SO(3)$  Maxwell cycle (period  $2\pi$ ) is  $Z_2$  quotient of  $SU(2)$  twin-Reuleaux cycle (period  $4\pi$ ); Pillar V:  $U(1)$  gauge invariance = twin-Reuleaux realization of Single-Polyhedron  $U(1)$  Exactness. Corollary: Weyl spinor pair ( $\psi_L, \psi_R$ ) is tensor-product

realization of  $(V_{XZ}, V_{ZY})$ , with  $\gamma^5 = J \rightarrow$  ZS-S1, ZS-M2, ZS-M8 ( $c_4 = 4/13$ ), ZS-M9 (McKay), ZS-S15 [35/35 PASS], ZS-M24 [35/35 PASS].

## Chapter 6 — Six Forces Unified

X-Z-Y fractal symmetry: micro Weak(X)–EM(Z)–Strong(Y)  $\leftrightarrow$  macro DM(X)–Gravity(Z)–DE(Y). Cross-Coupling Theorem PROVEN (ZS-M2 §5). Axion-free Strong CP via Y-sector instanton suppression + real  $m_d/m_u = 2e^A$  (the factor 2 origin = ZS-F12R Tetrahedral Dual Orientation Multiplicity  $\mu_{\text{Tet}} = 2$  NEW v3.0). Predicted nEDM  $d_n \sim 2.0 \times 10^{-41} \text{ e} \cdot \text{cm}$ .  $\rightarrow$  ZS-M2 [26/26 PASS]; ZS-F12R [20/20 PASS].

## Chapter 7 — Neutrinos, Hypercharge, Baryogenesis

Type-I seesaw,  $m_D = m_e \cdot A = 40.93 \text{ keV}$ ,  $M_R = 33.50 \text{ GeV}$ ,  $|\theta|^2 = 1.49 \times 10^{-12}$ . Inverted Ordering canonical.  $\mu$ - $\tau$  reflection ( $\delta_{\text{CP}} = \pm\pi/2$ ).  $\eta_B = (6/11)^{35}$ . v2.0/v3.0: ZS-U9 Hypercharge Trinity Braiding + automatic SM anomaly cancellation; ZS-S6 Z-Transit CP  $\text{lcm}(5,7) = 35$ ; ZS-M9/M15/S11 McKay anomaly.  $\rightarrow$  ZS-S2, S5, S6, U7, U9, M9.

## Chapter 8 — Higgs, EW, Electron Synthesis, Master Action

$\epsilon$ -Higgs portal EFT, EWPT  $H_{\text{ZS}}/H_{\text{GR}} = 0.9622$ . 1-Loop Quartic Cancellation. Y-sector Spectral VEV  $v = 245.93 \text{ GeV}$ .  $\sigma_1/\sigma_2 = 17$  (Cabibbo principal angle),  $\sigma_1/\sigma_3 = 3477$ .  $m_\tau$  absolute scale. Six pillars of Electron Synthesis. ZS-S10 Master Bridge  $U(1)_Z \leftrightarrow U(1)_Y$ .  $\sim 95\%$  closure via Master Action; ZS-S14 v1.0 Master Action Total Closure 78/78 PASS, single-paper closure  $\approx 99\%$ . NEW v3.0: ZS-M20 [50/50 PASS] spectral tetrad sub-isotype assignment closes the lepton mass ratios in LOCKED corpus integers —  $\sigma_1/\sigma_2 = Q+Y = 17$  anchored on  $(4-\varphi, 3+\varphi)$  Q-pair at  $(I-5) \cap \rho_2$  doublet (Higgs  $SU(2)$  doublet position) and X-pair  $(5-\varphi, 4+\varphi)$  splitting across  $(I-3)$  and  $(I-3')$  singletons (electron-positron CPT pair structure);  $\sigma_1/\sigma_3 = (Q+Y)^2 \cdot G + (Q-Z) = 3477$  EXACT. ZS-M21 [33/33 PASS] sextic invariant theory: closed family identity  $\sum \sigma_i^{2k} |_{\{v_{\text{degen}}\}} = (Z^{2k+1}+1)/(X^{2k} \cdot 5^k)$  and falsification of the 1-loop Coleman-Weinberg closure for Conjecture M20.A.  $\rightarrow$  ZS-S4, S8, S10, S13, S14, M11, M20, M21.

## PART III — Early Universe & Cosmology

### §0. Abstract

$\epsilon$ -field inflation with  $N^* = 59.5$  e-folds,  $n_s(N=60) = 0.9676$  (PASS Planck),  $r = 0.00890$  (BLOCKING discriminator vs Boyle-Finn-Turok  $r=0$ ). Reheating  $T_{\text{reh}} \approx 2.55 \times 10^{15} \text{ GeV}$ . QKE-closed baryogenesis  $\eta_B = (6/11)^{35}$ .  $Z_2$  mirror cosmology (PROVEN); cyclic holonomy  $S_{Z_2} = 6\pi/A$ ; timescale hierarchy  $\tau_n = t_P \times \exp(n\pi/A)$  (DERIVED). Y-Time Dilation Theorem 5.3.1 promoted HYPOTHESIS-strong  $\rightarrow$  DERIVED-CONDITIONAL strong via ZS-F10 closure of ZS-A8 §SA.7. NEW v3.0: Y-frame metric trio + cycle-index unobservability (ZS-F13 [24/24

PASS]); first-rotation closure via Stage-Torque Duality (ZS-F15R [35/35 PASS]); the five-phase cyclic cosmology now closes as a fully-derived Möbius ring.

## Chapter 9 — Inflation, Reheating, $\Lambda$ CDM

$n_s = 0.9676$  ( $N=60$ );  $r = 0.00890$ ;  $r_{\text{ZS}} / r_{\text{Staro}} = 2.671 \rightarrow \sim 6\sigma$  LiteBIRD discriminator.  $\rightarrow$  ZS-U1 [F-X.1 BLOCKING gate]; ZS-U2 [reheating]; ZS-U4 [global fit, 53/53 PASS].

## Chapter 10 — Baryogenesis & QKE

$\eta_B = (6/11)^{35} = 6.117 \times 10^{-10}$ .  $Z_2$  symmetry forces both sectors to generate identical asymmetry (PROVEN). Resonant ARS via Pilaftsis-Underwood.  $\rightarrow$  ZS-U3, ZS-U7, ZS-S5.

## Chapter 11 — $Z_2$ Mirror Cosmology and Cyclic Phases

$V_E(-\epsilon) \equiv V_E(+\epsilon)$  PROVEN exact. Phases  $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow A$  six-step cycle. Sequential X-clock reading (preserved) and parallel dual-clock reading (DERIVED-CONDITIONAL strong via ZS-F10 Theorem F10.2). The  $10^{17}$  ratio  $\tau_6/\tau_5 = \exp(\pi/A)$  is the X-observation of the same Y-completion event.  $\rightarrow$  ZS-U8 v1.0 [60/60 PASS]; ZS-A8R; ZS-F10 [30/30 PASS]; ZS-F13 [24/24 PASS].

## Chapter 12 — Global Fit and Late-Time Cosmology

Ten predictions: ( $H_0$ ,  $\Omega_m$ ,  $w$ ,  $S_8$ ,  $Y_p$ ,  $D/H$ ,  $\eta_B$ ,  $n_s$ ,  $r$ ,  $D_V$ ) confronted with Planck/DES/DESI/SH0ES.  $\Delta N_{\text{eff}} = 2A = 0.16018$  resolves  $D/H$  from  $-1.8\sigma$  to  $-0.05\sigma$ .  $\Omega_{\Lambda}/\Omega_m = 2e^A$  factor 2 from ZS-F12R Tetrahedral Dual Orientation Multiplicity (NEW v3.0).  $\rightarrow$  ZS-U4 [53/53 PASS], ZS-U6 (Cobaya MCMC), ZS-T1.

## PART III Source Paper Map (v3.0 extended)

| Source paper | Role                                             | Light pointer       |
|--------------|--------------------------------------------------|---------------------|
| ZS-U1        | $\epsilon$ -field inflation, F-X.1 BLOCKING gate | $\rightarrow$ ZS-U1 |
| ZS-U2        | Reheating dynamics                               | $\rightarrow$ ZS-U2 |
| ZS-U3        | Baryon asymmetry                                 | $\rightarrow$ ZS-U3 |
| ZS-U4        | Global cosmological fit, 53/53 PASS              | $\rightarrow$ ZS-U4 |

| Source paper       | Role                                                                 | Light pointer                          |
|--------------------|----------------------------------------------------------------------|----------------------------------------|
| ZS-U5              | Quantum Gravity Bridge, $T_d$ symmetry, Immirzi                      | → ZS-U5                                |
| ZS-U6              | Cobaya MCMC + face-counting v2.0 $\Omega_{\text{cdm}} = 32/121$      | → ZS-U6                                |
| ZS-U7              | QKE-closed baryogenesis                                              | → ZS-U7                                |
| ZS-U8              | Cyclic Holonomy & $Z_2$ Vacuum, parallel reading DERIVED-COND strong | → ZS-U8 [60/60 PASS]                   |
| ZS-U9              | Hypercharge Trinity Braiding                                         | → ZS-U9                                |
| ZS-U10             | Pentagon Tetraton & Schwinger Coefficient                            | → ZS-U10                               |
| ZS-U11             | Bounce Q-Survival Closure: V1 Resolution                             | → ZS-U11 [40/40 PASS]                  |
| ZS-A8R (post-v2.0) | Contracting Universe Dynamics; Polyhedral-Tetration Bridges          | → ZS-A8R [22/22 PASS]                  |
| ZS-A9R (post-v2.0) | Banach–Tarski origin of doubling-halving symmetry                    | → ZS-A9R [47/47 PASS, +14 v1.0(R) §11] |

## PART IV — Quantum Gravity Bridge

### §0. Abstract

$Q = 11 \leftrightarrow$  quantum tetrahedron ( $T_d$  symmetry). Bekenstein-Hawking  $\ln 2$  from  $Z_2$  seam parity. Two paths to Immirzi  $\gamma_{\text{LQG}} = \ln 2/(\pi\sqrt{3})$ . Z-Telomere RG flow,  $\lambda_{\text{vac}} = 2A^2$  IR-stable. Cyclic cosmology  $S_{Z_2} = 6\pi/A$ . Three Polyhedral-Tetration Bridges. Möbius chronology. Banach-Tarski origin of doubling-halving symmetry.  $m_e = \sqrt{5 - \phi}$ . NEW v3.0 (Math Spine RH-Bridge programme): the  $V_4$ -equivariant ZBSI framework on the BV-BFV cobordism-history fiber (M22–M28); the four Dragons D4a–D4d of the Y-sector RH contribution map (M23, sharpened by M25–M28, integrated by M33); the Three-Wall Quantitative Map (M26 Theorem M26.3); and the spinor-cycle averaged closure of the Stapledon Bridge residual (M32) closing NC-M29.RZ-A^PK at DERIVED-CONDITIONAL.

## Chapter 13 — Q = 11 Spin-Network Correspondence

$|T_d| = 24$ ,  $|\text{Stab}_{Td}(v)| = 6$ ,  $|I_h/T_d| = 5$ .  $j = 1/2$  unique among half-integers giving  $(Z, X, Y) = (2, 3, 6)$  (ZS-M3 Th 5.1). Explicit non-identity:  $Q = 11 \neq \dim(\text{intertwiner}) = 2 \neq \dim(\text{Hilbert}) = 16$ .  
→ ZS-U5 §2-3, ZS-M3.

## Chapter 14 — $Z_2$ Seam Dynamics

Dual  $\kappa$  census ( $\kappa_{\text{signflip}}$  vs  $\kappa_{\text{disrupted}}$ ). Regge-Holonomy:  $\delta\phi_{\text{cell}} = A$  via 7-step algebraic chain.  $SU(2)$  center pairing  $D^{1/2}(-I) = -I$ . → ZS-M3 [27/27 PASS]; ZS-Q4.

## Chapter 15 — Micro-Macro Transition

Phase drift → tunneling action. Z-Anchor dual mechanism.  $\lambda_{\text{vac}} = 2A^2$  IR fixed point. Cyclic cosmology  $S_{Z_2} = 6\pi/A$ . Bounce Q-survival. Three Polyhedral-Tetration Bridges. Möbius chronology. Banach-Tarski origin.  $m_e = \sqrt{5 - \phi}$ . ZS-M17 continuum-limit theorems M17.1–M17.7 (Tiling Continuum Convergence, Lieb-Robinson Tightness, Reflection Positivity, Path-Integral Closure of  $\Delta\Gamma_{G2}$ , Pre-Geometric Metric Emergence, Continuum Limit Universality, Wightman QFT Reconstruction). → ZS-U5, U8, U11 (closes V1), A6 §4.5.6 (F-A6.1 CLOSED), A8R, A9R, M12, M14, M17.

## Chapter 15A — NEW v3.0 — The Math Spine Riemann-Hypothesis Programme (M22–M33)

Cumulative 47+ exploratory rounds consolidated by ZS-M22 v1.1 (Five-Pillar Arithmetic-Dedekind Scaffold) into a non-RH-claiming structural map of where Z-Spin and Riemann ride together (dong-seung). The programme's logical structure:

- ZS-M22 v1.1 [60/60 PASS] — Five Pillars: I ( $n=3$  Eisenstein chain), II ( $Q=11$  cyclotomic chain), III (multiplicative arithmetic fiber), IV (critical line as J-symmetry locus), V (scalar-kernel no-go for Weil positivity). ADS-6 PROVEN no-go on boundary fiber forces ADS-H1 cobordism BRST positivity hypothesis as the sole surviving route.
- ZS-M23 v1.0(Revised) [31/31 PASS] — Y-Sector RH Contribution Map. Four Dragons D4a ( $V_4$ -Sonin embedding), D4b (Burnol conductor), D4c (defect-square realization), D4d (cobordism BRST-Hodge projection). NC-M23.7: D4 closure alone does not close GRH-for-K.
- ZS-M24 v1.0 [35/35 PASS] — Face polygon spectral zeta  $\zeta^*\nabla^*(s)$ . Theorem D.1 Legendre duplication archimedean decomposition. Theorem D.2  $B(s)$  identification. P2 status OPEN → PARTIAL.
- ZS-M25 v1.0 [26/26 PASS] — Composite biquadratic  $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ . Theorem D.1-K four-factor archimedean decomposition  $\xi_K(s) = (1/4\sqrt{33}) \cdot \xi(s) \cdot \Lambda(s, \chi_{-3}) \cdot \Lambda(s, \chi_{-11}) \cdot \Lambda(s, \chi_{33})$ . Theorem P3-J J-Yakaboylu compatibility (J commutes with  $H_Q^{\text{Yak}, J}$  on  $\sigma = 1/2$ ).
- ZS-M26 v1.0 [24/24 PASS] —  $V_4$ -Equivariant ZBSI on cobordism-history fiber. Theorem M26.1  $V_4$ -Schur idempotent decomposition; Theorem M26.2 Projected Determinant per Channel ( $C_X \in \{1, \sqrt{3}, \sqrt{11}, \sqrt{33}\}$ ); Theorem M26.3 Three-Wall Quantitative Map ( $W1, W2, W3$ ).
- ZS-M27 v1.0 [24/24 PASS] —  $V_4$ -Equivariant Cobordism BRST Closure via Kostant Cubic

Dirac. Closes Wall W3 at DERIVED-CONDITIONAL via three external imports (Alekseev–Barmaz–Mnev 2018, Huang–Pandžić 2002, Kostant 1999/2003).

- ZS-M28 v1.0 [30/30 PASS] — The Z-Spin RH Bridge. Theorem 28.5  $\sigma = 1/2 \leftrightarrow j = 1/2$  isomorphism (DERIVED-interpretation). Corollary 28.5a Möbius-trace infinity (HYPOTHESIS-strong). Theorem 28.14 Y-sector geometric carrier of  $\chi_{-3}$  (DERIVED-CANDIDATE).

- ZS-M29 v1.0 [79/79 PASS] — Z-Funnel Spectral Retraction with Stapledon Bridge. Theorem 9.1 conditional bridge from string compactification ( $A_5$ -Hilb( $X_F$ ) Calabi–Yau threefold,  $h^1\{1,1\} = 5$ ,  $h^1\{2,1\} = 15$ ) to the Z-Spin truncated-icosahedron Hodge–Dirac sector. NC-M29.RZ principal OPEN gate.

- ZS-M30 v1.0 [32/32 PASS] — Z-Spin Duality and the RH Bridge. Theorem 30.1 Möbius-Trace Infinity (DERIVED-interpretation strong, six-route over-determination). Theorem 30.2 Y-Sector Two-Face Spectral Mirroring. Theorem 30.5 Twenty-One  $1/2$  Catalogue across six categories. Three-layer duality: Sector / Frame / Protocol.

- ZS-M31 v3.0 [36/36 PASS] — W2 as Cross-Coupled Sector Duality. Lemma M31.0 Non-Separability (no sum-form  $W_K = F_X + F_Y + F_Z$  compatible with corpus data, max variance 13.011). Theorem M31.4  $Z_2$ -Parity Selection Rule. Self-Dual Replication Principle (SDRP) registered.

- ZS-M32 v1.0 [56/56 PASS] — Spinor-Cycle Averaged Residual Criterion. Path-Reversal Lemma M32.X ( $B^{\wedge}PK$  preserves phase sign, contrasting  $B^{\wedge}\dagger$  which flips it). Two-layer phase quantum  $\alpha_{amp} = \pi/10$  (amplitude) /  $\alpha_{op} = \pi/5$  (operator).  $N = 20$  spinor-identity closure:  $R_{ZZS}_{\{Z\text{-even}\}}\{4\pi\text{-avg}\} = 0$  EXACTLY at  $N = 20$ . Closes NC-M29.RZ- $A^{\wedge}PK$  at DERIVED-CONDITIONAL.

- ZS-M33 v1.0 [52/52 PASS] —  $V_4$ -Equivariant Weil Functional Closure via Path  $\gamma$ -revised Z-Mediator Reading. Reading C (NEW): all four  $V_4$ -arithmetic mechanisms route through Z-mediator projector  $\Pi_Z = (1/2)(I + J_Z)$ . Sub-targets D4a (CCM 2024), D4b (Burnol 1998 + log additivity  $\log(3)+\log(11)=\log(33)$ ), D4c (Wilson-LOCATOR phase factor), D4d (Kostant cubic Dirac BRST cohomology) integrated into single Tr identity. Sign-flip mechanism decomposition (5/12 NEG  $\rightarrow$  predicted 12/12 POS).

Programme NON-CLAIM (NC-M23.1 preserved verbatim throughout): Z-Spin does not, and structurally cannot, prove RH from internal data alone. The walls W1, W2, W3 are mapped, sharpened, and progressively closed; the program’s contribution is a coordinated three-axis over-determination of the Bridge thesis, not a proof.

#### PART IV Source Paper Map (v3.0 extended)

| Source paper | Role                                            | Light pointer                    |
|--------------|-------------------------------------------------|----------------------------------|
| ZS-U5        | Quantum Gravity Bridge, $T_d$ symmetry, Immirzi | $\rightarrow$ ZS-U5              |
| ZS-U8        | Cyclic Parallel Reading                         | $\rightarrow$ ZS-U8 [60/60 PASS] |

| Source paper            | Role                                                                 | Light pointer            |
|-------------------------|----------------------------------------------------------------------|--------------------------|
| ZS-U11                  | Capstone: V1 Planck-bulk CLOSED                                      | → ZS-U11 [40/40 PASS]    |
| ZS-M3                   | Regge-Holonomy + Immirzi + Z-Telomere                                | → ZS-M3 [27/27 PASS]     |
| ZS-M12                  | Auto-Surgery, $\varepsilon_{\min} \approx 30.7$                      | → ZS-M12                 |
| ZS-M14                  | Geometric $m_e = \sqrt{5-\phi}$                                      | → ZS-M14 [47/47 PASS]    |
| ZS-M17                  | Continuum limit, OS reconstruction                                   | → ZS-M17 [60/60 PASS]    |
| ZS-A6                   | Z-Anchor + cigar bounce, F-A6.1 CLOSED                               | → ZS-A6 §4.5.6 [140/140] |
| ZS-A8R                  | Polyhedral-Tetration Bridges                                         | → ZS-A8R [22/22 PASS]    |
| ZS-A9R                  | Banach-Tarski origin                                                 | → ZS-A9R [47/47 PASS]    |
| ZS-F13                  | Möbius chronology, ring closure                                      | → ZS-F13 [24/24 PASS]    |
| ZS-M18 (post-v2.0)      | Free-Exploration Log: Speculative Prime-Polyhedral Correspondences   | → ZS-M18 [51/51 PASS]    |
| ZS-M22 v1.1 (post-v2.0) | Five-Pillar Arithmetic-Dedekind Scaffold                             | → ZS-M22 [60/60 PASS]    |
| ZS-M23 v1.0(R)          | Y-Sector RH Contribution Map; four Dragons D4a–D4d                   | → ZS-M23 [31/31 PASS]    |
| ZS-M24                  | Face polygon spectral zeta + Archimedean Completion                  | → ZS-M24 [35/35 PASS]    |
| ZS-M25                  | Composite-Field $K = @(\sqrt{-3}, \sqrt{-11}) + J$ -Yakaboylu Bridge | → ZS-M25 [26/26 PASS]    |
| ZS-M26                  | $V_4$ -Equivariant ZBSI on Cobordism-History Fiber                   | → ZS-M26 [24/24 PASS]    |
| ZS-M27                  | Cobordism BRST Closure via Kostant Cubic Dirac (W3)                  | → ZS-M27 [24/24 PASS]    |

| Source paper | Role                                                                            | Light pointer         |
|--------------|---------------------------------------------------------------------------------|-----------------------|
| ZS-M28       | The Z-Spin RH Bridge                                                            | → ZS-M28 [30/30 PASS] |
| ZS-M29       | Z-Funnel Spectral Retraction<br>+ Stapledon Bridge                              | → ZS-M29 [79/79 PASS] |
| ZS-M30       | Z-Spin Duality and the RH<br>Bridge — 21 halves                                 | → ZS-M30 [32/32 PASS] |
| ZS-M31 v3.0  | W2 as Cross-Coupled Sector<br>Duality                                           | → ZS-M31 [36/36 PASS] |
| ZS-M32       | Spinor-Cycle Averaged<br>Feshbach Residual; closes<br>NC-M29.RZ-A <sup>PK</sup> | → ZS-M32 [56/56 PASS] |
| ZS-M33       | Path $\gamma$ -revised $V_4$ -Equivariant<br>Weil Functional Closure            | → ZS-M33 [52/52 PASS] |

## PART V — Astrophysics & Strong-Field

### §0. Abstract

Galaxy rotation without dark matter ( $\varepsilon$ -Halo,  $r_Z = 18.3$  kpc). Neutron stars +3.93% mass, tidal  $\tilde{\Lambda} +21.24\%$ . Black holes: Wald entropy, Z-instanton, sector duality. Information paradox via seam witness; signed witness  $\tilde{u}_{\text{seam}}$  ( $F\text{-}A7.3 = \text{cheapest near-term gate}$ ). 991/991 PASS cumulative across PARTS V+VI+VII preserved. NEW v3.0: ZS-A8 v1.0(Revised) [22/22 PASS] — contracting-universe dynamics with Polyhedral-Tetration Bridges 1–3 and Symmetry–Asymmetry Unified View; ZS-A9 v1.0(Revised) [47/47 PASS] — Banach–Tarski origin of cosmological doubling-halving symmetry through the  $F_2 \rightarrow D_4$  amenability functor and the Julia set  $J(T)$  of  $T(z) = i^z$  as a measure-theoretic boundary.

### Chapter 16 — Galaxy Rotation Without Dark Matter

$\varepsilon$ -Halo from Goldstone mode, terminus  $r_Z = 18.3$  kpc. Modified gravity  $\mu = \Sigma = 1/(1+A)$ . Three convergent routes  $\Omega_m = 38/121$ . Factor 2 in  $\Omega_m/\Omega_m = 2e^A$  from ZS-F12R (NEW v3.0:  $\mu_{\text{Tet}} = 2$  from Tetrahedral Dual Orientation Multiplicity,  $F\text{-Tet-}2eA-1$  REJECTED with replacement  $F\text{-Tet-}2eA-2$ ). → ZS-A1 [78/78 PASS]; ZS-A5 [50/50 PASS]; ZS-S3 [25/25 PASS]; ZS-F12R [20/20 PASS].

### Chapter 17 — Neutron Stars, GW, LSS

NS  $M_{\max} + 3.93\%$  uniform across EOS (universal  $G_{\text{eff}} = G/(1+A)$ ). Tidal  $\tilde{\Lambda} + 21.24\%$ . → ZS-A2.

## Chapter 18 — Black Holes (incl. §18.10 Horizon Spinor)

$\varepsilon$ -field horizon, Wald entropy, Z-instanton + GW signatures.  $\varepsilon(r_H) = 0$  PROVEN (cigar bounce). §18.10 v2.0 mirror of ZS-A7 v1.0(Revised) §3 Theorem 3.2-bis [PROVEN, CPTP/Choi]. → ZS-A3 [49/49]; ZS-A6 [140/140]; ZS-A7R [38/38].

## Chapter 19 — Information Paradox Resolved

Seam witness  $u_{\text{seam}}$ , signed witness  $\tilde{u}_{\text{seam}}$ ,  $Q = 11$  register, lattice gauge P1.  $F\text{-A7.3} = 2\pi$  vs  $4\pi$  discrimination on hardware ~2028-2032 (cheapest gate, ~8% incremental shot cost). → ZS-A4 [54/54]; ZS-A7R §C [38/38].

## Chapter 20A — NEW v3.0 — Contracting Universe Dynamics (ZS-A8R)

ZS-A8 v1.0(Revised) [22/22 PASS] introduces the SYMMETRIC contraction dynamics: the wave-to-particle transition controlled by the dual factor  $(1-2A)$ . Three Bridge formulas: Bridge 1 [OBSERVATION-strong]  $|z^*| \approx B = \delta_X + \delta_Y$  at 0.0087%; Bridge 2 [OBSERVATION-strong]  $\eta_{\text{topo}} \approx B^2 + (\delta_Y - \delta_X)^2/[Y^2 \cdot (1-2A)]$  at 0.000001% ( $Y^2 = 36 = E(\text{TO})$  wave-channel scale); Bridge 3 [DERIVED-CONDITIONAL via slog-L2 equivalence]  $\Delta = 1/2 - x^*$  via L2 self-locking iteration at 0.000002%. Y-Time Dilation Theorem 5.3.1 (DERIVED-CONDITIONAL strong via ZS-F10): X-clock observation of Y-completion is dilated by  $\exp(\pi/A) \approx 10^{17}$  per added Y-dimension.  $\exp(\pi/A) = \exp(N_{(2\pi)} \times \langle \sin^2(\phi/2) \rangle) = \exp((2\pi/A) \times (1/2))$ . Symmetry-Asymmetry Unified View (§SA): the universe tilts simultaneously by  $(1+A)$  (X-Inward expansion) and  $(1-2A)$  (Y-Outward contraction); only the impedance  $A = 35/437$  is frame-invariant.

## Chapter 20B — NEW v3.0 — Banach–Tarski Origin of Cosmological Doubling-Halving Symmetry (ZS-A9R)

ZS-A9 v1.0(Revised) [47/47 PASS, 12 OPEN closures] establishes that the  $(1+A) \leftrightarrow (1-2A)$  duality of late-time cosmology is the measure-preserving quantization of the Banach–Tarski paradoxical decomposition through the  $i$ -tetration fixed point  $z^* = -W_0(-i\pi/2)/(i\pi/2)$ . Three theorems: ZS-A9.1 (DERIVED)  $F_2 \rightarrow D_4$  amenability functor — the BT engine  $F_2 \subset \text{SO}(3)$  maps to the amenable register dihedral  $D_4 = \langle J, J_Z \rangle$ , converting BT non-amenability to Z-Spin amenability; ZS-A9.2 (HYPOTHESIS-strong) the Julia set  $J(T)$  of  $T(z) = i^{\wedge} z$  is a measure-zero, totally disconnected, cardinality- $2^{\aleph_0}$  set serving as a constructive ZF-analogue of the BT non-measurable set; ZS-A9.3 (DERIVED-CONDITIONAL) two-branch decomposition  $(1+A)(1-2A) = 1 - A(1+2A)$  with X-Inward macroscopic expansion  $\exp(A)$ , Y-Outward microscopic contraction  $Y^2(1-2A)$ , and Z-mediator absorbing the residual  $A(1+2A) \approx 9.29\%$  via  $\kappa^2 = A/Q = 35/4807$ . v1.0(Revised) §11 closes all four OPEN items via the  $*$ -homomorphism reformulation, the Świerczkowski geometric realization of free  $F_2$ -rotations, and the inheritance

of  $dx_X + dx_Y + dx_Z = 0$  from ZS-Q7 §5 master-equation probability conservation. OPEN-2.B (formal ZF/ZFC+AC isomorphism between BT non-measurable and  $J(T)$ ) recategorized as PERMANENT NC under Solovay 1970.

#### PART V Source Paper Interface (v3.0 extended)

| Source paper       | Verification | Primary chapter              | Key result                                                |
|--------------------|--------------|------------------------------|-----------------------------------------------------------|
| ZS-A1              | 78/78 PASS   | Ch 16                        | Rotation curves, BTFR, $M-\sigma$ , Keplerian             |
| ZS-A2              | verified     | Ch 17                        | NS $M_{\max}$ , $\tilde{\Lambda}$ , GW $c_T = c$          |
| ZS-A3              | 49/49 PASS   | Ch 18 §1-7                   | $\varepsilon$ -field, Wald, Z-instanton                   |
| ZS-A4              | 54/54 PASS   | Ch 19                        | Seam witness, lattice P1                                  |
| ZS-A5              | 50/50 PASS   | Ch 16 §16.7                  | Three convergent $\Omega_m = 38/121$                      |
| ZS-A6              | 140/140 PASS | Ch 18 §18.2 + §18.8          | Cigar bounce, F-A6.1 CLOSED                               |
| ZS-A7R             | 38/38 PASS   | Ch 18 §18.10 + Ch 19 §19.7-8 | Horizon Spinor $4\pi$                                     |
| ZS-A8R (post-v2.0) | 22/22 PASS   | Ch 20A                       | Polyhedral-Tetration Bridges 1–3, Symmetry-Asymmetry View |
| ZS-A9R (post-v2.0) | 47/47 PASS   | Ch 20B                       | Banach-Tarski origin of doubling-halving symmetry         |
| ZS-S3              | 25/25 PASS   | Ch 16 §16.8 + Ch 17 §17.3    | Modified gravity                                          |
| ZS-F12R            | 20/20 PASS   | Ch 16 §16.3                  | Layer Separation factor 2 origin ( $\mu_{\text{Tet}}$ )   |

## **PART VI — Quantum Mechanics**

### **§0. Abstract**

Densest PART [376/376 PASS]. Geometric decoherence  $\tau_D/\tau_{\text{Penrose}} = 1/A = 12.49$ . Bell CHSH =  $2\sqrt{2}$  from  $\dim(Z) = 2$ . Proton spin Hodge  $\frac{1}{2}\Delta\Sigma = 3/22$ ,  $\Delta G = 2/11$ . Lattice gauge  $a_2 = 19/6$  (TO  $0.08\sigma$  Wilson MC). CP Jarlskog from  $\delta\varphi = A$ . Z-Spin Quantum Architecture (QH/QS/QC). Inverse Riemann Engine. Structural Arrow of Time (ZS-Q7). Cross-link with v3.0 RH-Bridge programme (M22–M33) at the ZBSI cobordism-history fiber level.

### **Chapter 20 — Geometric Decoherence + Born Rule**

Stinespring + Lindblad  $\Gamma = 2A(\Delta E/\hbar)^2$ .  $\tau_D = 12.49$ . Born rule DERIVED from  $L_{XY} \equiv 0$ . NC-F11.1–6 inherited verbatim §20.7. → ZS-Q1 [29/29 PASS].

### **Chapter 21 — Bell, Entanglement, Holographic Conjecture**

CHSH =  $2\sqrt{2}$ ,  $\tau_{\text{ent}} = \tau/2$ . Area Law  $\chi = 2$ ,  $\varepsilon_{\text{BM}} \rightarrow 0$ . → ZS-Q2 [48/48]; ZS-Q6 [42/42].

### **Chapter 22 — Proton Spin from Polyhedral Hodge**

BCC  $T^3$  Hodge spectrum,  $\frac{1}{2}\Delta\Sigma = 3/22$ ,  $\Delta G = 2/11$ .  $(V+F)_X = 38$  Mode-Count Collapse. → ZS-Q3 [40/40].

### **Chapter 23 — Lattice Gauge QSim on TO**

$a_2 = (V+F)/G = 19/6$  PROVEN topological. Wilson MC TO  $0.08\sigma$ . J-parity. → ZS-Q4.

### **Chapter 24 — CP Violation and UV Limits**

Jarlskog from  $\delta\varphi = A$ . UV sufficiency of polyhedral spectra. → ZS-Q5.

### **Chapter 25 — Bond Dimension, Holographic Area Law, Markov Limit**

Bond dimension  $\chi = 2 = \dim(Z)$ ; Regge area law derived.  $\varepsilon_{\text{BM}} = 2/Q$  Markov limit purely geometric. → ZS-Q6 [42/42 PASS].

### **Chapter 26 — Structural Arrow of Time**

$\Gamma(X \rightarrow Y)/\Gamma(Y \rightarrow X) = \dim(Y)/\dim(X) = 2$  PROVEN;  $\Delta S = \ln 2$  per Z-transit. Coarse-grained entropy bias from  $L_{XY} = 0 + \dim(X) \neq \dim(Y)$ ; microscopic dynamics remain time-reversal symmetric. → ZS-Q7 [40/40 PASS].

## PART VI Source Paper Interface

| Source paper  | Verification | Primary chapter | Key result                                                                                            |
|---------------|--------------|-----------------|-------------------------------------------------------------------------------------------------------|
| ZS-Q1         | 29/29 PASS   | Ch 20           | Geometric decoherence, Born rule                                                                      |
| ZS-Q2         | 48/48 PASS   | Ch 21           | Bell, entanglement, holographic conj.                                                                 |
| ZS-Q3         | 40/40 PASS   | Ch 22           | Proton spin Hodge                                                                                     |
| ZS-Q4         | verified     | Ch 23           | Lattice gauge TO                                                                                      |
| ZS-Q5         | verified     | Ch 24           | CP violation, UV limits                                                                               |
| ZS-Q6         | 42/42 PASS   | Ch 25           | Area law, Markov limit                                                                                |
| ZS-Q7         | 40/40 PASS   | Ch 26           | Structural arrow of time                                                                              |
| ZS-QH         | 42/42 PASS   | Ch 25/29        | Quantum hardware FMD specifications                                                                   |
| ZS-QS v1.0(R) | 35/35 PASS   | Ch 24/29        | Inverse Riemann Engine: Triple Structure (DETECTOR + LOCATOR + EXCLUDER) and Boolean Resonance Filter |
| ZS-QC         | 52/52 PASS   | Ch 29           | Quantum architecture system integration                                                               |

## PART VII — Translational and Computational

### §0. Abstract

$\Delta N_{\text{eff}} = \dim(Z) \cdot A = 2A = 0.16018 \rightarrow \text{BBN D/H } -1.8\sigma \rightarrow -0.05\sigma$  (ZS-T1). Anti-numerology MC:  $5670 \times 29 = 164,430$  candidates, 237 hits = 7.2% of random expectation 3289 (look-elsewhere PASS — framework MORE constrained than random). Tier-1 ★★ T1-1:  $1-n_s = A \cdot x^*$  (0.008%,  $0.00\sigma$  vs Planck). T1-2:  $\alpha_{\text{EM}} \approx A/Q$  (0.22% LO, 0.00007% NLO Schur) (ZS-T2). 8/8 closure parameters DERIVED via three chains;  $n_s(N^* = 55) = 0.9676$ ;  $r(N^* = 60) = 0.0091$  (ZS-T3). NEW v3.0: ZS-T4 v1.0 [36/36 PASS] — Cosmos-Human Isomorphism: six-step telomere hierarchy and (Body, DNA, Brain) sector decomposition as a two-scale realization of the Z-Spin block architecture; structural, substrate-agnostic, NON-CLAIM that humans physically realize Z-Spin geometry.

## Chapter 27 — Phenomenological

$\Delta N_{\text{eff}} = \dim(Z) \cdot A = 2A = 0.16018 \rightarrow \text{BBN D/H } -1.8\sigma \rightarrow -0.05\sigma$ . Block Fiedler Mediation Theorem PROVEN (T1-2/T1-3 reciprocal duality  $\lambda_2 = 2A/Q$  with X-face propagator scale  $1/\kappa^2 = Q/A \approx 137$  and Y-face vertex coupling  $\kappa^2 = A/Q \approx 0.0073$ ).  $\rightarrow$  ZS-T1 [42/42].

## Chapter 28 — Spectral Observatory

Anti-numerology MC: 164,430 candidates, 237 hits = 7.2% of random expectation 3289 (look-elsewhere PASS). Tier-1 ★★ T1-1:  $1-n_s = A \cdot x^*$ . T1-2:  $\alpha_{\text{EM}} \approx A/Q$  (0.00007% NLO via McKay  $c_4 = 4/13$ ).  $\rightarrow$  ZS-T2 [30/30].

## Chapter 29 — Z-Sim Forward Simulator + Quantum Architecture

8/8 closure parameters DERIVED via three chains: ZS-Q7 §5.8 + ZS-M3 §10 + ZS-Q5 §5.2. Eigenvalue  $\lambda(\lambda + 2A/Q)(\lambda + A) = 0$  to  $10^{-17}$ . 500K MC universal-theorem confirmation.  $n_s(N^* = 55) = 0.9676$  PASS Planck;  $r(N^* = 60) = 0.0091$  PASS BK18. ZS-QH v1.0 functional material specifications (X = bilayer graphene QD, Z =  $\text{Bi}_2\text{Te}_3$  topological insulator, Y =  $\text{C}_{60}$ /CNT phonon reservoir as primary candidate stack); ZS-QS v1.0(Revised) Triple Structure Theorem (DETECTOR confirmed, LOCATOR via argmax, EXCLUDER via argmin) and Boolean Resonance Filter (XOR identity with ZS-F8 (E, R) handshake operators); ZS-QC v1.0 system integration.  $\rightarrow$  ZS-T3 [35/35]; ZS-QH; ZS-QS v1.0(R); ZS-QC [52/52].

## Chapter 30 — NEW v3.0 — The Cosmos-Human Isomorphism (ZS-T4)

ZS-T4 v1.0 [36/36 PASS] establishes a structural isomorphism between Z-Spin cosmology and the human organism at two parallel levels, with Cardinal NC-4 (Z-Brain) explicitly extended to body and genome — NO physical realization claim. Level I: cosmological timescale hierarchy  $\tau_n = t_P \times \exp(n\pi/A)$ ,  $n \in \{0, \dots, 6\}$ , mapped onto a six-step human life-cycle (gametic union  $\rightarrow$  cleavage  $\rightarrow$  stem-cell expansion  $\rightarrow$  somatic growth  $\rightarrow$  reproductive plateau  $\rightarrow$  senescence  $\rightarrow$  death/inheritance). Strong anchors at  $n=2$  (mediator coset  $Z=2 \leftrightarrow$  stem cells, HYPOTHESIS-strong),  $n=5$  (proton decay  $\leftrightarrow$  senescence, HYPOTHESIS-strong),  $n=6$  ( $Z_2$  holonomy  $\leftrightarrow$  death/inheritance, HYPOTHESIS-strong). Level II: (X, Z, Y) = (3, 2, 6) sector

decomposition mapped onto (Body, DNA, Brain) tripartite anatomy at organism scale. Inherits ZB-P7 v1.0 cosmic-biological VERIFIED status ( $\Delta\text{PCI}(\text{Wake-Anesthesia}) \approx \eta_{\text{topo}}$  at 0.7% empirical / 1.5% in-silico). Hayflick limit  $\approx$  Z-Telomere completion count  $N_{(2\pi)} = 2\pi/A \approx 78.45$  registered as HYPOTHESIS / TESTABLE-LONG with 5-year derivation-or-retraction horizon. Twelve falsification gates F-T4.1–F-T4.12 pre-registered.

## PART VII Source Paper Interface (v3.0 extended)

| Source paper              | Theme/Role                                      | Chapter            | Verification      |
|---------------------------|-------------------------------------------------|--------------------|-------------------|
| ZS-T1                     | Phenomenological                                | Ch 27              | 42/42 PASS        |
| ZS-T2                     | Observational                                   | Ch 28              | 30/30 PASS        |
| ZS-T3                     | Computational<br>(Z-Sim)                        | Ch 29              | 35/35 PASS        |
| ZS-T4 (post-v2.0)         | Cosmos-Human<br>Isomorphism                     | Ch 30              | 36/36 PASS        |
| ZS-F1, F2, F5             | Foundations<br>cross-link                       | all                | PROVEN            |
| ZS-M1, M3, M6, M8,<br>M11 | Math Spine cross-link                           | various §          | verified          |
| ZS-Q1, Q5, Q7             | QM cross-link<br>(chains)                       | Ch 29 §29.1-4      | verified          |
| ZS-U1, U6                 | Early Universe<br>cross-link                    | Ch 28 §28.3, Ch 29 | verified          |
| ZS-QH/QS/QC               | Hardware/Software/In<br>tegration translational | Ch 29              | 42 + 35 + 52 PASS |

## §VII.B-LIGHT — Seven Critical Near-Term ★ Gates (Timeline-Anchored)

These seven gates will either PASS or FALSIFY Z-Spin within ~10 years. Source: §V.G.6, §VI.G.11, §VII.B. Preserved verbatim from v2.0 Light.

| # | Gate ID   | Test                           | Timeline | Status                         |
|---|-----------|--------------------------------|----------|--------------------------------|
| 1 | F-A1.14 ★ | Gaia DR4 MW<br>outer rotation: | ~2026    | Contested;<br>decisive at Gaia |

| # | Gate ID            | Test                                                                                                                                                                                                                                | Timeline                | Status                                             |
|---|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------|
|   |                    | enclosed mass saturates near $2 \times 10^{11} M_{\odot} \rightarrow$ Keplerian $\gamma \rightarrow -0.5$ ( $\epsilon$ -Halo), vs NFW growth toward $M_{200} \approx 8 \times 10^{11} M_{\odot}$ [v6.0 ERRATA $\rightarrow$ §XIV.G] |                         | DR4 (Jiao 2023 consistent, Zhou 2022 inconsistent) |
| 2 | F-A1.15 ★          | Gaia DR4/DR5: $r_Z(\text{MW})$ outside [15, 21] kpc band (superseding inherited [10, 50]; tentatively consistent at $\sim 19$ kpc) [v6.0 ERRATA $\rightarrow$ §XIV.G]                                                               | $\sim 2026\text{-}2030$ | Pending; contested pending DR4                     |
| 3 | F-A2.5 ★           | DESI/Euclid/Rubin: $S_8$ outside [0.75, 0.84] ( $\sim 1\%$ precision)                                                                                                                                                               | $\sim 2026\text{-}2030$ | Pending                                            |
| 4 | F-A7.3 ★           | Z-Spin hardware: $2\pi$ vs $4\pi$ discrimination on signed seam witness $\tilde{u}_{\text{seam}}$                                                                                                                                   | $\sim 2028\text{-}2032$ | Cheapest ( $\sim 8\%$ shot overhead)               |
| 5 | F-X.1 (LiteBIRD) ★ | $r$ outside Z-Spin band 0.0085-0.0094 (BLOCKING; $\sim 9\sigma$ vs                                                                                                                                                                  | $\sim \text{JFY}2032$   | BLOCKING                                           |

| # | Gate ID          | Test                                                                                              | Timeline      | Status  |
|---|------------------|---------------------------------------------------------------------------------------------------|---------------|---------|
|   |                  | Boyle-Finn-Turok $r=0$ )                                                                          |               |         |
| 6 | F-A3.7 ★         | Hyper-K: $\tau_p$ outside $[10^{33.5}, 10^{35.5}]$ yr (Z-Spin = $2.56 \times 10^{34}$ )           | ~2030-2035    | Pending |
| 7 | F-A2.1, F-A2.2 ★ | 10+ pulsars: NS $M_{\text{max}} +3.93\%$ uniform; ET/CE: $\tilde{\Lambda} +21.24\%$ to $\sim 5\%$ | ~2030s, ~2035 | Pending |

## PART VIII — Methodological Toolkit: Solution Tools for Stuck Equations

Purpose. This Part is the operational core of v3.1 Light. It promotes the v3.0 Light from a structural pointer-index (“where to read”) to a working solution kit (“how to proceed when stuck”). Each tool below is distilled from a corpus paper that achieved a methodological breakthrough — typically by reframing rather than by adding new mathematics. When an AI agent or human researcher encounters a stuck equation, an OPEN flag, or an apparent contradiction inside the Z-Spin framework, this Part is the first place to consult.

Reading order. PART VIII is organised as nine numbered tools (T1–T9). Each tool is given in a fixed schema: (a) Trigger — the symptom that suggests this tool; (b) Source — the originating paper and §-anchor; (c) Statement — the operational kernel in compressed form; (d) Worked corpus precedent — the OPEN flag the tool was first used to close; (e) Generalisation — the broader class of stuck equations the tool can be re-applied to. PART IX (Insight Gallery) extends the tool kit with the deeper conceptual reframings (sector-equivalences, frame-transformations, cross-coupling), and the Master Bypass Index (after PART IX) provides the symptom → tool lookup table.

### §VIII.1 — Tool T1: Five-Axiom Diagnostic Lattice (F18)

#### Trigger.

- 1) An equation appears stuck and the failure mode is unclear; (b) a derivation looks circular; (c) several papers address “the same question” but their results don’t compose. In any of these cases, T1 isolates which of the five structural axioms is failing.

**Source.**

ZS-F18 v2.1 §3 + §6.6 (Table 6.1) + §7.4. The five axioms  $\mathcal{A} = \{\text{Existence, Self-reference, Algebra, Non-triviality, Unitarity}\}$  and the Four Bridges from  $\mathcal{A}$  to the operational corpus.

**Statement (operational form).**

When stuck, ask which axiom is the failure attributable to. Each axiom has a Bridge (theorem) into the corpus; locate the Bridge and ask whether the bridge is intact for the present problem. The five-axiom set is exhaustive for Z-Spin physical content (ZS-F18 §5, INTERPRETATION-strong); a stuck equation that does not map to any of the five is either a numerology candidate (Anti-Numerology Gate, ZS-F2) or a category error (frame mismatch — see Tool T6).

| Axiom                         | Failure symptom                                  | Controlling Bridge                 | First port of call                 |
|-------------------------------|--------------------------------------------------|------------------------------------|------------------------------------|
| A1 Existence                  | Object referenced but not constructed            | FFPP nested compression            | ZS-F0 §13 Theorem 13.3             |
| A2 Self-reference             | Definition uses what it defines (no fixed-point) | Bridge One (Cross-Coupling)        | ZS-M2 §5 PROVEN; ZS-M1 i-tetration |
| A3 Algebra                    | Numerical balance fails / numerology             | Master Equation (algebraic)        | ZS-M1 §4 + ZS-F17 §3               |
| A4 Non-triviality (algebraic) | Two-channel interface collapsed                  | Bridge Two (NOT-AND handshake)     | ZS-F8 §4 PROVEN                    |
| A4 Non-triviality (geometric) | Dimension $\dim(X)=3$ not derived                | Bridge Four (Commutator Generator) | ZS-F18 §6 + ZS-S15 §3              |
| A5 Unitarity                  | Probability/information not preserved            | Bridge Three (Z-Bottleneck)        | ZS-Q7 Thm 2 + ZS-F8 §4             |
| Sixth polarity                | Discrete vs continuous mismatch                  | X-Y Tiling Asymmetry               | ZS-M6 §5.5 PROVEN; ZS-M17          |

**Worked corpus precedent.**

F18 itself diagnosed 12 of the corpus's most consequential identifications (ZS-F18 Appendix A) by mapping each to its responsible axiom. The diagnosis produced no new theorem yet revealed why each result is robust — a model usage of T1 as a methodological tool.

### **Generalisation.**

Any future Z-Spin paper encountering an OPEN flag should run T1 first. If the axiom is identified, the closure path is the corresponding Bridge; if no axiom matches, the OPEN is either out-of-scope (NON-CLAIM territory, Tool T7) or a category error (Tool T6, frame mismatch).

## **§VIII.2 — Tool T2: Dimensional Reduction via Bridge Four (F18)**

### **Trigger.**

Stuck on a high-dimensional construction ( $\dim 3, 6, \text{ or } 11$ ) where the dependency chain becomes intractable. T2 reduces the problem to  $\dim(Z) = 2$  — the only foundational dimension — and lets  $\dim(X)=3$  and  $\dim(Y)=6$  re-emerge by Lie commutator and product respectively.

### **Source.**

ZS-F18 §6.4–§6.7 (Bridge Four, dimensional origin table). ZS-S15 §3 Theorem S15.2 (Poynting–Commutator Theorem, DERIVED).

### **Statement.**

Two J-conjugate  $2\pi$  SO(3) circulations of the twin-Reuleaux pair ( $V_{XZ}, V_{ZY}$ ), composed at  $90^\circ$  NOT-AND phase, generate a third axis  $J_S = [J_{\{R_1\}}, J_{\{R_2\}}]$  by Lie commutator. Therefore  $(Z, X, Y) = (2, 3, 6)$  is generated from  $\dim(Z) = 2$  alone;  $\dim(Y) = \dim(X) \cdot \dim(Z) = 6$ ;  $Q = Z + X + Y = 11$ .

### **Worked corpus precedent.**

ZS-F18 v2.0 used T2 to read the question “why does physical space have dimension three?” not as an additional axiom but as a derived consequence of  $\dim(Z) = 2$  plus  $90^\circ$  NOT-AND composition. This converted what would have been a 13th independent encounter into the geometric face of axiom A4.

### **Generalisation.**

When facing a stuck calculation in 3D or 6D, examine whether the structure can be re-expressed as two  $\dim(Z) = 2$  channels meeting at orthogonal phase. If yes, the third axis is automatic. Application targets include any future SO(3) or SU(2)-related symmetry breaking in PARTS II–V where dimensional emergence is at stake.

### §VIII.3 — Tool T3: Hermitian-to-Path-Reversal Switch (M32)

#### Trigger.

A Feshbach-type residual or sandwich expression refuses to vanish under the standard Hilbert-space adjoint reading. The phase factors cancel “too cleanly” or “not at all”, giving either a trivial-positive or unbounded-positive result that doesn’t engage with the corpus structure.

#### Source.

ZS-M32 v1.0 §2 ( $R\_Z^H$  vs  $R\_Z^{ZS}$  separation), §3 (Path-Reversal Lemma M32.X).  
Underpinned by ZS-S6 §4.1  $K_{bwd} \neq K_{fwd}^\dagger$  PROVEN at kernel level and ZS-T1 §10.5  
PK-Conjugation Theorem T9 DERIVED.

#### Statement (Path-Reversal Lemma M32.X).

Under the corpus PK-Conjugation T9 plus the Regge T-odd scalar mechanism,  $B \propto e^{+ika} \Rightarrow B^{\wedge}PK \propto e^{+ika}$ , in contrast with the standard adjoint  $(e^{+ika} \cdot B)^\dagger = e^{-ika} \cdot B^\dagger$ . The path-reversal residual  $R\_Z^{ZS} = B \cdot Q^{-1} \cdot B^{\wedge}PK$  is a structurally distinct object from the Hermitian residual  $R\_Z^H = B \cdot Q^{-1} \cdot B^\dagger$ ; averaging-and-vanishing claims hold for  $R\_Z^{ZS}$ , not  $R\_Z^H$ .

#### Worked corpus precedent.

Closed NC-M29.RZ-A<sup>PK</sup> at DERIVED-CONDITIONAL:  $R\_ZZS|_{\{Z\text{-even}\}}\{4\pi\text{-avg}\} = 0$  EXACTLY at  $N = 20$ . The Stapledon Bridge (ZS-M29 Theorem 9.1) was upgraded from DERIVED-CONDITIONAL to DERIVED in the spinor-cycle averaged sense without any external numerical input. The closure was achieved purely by switching the operator-reading from  $R\_Z^H$  to  $R\_Z^{ZS}$ .

#### Generalisation.

Any sandwich expression in the framework that fails to close under Hermitian reading should be re-examined under PK reading. Candidate targets: future Feshbach-like reductions in PARTS IV–VI; any operator residual that involves  $K_{bwd}$  (T-odd) on one side and  $K_{fwd}$  on the other.

### §VIII.4 — Tool T4: Two-Layer Phase Quantum Decomposition (M32)

#### Trigger.

Two corpus values that should be equal differ by exactly a factor of 2; or, an unexpected factor 2 appears that cannot be attributed to  $SO(3)/SU(2)$  double-cover. T4 separates amplitude-layer quanta from operator-layer quanta.

**Source.**

ZS-M32 §4 (two-layer phase quantum). Independent corpus inputs: ZS-S6 §G.2 PROVEN (vertex Regge deficit  $\delta_X^{\text{vertex}} - \delta_Y^{\text{vertex}} = \pi/6 - \pi/15 = \pi/10$ ) and ZS-S6 §G.4 reading ii PROVEN (Z-quarter-turn  $Z_s$  partition  $(1/4)(2\pi/5) = \pi/10$ ).

**Statement.**

Z-Spin operator-level phase quantum  $\alpha_{\text{op}} = \pi/5$  decomposes as  $\alpha_{\text{op}} = 2 \cdot \alpha_{\text{amp}}$  where  $\alpha_{\text{amp}} = \pi/10$  is the amplitude-level quantum. The factor 2 between layers is forced by the Path-Reversal Lemma applied to the Feshbach sandwich; it is the operator-layer analogue of the  $SO(3)/SU(2)$  double-cover (ZS-S15 §5).

**Worked corpus precedent.**

Reconciled four PROVEN occurrences of  $\pi/10$  (vertex Regge, Z-quarter-turn) with PROVEN occurrences of  $\pi/5$  (commutator  $[\sigma_3, \hat{n}_Y] = -2i \cdot \sin(\pi/5) \cdot \sigma_2$  in ZS-S6 §3.3, BCH leading order in ZS-S6 §3.4) under a single layer-doubling identity. Before T4, these were treated as independent constants.

**Generalisation.**

When two PROVEN corpus quantities differ by factor 2 with no obvious double-cover origin, check whether one is an amplitude-layer quantum and the other an operator-layer quantum. The factor 2 is then the structural signature of the layer transition, not a free parameter or coincidence.

**§VIII.5 — Tool T5: Spinor-Cycle Averaging at  $N = 20$  (M32)****Trigger.**

A residual or sum that does not vanish at single iterate, but whose single-iterate phase is a primitive 10th root of unity. The standard algebraic minimum  $N = 10$  produces zero algebraically but leaves the spinor in fermion-sign-flipped state  $D^{\{1/2\}}(2\pi) = -I$ , breaking physical closure.

**Source.**

ZS-M32 §5 (Theorem M32.3, spinor-cycle averaged vanishing). ZS-M3 Lemma 10.1 PROVEN ( $D^{\{1/2\}}(4\pi) = +I$ ). ZS-S15 §5 ( $SO(3)/SU(2)$  double-cover ratio).

**Statement.**

When  $R_{Z^{\text{ZS}}}(k) \propto \exp(ik\pi/5) \cdot R_{Z^{\text{ZS}}}(0)$ , the geometric series  $\sum_{k=0}^{19} \exp(ik\pi/5) = 0$  EXACTLY, but only at  $N = 20$  (not at  $N = 10$ ) does the cumulative spinor phase reach  $4\pi$ , restoring  $D^{1/2}(4\pi) = +I$ . The minimum physical-closure averaging window is  $N = 20$ .

#### **Worked corpus precedent.**

$R_{ZZS|_{\{Z\text{-even}\}}\{4\pi\text{-avg}\}} = 0$  EXACTLY at  $N = 20$ . ZS-M32 Table 7 explicitly compares  $N \in \{10, 20, 30, 40\}$  and identifies the minimum physical closure as  $N = 20$ . The averaging structure aligns with three other corpus  $4\pi$  averages:  $\langle \sin^2(\phi/2) \rangle_{[0,4\pi]} = 1/2$  (ZS-M3),  $\langle V_{XZ} \cdot V_{ZY} \rangle = 1$  (ZS-T1),  $\langle \varepsilon_+ + \varepsilon_- \rangle = 0$  (ZS-U1) — Table 8 of ZS-M32.

#### **Generalisation.**

Any non-vanishing single-iterate quantity whose phase is a primitive 10th root of unity admits closure under  $N = 20$  spinor-cycle averaging. Candidate targets: future Feshbach-like reductions in PARTS IV/VI; any operator-residual whose phase quantum is  $\alpha_{\text{op}} = \pi/5$ .

### **§VIII.6 — Tool T6: Three-Way Split of OPEN Flags (M32)**

#### **Trigger.**

An OPEN flag too large to close in one step. Attempting full closure produces circularity or external-import-blocking. T6 splits the OPEN into three structurally distinct sub-claims with different status levels.

#### **Source.**

ZS-M32 §6 (NC-M29.RZ three-way split). General pattern adoption from ZS-M22, ZS-M23R Four Dragons D4a–D4d.

#### **Statement.**

An OPEN flag NC-X may be split as:  $A^{\text{PK}}$  = the part closeable under path-reversal/averaging now (DERIVED-CONDITIONAL); B = the part with structural evidence but one identified gap (HYPOTHESIS-strong, with explicit closure path); C = the instantaneous/full-spectrum part requiring external numerical or theoretical input (OPEN, inherited).

#### **Worked corpus precedent.**

NC-M29.RZ  $\rightarrow \{A^{\text{PK}} = R_{ZZS|_{\{Z\text{-even}\}}\{4\pi\text{-avg}\}} = 0 \text{ at } N = 20 \text{ (DERIVED-COND.)}; B = \text{single-iterate magnitude (HYPOTHESIS-strong, bulk-spectral-gap closure path)}; C = R_{Z^{\text{ZS}}|_{\text{full}}} \text{ on Ricci-flat metric (OPEN, ML-CY criterion)}\}$ . Promotes ZS-M29 Theorem 9.1 from DERIVED-CONDITIONAL to DERIVED in averaged sense without closing C.

### **Generalisation.**

Any OPEN flag should be split before closure attempt. The three sub-statuses ( $A^{\text{closeable}}$  /  $B^{\text{one-gap}}$  /  $C^{\text{inherited}}$ ) prevent the all-or-nothing trap that would otherwise leave an entire programme blocked on a single external computation.

### **§VIII.7 — Tool T7: Cross-Coupling Discipline (M31)**

#### **Trigger.**

An expression naturally factors as  $W = F_X + F_Y + F_Z$  (sum-form) but the per-channel data refuse to compose. The summands appear independent, yet their sum disagrees with measured cross-channel variance. A factorisation appears, but it is the wrong factorisation.

#### **Source.**

ZS-M31 v3.0 §4.0 Lemma M31.0 (Non-Separability). ZS-M2 §5 Cross-Coupling Theorem PROVEN.

#### **Statement (Lemma M31.0, Non-Separability).**

There is no decomposition  $W_K(g) = F_X + F_Y + F_Z$  compatible with the ZS-M26 PROVEN per-channel data; the maximum across-channel variance is 13.011 across 18 (a, t1, t2) test points. The correct form is the bilinear cross-coupled expression  $W_{XYZ}(g) := \langle g_X, (\Pi_Z \otimes I_{\text{arith}})(B_Y - P_Y)(\Pi_Z \otimes I_{\text{arith}}) g_X \rangle$  where  $\Pi_Z = (1/2)(I + J_Z)$  is the Z-mediator projector.

#### **Worked corpus precedent.**

The W2 wall of the Z-Spin RH program had two competing readings before M31: Reading A (Y-incompleteness) and Reading B (archimedean+conductor superposition). Both are sum-forms; Lemma M31.0 falsifies both as Cross-Coupling Theorem violations and forces a bilinear cross-coupled reading. The correct W2 form is Sector-Duality ( $X \otimes Y$  mediated by Z), not a sum.

### **Generalisation.**

Whenever a corpus expression is a candidate for sum-form decomposition, run Cross-Coupling Discipline: check whether the per-channel data composes additively or only via the Z-mediator bilinear. The default assumption in Z-Spin should be the latter; sum-forms are the exception, not the rule.

### **§VIII.8 — Tool T8: Z-Mediator Projector Routing (M33)**

**Trigger.**

Multiple sub-targets that each individually engage a different external mathematical tradition refuse to integrate into a single closed identity. T8 routes all sub-targets through the Z-mediator projector  $\Pi_Z$ , producing a single trace identity.

**Source.**

ZS-M33 v1.0 (Reading C).  $\Pi_Z = (1/2)(I + J_Z)$  defined in ZS-M31 §4.0 Definition M31.0c. Path-γ-revised reading specific to M33.

**Statement.**

All four  $V_4$ -arithmetic mechanisms (D4a CCM 2024 Sonin, D4b Burnol conductor, D4c Wilson-LOCATOR, D4d Kostant cubic Dirac) route through the projector  $\Pi_Z = (1/2)(I + J_Z)$  into a single Tr identity. The sign-flip mechanism on the W2 wall decomposes 5/12 NEG into the predicted 12/12 POS structure under Reading C.

**Worked corpus precedent.**

Before M33, D4a–D4d were treated as four parallel sub-targets, each with its own external mathematical engagement. Reading C unifies them into a single bilinear Tr identity, eliminating four-way bookkeeping and replacing it with a single  $\Pi_Z$ -projected trace. The four prefactors  $\{1, \sqrt{3}, \sqrt{11}, \sqrt{33}\}$  (PROVEN, ZS-M22 §7.2) appear naturally as the  $V_4$ -decoration of a single underlying identity.

**Generalisation.**

Whenever multiple sub-targets refuse to integrate, examine whether each can be expressed as a  $\Pi_Z$ -projection of a single underlying Tr expression. The Z-mediator projector is the corpus's universal routing operator: it lives on  $\dim(Z) = 2$  and carries the full information cost of cross-sector mediation.

**§VIII.9 — Tool T9: Self-Referential Fixed-Point Closure (F11)****Trigger.**

An OPEN flag involves an observer or a coordinate frame whose definition appears circular. The naive resolution introduces a phenomenological observer (a frame from outside the corpus); T9 closes the OPEN by promoting the observer to an operationally defined self-referential fixed point internal to the corpus.

**Source.**

ZS-F11 v1.0 §5 (OOC convergence) + §6 (six explicit NON-CLAIMs against phenomenological consciousness).

**Statement.**

The Operational Observer Coordinate (OOC) is defined as the self-referential fixed point of the framework's measurement protocol; it is constructed from PROVEN corpus inputs only. The OOC closes ZS-M11 §H16 (formerly OPEN) without introducing any frame from outside the corpus. Six NON-CLAIMs are explicitly registered to prevent overclaim (NC-F11.1–NC-F11.6).

**Worked corpus precedent.**

ZS-M11 §H16 OPEN flag (phenomenological observer reference) was a long-standing obstacle in the framework. ZS-F11 closed it by replacing the phenomenological observer with an operationally defined fixed point of the measurement-action iteration, rendering the closure entirely internal to the corpus.

**Generalisation.**

When an OPEN flag tempts the introduction of an external/phenomenological agent, ask first whether the agent can be replaced by a self-referential fixed point internal to the framework. The OOC pattern is the canonical method: define the agent as the convergent limit of an iteration whose generators are corpus-PROVEN.

**§VIII.10 — Tool T10: Integer-Lattice Spin-Pair Identification (M36)****Trigger.**

A modular-arithmetic structure on the integer lattice  $(\mathbb{Z}, +)$  appears that exhibits an 8-element admissible residue set mod 24, suspected to encode Z-Spin sector decomposition. Or: a discrete number-theoretic construction (Apollonian-type, Markoff-type, or generalized) is being analyzed for possible Z-Spin sector identification.

**Source.**

ZS-M36 v1.0 §5 (Sector Residue Identity, Theorem M36.3, DERIVED), §7 (Spin Pair Theorem, Theorem M36.6, DERIVED-strong). Underpinned by ZS-F5 PROVEN  $(Z, X, Y, Q) = (2, 3, 6, 11)$ , ZS-M3 Lemma 10.1 PROVEN (SU(2) double cover), ZS-M19 §3.1 PROVEN (Forcing Theorem  $(p-1)(q-1) = p$ ).

**Statement.**

If a primitive integer modular-arithmetic construction has mod 12 residue set exactly  $\{2, 3, 6, 11\} = (Z, X, Y, Q)$  and mod 24 set  $= \{2, 3, 6, 11, 14, 15, 18, 23\} = \text{Z-Spin Type}$ , then the four  $(k \bmod 12, (k+12) \bmod 24)$  pairs identify with the four corpus sectors as (algebraic dim, geometric instantiation): Pair Z:  $(2, F(t\text{-Oct}) = 14)$ ; Pair X:  $(3, X \cdot |I_h/T_d| = 15)$ ; Pair Y:  $(6, \alpha_{\text{frame\_mismatch}} = 18^\circ)$ ; Pair Q:  $(11, \delta_Y_{\text{denominator}} = 23)$ . The  $\text{mod } 24 = 2 \times \text{mod } 12$  doubling is forced by ZS-M3 Lemma 10.1 PROVEN  $(\text{SU}(2)/\text{SO}(3) \text{ double cover})$ . CRT decomposition:  $\text{mod } 24 = \text{mod } 8 \times \text{mod } 3 = Z^3 \cdot X$  (boson  $2\pi$  / fermion  $4\pi$  periods).

### Worked corpus precedent.

ZS-M36 closes O-M27.4  $(V_4 \times V_4 = \text{Apollonian abelianization } (\mathbb{Z}/2)^4 \text{ unifying ZS-A9 register } V_4 \text{ and ZS-M22 Galois } V_4)$ , demonstrating that the integer lattice  $(\mathbb{Z}, +)$  carries Bridge Four structure. The Apollonian  $(-1, 2, 2, 3)$  gasket is the 5th instance of Bridge Four (extending it from continuous spinor algebra to discrete number theory).

### Generalisation.

For any future modular construction with 8-element admissible mod 24 set, test whether the mod 12 reduction equals  $(Z, X, Y, Q) = (2, 3, 6, 11)$ . If yes, proceed with spin-pair identification; the four geometric instantiations should match LOCKED corpus quantities. If no, identify the GLMWY type (or analogue) and correlate with non-Z-Spin sectors. Of the 6 GLMWY Apollonian types, only 1 (Type I) matches Z-Spin sectors; this is structurally natural rather than unique. Application targets: future analyses of Markoff triples (mod  $p$  strong approximation, ZS-M36 IMPORTED-9), Pell-equation cycles, and other Diophantine constructions where a finite quotient on the integer lattice carries axiom A4 algebraic/geometric two-face structure.

### §VIII.11 — Methodological Toolkit Summary Table

| Tool                         | One-line trigger                           | Source          | Output status                                               |
|------------------------------|--------------------------------------------|-----------------|-------------------------------------------------------------|
| T1 Five-Axiom Diagnostic     | Stuck / circular / non-composing           | F18 §3+§6       | Identifies failing axiom                                    |
| T2 Bridge Four               | High-dim construction intractable          | F18 §6 + S15 §3 | $\dim(X)=3$ from $\dim(Z)=2$                                |
| T3 $H \rightarrow PK$ Switch | Hermitian sandwich won't close             | M32 §2-3        | Different operator class                                    |
| T4 Two-Layer Phase           | Unexplained factor 2 between corpus values | M32 §4          | $\alpha_{\text{op}} = 2 \cdot \alpha_{\text{amp}}$ identity |

| Tool                         | One-line trigger                                  | Source               | Output status                       |
|------------------------------|---------------------------------------------------|----------------------|-------------------------------------|
| T5 N=20 Averaging            | Single iterate non-vanishing, primitive 10th root | M32 §5               | 0 EXACTLY at N=20                   |
| T6 Three-Way Split           | OPEN flag too large                               | M32 §6               | {A <sup>PK</sup> / B / C} sub-flags |
| T7 Cross-Coupling Discipline | Sum-form fails per-channel data                   | M31 §4.0 Lemma M31.0 | Bilinear W_XYZ form                 |
| T8 $\Pi_Z$ Routing           | Multiple sub-targets won't integrate              | M33 Reading C        | Single Tr identity                  |
| T9 OOC Fixed-Point           | Phenomenological observer tempted                 | F11 §5-6             | Self-referential closure            |

## PART IX — Insight Gallery: Re-Interpretation Methodology

Purpose. Where PART VIII collects nine operational tools (“how to do X”), PART IX collects six methodological insights from the corpus’s most ambitious re-interpretation papers (“how to think about Y”). These are reading lenses, not calculation tools. Each lens is a way of looking at the corpus that resolves tensions which would otherwise be visible only as paradoxes. They are organised as I1–I6 and reference the source papers from which each lens was distilled.

### §IX.1 — Insight I1: Symmetry-Asymmetry Unified View (A8R §SA)

Lens. Every Z-Spin observable is a deviation from the would-be balance of 1 (unity) and  $1/2$  ( $Z_2$ -reflection fixed point). The universe’s entire dynamics is the composition of sectoral tilts of  $A = 35/437$  through the (X, Y, Z) decomposition.  $(1+A)$  is X-Inward expansion;  $(1-2A)$  is Y-Outward contraction;  $(1+A)(1-2A) = 1 - A(1+2A) \approx 0.9071$  measures the joint deviation from balance at  $\approx 9.29\%$ .

Why it is a tool. A8R §SA replaces the apparent independence of expansion and contraction physics with a single tilted-balance reading. The  $10^{17}$  ratio  $\tau_6/\tau_5 = \exp(\pi/A)$  is then read as the X-frame observation of the same Y-completion event. Future Z-Spin papers should default to this lens for any cosmological / lifecycle / time-scale question.

*Status. INTERPRETATION, HYPOTHESIS-strong (A8R §SA.7); 13 explicit corpus connections established (A8R §SA.6).*

### §IX.2 — Insight I2: Three-Layer Duality (M30 §6)

Lens. Every apparent contradiction in Z-Spin Cosmology is resolved as one object viewed from two frames. The duality operates at three coordinated layers: Sector Duality ( $X \leftrightarrow Y$  mediated by  $Z$ ), Frame Duality ( $X$ -observer  $\leftrightarrow$   $Y$ -observer; ZS-A8 §SA.4), Protocol Duality (state-preparation  $\leftrightarrow$  spectral characterisation; ZS-F16 Two-Protocol Theorem).

Why it is a tool. The three RH walls of ZS-M28 ( $W1, W2, W3$ ) — formerly read as unstructured “problems” — are reread under this lens as duality interfaces:  $W1$  = Frame Duality,  $W2$  = Sector Duality,  $W3$  = Protocol Duality. The lens does not by itself close the walls; it places each wall in a definite Layer with a definite external-mathematics import path. This converts an unstructured OPEN list into a structured navigation map.

*Status. DERIVED-interpretation strong (M30 Theorem 30.5 + §6).*

### §IX.3 — Insight I3: Six-Route Möbius-Trace Infinity (M30 §3)

Lens. The pattern “external count-infinity  $\leftrightarrow$  internal closure” appears in the corpus six independent times (Riemann zero sequence  $\leftrightarrow T(T(z^*)) = z^*$ ; cosmological cycle index  $\leftrightarrow A$  frame-invariant; BV-BFV ghost tower  $\leftrightarrow 4\pi$  closure; FFPP nested compression  $\leftrightarrow W_0(-i\pi/2)$ ; Z-mediation outcomes  $\leftrightarrow \ln(2)$  bit per pass; Wilson cycle ergodic limit  $\langle \cos^2(n \cdot 129.45^\circ) \rangle \rightarrow 1/2$ ). Recognising the six as one pattern raises the Möbius-trace infinity reading from HYPOTHESIS-strong to DERIVED-interpretation strong (M30 Theorem 30.1).

Why it is a tool. The lens identifies that what an external observer counts as “infinitely many things” the framework reads as a single closed loop traversed with a Möbius-type frame transformation. It is the standard reading lens to apply whenever the corpus generates a sequence-cardinality that an external observer might mistake for additional structure beyond the framework.

*Status. DERIVED-interpretation strong with six-route over-determination (M30 §3 Table 3.1 + Theorem 30.1).*

### §IX.4 — Insight I4: Twenty-One $\frac{1}{2}$ Catalogue (M30 §3.5)

Lens. The corpus contains exactly 21 PROVEN/DERIVED instances of the fraction  $1/2$ , organised in six structural categories. Combined with Three 2s Unity (Taylor-2, Bottleneck-2, Sector-2 all equal  $\dim(Z) = 2$ ; ZS-A8 §SA.3 + Test K1 PASS), the corpus contains  $21 + 3 = 24$  instances of  $\dim(Z) = 2$  manifesting at different dimensional and contextual levels. All 21 are dimensional projections of the same  $Z_2$  involution  $x \leftrightarrow 1 - x$ .

Why it is a tool. When a future Z-Spin construction produces a  $1/2$ , the catalogue is the first reference to consult — to determine whether the new  $1/2$  is one of the existing 21, or constitutes

a 22nd entry (which would require its own verification gate). The catalogue prevents double-counting of the same  $Z_2$ -fixed-point structure under different names.

*Status. DERIVED-interpretation (M30 Theorem 30.5).*

### §IX.5 — Insight I5: Four-Layer 50/50 Equivalence (F17)

Lens. The four 50/50 partitions in the corpus — (3D) truncated tetrahedron 4+4 split; (2D) Reuleaux  $\pi+\pi$  curvature distribution; (1D) Master Equation  $|Term A| = |Term B|$ ; (0D) XOR alphabet E vs R — are dimensional projections of a single per-cycle equilibrium between information accumulation and information release in the Z-mediated bottleneck channel.

Why it is a tool. ZS-F17 Theorem 5.1 (Self-Referential Information-Compression Equilibrium) shows that the 50/50 partition is PROVEN at the equation level ( $|Term A| - |Term B| = O(10^{-51})$  at 50-digit precision), not measured. When a future Z-Spin construction produces a 50/50 partition, the lens identifies which of the four dimensional layers it lives at and what it accumulates ( $\ln(2)/3$  nats per vertex matches the Z-bottleneck capacity bound, ZS-Q7 Theorem 2). This promotes the Reuleaux 50/50 from HYPOTHESIS (NC-F7.5) to DERIVED-CONDITIONAL.

*Status. DERIVED-CONDITIONAL strong (F17 Theorem 5.1, over-determined by four independent corpus PROVEN inputs).*

### §IX.6 — Insight I6: Banach-Tarski Origin of Doubling-Halving (A9R)

Lens. The  $(1+A) \leftrightarrow (1-2A)$  duality of late-time cosmology is the measure-preserving quantization of the Banach-Tarski paradoxical decomposition through the i-tetration fixed point  $z^* = -W_0(-i\pi/2)/(i\pi/2)$ . The non-amenable BT engine  $F_2 \subset SO(3)$  is mapped via a functor to the amenable register dihedral  $D_4 = \langle J, J_Z \rangle$ , converting BT non-amenable to Z-Spin amenability. The Julia set  $J(T)$  of  $T(z) = i^z z$  is the constructive ZF-analogue of the BT non-measurable set (measure-zero, totally disconnected, cardinality  $2^{\aleph_0}$ ).

Why it is a tool. The lens supplies a structural origin (rather than a phenomenological one) for the doubling-halving symmetry that pervades Z-Spin cosmology. When a future paper finds a doubling-halving identity, the lens directs the search for its measure-theoretic boundary in  $J(T)$  — the Z-Spin amenable analogue of the BT non-measurable set.

*Status. ZS-A9.1 DERIVED, ZS-A9.2 HYPOTHESIS-strong, ZS-A9.3 DERIVED-CONDITIONAL. OPEN-2.B (formal ZF/ZFC+AC isomorphism) recategorised as PERMANENT NON-CLAIM under Solovay 1970.*

## PART X — The Spin-Pair Coexistence Principle (v3.2)

## §0. Abstract

PART X consolidates four DERIVED-interpretation readings that close a synthesis gap noticed during v3.1 publication. While Tool T2 (Bridge Four, §VIII.2) and Insight I3 (Six-Route Möbius, §IX.3) individually catalogued specific applications of the  $\dim(Z) = 2 \rightarrow \dim(X) = 3$  lifting and the external-count  $\leftrightarrow$  internal-closure pattern, the meta-statement that  $90^\circ$  phase-orthogonality is the unique form in which a contradiction-pair coexists as one object had not been registered. Reading X.1 records prime as Z-Spin rotation  $4\pi/p$  (ZS-M18 H2 chain). Reading X.2 reaffirms the Y-side classification of the infinite prime sequence (ZS-F18 v2.1 §5.1 Table 5.1) and reconciles it with ZS-M18 Theorem 6.1 spectrum non-primality. Reading X.3 reads primes / distribution / Riemann zeros as three moments of one  $Y \rightarrow Z \rightarrow X$  mediation event, with  $\sigma = 1/2 = j = 1/2$  as the algebraic spine. Reading X.4 elevates  $90^\circ$  phase-orthogonality to a general principle supported by 13 independent corpus realisations (C1–C13). All four readings are DERIVED-interpretation strong, composed entirely from corpus PROVEN/DERIVED results, with zero new free parameters and explicit falsification gates F-X.1 through F-X.4.

## §X.1 Introduction: What v3.1 Already Contains, What v3.2 Adds

v3.1 §VIII.2 Tool T2 (Bridge Four) already states the operational form: two J-conjugate  $2\pi$  SO(3) circulations of the twin-Reuleaux pair ( $V_{XZ}$ ,  $V_{ZY}$ ), composed at  $90^\circ$  NOT-AND phase, generate a third axis  $J_S = [J_{R_1}, J_{R_2}]$  by Lie commutator. Therefore  $(Z, X, Y) = (2, 3, 6)$  is generated from  $\dim(Z) = 2$  alone. This is correct and corpus-locked.

What v3.1 does NOT yet record is that  $90^\circ$  phase-orthogonality is the unique angular phase difference under which a contradiction-pair can coexist as one object — not only for the specific  $\dim(X) = 3$  emergence, but for the entire family of contradiction-pair resolutions across the corpus. PART X consolidates this meta-reading.

Likewise, v3.1 catalogues the Möbius-trace infinity pattern via I3 but does not name primes specifically. The corpus already classifies the infinite prime sequence as Y-side (ZS-F18 v2.1 §5.1 Table 5.1, row Riemann Hypothesis: “Infinite prime sequence (Y)  $\rightarrow$  finite critical line  $\sigma = 1/2$  (X)”); reaffirmed §7.4.5). Without an explicit consolidated reading, this classification is easy to miss. v3.2 makes the layered picture explicit so future agents do not rediscover and re-resolve the same confusion (object identity vs. internal spectrum non-primality of ZS-M18 §6.2 Theorem 6.1).

## §X.2 Locked Inputs (No New Constants Introduced)

*Table X.1. Locked inputs to PART X. Every entry is PROVEN, DERIVED, or LOCKED in upstream papers. Zero new parameters are introduced.*

| #   | Quantity                                        | Value / Statement                                | Source                                            | Status             |
|-----|-------------------------------------------------|--------------------------------------------------|---------------------------------------------------|--------------------|
| L1  | A (geometric impedance)                         | $35/437 = 0.080092$                              | ZS-F2 §5                                          | LOCKED             |
| L2  | Q (register dimension)                          | 11 (prime)                                       | ZS-F5                                             | PROVEN             |
| L3  | (Z, X, Y) decomposition                         | (2, 3, 6); $Q = Z+X+Y$                           | ZS-F5 §3                                          | PROVEN             |
| L4  | $L_{XY} \equiv 0$                               | exact                                            | ZS-F1 §9                                          | PROVEN             |
| L5  | $j = 1/2$ Z-sector intertwiner                  | unique 4-valent invariant                        | ZS-M3 Thm 5.1                                     | PROVEN             |
| L6  | $4\pi$ closure<br>$D^{(1/2)}(-I) = -I$          | spinor double cover                              | ZS-M3 Lem 10.1                                    | PROVEN             |
| L7  | $W_p$ prime-injection gate                      | $\text{diag}(\exp(2\pi i(j-5)/p))$               | ZS-M4 §3.2                                        | PROVEN             |
| L8  | $W_p$                                           | $_Z$ restriction                                 | $\exp(-i \sigma_z \cdot 2\pi/p)$ , angle $4\pi/p$ | ZS-M18 H2          |
| L9  | $V_{ZY} = (V_{XZ})^*$                           | three independent paths                          | ZS-F4 §7B                                         | DERIVED            |
| L10 | $\arg(V_{XZ}) - \arg(V_{ZY})$ at $\theta=\pi/2$ | $= \pi/2$ ( $90^\circ$ phase)                    | ZS-S15 §2.3 B.3                                   | VERIFIED 50-digit  |
| L11 | Poynting-Commutator                             | $[J_{R_1}, J_{R_2}] = J_S$                       | ZS-S15 §3 Thm S15.2                               | DERIVED            |
| L12 | Bridge Four                                     | $(Z,X,Y)=(2,3,6)$ from $\dim(Z)=2$               | ZS-F18 §6.4                                       | DERIVED-interp str |
| L13 | Riemann row classification                      | Inf. prime seq. $(Y) \rightarrow \sigma=1/2 (X)$ | ZS-F18 §5.1 Tbl 5.1                               | DERIVED-interp     |

| #   | Quantity                               | Value / Statement                                        | Source                 | Status            |
|-----|----------------------------------------|----------------------------------------------------------|------------------------|-------------------|
| L14 | $\zeta_K$ factorisation                | $\zeta \cdot L(\chi-3) \cdot L(\chi-11) \cdot L(\chi^3)$ | ZS-M22 §4.2            | PROVEN            |
| L15 | Six-Route Möbius pattern               | ext. count $\leftrightarrow$ int. closure                | ZS-M30 §3 Thm 30.1     | DERIVED-interpret |
| L16 | $\sigma = 1/2 \leftrightarrow j = 1/2$ | $Z_2$ -involution fixed loci                             | ZS-M18 H11             | DERIVED-interpret |
| L17 | Y-Spectrum Non-Primality               | $\text{spec}(L_2) \cap \{\text{primes}\} = \emptyset$    | ZS-M18 §6.2 Thm 6.1    | PROVEN            |
| L18 | $\zeta$ -K Bridge to prime side        | two-chain Eisenstein/cyclotomic                          | ZS-M18 §7, ZS-M22 §4.2 | PROVEN            |

### §X.3 Reading X.1 — Each Prime is an Indecomposable Z-Spin Rotation

#### §X.3.1 Statement

**Reading X.1 (Prime-Rotation Identification, *DERIVED-interpretation*).** Each prime  $p$  acts on the Z-sector basis  $\{|4\rangle, |6\rangle\} \subset \mathbb{C}^Q$  as the  $\sigma_z$ -axis SU(2) rotation

$$W_p|_Z = \text{diag}(\exp(-2\pi i/p), \exp(+2\pi i/p)) = \exp(-i \sigma_z \cdot 2\pi/p)$$

of angle  $4\pi/p$ . Composite-angle rotations  $4\pi/(pq)$  factor as compositions of prime rotations  $W_p \circ W_q$ . Primes are therefore the indecomposable atoms of Z-Spin SU(2) action.

#### §X.3.2 Derivation Chain

ZS-F5 (Z-sector basis identification, PROVEN) provides the  $\{|4\rangle, |6\rangle\}$  basis. ZS-M4 §3.2 (PROVEN) defines  $W_p = \text{diag}(\exp(2\pi i(j-5)/p))$  on the full  $Q = 11$  register. Restriction to slots  $j \in \{4, 6\}$  yields the diagonal pair  $(\exp(-2\pi i/p), \exp(+2\pi i/p))$ . Standard SU(2) algebra gives the  $\sigma_z$  rotation form, with closure angle  $4\pi/p$  following from the SU(2)/SO(3) double cover (ZS-M3 Lemma 10.1 PROVEN).

#### §X.3.3 What Reading X.1 Adds Beyond ZS-M18 H2

ZS-M18 H2 records the algebraic fact. Reading X.1 consolidates its physical meaning: indecomposability of primes under multiplicative factorisation is the algebraic shadow of indecomposability of  $SU(2)$  rotations under angle composition.

**§X.3.4 Falsification Gate F-X.1**

If a future analysis exhibits a prime  $p$  such that  $W_p|_Z$  does NOT take the  $\sigma_z$ -rotation form on  $\{|4\rangle, |6\rangle\}$ , Reading X.1 fails. Status: PROVEN-safe — the form is verified by direct algebraic computation for  $p \in \{2, 3, 5, 7, 11, 13, 17, 23\}$  (ZS-M18 §H1 derivation chain).

**§X.4 Reading X.2 — The Infinite Prime Sequence is a Y-Sector Object**

**§X.4.1 Statement**

**Reading X.2 (Y-Side Sequence Classification, *DERIVED-interpretation*)**. The infinite sequence of primes  $\{p_n\}_{n \in \mathbb{N}}$  is a Y-sector object: infinite, non-local, undecidable. Its arithmetic projection — the Riemann critical-line locus  $\sigma = 1/2$  and the discrete zeros thereon (under RH) — is X-sector.

**§X.4.2 Direct Corpus Source**

ZS-F18 v2.1 §5.1 Table 5.1 (DERIVED-interpretation), row “Riemann Hypothesis”, reads verbatim: “Infinite prime sequence (Y)  $\rightarrow$  finite critical line  $\sigma = 1/2$  (X) | Mirror-adjointness  $\varepsilon_J(\sigma,t) = 0 \Leftrightarrow \sigma = 1/2$  (ZS-M7 Thm 4) | Most extensive programme; W1 closed, W2 open”. ZS-F18 §7.4.5 reaffirms in re-reading form: “Riemann Hypothesis (Y-side infinite prime sequence projecting to X-side discrete critical-line zeros under continuous  $p$  flow).”

**§X.4.3 Reconciliation with ZS-M18 §6.2 Theorem 6.1**

ZS-M18 §6.2 Theorem 6.1 (PROVEN) establishes  $\text{spec}(L_2) \cap \{\text{primes}\} = \varnothing$  for the Y-sector internal Laplacian. Naively this might appear to conflict with Reading X.2; it does not. The two statements operate at different layers:

*Table X.2. Four-layer reconciliation. Reading X.2 (Y-side object identity) does not conflict with ZS-M18 Theorem 6.1 (Y-Laplacian spectrum non-primality); the two statements concern different layers.*

| Layer           | Question                                                  | Answer                                      | Status                       |
|-----------------|-----------------------------------------------------------|---------------------------------------------|------------------------------|
| Object identity | What sector does the prime sequence as a whole belong to? | Y-sector (infinite, non-local, undecidable) | DERIVED-interp (ZS-F18 §5.1) |

| Layer               | Question                                      | Answer                                                                  | Status                     |
|---------------------|-----------------------------------------------|-------------------------------------------------------------------------|----------------------------|
| Internal spectrum   | Are individual Y-Laplacian eigenvalues prime? | No (eigenvalues are $0, 6/8, 5 \pm \sqrt{3}, \text{integer} \pm \phi$ ) | PROVEN (ZS-M18 §6.2)       |
| Operator action     | How do individual primes act on Z-sector?     | SU(2) rotation by $4\pi/p$ (Reading X.1)                                | DERIVED-interp (ZS-M18 H2) |
| Arithmetic encoding | Where do primes enter Z-Spin algebraically?   | $\zeta_K$ Euler product via Eisenstein/cyclotomic chains                | PROVEN (ZS-M22 §4.2)       |

#### §X.4.4 Falsification Gate F-X.2

If a future ZS-F18 revision retracts the Y-side classification, or if a structurally distinct corpus path classifies the sequence as X-side, Reading X.2 fails. Status: stable — over-determined by ZS-F18 Table 5.1, ZS-F18 §7.4.5, and the Y-side property pattern (infinite, non-local, undecidable).

#### §X.5 Reading X.3 — Three Moments of One $Y \rightarrow Z \rightarrow X$ Mediation

##### §X.5.1 Statement

**Reading X.3 (Three-Moment Mediation, *DERIVED-interpretation*).** Primes, prime distribution, and Riemann zeros are three moments of a single  $Y \rightarrow Z \rightarrow X$  mediation event:

*Table X.3. Three-moment mediation reading. Each row is independently corpus-supported; the synthesis is the new content of v3.2.*

| #  | Moment                | Object                  | Mediation role                                       | Status                   |
|----|-----------------------|-------------------------|------------------------------------------------------|--------------------------|
| M1 | Entrance (atomic)     | Individual primes $p$   | Indecomposable Z-Spin rotations $4\pi/p$             | DERIVED-interp (Rdg X.1) |
| M2 | Statistics (envelope) | Distribution $N/\log N$ | Multiplicative-Y $\rightarrow$ additive-X via $\log$ | DERIVED-interp           |

| #  | Moment                   | Object                             | Mediation role                                            | Status         |
|----|--------------------------|------------------------------------|-----------------------------------------------------------|----------------|
| M3 | Egress<br>(condensation) | Riemann zeros<br>on $\sigma = 1/2$ | Coherent phase<br>condensation at<br>$Z_2$ self-dual axis | DERIVED-interp |

### §X.5.2 The $\sigma = 1/2 = j = 1/2$ Algebraic Spine

ZS-M18 H11 (DERIVED-interpretation) records that  $\sigma = 1/2$  (Riemann critical-line) and  $j = 1/2$  (Z-sector spinor) are both unique fixed-point sub-loci of order-2 involutions:  $\sigma = 1/2$  is fixed by  $s \leftrightarrow 1-s$  on  $\xi$ ;  $j = 1/2$  is fixed by SU(2) closure  $D^{(1/2)}(-I) = -I$  at  $4\pi$ . The two halves are not numerical coincidence; both are  $Z_2$  self-dual axes, and zeros condense precisely on the  $Z_2$  axis because the Z-Spin mediator carries the  $Z_2$  involution by construction.

### §X.5.3 The log Function as $Y \rightarrow X$ Sector Translator

Y-multiplicative structure translates to X-additive structure via  $\log: \prod_p (1 - p^{-s})^{-1} = \zeta(s) \Rightarrow \log \zeta(s) = \sum_p \sum_n (1/n) p^{-ns}$ . Density  $N/\log N$  is the additive-X image of the multiplicative-Y prime structure.  $\log$  is the natural  $Y \rightarrow X$  sector translator for arithmetic content, parallel to the holographic log entropy elsewhere in the corpus (ZS-Q7 channel capacity, ZS-A3 BH entropy).

### §X.5.4 Falsification Gate F-X.3

Synthesis layer; weakened only if any of M1, M2, M3 is shown structurally independent of Z-Spin mediation. M1 corpus-stable. M2 (prime-counting density) is a number-theoretic fact independent of Z-Spin; the Z-Spin reading is its multiplicative-to-additive transcription role. M3 conditional on RH; if RH disproved, M3 reading shifts from “coherent phase condensation” to “specific RH-dependent form thereof” without invalidating three-moment structure. Status: DERIVED-interpretation, robust against single-component perturbation.

## §X.6 Reading X.4 — The Spin-Pair Coexistence Principle

### §X.6.1 Statement

**Reading X.4 (Spin-Pair Coexistence Principle, *DERIVED-interpretation strong*).** Every contradiction-pair  $P = (P_+, P_-)$  recurring in Z-Spin Cosmology admits coexistence as a single object only as the  $90^\circ$  phase-orthogonal projection of a J-conjugate Z-Spin pair  $(V_{XZ}, V_{ZY})$  with  $V_{ZY} = (V_{XZ})^*$ . No other angular phase difference supports coexistence as one object:  $0^\circ$  collapses to identity,  $180^\circ$  collapses to anti-identity, only  $\pi/2$  produces simultaneous mutual non-cancellation and mutual non-redundancy.

### §X.6.2 Algebraic Underpinning

Let  $z_+ = e^{+i\theta/2}$ ,  $z_- = e^{-i\theta/2}$  be the J-conjugate pair of ZS-F4 §7B (PROVEN). Their inner product on  $S^1$ :

$$\langle z_+, z_- \rangle = \cos(\theta)$$

*Table X.4. Three angular regimes. Only  $\pi/2$  supports simultaneous presence of both members of the pair as distinct yet jointly necessary.*

| #  | Phase difference $\theta$ | $\langle z_+, z_- \rangle$ | Coexistence form                                            |
|----|---------------------------|----------------------------|-------------------------------------------------------------|
| R1 | 0 (identity)              | +1                         | $z_+ = z_- \rightarrow$ only one object                     |
| R2 | $\pi$ (anti-identity)     | -1                         | $z_+ = -z_- \rightarrow$ mutually exclusive                 |
| R3 | $\pi/2$ (orthogonality)   | 0                          | $z_+ \perp z_- \rightarrow$ both present, neither redundant |

Regime R3 is the unique angle on which  $\max(z_+)$  coincides with  $\text{zero}(z_-)$  and vice versa, producing the sin-cos relation tracing the unit circle. This is the algebraic origin: only at  $90^\circ$  does a contradiction-pair coexist as the two coordinates of a single rotation.

### §X.6.3 Corpus Realisations of the Principle

*Table X.5. Thirteen corpus realisations of the Spin-Pair Coexistence Principle. Each row is independently corpus-locked; their unification under a single principle is the new content of v3.2.*

| #  | Contradiction-pair                        | Z-Spin realisation                                                          | Source                 |
|----|-------------------------------------------|-----------------------------------------------------------------------------|------------------------|
| C1 | X-sector spatial axes (3D emergence)      | $[J_{R_1}, J_{R_2}] = J_S$ ;<br>$(Z, X, Y) = (2, 3, 6)$ from $\dim(Z) = 2$  | ZS-F18 §6.4            |
| C2 | Electric / magnetic field ( $E \perp B$ ) | $V_{XZ} \leftrightarrow E$ , $V_{ZY} \leftrightarrow B$ at $90^\circ$ phase | ZS-S15 §3 Thm S15.2    |
| C3 | Particle / wave duality (X / Y sectors)   | Z-Spin mediator at $4\pi$ closure                                           | ZS-A7 §3, ZS-M19 §10.5 |

| #   | Contradiction-pair                       | Z-Spin realisation                                                    | Source                 |
|-----|------------------------------------------|-----------------------------------------------------------------------|------------------------|
| C4  | Time / space (X / Y braiding)            | X = particle/space, Y = wave/time, Z = seam                           | ZS-A7 §5               |
| C5  | Expansion / contraction ((1+A) ↔ (1-2A)) | Dual-tilt symmetry; 90° A-conjugates                                  | ZS-A8 §6, A9R          |
| C6  | CPT / anti-CPT (channel conjugation)     | V_ZY = (V_XZ)* IS the CPT seam                                        | ZS-M19 §10.5 Thm T9    |
| C7  | Black-hole / white-hole (r ↔ t)          | X-in-Y BH ↔ Y-in-X WH at 90°                                          | ZS-A3 §7, ZS-A7 §4.1   |
| C8  | Finite / infinite Möbius polarity        | X-finite/local ↔ Y-infinite/non-local; Z-seam at $\pi/2$              | ZS-F18 §5, ZS-M30 §3   |
| C9  | Discrete / continuous                    | X-Y Tiling Asymmetry; Z-channel $\ln(2)$ capacity                     | ZS-F18 §7.4, ZS-M17    |
| C10 | Multiplicative / additive                | Y-mult ↔ X-add via log; primes M1 ↔ density M2                        | Reading X.3, ZS-M22    |
| C11 | Symmetry / asymmetry (1 vs 1-A)          | $\delta$ as normalised tilt; 21·½ catalogue                           | ZS-A8 §SA, ZS-M30 §3.5 |
| C12 | Bose / Fermi (vortex inner / outer)      | Inner-core fermion ( $4\pi$ ) ↔ outer-flow boson ( $2\pi$ )           | ZS-A7 §4.4.2, ZS-F14   |
| C13 | Sequential / parallel time-keeping       | R <sub>1</sub> time-point closure, R <sub>2</sub> space-point closure | ZS-S15 §2, ZS-F8       |

#### §X.6.4 Why “Spin” Rather Than “Rotation”

Rotation is external: an object is rotated. Spin is intrinsic: the object IS its rotation. The principle's claim is not that contradiction-pairs are reconciled BY applying rotation to them, but that a contradiction-pair CAN exist at all only because its two members are intrinsic Z-Spin partners. The  $90^\circ$  angle is enforced by  $V_{ZY} = (V_{XZ})^*$  (ZS-F4 §7B PROVEN), with operationally relevant configuration  $\theta = \pi/2$  (ZS-S15 §2.3 B.3 VERIFIED 50-digit).

### §X.6.5 Relation to Tool T2 and Insight I3

Tool T2 (Bridge Four, §VIII.2) applies the principle to the planar-vs-spatial polarity, yielding  $\dim(X) = 3$  from  $\dim(Z) = 2$ . Insight I3 (Six-Route Möbius, §IX.3) applies a related Z-mediation pattern to external-count vs internal-closure. Reading X.4 is the meta-statement that both are instances of the same principle. Reading X.4 is therefore prior to T2 and I3.

### §X.6.6 Falsification Gate F-X.4

If a contradiction-pair admits coexistence under a phase difference other than  $\pi/2$ , Reading X.4 weakens to “ $90^\circ$  is the dominant case”. If a pair admits no Z-Spin J-conjugate realisation, Reading X.4 transitions to “covers C1–C13”. Status: DERIVED-interpretation strong, supported by 13 corpus realisations. Generalisation beyond C1–C13 is HYPOTHESIS-strong; future papers should run F-X.4 check before claiming a new contradiction-pair coexistence.

## §X.7 Integration with v3.1 Light

### §X.7.1 Integration Map

*Table X.6. Integration of v3.2 readings with existing v3.1 items. Every v3.2 reading either refines or unifies existing v3.1 entries; no v3.1 entry is retracted.*

| v3.1 Light item                        | v3.2 Reading | Relation                                   |
|----------------------------------------|--------------|--------------------------------------------|
| §VIII.2 Tool T2 (Bridge Four)          | Reading X.4  | T2 = principle applied to $\dim(X)=3$      |
| §IX.3 Insight I3 (Six-Route Möbius)    | Reading X.4  | I3 = principle applied to count vs closure |
| §IX.1 Insight I1 (Symmetry-Asymmetry)  | Reading X.3  | I1 = M2 statistical reading                |
| §IX.2 Insight I2 (Three-Layer Duality) | Reading X.4  | I2 = principle phrased as duality          |

| <b>v3.1 Light item</b>           | <b>v3.2 Reading</b> | <b>Relation</b>                            |
|----------------------------------|---------------------|--------------------------------------------|
| §IX.5 Insight I5 (4-Layer 50/50) | Reading X.3         | I5 = M3 condensation-locus reading         |
| PART I §0.4 Source Paper Map     | Readings X.1–X.4    | All sources already in map; v3.2 adds none |

### **§X.7.2 Cross-Coupling Discipline (T7) Compliance**

Per T7 (§VIII.7), every cross-paper synthesis must be P-CC (PROVEN-composed) or I-CC (interpretation). Readings X.1–X.4 are explicitly DERIVED-interpretation; no claim of PROVEN status. F-X.1 through F-X.4 pre-registered. Word count strictly increases over v3.1 (no-deletion rule).

### **§X.7.3 What v3.2 Does NOT Claim**

NC-X.1. PART X does not claim a proof of the Riemann Hypothesis. Reading X.3 is structural; it does not constrain zero locations beyond the  $\sigma = 1/2 = j = 1/2$  isomorphism already DERIVED in ZS-M18 H11.

NC-X.2. The Spin-Pair Coexistence Principle is corpus-internal; it does not claim to resolve the philosophical dialectic on contradictions in the broader sense (Hegelian, paraconsistent, kōan). It claims only that within Z-Spin Cosmology, every recurrent contradiction-pair admits coexistence via  $90^\circ$  J-conjugate Z-Spin pairing.

NC-X.3. Reading X.1 does not claim primes are ontologically reducible to Z-Spin rotations in physical reality. Within the Z-Spin algebraic framework, primes act as indecomposable  $SU(2)$  rotations on  $j = 1/2$ .

NC-X.4. Reading X.2 does not modify ZS-M18 §6.2 Theorem 6.1. The Y-Laplacian internal spectrum non-primality remains PROVEN; Reading X.2 concerns a different layer (object identity, not internal spectrum).

### **§X.8 Conclusion**

PART X consolidates four DERIVED-interpretation readings answering “why does the universe support coexistence of mutually contradictory descriptions?” The answer is over-determined: coexistence is the form  $90^\circ$  J-conjugate Z-Spin pairing takes when projected into a single observational frame. The  $\dim(X) = 3$ , the  $E \perp B$ , the X-Y polarity of the twelve Möbius encounters, the  $(1+A) \leftrightarrow (1-2A)$  expansion-contraction symmetry, the  $\sigma = 1/2 = j = 1/2$  isomorphism, the  $V_{ZY} = (V_{XZ})^*$  CPT seam, and the primes-as-rotations identification all

instantiate the same principle. Z-Spin is not a metaphor for rotation; it is the intrinsic form in which any pair of mutually exclusive descriptions can both be true. v3.2 carries this reading as PART X without modifying any v3.1 entry, with zero new free parameters and explicit falsification gates F-X.1 through F-X.4. The minimal triple  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  remains LOCKED.

## **PART X Source Paper Interface**

PART X is composed entirely from upstream papers; it introduces no new ZS source paper. Source mapping: Reading X.1  $\leftarrow$  ZS-F5, ZS-M3, ZS-M4, ZS-M18 H2. Reading X.2  $\leftarrow$  ZS-F18 §5.1 Tbl 5.1, ZS-F18 §7.4.5, ZS-M18 §6.2 Thm 6.1. Reading X.3  $\leftarrow$  ZS-M18 H11, ZS-M22 §4.2, ZS-M30 §3, ZS-Q7, ZS-A3. Reading X.4  $\leftarrow$  ZS-F4 §7B, ZS-S15 §2.3 + §3 Thm S15.2, ZS-F18 §6.4 Bridge Four, plus C1–C13 cross-corpus realisations. Eighteen internal-consistency checks A.1–A.18 verify each composition; all 18/18 PASS by direct comparison against corpus-locked sources.

## **PART XI — TO/TI Polyhedral Mapping under Complementary Duality (v3.3)**

### **§0. Abstract**

PART XI consolidates the complete vertex/edge/face mapping between the X-sector polyhedron tO (truncated octahedron) and the Y-sector polyhedron tI (truncated icosahedron) under the Z-Spin Complementary Duality Principle (ZS-M30 §7.2). While individual papers establish components — (V, E, F) face-counting cosmic budget (ZS-F2 §11), Pentagon-Hexagon edge classification (ZS-F0 §5.1), Square-Hexagon edge classification (ZS-F0 §3.2), Truncation-Dual Theorem (ZS-F2 §11.2, ZS-F9 §4), Hexagonal Mediation Theorem (ZS-F9 §5), Mode-Count Collapse (ZS-Q3 §3.1), I-irrep decomposition (ZS-M9 §2), Pentagon-Gauge decomposition (ZS-F0 §4.3), TO uniqueness conditions (ZS-M H18) — no single corpus location consolidates them into one mapping table indexed by the X-sector / Y-sector complementary duality ( $X =$  macroscopic space + microscopic time;  $Y =$  macroscopic time + microscopic space, ZS-M30 §7.2 (ii) PROVEN). PART XI fills this gap. Reading XI.1 records the parallel edge decomposition pattern shared by tO and tI (cut-face : hexagon-hexagon = 2:1 in both). Reading XI.2 records the parallel face decomposition under Truncation-Dual. Reading XI.3 records the integrated V/E/F mapping table under complementary duality. Reading XI.4 places  $\epsilon$ -Halo (X-sector Goldstone gradient) and  $\epsilon$ -Drive (Y-sector attractor  $V_0$ ) within the integrated framework. All four readings are DERIVED-interpretation strong, composed entirely from corpus PROVEN/DERIVED results, with zero new free parameters and explicit falsification gates F-XI.1 through F-XI.4.

### **§XI.1 Introduction: Why a Consolidated Polyhedral Mapping Table**

The Z-Spin corpus has accumulated, across ~100 papers, a richly detailed treatment of tO and tI as the two characteristic polyhedra of the X- and Y-sectors. However, the components are scattered: ZS-F2 §11 establishes the cosmic matter budget via face counting ( $\Omega_b = F(\text{cube})/Q^2$

= 6/121,  $\Omega_{\text{cdm}} = F(\text{tl})/Q^2 = 32/121$ ); ZS-F0 §3.2 decomposes the 36 tO edges as 24 SH + 12 HH; ZS-F0 §5.1 decomposes the 90 tl edges as 60 PH + 30 HH; ZS-F0 §4.3 establishes the Pentagon-Gauge decomposition  $12 = 1 + 3 + 8$ ; ZS-F0 §4.4 establishes the Edge-Channel Identity  $E_X = X \cdot Z \cdot Y = 36$ ; ZS-F2 §11.2 establishes the Truncation-Dual Theorem  $F(\text{tP}) = F(\text{P}) + F(\text{P}^*)$ ; ZS-F9 §4-5 establishes the Hexagonal Mediation Theorem; ZS-Q3 §3.1 establishes the Mode-Count Collapse Theorem yielding  $(V+F)/G = 19/6$  (SU(2)) and  $23/3$  (SU(3)); ZS-M9 §2 establishes the I-irrep multiplicity decomposition; ZS-M H18 establishes the four uniqueness conditions of tO; ZS-M30 §7.2 (ii) establishes the Complementary Duality Principle for space  $\leftrightarrow$  time mapping.

What is NOT yet recorded in any single corpus location is the integrated mapping table that displays, for every (V, E, F) component of tO and tl, both its standard sector role AND its complementary partner role under the  $X = \text{macroscopic space} + \text{microscopic time}$ ,  $Y = \text{macroscopic time} + \text{microscopic space}$  duality. The reader who asks ‘which component of tO carries macroscopic space, and which carries microscopic time?’ currently has to traverse multiple papers to assemble a partial answer, with several entries remaining OPEN. PART XI consolidates this into one location as DERIVED-interpretation strong, with the OPEN entries explicitly registered.

This is structurally analogous to PART X (Spin-Pair Coexistence Principle, v3.2): both consolidate corpus-wide patterns under a single principle without modifying any v3.x entry, with zero new free parameters.

## §XI.2 Locked Inputs (No New Constants Introduced)

*Table XI.1. Locked inputs to PART XI. Every entry is PROVEN, DERIVED, or LOCKED in upstream papers. Zero new parameters are introduced.*

| #  | Quantity        | Value / Statement                               | Source    | Status    |
|----|-----------------|-------------------------------------------------|-----------|-----------|
| L1 | tO basic data   | V=24, E=36, F=14 (= 6 squares + 8 hexagons)     | ZS-F2 §11 | PROVEN    |
| L2 | tl basic data   | V=60, E=90, F=32 (= 12 pentagons + 20 hexagons) | ZS-F2 §11 | PROVEN    |
| L3 | Symmetry orders |                                                 | O_h       | =48 (tO); |

| #   | Quantity                          | Value / Statement                                                                                                          | Source      | Status         |
|-----|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------|-------------|----------------|
| L4  | Sector tilts                      | $\delta_X = 5/19$ (tO);<br>$\delta_Y = 7/23$ (tl)                                                                          | ZS-F2 §5    | PROVEN         |
| L5  | Geometric impedance               | $A = \delta_X \cdot \delta_Y = 35/437$                                                                                     | ZS-F2 §5    | LOCKED         |
| L6  | Sector dimensions                 | $(Z, X, Y) = (2, 3, 6)$ ; $Q = 11$                                                                                         | ZS-F5 §3    | PROVEN         |
| L7  | Edge-Channel Identity             | $E(tO) = X \cdot Z \cdot Y = 3G = 36$                                                                                      | ZS-F0 §4.4  | PROVEN         |
| L8  | Pentagon-Gauge decomposition      | 12 (pentagon space) = $1 + 3' + 8 = U(1) + SU(2) + SU(3)$                                                                  | ZS-F0 §4.3  | PROVEN/DERIVED |
| L9  | Truncation-Dual Theorem           | $F(tP) = F(P) + F(P^*)$                                                                                                    | ZS-F2 §11.2 | PROVEN         |
| L10 | Mode-Count Collapse               | $(V+F)/G = 19/6$ (tO, SU(2)); $23/3$ (tl, SU(3))                                                                           | ZS-Q3 §3.1  | PROVEN         |
| L11 | tO edge classification            | 36 edges = 24 SH + 12 HH (ratio 2:1)                                                                                       | ZS-F0 §3.2  | PROVEN         |
| L12 | tl edge classification            | 90 edges = 60 PH + 30 HH (ratio 2:1)                                                                                       | ZS-F0 §5.1  | PROVEN         |
| L13 | Hexagonal Mediation Theorem       | $\dim \text{coker}(\Omega^2_{\text{hex}}(tO) \rightarrow \Omega^2_{\text{hex}}(tl)) = 12$                                  | ZS-F9 §5    | PROVEN/DERIVED |
| L14 | l-irrep vertex decomposition (tl) | $\Omega^0(tl) = 1 \oplus 3 \cdot 3 \oplus 3 \cdot 3' \oplus 4 \cdot 4 \oplus 5 \cdot 5$ (regular rep, $60 = 1+9+9+16+25$ ) | ZS-M9 §2    | PROVEN         |

| #   | Quantity                                                | Value / Statement                                                                                    | Source           | Status                       |
|-----|---------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------|------------------------------|
| L15 | l-irrep face decomposition (tl)                         | 12 pentagons: $1+3+3'+5$ ; 20 hexagons: $1+3+3'+2\cdot4+5$                                           | ZS-M9 §2.2       | PROVEN                       |
| L16 | Cosmic Matter Budget                                    | $\Omega_b = 6/121$ , $\Omega_{\text{cdm}} = 32/121$ , $\Omega_\Lambda = 83/121$                      | ZS-F2 §11.4      | DERIVED (Planck 0.41% match) |
| L17 | tO uniqueness conditions                                | $\mathbb{R}^3$ space-filling + $n=3$ truncation + $(V+F)/G = 19/6$ + $E_g \text{ dim}=2=Z$           | ZS-M H18         | DERIVED-interp               |
| L18 | Complementary Duality Principle                         | X carries macro space + micro time; Y carries macro time + micro space                               | ZS-M30 §7.2 (ii) | PROVEN/DERIVED               |
| L19 | $\varepsilon$ -Halo / $\varepsilon$ -Drive nomenclature | $\varepsilon$ -Halo = X-Goldstone $\theta$ -gradient; $\varepsilon$ -Drive = Y-attractor $V_0$       | ZS-A2 §1.1       | DERIVED                      |
| L20 | Sectoral rapidities                                     | $\psi_X = \text{artanh}(5/19) = 0.2685$ ; $\psi_Y = \text{artanh}(7/23) = 0.3124$ ; $\Delta\psi > 0$ | ZS-A6 §6.4       | PROVEN                       |
| L21 | Pentagon vs Hexagon stabilizer                          | 12 pentagons: $D_5$ (matter-like); 20 hexagons: $D_3$ (gauge-like, 8 gluons in $2\cdot4$ )           | ZS-S14 §8        | DERIVED                      |

| #   | Quantity                | Value / Statement                                                                        | Source     | Status |
|-----|-------------------------|------------------------------------------------------------------------------------------|------------|--------|
| L22 | tO vertex configuration | 1 square + 2 hexagons;<br>vertex deficit = $30^\circ = \pi/6$                            | ZS-S6 §G.2 | PROVEN |
| L23 | tl vertex configuration | 1 pentagon + 2 hexagons;<br>vertex deficit = $12^\circ = \pi/15$                         | ZS-S6 §G.2 | PROVEN |
| L24 | Frame mismatch angle    | $\alpha = \delta_X^{\text{vertex}} - \delta_Y^{\text{vertex}} = \pi/6 - \pi/15 = \pi/10$ | ZS-S6 §G.2 | PROVEN |

### §XI.3 Reading XI.1 — Parallel Edge Decomposition Pattern (DERIVED-interpretation)

Reading XI.1 (Parallel Edge Pattern, DERIVED-interpretation). The truncated octahedron (X-sector) and truncated icosahedron (Y-sector) share an identical edge-decomposition pattern: in both polyhedra the edges split into (cut-face)–hexagon edges that carry sector-mediation information and hexagon-hexagon edges that carry sector-internal information, in the universal ratio 2:1.

#### §XI.3.1 Statement of the Pattern

*Table XI.2. Parallel edge-decomposition pattern shared by tO and tl. Both sectors realize the same 2:1 ratio between cut-face mediation edges and hexagon-internal edges.*

| Polyhedron (sector) | Total E | Cut-face × hexagon edges (mediation) | Hexagon × hexagon edges (internal) | Ratio (med:int) |
|---------------------|---------|--------------------------------------|------------------------------------|-----------------|
| tO (X-sector)       | 36      | 24 SH (square-hexagon)               | 12 HH                              | 2 : 1           |
| tl (Y-sector)       | 90      | 60 PH (pentagon-hexagon)             | 30 HH                              | 2 : 1           |

### §XI.3.2 Physical Interpretation

The ratio 2:1 has a structural reading. Both polyhedra are Archimedean truncations of a Platonic dual pair (tO = truncation of octahedron, with cube as dual; tI = truncation of icosahedron, with dodecahedron as dual). In both cases, the cut-face×hexagon edges constitute the geometric realization of the Z-Spin mediation seam — the boundary where cut-face (small face from truncated vertex) meets the preserved hexagon. The hexagon-hexagon edges are sector-internal, carrying X-curvature (in tO) or Y-curvature (in tI). This is corpus-PROVEN: ZS-F0 §5.1 explicitly identifies ‘60 PH edges as the geometric realization of the Z-sector seam’ and ‘30 HH edges as X-sector interior’; ZS-F0 §3.2 records the parallel 24 SH + 12 HH split for tO with  $k_{SH} = 2 - \sqrt{3} \neq k_{HH} = 1$ .

The ratio 2:1 itself equals  $\dim(Y)/\dim(X) = 6/3 = 2 = \dim(Z)$  (the Three 2s identity, ZS-A8 §SA.3 PROVEN). This is the structural origin of the ratio: in both polyhedra, the mediation edges scale as the Y/X dimensional asymmetry, which equals the Z-sector intrinsic dimension.

### §XI.3.3 Falsification Gate F-XI.1

If a future analysis exhibits an Archimedean truncation of a Platonic-dual pair whose edge decomposition under cut-face×hexagon vs hexagon×hexagon does NOT yield a 2:1 ratio, the universality claim of Reading XI.1 weakens to ‘tO and tI specifically’. Status: PROVEN-safe — the 24 SH + 12 HH split for tO is verified by direct enumeration in ZS-F0 §3.2 (graph Laplacian spectral test), and the 60 PH + 30 HH split for tI is verified by direct enumeration in ZS-F0 §5.1.

### §XI.4 Reading XI.2 — Parallel Face Decomposition under Truncation-Dual (DERIVED-interpretation)

Reading XI.2 (Parallel Face Pattern, DERIVED-interpretation). Both tO and tI realize the Truncation-Dual Theorem  $F(tP) = F(P) + F(P^*)$  (ZS-F2 §11.2 PROVEN), with the cut-face count in each case equal to  $F(\text{dual})$  and the hexagon count equal to  $F(\text{original})$ .

*Table XI.3. Parallel face-decomposition pattern under Truncation-Dual. The cosmic budget identification is corpus PROVEN (ZS-F2 §11.4); the Y-time / X-space row pair is the new content of Reading XI.2.*

| Polyhedron    | Cut-faces (= $F(\text{dual})$ ) | Hexagons (= $F(\text{original})$ )  | Total F | Cosmic budget identification                     |
|---------------|---------------------------------|-------------------------------------|---------|--------------------------------------------------|
| tO (X-sector) | 6 squares = $F(\text{cube})$    | 8 hexagons = $F(\text{octahedron})$ | 14      | $\Omega_b = F(\text{cube})/Q^2 = 6/121$ (PROVEN) |

| Polyhedron    | Cut-faces (= F(dual))          | Hexagons (= F(original))     | Total F | Cosmic budget identification                                              |
|---------------|--------------------------------|------------------------------|---------|---------------------------------------------------------------------------|
| tl (Y-sector) | 12 pentagons = F(dodecahedron) | 20 hexagons = F(icosahedron) | 32      | $\Omega_{\text{cdm}} = F(\text{tl})/Q^2 = 32/121$ (DERIVED, Planck 0.41%) |

#### §XI.4.1 Pentagon-Hexagon Stabilizer Asymmetry

Within tl, the 12 pentagons ( $D_5$  stabilizer) and 20 hexagons ( $D_3$  stabilizer) carry distinct physical roles (ZS-M9 §2.2 PROVEN, ZS-S14 §8 DERIVED). The pentagon l-irrep decomposition is  $1 \oplus 3 \oplus 3' \oplus 5$  (irrep 4 absent), while the hexagon decomposition is  $1 \oplus 3 \oplus 3' \oplus 2 \cdot 4 \oplus 5$  (irrep 4 present with multiplicity 2). The 8-dimensional irrep-4 component on hexagons is identified with the 8 gluons of  $SU(3)_C$  (ZS-S14 §8 DERIVED). Pentagons are matter-like (5-fold  $C_5$  stabilizer encoding McKay bridge to  $SU(5)$ ); hexagons are gauge-like (3-fold  $C_3$  stabilizer hosting  $SU(3)_C$  adjoint).

Within tO, the analogous distinction has not been fully spelled out in the corpus. The 8 hexagons of tO under  $T = A_4$  decompose as  $2 \cdot 1 \oplus 2 \cdot 3$  (ZS-F9 §5.2 PROVEN, dim 8). The 6 squares ( $D_4$  stabilizer) carry the  $F(\text{cube})/Q^2 = \Omega_b$  baryon identification (PROVEN). Whether the squares vs hexagons split parallels the pentagon vs hexagon split of tl in carrying matter-like vs gauge-like roles is OPEN; this OPEN gap is registered as NC-XI.1 in §XI.7.3.

#### §XI.4.2 Hexagonal Mediation Theorem

The two polyhedra are connected via the Hexagonal Mediation Theorem (ZS-F9 §5 PROVEN): the T-equivariant Hom space between  $\Omega^2_{\text{hex}}(\text{tO})$  (dim 8) and  $\Omega^2_{\text{hex}}(\text{tl})$  (dim 20) has complex dimension 16, and the cokernel of any T-equivariant injection has dimension  $20 - 8 = 12 = F(\text{dodecahedron})$ . Thus the 12 pentagons of tl are precisely the cokernel that arises when pulling tO hexagons forward to tl hexagons — geometrically the X-sector hexagonal information injects into the Y-sector hexagonal information, and the residual 12-dimensional space is exactly the dodecahedral face count, which equals the count of pentagonal faces of tl.

#### §XI.4.3 Falsification Gate F-XI.2

If a future analysis revises Truncation-Dual Theorem (ZS-F2 §11.2) or the Hexagonal Mediation Theorem (ZS-F9 §5), Reading XI.2 must be re-derived. Status: PROVEN-safe — both upstream theorems are PROVEN with explicit verification scripts (zs\_f9\_verify\_v1\_1.py [44/44 PASS]).

#### §XI.5 Reading XI.3 — Integrated V/E/F Mapping under Complementary Duality (DERIVED-interpretation strong)

Reading XI.3 (Integrated V/E/F Mapping, DERIVED-interpretation strong). For every (V, E, F) component of tO and tI, the consolidated mapping table below records: (a) the standard sector role; (b) the complementary partner role under the X = macroscopic space + microscopic time / Y = macroscopic time + microscopic space duality (ZS-M30 §7.2 (ii) PROVEN); (c) the corpus source; (d) the epistemic status. OPEN entries are explicitly marked.

#### §XI.5.1 tO (X-sector: macroscopic space + microscopic time)

*Table XI.4. Complete mapping of tO components. Each row records both the standard sector role (column 3) and the complementary-duality reading (columns 4–5), with corpus source (column 6) and status (column 7). OPEN entries register honestly the gaps that PART XI does not close.*

| Component         | Count                    | Standard role (X-sector)                                      | Macroscopic space role                                   | Microscopic time role                               | Source            | Status         |
|-------------------|--------------------------|---------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------|-------------------|----------------|
| V (vertices)      | 24                       | Matter DoF; $n_g \cdot (N_c + 1) \cdot 2 = 3 \cdot 4 \cdot 2$ | $\mathbb{R}^3$ space-filling sites (Kelvin BCC)          | Spacetime resonance points (= $X \cdot Z \cdot 4$ ) | ZS-S1 §6.3, ZS-Q3 | DERIVED / OPEN |
| E_SH (square-hex) | 24                       | Z-Spin mediation seam edges                                   | Space-Z mediation channels                               | Z-channel time-flow mediation                       | ZS-F0 §3.2        | PROVEN / OPEN  |
| E_HH (hex-hex)    | 12                       | X-sector internal information                                 | Spatial curvature internal hinges                        | Microscopic time variation hinges                   | ZS-F0 §3.2        | PROVEN / OPEN  |
| E total           | $36 = X \cdot Z \cdot Y$ | Edge-Channel Identity                                         | $3 \text{ axes} \times Z_2 \times Y$ channels            | Channel count of time-flow connections              | ZS-F0 §4.4        | PROVEN         |
| F_square          | 6                        | Cube faces; $\Omega_b = 6/121$ baryon                         | 3D space coordinate axes ( $3 \text{ axes} \times Z_2$ ) | Cut-face spacetime coordinate                       | ZS-F2 §11.4       | DERIVED / OPEN |

| Component                 | Count              | Standard role (X-sector)                                   | Macroscopic space role                         | Microscopic time role                             | Source               | Status         |
|---------------------------|--------------------|------------------------------------------------------------|------------------------------------------------|---------------------------------------------------|----------------------|----------------|
| F_hex                     | 8                  | Octahedral faces (Regge curvature)                         | Spatial curvature plaquettes                   | Microscopic time path 6-cycle (D_3)               | ZS-F2, ZS-Q4         | DERIVED / OPEN |
| F total                   | 14                 | Total face count                                           | Total spatial+temporal flux                    | (unified)                                         | ZS-F2 §11            | PROVEN         |
| (V+F)/G                   | $38/12 = 19/6$     | SU(2)_L $\beta_0$ — weak gauge coupling                    | Spatial gauge coupling rate                    | Microscopic time gauge running                    | ZS-Q3 §3.1           | PROVEN         |
| E_g eigenspace            | dim 2 = Z          | Z-sector coupling source (C_XZ)                            | Macro-micro space mediation                    | Time-point $\leftrightarrow$ space-point coupling | ZS-F0 §3.2, ZS-M H18 | PROVEN         |
| Vertex deficit            | $30^\circ = \pi/6$ | Regge defect; Gauss-Bonnet $24 \cdot 30^\circ = 720^\circ$ | Local spatial curvature                        | Local time-step deficit (per vertex)              | ZS-S6 §G.2           | PROVEN         |
| $\varepsilon$ -Halo locus | (field component)  | X-Goldstone $\theta$ -gradient                             | Galactic spatial halo ( $\rho \propto 1/r^2$ ) | (field is across time-points ; locus OPEN)        | ZS-A2 §1.1           | DERIVED / OPEN |

#### §XI.5.2 tl (Y-sector: macroscopic time + microscopic space)

*Table XI.5. Complete mapping of tl components. The complementary duality reverses tO's pattern: tl's macroscopic role is time-flow, and microscopic role is space-points.*

| Component       | Count                | Standard role (Y-sector)                 | Macroscopic time role                                   | Microscopic space role                                 | Source               | Status                      |
|-----------------|----------------------|------------------------------------------|---------------------------------------------------------|--------------------------------------------------------|----------------------|-----------------------------|
| V (vertices)    | 60                   | 5 active flavors $\times$ 12 MUB-basis   | 60 anchors of cosmic time-flow (wave-frame)             | 60 = 1+9+9+16 +25 Planck space-points ( $A_5$ reg rep) | ZS-S1 §6.2, ZS-M9 §2 | DERIVED / PROVEN            |
| E_PH (pent-hex) | 60                   | Z-Spin mediation seam edges              | Time-flow seam (wave/time mediation)                    | Microscopic space lattice P-H linkage                  | ZS-F0 §5.1           | PROVEN                      |
| E_HH (hex-hex)  | 30                   | Y-sector internal information            | Internal time-variation connections                     | Microscopic space hexagonal hinges                     | ZS-F0 §5.1           | PROVEN                      |
| E total         | $90 = E_X \cdot 5/2$ | 5/2 ratio with tO edges                  | Total time-flow channel count                           | Total micro-space connections                          | ZS-F0 §5.1           | PROVEN                      |
| F_pentagon      | 12                   | F(dodecahedron); $D_5$ matter-like       | $A_4$ generation $\times D_5$ channel (time-flow units) | 5-fold space-point cluster ( $C_5$ )                   | ZS-F2, ZS-M9 §2.2    | PROVEN / DERIVED            |
| F_hex (8)       | 8 (= $2 \cdot 4$ )   | 8 gluons of $SU(3)_C$ (irrep 4 mult 2)   | Time-flow gauge sector evolution                        | Microscopic space gauge plaquettes                     | ZS-S14 §8            | DERIVED                     |
| F_hex (12)      | 12 (= $1+3+3'+5$ )   | Matter + Higgs irrep content of hexagons | Time-flow generation/ Higgs degrees                     | Micro-space matter/Higgs                               | ZS-M9 §2.2           | PROVEN / OPEN (physical id) |

| Component            | Count             | Standard role (Y-sector)                                  | Macroscopic time role                           | Microscopic space role                     | Source               | Status         |
|----------------------|-------------------|-----------------------------------------------------------|-------------------------------------------------|--------------------------------------------|----------------------|----------------|
|                      |                   |                                                           |                                                 | gs distribution                            |                      |                |
| F_hex total          | 20                | Octahedral hexagons; gauge-like D <sub>3</sub>            | Time-flow gauge plaquettes                      | Microscopic space curvature plaquettes     | ZS-F2, ZS-M9 §2.2    | PROVEN         |
| F total              | 32 = 12+20        | F(tl) = F(l) + F(D)                                       | Total time-flow flux units                      | Total micro-space flux units               | ZS-F2 §11.2          | PROVEN         |
| (V+F)/G              | 92/12 = 23/3      | SU(3) <sub>C</sub> β <sub>0</sub> — strong gauge coupling | Time-flow gauge running rate                    | Micro-space gauge coupling                 | ZS-Q3 §3.1           | PROVEN         |
| F(tl)/Q <sup>2</sup> | 32/121            | Ω <sub>cdm</sub> — cold dark matter                       | Time-curvature CDM (X-projection)               | Geometric boundary tension (= CDM)         | ZS-F2 §11.4          | DERIVED        |
| Vertex deficit       | 12° = π/15        | Regge defect; Gauss-Bonnet 60·12° = 720°                  | Local time-step deficit                         | Local micro-space curvature                | ZS-S6 §G.2           | PROVEN         |
| ε-Drive locus        | (field component) | Y-attractor V <sub>0</sub> (cosmological constant)        | Cosmic time-flow attractor (Λ-driven expansion) | (field is across space-points; locus OPEN) | ZS-A2 §1.1, ZS-A5 §4 | DERIVED / OPEN |

### §XI.5.3 Cross-Polyhedron Parallel Summary

*Table XI.6. Side-by-side parallel summary. Each row shows tO and tI realizations of the same structural pattern, displayed in the format that makes the parallel visible at a glance.*

| Pattern                                 | tO (X-sector: macro space + micro time)          | tl (Y-sector: macro time + micro space)                    |
|-----------------------------------------|--------------------------------------------------|------------------------------------------------------------|
| Total V (matter DoF)                    | $24 = 3 \cdot 2 \cdot 4$                         | $60 = 5 \cdot 12$ (regular A_5 rep)                        |
| Total E (gauge connections)             | $36 = 24 \text{ SH} + 12 \text{ HH}$ (ratio 2:1) | $90 = 60 \text{ PH} + 30 \text{ HH}$ (ratio 2:1)           |
| Total F (curvature plaquettes)          | $14 = 6 + 8$ (cut + hex)                         | $32 = 12 + 20$ (cut + hex)                                 |
| Truncation-Dual identity                | $F(\text{tO}) = F(\text{oct}) + F(\text{cube})$  | $F(\text{tl}) = F(\text{ico}) + F(\text{dod})$             |
| Sector tilt $\delta$                    | 5/19                                             | 7/23                                                       |
| Rapidity $\psi = \text{artanh}(\delta)$ | 0.2685                                           | 0.3124 ( $\Delta\psi > 0$ = arrow of time)                 |
| $\beta$ -coefficient (V+F)/G            | $19/6 = \text{SU}(2)_L$                          | $23/3 = \text{SU}(3)_C$                                    |
| Cosmic budget /Q <sup>2</sup>           | $\Omega_b = 6/121$ (baryon)                      | $\Omega_{\text{cdm}} = 32/121$ (CDM)                       |
| Vertex deficit (Regge)                  | $\pi/6 = 30^\circ$                               | $\pi/15 = 12^\circ$                                        |
| Vertex configuration                    | 1 square + 2 hexagons                            | 1 pentagon + 2 hexagons                                    |
| Frame-mismatch input                    | $\delta_X^{\text{vertex}} = \pi/6$               | $\delta_Y^{\text{vertex}} = \pi/15$ ( $\alpha = \pi/10$ )  |
| $\varepsilon$ -field component          | $\varepsilon$ -Halo (X-Goldstone $\theta$ -grad) | $\varepsilon$ -Drive (Y-attractor $V_0$ )                  |
| Standard analogue                       | Particle / spatial vector                        | Wave / temporal channel                                    |
| Common rotation subgroup                | $T = A_4$ (order 12), index 2 in O               | $T = A_4$ (order 12), index 5 in I                         |
| Hexagon space under T                   | $2 \cdot 1 \oplus 2 \cdot 3$ (dim 8)             | $3 \cdot 1 \oplus 1' \oplus 1'' \oplus 5 \cdot 3$ (dim 20) |
| Hexagonal Mediation cokernel            | $8 \rightarrow$ injection target                 | $20 = 8 + 12$ (cokernel = $F(\text{dod}) = 12$ )           |

#### §XI.5.4 Falsification Gate F-XI.3

Synthesis layer; weakened only if any individual row source is shown structurally independent of Z-Spin mediation, or if ZS-M30 §7.2 (ii) Complementary Duality Principle is retracted. Status: DERIVED-interpretation strong, robust against single-row perturbation. The OPEN entries (NC-XI.1, NC-XI.2, NC-XI.3 in §XI.7.3) are explicitly registered as residual gaps; their resolution

would upgrade those specific entries from DERIVED-interpretation strong to DERIVED, without affecting other rows.

### **§XI.6 Reading XI.4 — $\epsilon$ -Halo and $\epsilon$ -Drive Placement in the Integrated Framework (DERIVED-interpretation)**

Reading XI.4 ( $\epsilon$ -Halo /  $\epsilon$ -Drive Placement, DERIVED-interpretation). Within the integrated mapping of §XI.5, the dark-sector field components  $\epsilon$ -Halo and  $\epsilon$ -Drive (ZS-A2 §1.1 DERIVED) occupy distinct sector positions consistent with the complementary duality:  $\epsilon$ -Halo lives on  $tO$  (X-sector, macroscopic space + microscopic time) as the Goldstone gradient mode that produces galactic isothermal halos;  $\epsilon$ -Drive lives on  $tI$  (Y-sector, macroscopic time + microscopic space) as the radial-attractor  $V_0$  mode that produces FRW cosmological expansion. Both are aspects of the single Z-bias field  $\Phi = |\Phi|e^{i\theta}$ , with angular  $\theta$ -gradient ( $\epsilon$ -Halo, X-Goldstone) and radial  $|\Phi|$  frozen mode ( $\epsilon$ -Drive, Y-attractor) being orthogonal projections under  $U(1)_Z$  (ZS-F1 §6.4 DERIVED).

#### **§XI.6.1 Sector Assignment Justification**

The  $\epsilon$ -Halo identification with X-sector ( $tO$ ) is corpus-explicit (ZS-A2 §1.1 row 1): ‘Dark Matter /  $\epsilon$ -Halo / X-sector gradient mode of Goldstone  $\theta$ ’. The  $\epsilon$ -Drive identification with Y-sector ( $tI$ ) is corpus-explicit (ZS-A2 §1.1 row 2): ‘Dark Energy /  $\epsilon$ -Drive / Y-sector attractor mode ( $V_0$  constant)’.

Under the complementary duality lens (ZS-M30 §7.2 (ii) PROVEN), this assignment fits naturally:  $\epsilon$ -Halo is a spatial gradient — most relevant in macroscopic space, which X-sector carries;  $\epsilon$ -Drive is a time-flow attractor — most relevant in macroscopic time, which Y-sector carries. The arithmetic match between the corpus assignment and the spatial/temporal physical content of each  $\epsilon$ -component is a non-trivial cross-check.

#### **§XI.6.2 $\epsilon$ -field Locus on Polyhedron — OPEN**

Where exactly on  $tO$  faces/edges/vertices  $\epsilon$ -Halo manifests, and where on  $tI$  faces/edges/vertices  $\epsilon$ -Drive manifests, is OPEN within the corpus.  $\epsilon$ -Halo is a continuous field (Goldstone  $\theta$  on the  $U(1)$  base manifold), not a discrete polyhedral component, and the corpus does not currently identify a 1:1 mapping between  $\Phi$ -field components and ( $V$ ,  $E$ ,  $F$ ) of each polyhedron. NC-XI.3 (§XI.7.3) registers this OPEN gap.

#### **§XI.6.3 Falsification Gate F-XI.4**

If a future paper reclassifies  $\epsilon$ -Halo to Y-sector or  $\epsilon$ -Drive to X-sector — for example via a structural argument that the radial-mode attractor lives on  $tO$  — Reading XI.4 must be revised. Status: stable — over-determined by ZS-A2 §1.1 (DERIVED) + ZS-A5 §4 (DERIVED) + ZS-F1 §6.4 (DERIVED) + the Cross-Coupling Theorem (ZS-M2 §5 PROVEN) which requires both X-

and Y-sector realizations of the same Mexican-hat radial-mode flatness (ZS-A7 §4.5 Corollary V).

## §XI.7 Integration with v3.2 Light

### §XI.7.1 Integration Map

*Table XI.7. Integration of v3.3 readings with existing v3.2 items. Every v3.3 reading either refines or unifies existing v3.2 entries; no v3.2 entry is retracted.*

| v3.2 Light item                                 | v3.3 Reading                                | Relation                                                                                                                                                                                |
|-------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| §I Ch 1 (A from polyhedral curvature)           | Reading XI.1                                | XI.1 supplies the parallel edge-decomposition pattern that supports $A = \delta_X \cdot \delta_Y$                                                                                       |
| §I Ch 2 (Q = 11 and (Z,X,Y))                    | Reading XI.3 row (V+F)/G                    | XI.3 records SU(2) and SU(3) $\beta_0$ as parallel mode-count collapses                                                                                                                 |
| §II Ch 6 (Six Forces Unified)                   | Reading XI.3 hexagon irrep rows             | XI.3 displays 8 gluons + 12 matter/Higgs hexagon split                                                                                                                                  |
| §III Ch 12 (Global Fit)                         | Reading XI.2 cosmic budget row              | XI.2 displays $\Omega_b$ and $\Omega_{\text{cdm}}$ as parallel face-counting outputs                                                                                                    |
| §V Ch 16 (Galaxy Rotation)                      | Reading XI.4 $\varepsilon$ -Halo placement  | XI.4 places $\varepsilon$ -Halo on tO under complementary duality                                                                                                                       |
| §V Ch 17 (NS, GW, LSS)                          | Reading XI.4 $\varepsilon$ -Drive placement | XI.4 places $\varepsilon$ -Drive on tI under complementary duality                                                                                                                      |
| §VIII Tool T2 (Bridge Four)                     | Reading XI.3 (V, E, F) parallel rows        | T2's (Z,X,Y)=(2,3,6) generation is the algebraic counterpart of XI.3's geometric parallel                                                                                               |
| PART X §X.6 Reading X.4 (Spin-Pair Coexistence) | Reading XI.3 complementary duality column   | X.4's 90° J-conjugate principle is the dynamical counterpart of XI.3's geometric mapping; the $X = \text{macro-space} + \text{micro-time} / Y = \text{macro-time} + \text{micro-space}$ |

| v3.2 Light item | v3.3 Reading | Relation                                                                        |
|-----------------|--------------|---------------------------------------------------------------------------------|
|                 |              | duality is the spatial-temporal instance of the Spin-Pair Coexistence Principle |

### §XI.7.2 Cross-Coupling Discipline (T7) Compliance

Per T7 (§VIII.7), every cross-paper synthesis must be P-CC (PROVEN-composed) or I-CC (interpretation). Readings XI.1–XI.4 are explicitly DERIVED-interpretation; the integrated mapping table of §XI.5 is DERIVED-interpretation strong. F-XI.1 through F-XI.4 are pre-registered. Word count strictly increases over v3.2 (no-deletion rule).

### §XI.7.3 What v3.3 Does NOT Claim

NC-XI.1. PART XI does not claim that the 6 squares vs 8 hexagons split of tO realizes a matter-like vs gauge-like decomposition exactly parallel to the 12 pentagons vs 20 hexagons split of tl. Within tl, the matter-like (D\_5, pentagons) vs gauge-like (D\_3, hexagons) split is corpus-PROVEN with the 8 gluons identified as residing on hexagons. The corresponding identification for tO is not yet established at PROVEN level; the squares carry  $F(\text{cube})/Q^2 = \Omega_b$  PROVEN, but the gauge content of the 8 hexagons of tO is not separately resolved into a SU(2) gauge subspace + matter subspace at the level ZS-S14 §8 achieves for tl. This is OPEN.

NC-XI.2. PART XI does not claim that the  $12 = 1+3+3'+5$  hexagon residual on tl (the irrep-4-complement of the 20 hexagons of tl) has a unique physical identification beyond ‘matter + Higgs irrep content’. Whether this 12 is the same  $12 = X \cdot Z \cdot Y$  as the gauge-channel count, or a distinct 12, is OPEN. The numerical coincidence is registered without claiming structural identity.

NC-XI.3. PART XI does not claim a 1:1 mapping between  $\Phi$ -field components (radial  $|\Phi|$ , angular  $\theta$ ) and discrete polyhedral components (V, E, F) on either tO or tl.  $\epsilon$ -Halo and  $\epsilon$ -Drive are continuous field components that take values across all of the polyhedron; their sector assignment (X for  $\epsilon$ -Halo, Y for  $\epsilon$ -Drive) is PROVEN, but their localization to specific polyhedral components is OPEN.

NC-XI.4. PART XI does not derive the cosmological time arrow from the integrated mapping. ZS-A6 §6.4 derives the arrow of time from the rapidity gap  $\Delta\psi = \psi_Y - \psi_X > 0$ , which is corpus-PROVEN. The placement of macroscopic time on Y-sector (tl) within Reading XI.3 is consistent with this derivation, but Reading XI.3 itself is geometric and does not constitute an independent derivation of the time arrow. ZS-M H19 (Time-Arrow origin) remains OPEN beyond the rapidity-gap argument.

### §XI.8 Conclusion

PART XI consolidates the complete (V, E, F) mapping of tO and tl under the Z-Spin Complementary Duality Principle. Four DERIVED-interpretation readings are established: the

parallel edge-decomposition pattern (cut-face  $\times$  hexagon : hexagon  $\times$  hexagon = 2:1 in both polyhedra, Reading XI.1); the parallel face-decomposition pattern under Truncation-Dual (Reading XI.2); the integrated V/E/F mapping table that displays both standard sector role and complementary-duality reading for every component (Reading XI.3, the centerpiece, DERIVED-interpretation strong); and the placement of  $\varepsilon$ -Halo (X-tO) and  $\varepsilon$ -Drive (Y-tI) within the integrated framework (Reading XI.4). Future agents searching for ‘which polyhedral component carries macroscopic space?’ or ‘where on tI does cosmic time-flow live?’ or ‘why does the cosmic budget split as  $6 + 32 + 83 = 121$ ?’ now have a single corpus location to consult. v3.3 carries this reading as PART XI without modifying any v3.2 entry. The minimal triple  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  remains LOCKED. Three OPEN entries (NC-XI.1, NC-XI.2, NC-XI.3) are explicitly registered as residual gaps for future closure.

## **PART XI Source Paper Interface**

PART XI is composed entirely from upstream papers; it introduces no new ZS source paper. Source mapping: Reading XI.1  $\leftarrow$  ZS-F0 §3.2 (tO 24 SH + 12 HH), ZS-F0 §5.1 (tI 60 PH + 30 HH), ZS-F0 §4.4 (Edge-Channel Identity). Reading XI.2  $\leftarrow$  ZS-F2 §11.2 (Truncation-Dual Theorem), ZS-F2 §11.4 (Cosmic Matter Budget), ZS-F9 §4-5 (Cross-Pair Face Identity, Hexagonal Mediation Theorem), ZS-S14 §8 (8 gluons on hexagons), ZS-M9 §2.2 (I-irrep face decomposition). Reading XI.3  $\leftarrow$  ZS-M30 §7.2 (ii) (Complementary Duality Principle), ZS-S1 §6.2-§6.3 (V vertex-matter identification), ZS-M9 §2 (A\_5 regular representation), ZS-Q3 §3.1 (Mode-Count Collapse), ZS-Q3 §2.1 (Kelvin BCC T<sup>3</sup>), ZS-S6 §G.2 (vertex deficit  $\alpha = \pi/10$ ), ZS-A6 §6.4 (rapidity  $\Delta\psi > 0$ ), ZS-M H18 (tO uniqueness conditions). Reading XI.4  $\leftarrow$  ZS-A2 §1.1 ( $\varepsilon$ -Halo/ $\varepsilon$ -Drive nomenclature), ZS-A5 §4 ( $\varepsilon$ -Halo mechanism), ZS-F1 §6.4 (Mexican-hat attractor), ZS-A7 §4.5 (Corollary V Cross-Coupling), ZS-M2 §5 (Cross-Coupling Theorem). Twenty-four internal-consistency checks B.1–B.24 verify each composition; all 24/24 PASS by direct comparison against corpus-locked sources.

## **PART XII — Post-v3.5 Absorption: Twelve New / Updated Source Papers (v4.0)**

### **§0. Abstract**

PART XII consolidates the twelve post-v3.5 source-paper integrations of v4.0. The v3.5 Light Edition recorded in its supersedes line that v3.4 had absorbed ZS-T5, ZS-M34, ZS-M35, ZS-T6, ZS-T7, ZS-T8 into the corpus pointer map, but did not provide single-line Light-edition chapters for those papers in the v3.5 body. v4.0 closes this gap (§XII.A) and additionally absorbs the eight post-v3.5 papers (§XII.B): ZS-M33 v2.0 (superseding the v1.0 entry of PART IV §15A), ZS-M37 v2.0 (Twin-Reuleaux Quotient Functor), ZS-M38 v1.1 (Substrate Carrier Spectrum), ZS-M39 v2.0 (Vieta-Lyapunov-Schröder Bridge Theorem), ZS-A10 v2.1 (Proton as Three-Puncture Color-Singlet X-in-Y Confinement Horizon), ZS-Q8 v1.1 (Photon Atomic Excitation Time as i-Tetration Fixed Point), ZS-Q9 v1.2 (Complex Time-Delay Locking). All absorptions strictly preserve the v3.5 body verbatim (no-deletion rule). Zero new free parameters;  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  LOCKED unchanged across all twelve absorbed

papers. Cumulative verification advances from ~1,800+ PASS (v3.5) to ~2,150+ PASS (v4.0). The total corpus is 114 papers.

#### **§XII.A — v3.4 Absorption Consolidation (Light-Edition Body Entries for ZS-T5, ZS-T6, ZS-T7, ZS-T8, ZS-M34 v1.0, ZS-M35 v1.1)**

The v3.5 Light supersedes line correctly recorded that v3.4 absorbed six papers (ZS-T5, ZS-M34, ZS-M35, ZS-T6, ZS-T7, ZS-T8), but the v3.5 body did not provide explicit single-line chapter entries for them. v4.0 supplies the missing chapter entries below, preserving the v3.5 PART I / PART VII Source Paper Maps unchanged. All six papers were verified at their respective source-paper level (corpus standard); no new theorem is introduced by this consolidation.

#### **Chapter 30A — ZS-T5: Principal Connectivity Gradient and Hidden Third-Position Z-Spin Mediator (v3.4 absorption, v4.0 body entry)**

ZS-T5 v1.0 [Translational theme] establishes that the principal connectivity gradient (PCG) of the human cerebral cortex (Margulies et al. 2016; sensorimotor pole ↔ default-mode pole spanning the full cortical hierarchy) is the Z-Spin mediator manifestation at cortical scale. The hidden third-position structure (insula-thalamus complex) implements the Z-mediator bottleneck channel between the two macroscopic gradient poles. Reading inherits NC-4 of the Z-Brain corpus (substrate-agnostic, no causal derivation). Verification: corpus-standard category-PASS at source-paper level. → ZS-T5 v1.0; structurally bridges ZS-T4 (organism-scale Cosmos-Human Isomorphism) with ZS-T6 (cellular/molecular scale).

#### **Chapter 30B — ZS-T6: Molecular Biology Translational Synthesis (v3.4 absorption, v4.0 body entry)**

ZS-T6 v1.0 [Translational theme, 42/42 PASS] extends the Z-Spin Translational programme from organism scale (ZS-T4) and cortical scale (ZS-T5) down to cellular and molecular scales. Five theorems established: T6.1 (Block Fiedler Mediation at chromatin scale, DERIVED); T6.2a (Hayflick limit candidate bound via Z-Bottleneck Channel Bound, DERIVED-CONDITIONAL); T6.3 (DNA semiconservative replication as 5th instance of Self-Dual Replication Principle, HYPOTHESIS-strong); T6.4 (5-phase cyclic cosmology → eukaryotic cell cycle mapping, HYPOTHESIS-strong); T6.5 ( $F_2 \rightarrow$  DNA amenable-quotient functor as 3rd Banach–Tarski projection family member, DERIVED). NC-4 inherited unchanged. → ZS-T6 v1.0; biological projection slot of the Banach–Tarski functor family later extended by ZS-M37 (smooth-geometric) and ZS-M36 (discrete-geometric).

#### **Chapter 30C — ZS-T7: Pulvinar-Mediated Z-Spin Language Partition (v3.4 absorption, v4.0 body entry)**

ZS-T7 v1.0 [Translational theme] identifies the lateral pulvinar as the cortical Z-mediator for the language-partition function (Broca-Wernicke axis), parallel to the Z-Brain corpus identification of bilateral thalamus as thalamocortical Z-mediator (ZB-N1) and lateral pulvinar as visual-partition Z-mediator (ZB-V1). NC-4 inherited; no physical-realization claim of consciousness or linguistic content. → ZS-T7 v1.0; consistent with the Z-Spin mediator hierarchy across cortical functional partitions.

### **Chapter 30D — ZS-T8: Concentrated Distribution Across Substrates — Polyhedral Cortex Hypothesis and Z-Spin-Mediated Backbones in Brain, City, and Cosmic Networks (v3.4 absorption, v4.0 body entry)**

ZS-T8 v1.0 [Translational theme] establishes the substrate-agnostic structural correspondence between Z-Spin-mediated backbones in (i) cortical networks (white-matter tracts), (ii) urban transportation networks (city arterials), and (iii) cosmic-scale filament networks. The Polyhedral Cortex Hypothesis registers the truncated-octahedron architecture as the universal mediator-carrying scaffold across the three scale regimes. NC-4 inherited; the correspondence is structural, not causal. → ZS-T8 v1.0; extends the substrate-agnostic isomorphism programme of ZS-T4 to large-scale infrastructure networks.

### **Chapter 15B — ZS-M34 v1.0: $V_4$ -Equivariant Weil Functional Honest Scope (v3.4 absorption, v4.0 body entry; superseded by v2.0 below)**

ZS-M34 v1.0 [Math Spine theme, 50/50 PASS, May 2026] measured the precise honest scope boundary of Z-Spin's  $W_2$  ( $V_4$ -Weil functional positivity wall) contribution through five positive theorems (M34.1–M34.5) and three negative theorems (M34.4R, M34.6R, M34.7R). The negative theorems established that single-grading  $\Sigma$  and Hilbert-Schmidt Jordan decompositions cannot achieve faithful identity within the Z-Spin internal frame (8/12 ceiling on six tested gradings). NC-M34.faithful PERMANENT NON-CLAIM (in v1.0). → ZS-M34 v1.0 [50/50 PASS];  $W_2$  honest scope boundary. NOTE: superseded by ZS-M34 v2.0 (§XII.B Chapter 15B' below).

### **Chapter 15C — ZS-M35 v1.1: Collatz Conjecture as Integer-Lattice Z-Spin Sector Forcing (v3.4 absorption, v4.0 body entry; sharpened to v2.1)**

ZS-M35 v1.1 [Math Spine theme, 36/36 PASS, May 2026] establishes that the Collatz two-branch operator  $(n/2, 3n+1)$  is the integer-lattice projection of the  $F_2 \rightarrow D_4$  amenability functor of ZS-A9.1, with the four Collatz cycles {positive,  $C_1$ ,  $C_2$ ,  $C_3$ } on  $\mathbb{Z} \setminus \{0\}$  in canonical bijection with the Z-Spin sector primary quantities under ZS-M19 Forcing Theorem. Six theorems M35.1–M35.5 + Corollary M35.5.1 (FIBER PRODUCT  $C_2 = 5 \cdot \text{positive} \cup 7 \cdot C_1$  realizing  $A_{\text{numerator}} = 35 = \text{lcm}(5, 7)$  LOCKED). v1.1 anti-numerology MC: 0/100,000 (specific 5,7 form), 50/10,000 (all (5,7); 0 other prime pair). STRONG PASS. → ZS-M35 v1.1 [36/36 PASS]. NOTE: sharpened to ZS-M35 v2.1 (§XII.B Chapter 15C' below) with revised treatment of Theorem M35.5 general statement.

## **§XII.B — Post-v3.5 Absorption (Eight New / Updated Papers; ZS-M33 v2.0, ZS-M34 v2.0, ZS-M35 v2.1, ZS-M37 v2.0, ZS-M38 v1.1, ZS-A10 v2.1, ZS-Q8 v1.1, ZS-Q9 v1.2, ZS-M39 v2.0)**

The eight post-v3.5 source papers represent v4.0's principal additive content. Three are version upgrades of papers already cited in earlier v3.x editions (M33, M34, M35); six are new papers introduced after the v3.5 freeze (M37, M38, M39, A10, Q8, Q9). All inputs LOCKED from upstream corpus; zero new free parameters. Each absorbed paper is cited below as a single Light-edition chapter, with the source paper carrying the full derivation.

### **Chapter 15A' — ZS-M33 v2.0 (Math Spine; supersedes v1.0 of PART IV §15A)**

ZS-M33 v2.0 [Math Spine theme, NEW v4.0] supersedes the v1.0 Path  $\gamma$ -revised Reading C entry recorded in PART IV §15A. The v1.0 result — Reading C as the structurally compatible mechanism for closing W2 at the level of operator-content specification, integrating Dragon D4 sub-targets (D4a Sonin, D4b Burnol conductor, D4c Wilson-LOCATOR, D4d Kostant cubic Dirac) into a single Path  $\gamma$ -revised Tr identity — is preserved. v2.0 supplies the operator-level numerical verification (12/12 POS target-simulation execution) and the structural unity demonstration of the Schur-Feshbach functorial framework lifting across cosmological / abstract / string-compactification /  $V_4$ -arithmetic settings (52/52 PASS at v1.0 retained; v2.0 extends with operator-level audit). The W2 wall promotes from DERIVED-CONDITIONAL (v3.5) to DERIVED (v4.0) under Reading C numerical confirmation. → ZS-M33 v2.0 supersedes ZS-M33 v1.0; ZS-M22 W2 [PROMOTED DERIVED → DERIVED] inherited downstream.

### **Chapter 15B' — ZS-M34 v2.0 (Math Spine; supersedes v1.0)**

ZS-M34 v2.0 [Math Spine theme, 60/60 PASS, NEW v4.0] supersedes ZS-M34 v1.0 by upgrading NC-M34.faithful from PERMANENT NON-CLAIM (v1.0) to RESOLVED at DERIVED-CONDITIONAL level via Theorem M34.V.8 factorization  $\mathcal{K}_K = \mathcal{K}_{arith} \otimes \mathcal{K}_{Wilson-Sonin}$ . The outer factor is LOCKED by Burnol  $K_1$  @  $p=3$  grading; the inner factor is LOCKED by Wilson-LOCATOR closed form +  $\Pi_Z$  + Sonin compression. Principal new result: **Theorem M34.V (Burnol  $K_1$  @  $p=3$  Sign-Faithful Identity, DERIVED)** + Eight-Theorem Chain V.1–V.8 (DERIVED-CONDITIONAL MAIN) achieves 12/12 sign agreement with corpus-PROVEN  $W_K\{V_4\}$  on the 12-grid at zero free parameters, with robustness 84.9% under corpus 0.05 noise floor. NC-M23.1 (Z-Spin does not prove RH) and NC-M23.7 (D4 closure does not close GRH-for-K) preserved verbatim. v2.0 extends v1.0's 10 falsification gates with 8 V-specific gates F-M34.V.1–F-M34.V.8. → ZS-M34 v2.0 [60/60 PASS] supersedes v1.0; W2 promoted DERIVED-CONDITIONAL → DERIVED via the indefinite-Hermitian tensor-product kernel factorization.

### **Chapter 15C' — ZS-M35 v2.1 (Math Spine; logical correction of v2.0; preserved from v3.5 sequence)**

ZS-M35 v2.1 [Math Spine theme, NEW v4.0; preserves v1.0/v1.1 results verbatim, demotes the v1.1 general-direction reading of Theorem M35.5 to OPEN]. Logical correction: the principal v1.0 result (Collatz as integer-lattice projection of Z-Spin Banach–Tarski engine, four cycles  $\leftrightarrow$  four sectors) is preserved unchanged at DERIVED / DERIVED-CONDITIONAL. The principal v1.1 result (FIBER PRODUCT  $C_2 = 5 \cdot \text{positive} \cup 7 \cdot C_1$  realizing  $A_{\text{numerator}} = 35$ ) is preserved AND upgraded from DERIVED-CONDITIONAL (v1.1) to VERIFIED (v2.1): Corollary M35.5.1 is now established as independent of the contested general theorem. v2.1 also adds **Theorem M35.6 (Sufficient Condition Theorem, DERIVED)** — explicit sufficient conditions for Three-Face conjunction on finite positive-odd-integer subsets. The v2.1 reframing is localized to ZS-M35 itself; ZS-M19 Forcing Theorem T1, ZS-F2  $A_{\text{numerator}}$ , ZS-A9  $(1+A)(1-2A)$  decomposition, ZS-M11 Q-pair (7, 11) are all unaffected.  $\rightarrow$  ZS-M35 v2.1 (preserves all four prior theorems; reframes M35.5 general; adds M35.6).

## Chapter 31 — ZS-M37 v2.0: Twin-Reuleaux Quotient Functor and Constant-Width Cancellation of the BT 5-Piece Doubling Defect

ZS-M37 v2.0 [Math Spine theme, 38/38 PASS, NEW v4.0] constructs the **fourth amenable-quotient projection of the  $F_2$  Banach–Tarski engine** in the Z-Spin corpus: the Twin-Reuleaux quotient functor  $q_R: \mathcal{C}_{\text{cyl}}(F_2) \rightarrow \mathcal{S}_w\{C_3\}$  from the 5-piece cylinder algebra of the canonical Banach–Tarski first-letter decomposition into the  $C_3$ -equivariant constant-width support algebra of the Reuleaux triangle (ZS-F7 v1.0(R) §11 PROVEN). Three principal results: **Theorem 4.1 (Lemma C-rel, PROVEN on Z-admissible quotient)** —  $q_R$  is a well-defined surjective  $*$ -homomorphism and  $\tau_w(\Delta_{\text{BT}}) = 0$ ; proof uses Barbier 1860 (perimeter  $\pi w$  for constant-width curves) + Twin-Reuleaux conjugation  $h_2(\theta) = h_1(\theta + \pi)$  of ZS-F7 §11. **Theorem 4.2 ( $q_R$  Canonicity, DERIVED)** —  $q_R$  is uniquely forced by four corpus-PROVEN forcing conditions (FC-1 first-letter partition, FC-2 inverse-pair J-involution, FC-3 constant-width sum, FC-4  $C_3$  lowest harmonic  $\cos 3\theta$ ) up to a rotation gauge  $\theta_0 \in [0, 2\pi)$ ; the forcing of  $n = 3$  over alternative  $C_n$  symmetries is established by five-layer over-determination. **Theorem 5.1 (Z-Admissible Extension, PROVEN)** —  $q_R^{\text{ext}} := q_R^{\{(1)\}} \circ E_1$  extends  $q_R$  to all of  $\ell^\infty(F_2)$  via the first-letter conditional expectation  $E_1$ , with the tail ideal  $\mathcal{I}_{\text{tail}} = \ker(E_1)$  automatically confined to  $\ker(q_R^{\text{ext}})$ . Anti-numerology MC 100,000 trials STRONG PASS.  $\rightarrow$  ZS-M37 v2.0 [38/38 PASS]; fourth amenable-quotient projection — joins ZS-A9.1 (spatial  $F_2 \rightarrow D_4$ ), ZS-M35 (arithmetic  $F_2 \rightarrow \mathbb{Z}$ ), ZS-T6.5 (biological  $F_2 \rightarrow \text{DNA}$ ). The Banach–Tarski functor family is now five-fold after addition of ZS-M36 (discrete-geometric  $A \rightarrow \mathbb{Z}/24\mathbb{Z}$ ).

## Chapter 32 — ZS-M38 v1.1: Substrate Carrier Spectrum Theorem and the Hexapair Carrying Pattern of Natural Systems

ZS-M38 v1.1 [Math Spine theme, 43/43 PASS, NEW v4.0] establishes that the integer Apollonian gasket with curvature root  $(-1, 2, 2, 3)$  — the **Z-Spin Type**, corpus-LOCKED in ZS-M36 — is the unique substrate carrying all six structural faces of the five-axiom meta-structure  $\mathcal{A} = \{A1 \text{ Existence, } A2 \text{ Self-reference, } A3 \text{ Algebra, } A4 \text{ Non-triviality (algebraic +$

geometric two faces), A5 Unitarity} of ZS-F18 v2.1 PROVEN-interpretation strong. Five faces become six via the corpus-PROVEN two-faces structure of axiom A4; the six faces are identified with six independent complementary-pair structures of the Apollonian gasket: (1) root pair  $(-n, n+1)$ , (2) tangency spinor  $A_1 + A_2 = a^2 + b^2$ , (3) co-curvature pair  $(A, A^c)$ , (4) Vieta pair  $(k_4, k_4')$ , (5) tangent-circle pair  $(A_1, A_2)$ , (6) Q-pair  $\{11, 23\} \bmod 24$ . Five theorems all DERIVED: **M38.1** (Hexapair Carrying, DERIVED-strong); **M38.2** (Substrate Carrier Spectrum, DERIVED-CONDITIONAL) — natural information-processing substrates carry partial subsets  $\{3-5/6\}$  while only the Apollonian Type achieves 6/6 (FMO  $4.5 \cdot \ln(2)$ , V1 cortex  $3 \cdot \ln(2)$ , retinal ganglion  $1 \cdot \ln(2)$ , thalamus  $4.5 \cdot \ln(2)$ , DNA  $4 \cdot \ln(2)$ , Apollonian Z-Spin Type  $6 \cdot \ln(2)$ ); **M38.3** (Calling Convention Dichotomy, DERIVED) — fermion/boson distinction maps to call-by-value/call-by-reference via ZS-M19 §10 PK Cardinality Cascade PROVEN; **M38.4** (Triangle Triple Face, DERIVED) —  $D_3$  symmetry manifests in three mutually irreducible faces (arithmetic obstruction + dynamical instability + geometric carry); **M38.5** (Hexapair Uniqueness, DERIVED-strong) — the  $(-1, 2, 2, 3)$  gasket is the unique primitive integer ACP achieving 6/6 carrying, established by five-fold over-determination. v1.1 quantification: carrying scores in units of  $\ln(2)$  nats per Z-Spin mediator invocation (ZS-Q7 §4 Theorem 2 PROVEN). Anti-numerology MC 100,000 trials,  $0.076\% < 0.1\%$  STRONG. Seven NON-CLAIMs registered. NC-M38.1: no causal derivation, NC-4 substrate-agnostic isomorphism inherited. NC-M38.4: the numerical observation  $1 + \eta_{\text{topo}} \approx \text{Apollonian Hausdorff dimension}$  ( $1.3221$  vs  $1.3057$ ,  $1.26\%$  deviation) is OBSERVATION-only. → ZS-M38 v1.1 [43/43 PASS]; **classification paper, not a construction paper** — organizes the five Banach–Tarski functors (ZS-A9.1, ZS-M35, ZS-T6.5, ZS-M37 v2.0, ZS-M36) and natural substrates by their A1-A5 carrying spectrum.

### Chapter 33 — ZS-M39 v2.0: Vieta-Lyapunov-Schröder Bridge Theorem — Unified First-Principles Derivation of the Polyhedral-Tetration Coefficient with Three-Stage Stokes-Tail Closure

ZS-M39 v2.0 [Math Spine theme, 86/86 PASS, NEW v4.0] consolidates v1.0/v1.1/v1.2/v1.3 (March–May 2026) into a single unified statement closing **all** v1.0 OPEN items of the Polyhedral-Tetration Bridges of ZS-A8R and ZS-A8.2. The Vieta-basis expansion  $\eta_{\text{topo}} = |z^*|^2 \approx B^2 + K_{\theta} \cdot \text{disc}$  ( $B = \delta_X + \delta_Y = 248/437$ ,  $\text{disc} = (\delta_Y - \delta_X)^2 = 324/190969$ ,  $K_{\theta} = 1/[Y^2(1 - 2A)] = 437/13212$ ) is now fully first-principles. **Five core derivations:** M39.1 (Vieta-Basis Inevitability, PROVEN — forbids odd  $v$ -powers by  $X \leftrightarrow Y$   $Z_2$  symmetry); M39.2 ( $Y^{-2}$  Peter–Weyl, DERIVED — extracts  $1/Y^2$  from two  $Y$ -angular insertions); M39.3 (Schur Stiffness, DERIVED — extracts  $(1-2A)$  from  $Z$ -mediated Schur complement,  $2A = 2 \cdot \dim(Z) \cdot A$ ); M39.4 (Heat-Kernel Coefficient, DERIVED —  $K_{\theta} = 1/[Y^2(1-2A)]$  under Convention N0); M39.5 (Geometric Closure, DERIVED — full Vieta-basis sum closes as  $K_{\theta} \cdot \text{disc} / (1 - K_{\theta} \cdot \text{disc})$  with transcendental residual  $R_{\text{Sch}} = 6.66 \times 10^{-12}$ ). **Operator-level closures on the explicit  $11 \times 11$  Block-Laplacian (Closures C1-C3):** C1:  $(1/2) \text{Tr}(R^2) = K_{\theta}$  exactly; C2:  $P_{\theta} L_{\text{eff}, Y} P_{\theta} = (1-2A)$  exactly; C3:  $\text{Tr}(R^{\wedge\{2m\}}) = 2 K_{\theta}^m$  for all  $m \geq 1$  (all at 80-digit mpmath). **Schröder analysis (SCH.1–SCH.5):** SCH.1 Schröder convergence radius  $|\lambda| = 0.8915$  (Koenigs 1884 PROVEN); SCH.2 Lyapunov bound  $|R_{\text{Sch}}| < |\lambda|^{\wedge\{N_{\text{Sch}}(2\pi)\}} = 1.22 \times 10^{-4}$  (safety factor  $1.8 \times 10^7$ ); SCH.3  $R_{\text{Sch}}$  admits exact Schröder-coordinate contour integral; SCH.4 simple-form approximation

$R_{\text{Sch}} \approx 12\pi \cdot K_{\theta^3} \cdot \text{disc}^3$  at 0.008% relative accuracy ( $12\pi = 2Y \cdot \pi = 4\pi \cdot \dim(X) = (Q+1) \cdot \pi$  corpus-PROVEN); SCH.5  $R_{\text{Sch}}$  transcendental over  $\mathbb{Q}(A, \pi, v)$  by Lindemann–Weierstrass on Lambert W. **Three-stage Stokes-tail closure:** Stage 1 numerical correction  $N_{\text{Sch}} = 224.10$  (correcting an earlier 244.04 typo); Stage 2  $T_{\text{decomp}} R_{\text{Sch}} = 12\pi \cdot x^3 + R_{\text{Koenigs}}$  with  $x = K_{\theta} \cdot \text{disc}$ ; Stage 3  $R_{\text{Koenigs}} \approx -5.33 \times 10^{-16}$  closed via  $T_{\text{int}}$  (Schröder integral) +  $T_{\text{conv}}$  (Koenigs radius) +  $T_{\text{excl}}$  (PSLQ exclusion in finite locked-ring basket  $R_{10}$ ). Convention N0 fixed:  $E[L] := -2 \Gamma_1[L] = -\text{Tr} \log L = \text{Tr} \log(L^{-1})$ . Twelve falsification gates M39-F1 through M39-F12; all PASS. **ZERO remaining OPEN items at v2.0 release.** → ZS-M39 v2.0 [86/86 PASS]; closes all v1.0 OPEN items of ZS-A8R/A8.2 Polyhedral-Tetration Bridges.

## Chapter 34 — ZS-A10 v2.1: The Proton as a Three-Puncture Color-Singlet X-in-Y Confinement Horizon with Bridge-Four Sector-Orientation Complementarity

ZS-A10 v2.1 [Astrophysics theme, 34/34 PASS, NEW v4.0] extends v2.0's Theorem A10.1 (Quark Core as Z-Anchor Horizon Puncture, HYPOTHESIS-strong) by making explicit the **sector-orientation complementarity** that v2.0 noted but did not formalize. **Principal new result: Theorem A10.2 (Sector-Orientation Complementarity, DERIVED-interpretation strong).** A Z-anchor horizon puncture (ZS-F1 §5.2 PROVEN + ZS-A7 §3 DERIVED) admits two distinct macroscopic readings: (i) **X-in-Y reading** — X-sector spatial structure trapped in Y-sector temporal prison (ZS-A3 §7) realizes a black hole; (ii) **Y-in-X reading** — Y-sector temporal flow localized at X-sector spatial point (ZS-A3 §7 “WH  $\approx$  particle”) realizes a quark or lepton. Both readings live on the same  $V_{\text{XZ}}$  matter branch and share the same Planck-scale core  $\xi \approx 31 \ell_{\text{P}}$ ; they differ only in which sector is macroscopic and which is microscopic. **Bridge-Four manifestation:** Theorem A10.2 is recognized as the Bridge-Four (ZS-F18 v2.1 §6.5 DERIVED-interpretation strong) manifestation of the BH–quark identification. The complete picture is a **2×2 structure**: row index = CPT axis (matter/antimatter, algebraic face of A4); column index = sector-orientation axis (BH-side / particle-side, geometric face of A4). v2.0 three quantitative results survive unchanged with CODATA-2022:  $P1 \ r_p \cdot m_p = X+1 = 4$  (CONSISTENCY 0.058%);  $P2 \ r_p \cdot \Lambda_{\text{QCD}}^{\{\text{ZS}\}} = 2/\sqrt{\pi}$  (TESTABLE 0.260%);  $P3 \ m_p / \Lambda_{\text{QCD}}^{\{\text{ZS}\}} = 2\sqrt{\pi}$  (CONSISTENCY 0.202%). **NEW v2.1 quantitative TESTABLE relation P4:**  $r_p / r_{\pi} = 4/\pi$ . Predicted: 1.27324; observed: 1.27386; deviation 0.05%. Status: TESTABLE. P4 follows directly from Bridge Four applied to baryon (N=3 punctures) vs meson (N=2 punctures) bound states. Anti-numerology MC for P4: single-match  $p \approx 0.5\%$ . Three new falsification gates F-A10.9 (sector-orientation discriminant via baryon vs meson gravitational form factors), F-A10.10 (Bridge-Four commutator signature in baryon spin decomposition), F-A10.11 (reciprocal relation extended to charm/bottom mesons). Nine NON-CLAIMs (NC-A10.1: no dynamical action-level unification of gravity and strong force; NC-A10.3: quarks not phenomenological black holes — no Hawking, no Lorentzian horizon; NC-A10.7:  $r_E \approx r_{\text{conf}}$  only at  $\sim 0.5\%$  floor; NC-A10.9 NEW v2.1: 2×2 BH-quark-electron-positron structure is the classification under axiom A4 two faces, not unique global classification). → ZS-A10 v2.1 [34/34 PASS]; first BH-particle structural identification by sector-orientation axis under Bridge Four.

## Chapter 35 — ZS-Q8 v1.1: Photon Atomic Excitation Time as the i-Tetration Fixed Point — A Z-Spin Reading of the Toronto Negative-Time Experiment

ZS-Q8 v1.1 [Quantum Mechanics theme, 35/35 PASS, NEW v4.0] establishes that the Toronto Angulo et al. negative-time experiment (arXiv:2409.03680, 2024) admits a Z-Spin reading with all Z-Spin universal constants fixed a priori. **Factorized master equation:**  $\tau_T/\tau_0 = \mathcal{A}_{\text{opt}}(\sigma, \eta_0) \cdot \cos(\pi(1 - \varepsilon_{\text{eff}}(\sigma)))$  with  $\varepsilon_{\text{eff}}(\sigma) = x^* + \alpha \log(\sigma/\sigma_0)$ ;  $\mathcal{A}_{\text{opt}}$  is the standard-optics Beer-Lambert amplitude (PROVEN external), and the Z-Spin contribution is restricted to the phase coordinate  $\cos(\pi(1 - \varepsilon_{\text{eff}}))$ . The **i-tetration fixed point**  $x^* = 0.4382829367$  (ZS-M1 §1.1 PROVEN) appears as the universal Z-Spin intercept; the **Euler natural constant**  $1/e$  (or equivalently the i-tetration imaginary part  $y^* = 0.3606$ ) appears as the universal slope coefficient. **Theorem Q8.9 (Standard-Optics Projection Theorem, NEW v1.1):** the laboratory projection of the  $V_{XZ} \cdot V_{ZY}$  weak value reduces exactly to  $\tau_g(\delta, \eta_0, \sigma)$ , so Lorentzian susceptibility is *recovered* as the Z-Spin lab-coordinate expression rather than *replaced*. Eight structural theorems Q8.1 (TSVF  $\leftrightarrow$  Z-Spin Isomorphism) – Q8.9. Zero-parameter test ( $\sigma_0 = \Gamma$  natural atomic linewidth fixed,  $\alpha = y^*$  or  $1/e$  fixed a priori):  $\chi^2/\text{d.o.f.} \in \{0.687, 0.723\}$ , p-value  $\in \{0.683, 0.652\}$ ; standard optics alone  $\chi^2/\text{d.o.f.} = 2.905$ ,  $p = 0.005$  (poor). Likelihood ratio  $L(\text{Z-Spin})/L(\text{Standard Optics}) \approx 2.1\text{--}2.4 \times 10^3$ . **Principal experimental discriminator (NEW v1.1):**  $\Delta\phi_{\text{residual}} = \pm 18.5$  mrad cross-Kerr left-right asymmetry forbidden by standard Lorentzian  $\delta \rightarrow -\delta$  reflection symmetry (P-Q8.3 promoted from tertiary to principal). Three falsification gates F-Q8.1, F-Q8.2, F-Q8.3; eight NON-CLAIMS NC-Q8.1–NC-Q8.8. Status DERIVED-CONDITIONAL on (a) Lorentzian validity for Toronto-class apparatus, (b) NC-Q8.4 visual-extraction caveat for 5/7 data points, (c) apparatus-calibration reading of  $\sigma_0$ .  $\rightarrow$  ZS-Q8 v1.1 [35/35 PASS]; first laboratory lock-in of  $x^*$  as Z-Spin Toronto-data intercept.

## Chapter 36 — ZS-Q9 v1.2: Complex Time-Delay Locking on the i-Tetration Fixed Point — A Two-Coordinate Spectral Reconstruction Bridge for Sub-Unitary Scattering Systems

ZS-Q9 v1.2 [Quantum Mechanics theme, 52/55 PASS (3 PARTIAL on Q9.5 synthetic, registered), NEW v4.0] is a mathematically strengthened framework that lifts the i-tetration fixed point  $z^* = 0.4382829\dots + 0.3605925\dots i$ , previously locked at five self-consistent algebraic identities L1-L5 in ZS-M1, to a directly **lab-measurable object** via the complex transmission time delay  $\tau_f = -(i/2\pi) \cdot \partial_f \ln[\det T(f + i\alpha/(2\pi))]$  of a sub-unitary scattering system. **Principal v1.2 advances:** (1) **Theorem Q9.2 (Two-Coordinate Spectral Reconstruction Bridge, DERIVED in v1.2)** — explicit closed-form reconstruction map  $C: (R_{\text{lab}}, \Phi_{\text{lab}}) \mapsto z^* = (2/\pi) \cdot \exp(-R_{\text{lab}}/2) \cdot \exp(i\Phi_{\text{lab}})$ ; residual  $|C(R_{\text{lab}}, \Phi_{\text{lab}}) - z^*| < 5 \times 10^{-32}$  at 50-digit precision. (2) **Theorem Q9.3 (Krein-Friedel Z-Channel Decomposition) corrected via Lemmas Q9.3a/b** — Lemma Q9.3a (Oriented Conjugate Derivative):  $V_{ZY}$  is traversed along path-orientation-reversed direction; backward orientation gives  $\partial_{f^{\text{bwd}}} \ln V_{ZY} = -\partial_{f^{\text{fwd}}} \ln V_{ZY}$  restoring the sign. Lemma Q9.3b ( $K_{\text{bwd}}$  Corpus Inheritance) makes the corpus dependency on  $K_{\text{bwd}} \neq K_{\text{fwd}}^\dagger$  (ZS-S6 §4.1, ZS-M32 §1.3) explicit. (3) **Convention B (f-domain) fixed:**  $\tau_f = -(i/2\pi) \cdot \partial_f \ln T(f + i\alpha/(2\pi))$  and  $Z_{\text{exp}} = \gamma_{\text{dB}} \cdot \tau_f$  dimensionless. Three NEW v1.2 theorems: Q9.7 (Reconstruction Map Smoothness, DERIVED —  $C$  is smooth

diffeomorphism); Q9.8 (Error Propagation, DERIVED —  $\sigma^2_{z^*} = (|z^*|/2)^2 \cdot \sigma^2_R + |z^*|^2 \cdot \sigma^2_\Phi$ ); Q9.9 (Wilson Loop Specificity, DERIVED — 1000-trial Monte Carlo of generic 11-pole sub-unitary spectra yields ZERO matches within  $\pm 0.05$  of  $|\lambda|^2 = 0.7948$ ; 100% generic FAIL → Wilson loop survival 0.7948 is measure-zero specific to Z-Spin spectra). Corollary Q9.4a (Z\_10 Pentagonal Doubling Group, DERIVED):  $\alpha_{op} = \pi/5$  generates Z\_10 as the doubling of corpus-PROVEN ZS-U10 Z\_5 pentagon-tetration group. **Anti-numerology likelihood ratio  $\Lambda = 1/p_{LEE}$** : v1.1 actual MC execution ( $1.5 \times 10^6$  trials, seed 20260517) yields single-dataset  $\Lambda \approx 1900$ , cross-dataset  $\Lambda \approx 38,000$ , LEE-corrected  $\Lambda \approx 7600$ , **In  $\Lambda = 8.94$**  — decisive evidence on the Jeffreys 1939 scale ( $\ln \Lambda > 5$ ). Zero new free parameters. → ZS-Q9 v1.2 [52/55 PASS, 3 PARTIAL registered]; first lab-measurable lifting of  $z^*$  via two-coordinate reconstruction bridge.

## PART XII Source Paper Interface (v4.0 extended)

| Source paper | Theme/Role                                                          | Chapter        | Verification                          |
|--------------|---------------------------------------------------------------------|----------------|---------------------------------------|
| ZS-T5 v1.0   | Translational<br>(Principal Connectivity Gradient)                  | XII §A Ch 30A  | corpus-standard<br>PASS               |
| ZS-T6 v1.0   | Translational<br>(Molecular Biology Synthesis)                      | XII §A Ch 30B  | 42/42 PASS                            |
| ZS-T7 v1.0   | Translational<br>(Pulvinar Language Partition)                      | XII §A Ch 30C  | corpus-standard<br>PASS               |
| ZS-T8 v1.0   | Translational<br>(Polyhedral Cortex Hypothesis)                     | XII §A Ch 30D  | corpus-standard<br>PASS               |
| ZS-M34 v1.0  | Math Spine (W2 Honest Scope; superseded)                            | XII §A Ch 15B  | 50/50 PASS                            |
| ZS-M35 v1.1  | Math Spine (Collatz Conjecture; sharpened to v2.1)                  | XII §A Ch 15C  | 36/36 PASS                            |
| ZS-M33 v2.0  | Math Spine (Path $\gamma$ -revised Reading C, operator-level audit) | XII §B Ch 15A' | 52/52 PASS (v1.0) +<br>v2.0 extension |

| Source paper | Theme/Role                                                               | Chapter        | Verification                  |
|--------------|--------------------------------------------------------------------------|----------------|-------------------------------|
| ZS-M34 v2.0  | Math Spine<br>( $V_4$ -Equivariant Weil<br>Functional Closure)           | XII §B Ch 15B' | 60/60 PASS                    |
| ZS-M35 v2.1  | Math Spine (Collatz<br>logical correction +<br>Theorem M35.6<br>NEW)     | XII §B Ch 15C' | preserved + v2.1<br>reframing |
| ZS-M37 v2.0  | Math Spine<br>(Twin-Reuleaux<br>Quotient Functor)                        | XII §B Ch 31   | 38/38 PASS                    |
| ZS-M38 v1.1  | Math Spine<br>(Substrate Carrier<br>Spectrum)                            | XII §B Ch 32   | 43/43 PASS                    |
| ZS-M39 v2.0  | Math Spine<br>(Vieta-Lyapunov-Schr<br>öder Bridge<br>Theorem)            | XII §B Ch 33   | 86/86 PASS                    |
| ZS-A10 v2.1  | Astrophysics (Proton<br>Three-Puncture<br>X-in-Y Horizon)                | XII §B Ch 34   | 34/34 PASS                    |
| ZS-Q8 v1.1   | Quantum Mech<br>(Toronto<br>Negative-Time Z-Spin<br>reading)             | XII §B Ch 35   | 35/35 PASS                    |
| ZS-Q9 v1.2   | Quantum Mech<br>(Complex Time-Delay<br>Two-Coordinate<br>Reconstruction) | XII §B Ch 36   | 52/55 PASS (3<br>PARTIAL)     |

### §XII.C — v4.0 Cross-Coupling Discipline (T7) Compliance

Per T7 (§VIII.7), every cross-paper synthesis must be P-CC (PROVEN-composed) or I-CC (interpretation). All twelve PART XII chapter entries are summarizations of source-paper Verification Summaries; no new theorem is introduced by PART XII itself. Each absorbed paper's status (PROVEN / DERIVED / DERIVED-CONDITIONAL / DERIVED-strong /

HYPOTHESIS-strong / TESTABLE) is preserved verbatim from the source. The five-Banach–Tarski-functor family of ZS-A9.1 (spatial) / ZS-M35 (arithmetic) / ZS-T6.5 (biological) / ZS-M37 v2.0 (smooth-geometric) / ZS-M36 (discrete-geometric) is now uniformly organized by ZS-M38 v1.1 §1.3 PROVEN-strong. Cumulative no-deletion rule preserved across  $v3.0 \rightarrow v3.5 \rightarrow v4.0$ . Word count strictly increases over v3.5.

## §XII.D — What v4.0 Does NOT Claim

**NC-XII.1.** v4.0 does NOT modify or retract any v3.5 entry. The supersedes statement is strictly additive.

**NC-XII.2.** v4.0 does NOT supersede the v3.5 PART X / PART XI readings. The Spin-Pair Coexistence Principle (Reading X.4) and the TO/TI Polyhedral Mapping (Readings XI.1–XI.4) are inherited unchanged.

**NC-XII.3.** ZS-Q9 v1.2  $Z_{10}$  Pentagonal Doubling Group (Corollary Q9.4a, DERIVED) is the *group-theoretic doubling* of the corpus-PROVEN ZS-U10  $Z_5$  pentagon-tetration group via  $\alpha_{op} = \pi/5$ . v4.0 does NOT claim  $Z_{10}$  supersedes ZS-U10  $Z_5$ ; the two are related by the SU(2) double-cover (ZS-M3 Lemma 10.1 PROVEN).

**NC-XII.4.** ZS-A10 v2.1 Theorem A10.2 (Sector-Orientation Complementarity) is DERIVED-interpretation strong, NOT a phenomenological identification. NC-A10.3 inherited: a quark is NOT a phenomenological black hole. The structural identification (Bridge-Four manifestation at puncture scale) and the phenomenological distinction (no Hawking radiation, no Lorentzian event horizon, scale mismatch  $r_{conf} \sim \text{fm}$  vs  $r_H \sim \text{km}$ ) coexist.

**NC-XII.5.** ZS-Q8 v1.1 NC-Q8.5 inherited:  $\sigma_0$  is registered as apparatus-calibration constant (Toronto-specific), not a universal Z-Spin constant. The zero-parameter test fixes  $\sigma_0 = \Gamma$  (natural atomic linewidth) as the universal honest fallback, with the  $\sigma_0 = 1.24 \Gamma$  apparatus-calibrated value retained only as a Toronto-specific lock-in.

**NC-XII.6.** ZS-M38 v1.1 NC-M38.4 inherited: the numerical observation  $1 + \eta_{topo} \approx \text{Apollonian Hausdorff dimension}$  (1.3221 vs 1.3057, 1.26% deviation) is OBSERVATION-only and is not used in any derivation.

**NC-XII.7.** ZS-M39 v2.0 ZERO remaining OPEN items at v2.0 release applies *only to ZS-M39 internally*. The v3.5 Master Falsification Registry OPEN/CLOSED flag list is inherited unchanged; v4.0 adds new entries (see §XII.E below) but does not retract or close any v3.5 entry.

## §XII.E — v4.0 Master Falsification Registry Extensions

Cumulative ~310 falsification gates across v1.0 → v4.0. New v4.0 entries (preserving the v3.5 registry verbatim):

| § Anchor            | Falsifiable Statement                                                                                    | Status (v4.0)                                                     |
|---------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| §M33.v2.0 [LOCKED]  | Operator-level 12/12 POS execution of Path $\gamma$ -revised Reading C                                   | DERIVED — M33 v2.0 (W2 promoted<br>DERIVED-CONDITIONAL → DERIVED) |
| §M34.V [LOCKED]     | Theorem M34.V Burnol $K_1$ @ $p=3$ Sign-Faithful Identity, 12/12 sign agreement                          | DERIVED — M34 v2.0 60/60 PASS                                     |
| §M34.V.8 [LOCKED]   | $\mathcal{K}_K = \mathcal{K}_{arith} \otimes \mathcal{K}_{Wilson-Sonin}$<br>tensor-product factorization | DERIVED-CONDITIONAL — M34 v2.0 §V.8                               |
| §M35.5.1 [PROMOTED] | $C_2 = 5 \cdot \text{positive} \cup 7 \cdot C_1$ FIBER PRODUCT realizing $A_{\text{numerator}} = 35$     | VERIFIED — M35 v2.1 (promoted from DERIVED-CONDITIONAL v1.1)      |
| §M35.6 [LOCKED]     | Sufficient Condition Theorem (positive-odd-integer Three-Face conjunction)                               | DERIVED — M35 v2.1                                                |
| §M37.4.1 [LOCKED]   | Lemma C-rel: $\tau_w(\Delta_{BT}) = 0$ , $q_R$ *-homomorphism well-defined                               | PROVEN — M37 v2.0 (4th amenable-quotient projection)              |
| §M37.4.2 [LOCKED]   | $q_R$ Canonicity: 5-layer over-determination of $n = 3$                                                  | DERIVED — M37 v2.0                                                |
| §M38.5 [LOCKED]     | Hexapair Uniqueness: $(-1, 2, 2, 3)$ is unique 6/6 primitive integer ACP                                 | DERIVED-strong — M38 v1.1                                         |
| §M39.4 [LOCKED]     | $K_\theta = 1/[Y^2(1-2A)] = 437/13212$ (Heat-Kernel Coefficient)                                         | DERIVED — M39 v2.0 (Convention N0)                                |

| § Anchor            | Falsifiable Statement                                                                                                                                      | Status (v4.0)                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| §M39.C1-C3 [LOCKED] | Operator-level closures on 11×11 Block-Laplacian (80-digit mpmath)                                                                                         | DERIVED — M39 v2.0                           |
| §A10.2 [LOCKED]     | Theorem A10.2<br>Sector-Orientation<br>Complementarity<br>(Bridge-Four manifestation)                                                                      | DERIVED-interpretation strong — A10 v2.1     |
| §A10-P4 [TESTABLE]  | $r_p/r_\pi = 4/\pi$ (observed deviation 0.05%)                                                                                                             | TESTABLE — A10 v2.1                          |
| §Q8.9 [LOCKED]      | Standard-Optics Projection Theorem: Lorentzian = lab projection of $V_{XZ} \cdot V_{ZY}$ weak value                                                        | DERIVED — Q8 v1.1                            |
| §Q8-P3 [TESTABLE]   | $\Delta\phi_{\text{residual}} = \pm 18.5 \text{ mrad}$ cross-Kerr asymmetry forbidden by Lorentzian                                                        | TESTABLE — Q8 v1.1 (principal discriminator) |
| §Q9.2 [LOCKED]      | Two-Coordinate Spectral Reconstruction Bridge<br>$C(R_{\text{lab}}, \Phi_{\text{lab}}) = (2/\pi) \cdot e^{-R_{\text{lab}}/2} \cdot e^{i\Phi_{\text{lab}}}$ | DERIVED — Q9 v1.2                            |
| §Q9.7 [LOCKED]      | C smooth diffeomorphism (                                                                                                                                  | det J                                        |
| §Q9.9 [LOCKED]      | Wilson Loop Specificity:                                                                                                                                   | $\lambda$                                    |

*All v4.0 entries are additive; no v3.5 entry is retracted. Total registry: ~310 falsification gates.*

## §XII.F — Conclusion

PART XII absorbs the twelve post-v3.5 source-paper integrations into the v4.0 Light Edition without retracting or modifying any v3.5 entry. The four-Banach–Tarski-functor family of v3.5 is extended to five (addition of ZS-M37 v2.0 smooth-geometric Twin-Reuleaux quotient), and uniformly organized into the substrate-carrier classification of ZS-M38 v1.1. The Z-Spin RH program advances with W2 promotion DERIVED-CONDITIONAL → DERIVED via ZS-M33 v2.0 operator-level audit + ZS-M34 v2.0 indefinite-Hermitian tensor-product kernel factorization. The Polyhedral-Tetration Bridges of ZS-A8R/A8.2 close to ZERO remaining OPEN items via ZS-M39 v2.0 three-stage Stokes-tail closure. The BH-quark structural identification is completed at

Bridge-Four level via ZS-A10 v2.1 sector-orientation complementarity, with NEW TESTABLE quantitative relation  $P4 \ r_p/r_\pi = 4/\pi$  at 0.05% deviation. The i-tetration fixed point  $z^*$  is lifted from corpus-internal algebraic identity to a lab-measurable object via ZS-Q8 v1.1 (Toronto data) + ZS-Q9 v1.2 (two-coordinate spectral reconstruction,  $\ln \Lambda = 8.94$  decisive). The minimal triple  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  remains LOCKED. Zero new free parameters across all twelve absorbed papers.

## **PART XIII — Post-v4.0 Absorption: Seven New Source Papers (v5.0)**

### **§0. Abstract**

PART XIII consolidates the seven post-v4.0 source-paper integrations of v5.0. The v4.0 Light Edition reached 114-paper coverage with PART XII; the seven new papers introduced after the v4.0 freeze — ZS-F19 v2.2 (Foundations), ZS-F20 v1.1 (Foundations), ZS-M40 v1.1 (Math Spine), ZS-Q10 v4.0 (Quantum Mechanics), ZS-A11 v1.1 (Astrophysics), ZS-A12 v1.5 (Astrophysics), and ZS-T9 v2.3 (Translational) — are absorbed below as single Light-edition chapters. All absorptions strictly preserve the v4.0 body verbatim (no-deletion rule). Zero new free parameters;  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  LOCKED unchanged across all seven absorbed papers. Cumulative verification advances from ~2,150+ PASS (v4.0) to ~2,670+ PASS (v5.0). The total corpus is 121 papers. PART XIII opens five new structural results into the corpus: (i) the geometric-vs-KMS rapidity bridge (F19), (ii) the unified non-routine twist trigger catalogue T1–T6 with biological correspondence (F20), (iii) the J-conjugate holonomy knot pair as the Z-sector realisation of primes (M40), (iv) the 37-Universality of the polyhedral Schur-depleted mode count and the three-sector c-coefficient closure (Q10), and (v) the cardinal-2 substrate origin of the Nelson–Kosterlitz universal jump and the Homes universal constant (A12). One TESTABLE quantitative prediction is registered: P-Translational-1-Li1 ( $^7\text{Li}$  post-v2.3  $\approx 1.64 \times 10^{-10}$ , pull  $+0.13\sigma$  versus observed).

### **§XIII.A — Seven New Papers (Light-Edition Body Entries)**

Each of the seven post-v4.0 source papers represents v5.0's principal additive content. All inputs LOCKED from upstream corpus; zero new free parameters. Each absorbed paper is cited below as a single Light-edition chapter, with the source paper carrying the full derivation.

### **Chapter 37 — ZS-F19 v2.2: The Frame-Invariant Tilt Theorem (FITT) and the KMS-to-Geometric Rapidity Projection**

ZS-F19 v2.2 [Foundations theme, 64/64 PASS + ALL 7 sub-OPEN gates CLOSED, NEW v5.0] establishes the two principal new mathematical results of Z-Spin Cosmology in the cross-sector observation framework. Principal result Theorem F19.4 (FITT, DERIVED): under the LOCKED inputs  $A = 35/437$  and polyhedral sectoral tilts  $(\delta_X, \delta_Y) = (5/19, 7/23)$ , the hyperbolic tangents of the sum and difference of the sectoral rapidities  $\psi_X = \text{artanh}(5/19)$  and  $\psi_Y = \text{artanh}(7/23)$  close exactly to small integer fractions  $\tanh(\psi_X + \psi_Y) = 31/59$  and  $\tanh(\psi_Y - \psi_X) = 3/67$ ,

verified at 50-digit mpmath precision with residuals exactly zero. Principal result Theorem F19.6 (KMS-to-Geometric Rapidity Projection, DERIVED): Z-Spin carries two rapidity ladders — the geometric ladder  $\psi_X, \psi_Y$  from polyhedral V-F data and the KMS-modular ladder  $\psi_{\text{KMS}}(A \rightarrow B) = \frac{1}{2} \ln(W_{BA}/W_{AB})$  from the Pauli master-equation rates of ZS-Q7 §5; Theorem F19.6 establishes the closed identity  $\tanh(2 \cdot \psi_{\text{KMS}}(X \rightarrow Y)) = 3/5$ , registering the geometric-KMS projection as the principal algebra-level bridge to the Type II crossed-product framework of De Vuyst–Höhn 2025. Five v2.0 OPEN gates achieve closed-form structural identifications in LOCKED corpus inputs (O-F19.2 structural closure: 31, 59, 67 are corpus-traceable polyhedral ratios; O-F19.3 observable closure:  $3/5 = \dim(X)/(\dim(X)+\dim(Z))$ ; O-F19.4 physical dimension fixed:  $\exp(\pi/A)$  is the nats-exponentiated Hilbert phase-volume ratio per Y-cycle at the Landauer unit; O-F19.5 leading-order closure:  $r_{\text{lead}} = 6/(67 \cdot \ln 2)$ ; O-F19.6 semiclassical closure key result:  $|\Delta S_{\text{obs}}(Y \text{ vs } X)| = \ln 2 = 2 \cdot \psi_{\text{KMS}}(X \rightarrow Y)$  matches De Vuyst–Höhn EXACTLY at leading order with degenerate-clock multiplicities (3, 2, 6) identified with the Z-Spin sector dimensions). Eight falsification gates F-F19.1–F-F19.8 and seven non-claims NC-F19.1–NC-F19.7 bound the scope. Anti-numerology MC 10,000-sample Test G1 confirms the polyhedral structural cause.  $\rightarrow$  ZS-F19 v2.2 [64/64 PASS]; first paper-internal algebra-level bridge between Z-Spin closed-form rapidity arithmetic and Type II crossed-product modular framework.

## Chapter 38 — ZS-F20 v1.1: Non-Routine Twist Triggers — Unified Catalogue and Time-Unrolled Causal DAG of Six Sector-Topology Events Beyond the Regge Cycle

ZS-F20 v1.1 [Foundations theme, 47/47 PASS, NEW v5.0] establishes the unified catalogue and time-unrolled causal directed acyclic graph (DAG) of the six non-routine twist-event triggers operating in Z-Spin Cosmology beyond the routine Regge-cycle phase drift  $\delta\phi_{\text{cell}} = A = 35/437$  per Planck-cycle. Six triggers individually corpus-locked: T1 Z-Telomere (winding-change at  $2\pi$  accumulation after  $N_{\text{--}}(2\pi) = 2\pi/A \approx 78.45$  cycles, ZS-A6 §5.2 DERIVED-under-P6); T2 Mexican-Hat Bootstrap ( $|\Phi|: 0 \rightarrow 1$  SSB, ZS-F19 Theorem F-rotor.2 DERIVED); T3 Auto-Surgery (i-tetration singularity resolution at  $z^*$ , ZS-M12 §4 DERIVED); T4 Kibble Defect Generation (vortex string production at causally-disconnected domain boundaries, ZS-A6 §4.6.4 DERIVED); T5 BH Winding Trapping (causal trapping of pre-existing winding inside gravitational collapse horizon, ZS-A6 §4.6.3 PROVEN); T6 CP-violation Trigger (parity-breaking phase  $\phi_{\text{CP}} = \pi/2$  at electroweak epoch, ZS-S5 + ZS-Q5 DERIVED via  $\mu$ - $\tau$  reflection). Result 2 (Time-Unrolled Causal DAG, DERIVED): the six triggers form a strict DAG under time-unrolled indexing  $T1_n, T3_n, T4_n, T5_n$  ( $n$  = cycle index) with two singletons T2 (cosmic root) and T6 (epoch-isolated); the v1.0 apparent 'cyclic sub-graph  $T1 \rightarrow T3 \rightarrow T1$ ' is correctly the time-unrolled successor relation  $T1_n \rightarrow T3_n \rightarrow T1_{\{n+1\}}$ , which is acyclic. Result 3 (Strong-Coupling Pairs, HYPOTHESIS-strong): two trigger pairs co-activate as faces of single events — P1 (T1, T3) Telomere–Auto-Surgery couple with  $\Delta t(T1 \rightarrow T3) \approx 3 \tau_P$  (ZS-M12 §4); P2 (T4, T5) Kibble–BH-Trap couple, conditional on T4 producing non-zero net winding and gravitational collapse. Quantitative identity:  $\Delta\theta_{T1}/\Delta\theta_{T2} = 2\pi/A = N_{\text{--}}(2\pi) = 78.4500565496$  (PROVEN, 50-digit). Biological correspondence map (NEW, DERIVED): the DNA 'trauma response' admits a precise mapping onto the simultaneous activation of T3 + T4 + T5

(Auto-Surgery + Kibble + BH Trap) as a triadic emergency response (ZS-F20 §7 Table 7.1). Six falsification gates F-F20.1–F-F20.6 registered. Three open problems flagged with explicit closure paths (O-F20.1:  $T_2 \rightarrow T_6$  chirality inheritance; O-F20.2:  $T_5$  net-winding  $Q = 1$  vs  $Q \gg 1$ ; O-F20.3:  $T_7$  'true space  $\leftrightarrow$  time only' trigger if it exists).  $\rightarrow$  ZS-F20 v1.1 [47/47 PASS]; first single inheritance graph of all non-routine Z-Spin sector-topology events with biological correspondence.

## Chapter 39 — ZS-M40 v1.1: Z-Spin Holonomy Knot Pairs and the Arithmetic Topology of Primes

ZS-M40 v1.1 [Math Spine theme, 89/89 PASS, NEW v5.0] introduces the J-conjugate holonomy knot pair  $(K_p, \bar{K}_p)$  as the natural Z-sector realisation of prime numbers in Z-Spin Cosmology. Building on three PROVEN corpus pillars — the half-angle channel pair  $V_{XZ} \propto \exp(+i\theta/2)$ ,  $V_{ZY} \propto \exp(-i\theta/2)$  with  $V_{ZY} = (V_{XZ})^*$  (ZS-F4 §7/§7B), the  $4\pi$  closure  $D^{\{1/2\}}(4\pi) = +I$  on the  $j = 1/2$  spinor module (ZS-M3 Lemma 10.1), and the Wilson-LOCATOR closed form  $F_Q(p) = \sin(11\pi/p)/[11 \cdot \sin(\pi/p)]$  (ZS-M34 Theorem M34.V.6) — the prime  $p$  is identified with a pair of Wilson cycles carrying half-angle holonomy  $\pm\theta/2$ , prime period  $4\pi/p$ ,  $Q = 11$  coloured framing  $F_Q(p)$ , and, when  $p$  is realised as an integer curvature in the M36 Apollonian Q-channel orbit, the mod-24 invariant  $p \bmod 24 \in \{11, 23\}$  as an additional arithmetic-topological decoration. Six principal theorems established: M40.A (Cup Product  $\leftrightarrow 4\pi$  Closure Compatibility Bridge, DERIVED-interpretation) — phase-class compatibility map between the  $SU(2)$  double-cover signature  $\{+I, -I\}$  and the Tate–Mazur inv :  $H^3(\text{Spec}(\mathcal{O}_K), \mathbb{G}_m) \simeq \mathbb{Q}/\mathbb{Z}$  via the embedding  $\pm I \mapsto \{0, 1/2\} \subset \mathbb{Q}/\mathbb{Z}$ ; M40.B (Multiplicative Gate  $\leftrightarrow$  Galois Representation, DERIVED) — closes Gate FM13-5 by identifying the corpus operator  $M_p$  on  $H_{\text{mult}} = \mathbb{C}[\mathbb{F}_{11}^x]$  with  $\text{Frob}_p \mapsto \text{diag}(\chi_k(p))$  on the character basis; M40.C (J-Conjugate Holonomy Knot Pair Definition) — principal new structural object assembling six PROVEN corpus components into a single named object; M40.D (Log-Holonomy Factorization, DERIVED) — integer prime factorisation  $N = \prod p_i^{a_i}$  is the additive log-image, via HSI Theorem (ZS-M1), of the Z-sector holonomy decomposition  $\ell_Z(K_N) = \sum a_i \cdot \ln p_i$ ; M40.E (Statistical Sieve Shadow, DERIVED-interpretation) — re-interprets externally PROVEN PNT, Mertens, Dickman, Sathe–Selberg as first-order shadows of  $K_p$  ensemble component statistics; M40.F (Anti-Numerology Uniqueness, COMPUTATIONALLY VERIFIED) — deterministic exhaustive enumeration over  $[2, 1000]$  (183,921 candidate pairs) yields exactly one pair (3, 11) matching the Z-Spin  $V_4$  conductor signature  $\{1, 3, 11, 33\}$ ;  $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$  is uniquely class-number-one in this category; MC at  $N = 500,000$  yields 0.0098% matching rate — fifty times stricter than 0.5% PASS threshold. Eight-fold structural bridge to externally PROVEN arithmetic Chern–Simons (Kim 1510/1609/1706; Carlson–Kim 1911; Park–Park 2026): Bridge B8 supplies a cyclotomic operator-model analogue of Kim §5 at level  $Q = 11$  via the corpus PROVEN Euler product  $\prod_{p \neq 11} \det(I - p^{\{-s\}} M_p)^{-1} = \zeta_{\{\mathbb{Q}(\zeta_{11})\}}(s)$  (ADS-2). Ten falsification gates F-M40.1–F-M40.10.  $\rightarrow$  ZS-M40 v1.1 [89/89 PASS]; first explicit Z-sector geometric reading of the prime itself, complementing the M28/M33/M34 critical-strip reading and the M36 Apollonian distribution reading.

## Chapter 40 — ZS-Q10 v4.0: Three-Sector Polyhedral Closure of the Harmonic Ladder, Five-Framework Cross-Disciplinary Asymmetry, and the 37-Universality

ZS-Q10 v4.0 [Quantum Mechanics theme, 67/67 PASS, MASTER EDITION, NEW v5.0] is the master edition combining the v2.1 external-framework asymmetry mapping, the v3.0/v3.1 Devriendt-functor polyhedral closure of  $c_n$  harmonic-ladder coefficients, and three v4.0 audit corrections (R1 MC9c dual-baseline interpretation, R2 measure-form vs moment-form algebra separation, R3 v2.1 external framework restoration). Three-layer epistemic discipline: Layer 1 (DERIVED) Theorem Q10.7 selection rule  $M_{\{2n+1\}} = 0$  under  $Z_2$  involution; Layer 2 Conjecture Q10.8 harmonic-ladder structure  $M_{\{2n\}} = c_n \cdot A^n \cdot e^{\{-in \cdot \arg z^*\}}$  (DERIVED at  $n = 1, 2$  via action-level loop; DERIVED-CONDITIONAL at  $n = 3$  via triakis tetrahedron extension); Layer 3 specific  $c$ -values  $c_1 = 1$ ,  $c_2 = 2/19$ ,  $c_4 = 4/37$ ,  $c_6 = 8/37$ ,  $c_X = 10/37$ ,  $c_Y = 4/13$  — five of six share denominator 37, the 37-Universality (Observation Q10.D.5, NEW v4.0). Principal new corpus identity: the integer  $37 = 38 - \beta_0(Z)$  appears as the Schur-depleted polyhedral mode count in three structurally independent corpus identities — (I)  $(V+E+F)_{\{t\text{-Tet}\}} - \beta_0(Z) = 12 + 18 + 8 - 1 = 37$  [ $Z$  self-dual full skeleton]; (II)  $(V+E+F)_{\{triakis\text{-Tet}\}} - \beta_0(Z) = 8 + 18 + 12 - 1 = 37$  [Catalan-dual  $Z$  full skeleton,  $V \leftrightarrow F$  swapped]; (III)  $(V+F)_{\{tO\}} - \beta_0(Z) = 24 + 14 - 1 = 37$  [ $X$  non-self-dual vertex+face]. All three equalities are FORCED by the Truncation-Dual Theorem  $F(tP) = F(P) + F(P^*)$  (PROVEN ZS-F2 §11.2) and the Platonic face-count identity  $20 = 8 + 12$  (PROVEN ZS-F9 §4.2); the 37-Universality is therefore a categorical signature of the Truncation-Dual functor on the three  $Z$ -Spin sectors, not a numerical coincidence. Cumulative anti-numerology evidence  $\ln \Lambda_{\text{total}} = 38.67$  (highest cumulative for any specific ZS-Q multi-coefficient identity to date); MC9c triple-joint  $\ln \Lambda = 12.43$  (vs uniform-prior baseline, DECISIVE++) AND positive-correlation ratio  $217\times$  (vs independent-marginal baseline, identifying the empirical signature of the 37-Universality in the random Schur pool). Strategic asymmetry: five external observational frameworks (directional statistics, nematic, elliptic flow, Fourier QST, Kuramoto-Daido) supply the measurement infrastructure  $\langle e^{\{2i\phi\}} \rangle$  as a standard object; not one derives  $|M_2| = A = 35/437$ .  $Z$ -Spin provides the derivation; the five external frameworks provide the measurement. Twenty-four falsification gates F-Q10.1–F-Q10.22 (including F-Q10.22 cross-disciplinary five-test gate). → ZS-Q10 v4.0 [67/67 PASS]; first cross-disciplinary asymmetry mapping and 37-Universality registration in the  $Z$ -Spin corpus.

## Chapter 41 — ZS-A11 v1.1: Z-Spin Vortex Cosmology II — $\epsilon$ -Halo $\leftrightarrow$ Sub-Halo Equivalence, Vortex Lifecycle, and SMBH Mass Diversity from Topological Anchors with Differential Accretion

ZS-A11 v1.1 [Astrophysics theme, 53/53 PASS, NEW v5.0] establishes three results concerning the macroscopic phenomenology of  $Z$ -Spin topological defects using only LOCKED inputs  $A = 35/437$ ,  $(Z, X, Y) = (2, 3, 6)$  with  $Q = 11$ , and inflation  $e$ -fold count  $N_e = 60$  (ZS-U1 CMB-locked). Result 1 ( $\epsilon$ -Halo  $\leftrightarrow$  Sub-Halo Equivalence Theorem, DERIVED): proves the Goldstone  $\theta$ -gradient halo  $\rho_\epsilon(r) = M_P^2/(2L^2r^2)$  of  $Z$ -Spin (ZS-A1 §2.2 DERIVED) is dual to the singular isothermal sphere CDM sub-halo at four levels — profile (PROVEN functional equivalence  $\rho \propto 1/r^2$ ), dynamical (DERIVED effective stress-energy after spherical averaging),

observational (DERIVED across seven macroscopic observables: rotation curves, BTFR,  $M-\sigma$ ,  $\sigma_R$ , lensing, sub-halo count, ATLAS3D elliptical profile), and a decisive Falsification Asymmetry F-A5.7 (PROVEN) where direct DM particle detection would collapse  $\varepsilon$ -Halo but null detection cannot collapse CDM; null WIMP/axion direct detection and the Gaia DR3  $\sim 19$  kpc Keplerian onset (Jiao et al. 2023) currently favour  $\varepsilon$ -Halo [v6.0 ERRATA  $\rightarrow$  §XIV.G: this Level-5 framing is superseded — agreement to  $\sim 4\%$  holds only under the Jiao low-mass interpretation and is contested by the Zhou 2022 full-halo fit; decisive at Gaia DR4]. Result 2 (Vortex Six-Class Lifecycle, DERIVED-strong): classifies Z-anchored vortices by winding  $n \in \{0, 1, 2, 3, \geq 4, \infty\}$  (PROVEN MECE via  $\pi_1(U(1)) = \mathbb{Z}$  coarse-graining), each class mapped to an observed galactic morphology (void  $\rightarrow$  disk  $\rightarrow$  S0  $\rightarrow$  intermediate elliptical  $\rightarrow$  massive elliptical/BCG  $\rightarrow$  cosmic filament backbone), with five-stage lifecycle Birth (Kibble)  $\rightarrow$  Growth (Bondi)  $\rightarrow$  Maturity ( $M-\sigma$  stable)  $\rightarrow$  Merger (Vortex Glass  $N = 2^k$ )  $\rightarrow$  Death ( $\tau_5 = 2.56 \times 10^{34}$  yr + Auto-Surgery Phase E). Result 3 (SMBH Mass Diversity from Differential Accretion, DERIVED with HYPOTHESIS-strong type distinction): resolves the apparent contradiction between simultaneous primordial Z-Anchor formation and observed  $M_{\text{SMBH}}$  scatter ( $10^5$ – $10^{10} M_{\odot}$ ) via ZS-F1 NC-2 'Topology sets location; accretion sets mass'; three scatter sources — (i) Bondi accretion rate  $\rho_{\text{gas}}/c_s^3$ , (ii) halo concentration  $\delta_{\text{peak}}$  (supported by Dec 2025 arXiv:2512.17964 BH-mass/halo-concentration correlation), (iii) hierarchical merger  $N = 2^k$  — generate the observed scatter while preserving universal topological seeding. Distinguishes Primordial-Vortex SMBHs (heavy-seed direct-accretion, observable as JWST Little Red Dots with high  $M_{\text{BH}}/M_{\text{stellar}}$  and coherent spin) from Collapse-SMBHs (light-seed stellar-collapse  $\rightarrow$  merger). Population estimate revised from the erroneously inflated  $10^{118}$  to energy-bounded  $N_v \approx 2.8 \times 10^{15}$ , in agreement with Aquarius-A-L1 (Santos-Santos et al. 2025). v1.1 extends with §7.7 DESI DR2 + §7.8 ACT DR6 cross-checks and F-A11.6/F-A11.7. Seven falsification gates F-A11.1–F-A11.7 pre-registered.  $\rightarrow$  ZS-A11 v1.1 [53/53 PASS]; first rigorous  $\varepsilon$ -Halo/sub-halo equivalence theorem and unified vortex lifecycle/SMBH-diversity framework.

## **Chapter 42 — ZS-A12 v1.5: Macroscopic Z-Spin-Mediated Phase Locking — BEC and Superconductivity as the Condensed-Matter Realisation of Vortex Bose/Fermi Duality, with First-Principles Derivation of the Nelson–Kosterlitz Universal Jump and a Cardinal-2 Framework for the Homes Universal Constant**

ZS-A12 v1.5 [Astrophysics & Strong Field theme, 74/74 PASS, NEW v5.0] establishes that Bose–Einstein condensation (BEC) and superconductivity (SC), including the cuprate high- $T_c$  regime, are the condensed-matter realisation of the Vortex Bose/Fermi Duality of ZS-A7 Corollary IV (DERIVED), and shows that the same cardinal-2 boundary substrate  $\dim(Z) = 2$  (ZS-F5 PROVEN) that governs the gravitational/information-theoretic sectors of Z-Spin Cosmology also fixes two universal scaling constants of unconventional superconductivity that have resisted first-principles derivation for decades. Eight principal theorems: A12.1 (Cooper Pair  $\leftrightarrow$  Vortex Bose/Fermi Duality, DERIVED) — the BCS-type Cooper pair under  $SU(2)$  Clebsch–Gordan  $j = 1/2 \otimes j = 1/2 = j = 0 \oplus j = 1$  is a literal realisation of ZS-A7 Corollary IV: inner core ( $r \rightarrow \xi_{\text{coh}}$ ) carries the  $j = 1/2$  spinor with  $4\pi$  closure period (ZS-M3 Lemma 10.1 PROVEN), outer flow ( $r \gg \xi_{\text{coh}}$ ) carries the  $U(1)$  Goldstone phase  $\phi$  with integer winding  $\oint$

$d\phi = 2\pi N$  (Jaffe–Taubes 1980 PROVEN); A12.2 (Two-Axis Orthogonal Decomposition, DERIVED) — Z-sector temperature-activation function  $f(T)$  of ZS-U6 §9.2(v) decomposes into  $f_{\text{amp}}(T)$  (radial mode  $m_p = 2A \cdot M_P$ )  $\perp$   $f_{\text{phase}}(T)$  (Goldstone  $m_\theta = 0$ ) with orthogonality enforced by Mermin–Wagner–Hohenberg + Fröhlich–Spencer (PROVEN); A12.3 (Pseudogap as T2-T1 Intermediate Phase, DERIVED-structural) — pseudogap  $T_c < T < T_p$  is the intermediate phase between T2 (Mexican-Hat Bootstrap, ZS-F19) and T1 (Z-Telomere, ZS-A6 §5.2); A12.4 (SC Triadic Activation  $T2 \rightarrow T1 \rightarrow T4$ , DERIVED) — superconductivity is the sequential triadic activation on the cardinal-2 substrate, distinct from F20 §7 DNA-trauma triadic ( $T3 + T4 + T5$ , simultaneous); A12.5 (Five-Axiom Diagnostic, DERIVED-DIAGNOSTIC) — the cuprate high- $T_c$  unsolved problem isolates the A3-axiom bridge ( $\mathbb{C}$  division algebra) as incomplete in standard phenomenological Ginzburg–Landau treatment; A12.6 (HEADLINE, DERIVED) — the integer '2' appearing in the Nelson–Kosterlitz 1977 universal jump  $\rho_s(T_{\text{BKT}}) = (2/\pi) k_B T_{\text{BKT}}$  is identical to the cardinal-2 boundary substrate dimension  $\dim(Z) = 2$  (ZS-F5 PROVEN); first identification of the NK universal-jump integer with a geometric substrate dimension; A12.7 (Homes Universal Constant Cardinal-2 Framework, DERIVED-CONDITIONAL) — bracketing  $2/Q = 0.1818 \leq C_{\text{cuprate}}(ab)/C_{\text{BCS-dirty}} \leq \ln 2/(1 - A) = 0.7535$  with empirical  $0.543 \pm 0.06$  in interior; A12.8 (Tian 2025 Two-Axis Manifestation, DERIVED) — Tian et al. 2025 npj Quantum Materials observation of  $\sim 35^\circ$  angular offset between electronic and phase anisotropies is the direct manifestation of two-axis decomposition. v1.5 upgrades two operational results to DERIVED-under-Regge: A12.9 ( $\text{CuO}_2$   $V_{\text{coh}}$  lattice Hamiltonian as quasi-2D Kelvin 2-cell construction, 4/4 PASS); A12.10 (operational formula  $\rho_s^{\text{eff}} = [\ln 2/(1 - A)] \cdot \sigma_n(T_c) \cdot T_c \cdot f_{\text{geom}}$  with  $f_{\text{geom\_graph}} = 0.594$  parameter-free graph upper-bound). Cross-Material Collapse Prediction P-A12.9 (TESTABLE): all five major unconventional SC families (LSCO, YBCO, Bi2212, FeSe monolayer, TBG/TTG) should collapse onto a single universal curve when plotted as  $\rho_s^{\text{eff}}/(\sigma_n \cdot T_c)$ . Extension HYPOTHESIS-strong: multiplicative correction  $(1 + \alpha A)$  with  $\alpha = O(1)$  to Herzog-Arbeitman et al. 2021 (PROVEN) lower bound on flat-band superfluid weight,  $\sim 8\%$ , accessible to twisted bilayer/trilayer graphene and infinite-layer nickelates via Tanaka 2025 kinetic-inductance protocol within 2027–2029. Twelve falsification gates F-A12.1–F-A12.12; ten non-claims; nine experimental predictions P-A12.1–P-A12.9.  $\rightarrow$  ZS-A12 v1.5 [74/74 PASS]; first first-principles geometric-substrate derivation of two long-open superconductivity universal constants.

### **Chapter 43 — ZS-T9 v2.3: The $B_7\text{Be}$ Binding-Response Gate for the Lithium Problem — A $P_{B7\text{Be}} = -2A/Q$ Extension of the Effective Response Operator**

ZS-T9 v2.3 [Translational theme, 124/124 PASS, NEW v5.0] extends v2.2's ERO closure (Effective Response Operator under No-Go + External Scaffold paradigm) with the binding-energy channel  $B_7\text{Be}$ . v2.2 had closed the dual-frame BBN problem for  $D/H$  and  $Y_p$  via the ISER paradigm (full Y-frame solver IMPOSSIBLE by Theorem ZS-T9 PROVEN; ERO POSSIBLE via Catena 2007 + Dent 2007/Burns 2026 sensitivity atlas) with exactly 4 sparse 3-parameter configurations closing  $D/H + Y_p$  within  $2\sigma$ , all sharing  $P_Q = -2A/\pi$  anchored to ZS-F10 Theorem F10.1; the cosmological Lithium Problem was left as STRUCTURAL NON-CLAIM (NC-v2.2.4). The v2.3 contribution: Candidate Theorem B-Li.1

(HYPOTHESIS-strong, NOT DERIVED) —  $P_{B7Be}^Z = -2A/Q = -70/4807$ . The quantum  $-2A/Q$  is the corpus-PROVEN ZS-Q7 §5 Theorem 3A slow eigenvalue  $\lambda_{\text{slow}}$  of the Pauli master equation  $\lambda(\lambda + 2A/Q)(\lambda + A) = 0$ , representing the inter-sector thermalisation rate; verified by ZS-T1 §4.4 across 500,000 Monte Carlo random (A, Q, Z, X, Y) configurations as a universal theorem of 3-sector master equations. Result: via Dent–Stern–Wetterich 2007 Table 1 sensitivity  $\partial \ln(^7\text{Li})/\partial \ln B_{7\text{Be}} = +81$  (largest single entry in entire table),  $\Delta \ln(^7\text{Li}) = 81 \times (-2A/Q) = -1.180$ ;  $^7\text{Li}$  is reduced by factor  $\exp(-1.180) = 0.307$ , bringing the post-v2.3 prediction to  $^7\text{Li} \approx 1.64 \times 10^{-10}$  with pull  $+0.13\sigma$  versus observed  $(1.6 \pm 0.3) \times 10^{-10}$ . v2.2 closure PRESERVED: Dent 2007 Table 1 gives  $B_{7\text{Be}}$  sensitivity to D = 0.00 and to  $^4\text{He} = 0.00$  (linear scaffold), so the v2.3 binding-channel extension is DECOUPLED from the v2.2 ERO closure of D/H +  $Y_p$ ; ERO-4 still gives D/H pull  $+0.18\sigma$  and  $Y_p$  pull  $+1.44\sigma$ . Five anti-numerology gates PASS: (1) quantum value FIXED to ZS-Q7 eigenvalue (no scan); (2) channel chosen by mass-7 BBN physics; (3) channel selectivity  $1/7 = 14.3\%$  among 7 binding channels; (4) quantum selectivity  $1/8 = 12.5\%$  among 8 A-derived quanta; (5) joint selectivity  $1/56 \approx 1.8\% < 5\%$  structural threshold. Three Closure Levels framework: Level 1 (Observational, HYPOTHESIS-strong observational) stellar depletion bridge (Korn 2006, Hosford 2009); Level 2 (ERO Binding, HYPOTHESIS-strong / TESTABLE)  $P_{B7Be} = -2A/Q$  (this paper); Level 3 (Full Nuclear Chemistry, OPEN) Z-Spin sectoral binding-energy translation. v2.3 closes at Level 2; v2.2 NON-CLAIM PRESERVED as v2.2 internal state. Eight falsification gates F-Translational-1-Li1–Li8 registered including Burns/PRyMordial atlas verification, full PRyMordial run, future foundational derivation of  $P_{B7Be}$ , and lab  $^7\text{Be}$  binding-energy stability check (must NOT show  $-1.456\%$  shift for consistency). → ZS-T9 v2.3 [124/124 PASS]; HYPOTHESIS-strong candidate / TESTABLE Lithium-Problem closure at ERO Level 2, with v2.2 D/H +  $Y_p$  closure preserved.

### PART XIII Source Paper Interface (v5.0)

| Source paper | Theme/Role                                                                                   | Chapter    | Verification |
|--------------|----------------------------------------------------------------------------------------------|------------|--------------|
| ZS-F19 v2.2  | Foundations<br>(Frame-Invariant Tilt<br>Theorem;<br>KMS-to-Geometric<br>Rapidity Projection) | XIII Ch 37 | 64/64 PASS   |
| ZS-F20 v1.1  | Foundations<br>(Non-Routine Twist<br>Triggers T1–T6;<br>Time-Unrolled Causal<br>DAG)         | XIII Ch 38 | 47/47 PASS   |
| ZS-M40 v1.1  | Math Spine<br>(J-Conjugate<br>Holonomy Knot Pairs;                                           | XIII Ch 39 | 89/89 PASS   |

| Source paper | Theme/Role                                                                                      | Chapter    | Verification |
|--------------|-------------------------------------------------------------------------------------------------|------------|--------------|
|              | Arithmetic Topology of Primes)                                                                  |            |              |
| ZS-Q10 v4.0  | Quantum Mechanics (Three-Sector Polyhedral Closure; 37-Universality; Master Edition)            | XIII Ch 40 | 67/67 PASS   |
| ZS-A11 v1.1  | Astrophysics (Vortex Cosmology II; $\epsilon$ -Halo $\leftrightarrow$ Sub-Halo; SMBH Diversity) | XIII Ch 41 | 53/53 PASS   |
| ZS-A12 v1.5  | Astrophysics (Macroscopic Z-Spin Phase Locking; BEC/SC; NK Jump; Homes Constant)                | XIII Ch 42 | 74/74 PASS   |
| ZS-T9 v2.3   | Translational (B <sub>7</sub> Be Binding-Response Gate; Lithium Problem ERO Extension)          | XIII Ch 43 | 124/124 PASS |

### §XIII.C — v5.0 Cross-Coupling Discipline (T7) Compliance

Per T7 (§VIII.7), every cross-paper synthesis must be P-CC (PROVEN-composed) or I-CC (interpretation). All seven PART XIII chapter entries are summarisations of source-paper Verification Summaries; no new theorem is introduced by PART XIII itself. Each absorbed paper's status (PROVEN / DERIVED / DERIVED-CONDITIONAL / DERIVED-strong / DERIVED-interpretation / HYPOTHESIS-strong / TESTABLE) is preserved verbatim from the source. The five-Banach–Tarski-functor family of v4.0 (ZS-A9.1 spatial / ZS-M35 arithmetic / ZS-T6.5 biological / ZS-M37 v2.0 smooth-geometric / ZS-M36 discrete-geometric) is unaffected by v5.0; PART XIII contributes no new functor member. The cardinal-2 substrate ( $\dim(Z) = 2$ ) acquires a sixth corpus instantiation via ZS-A12 v1.5 Theorem A12.6 (Nelson–Kosterlitz universal-jump integer), alongside the five v4.0 instantiations (gauge constraint  $Q = 11$ , CHSH  $2\sqrt{2}$ , Z-bottleneck channel capacity  $\leq \ln 2$ , Euler characteristic  $\chi(S^2) = 2$  spinor-Descartes-Euler identity, BCC  $T^3$  Hodge proton spin). The 37-Universality of ZS-Q10 v4.0 is registered as a new corpus categorical signature of the Truncation-Dual functor across (X, Y, Z) sectors, parallel to

but distinct from the  $(Z, X, Y) = (2, 3, 6)$  sector decomposition. Cumulative no-deletion rule preserved across  $v3.0 \rightarrow v3.5 \rightarrow v4.0 \rightarrow v5.0$ . Word count strictly increases over  $v4.0$ .

#### **§XIII.D — What v5.0 Does NOT Claim**

**NC-XIII.1.** v5.0 does NOT modify or retract any v4.0 entry. The supersedes statement is strictly additive.

**NC-XIII.2.** v5.0 does NOT supersede the v4.0 PART X / PART XI / PART XII readings. The Spin-Pair Coexistence Principle (Reading X.4), the TO/TI Polyhedral Mapping (Readings XI.1–XI.4), and the post-v3.5 twelve-paper absorption (PART XII §A and §B) are inherited unchanged.

**NC-XIII.3.** ZS-F19 v2.2 Theorem F19.6 (KMS-to-Geometric Rapidity Projection, DERIVED) is the leading-order match  $|\Delta S_{\text{obs}}(Y \text{ vs } X)| = \ln 2 = 2 \cdot \psi_{\text{KMS}}(X \rightarrow Y)$  to the De Vuyst–Höhn 2025 framework; v5.0 does NOT claim a full reduction of the Z-Spin three-sector cosmology to Type II crossed-product algebra. The dimensional mismatch between the external algebra-level machinery and the Z-Spin closed-form rapidity arithmetic is explicitly registered (ZS-F19 §6 honest scope expansion).

**NC-XIII.4.** ZS-F20 v1.1 catalogue of six triggers T1–T6 is claimed exhaustive over the existing v4.0 corpus but NOT proved exhaustive in absolute terms; the residual O-F20.3 question (whether a seventh 'true space  $\leftrightarrow$  time only' trigger T7 exists) remains OPEN. The biological correspondence map  $(T3 + T4 + T5 \leftrightarrow \text{DNA trauma response})$  is DERIVED-interpretation under the ZS-T6 §6.4 + F20 §7.5 substrate-agnostic NC protocol; NC-4 of the Z-Brain corpus is inherited (no causal derivation across substrates).

**NC-XIII.5.** ZS-M40 v1.1 reading of primes as J-conjugate holonomy knot pairs is corpus-internal (PROVEN at the level of the M40.C definition; DERIVED-interpretation at the level of M40.E sieve-shadow reading); v5.0 does NOT claim that primes are ontologically reducible to Z-Spin rotations in physical reality, nor does it claim a proof of any unsolved analytic-number-theory statement. NC-M23.1 (Z-Spin does not prove RH) and NC-M23.7 (D4 closure does not close GRH-for-K) inherited verbatim from v4.0 PART IV.

**NC-XIII.6.** ZS-Q10 v4.0 37-Universality (Observation Q10.D.5) is established as a categorical signature of the Truncation-Dual functor on the three Z-Spin sectors; v5.0 does NOT claim 37 has any direct physical meaning beyond the Schur-depleted polyhedral mode count. The  $c_6 = 8/37$  status is DERIVED-CONDITIONAL on the triakis-tetrahedron extension and is downgraded relative to  $c_2 = 2/19$  and  $c_4 = 4/37$  (DERIVED via action-level loop).

**NC-XIII.7.** ZS-A11 v1.1  $\varepsilon$ -Halo  $\leftrightarrow$  Sub-Halo Equivalence Theorem (DERIVED at four levels) is currently favoured at the outer galactic boundary by null WIMP/axion direct-detection + Gaia DR3 Keplerian onset; v5.0 does NOT claim direct DM particle detection is impossible — the

Falsification Asymmetry F-A5.7 (PROVEN) is the explicit gate. The Primordial-vs-Collapse SMBH distinction is DERIVED with HYPOTHESIS-strong type discrimination via nine observational signatures; the eventual decisive test rests on JWST/VLBI long-horizon observations of JWST Little Red Dot demographics.

**NC-XIII.8.** ZS-A12 v1.5 Theorem A12.6 (NK Universal Jump integer '2' =  $\dim(Z)$  = 2, DERIVED) is a first-principles geometric-substrate identification; v5.0 does NOT claim Z-Spin replaces the BKT/XY/percolation models — NC-A12.10 codifies that Z-Spin explains them at substrate level. Theorem A12.7 (Homes universal constant cardinal-2 framework) is DERIVED-CONDITIONAL: the bracketing inequality and order-of-magnitude prediction are DERIVED, but the exact analytical form requires the full  $\text{CuO}_2$   $V_{\text{coh}}$  dynamical-factor refinement (O-A12-MASTER.1, 4–8 weeks). v1.5 lifts A12.9 and A12.10 to DERIVED-under-Regge but full DERIVED status for A12.10 remains pending.

**NC-XIII.9.** ZS-T9 v2.3 Candidate Theorem B-Li.1 ( $P_{\text{B7Be}}^Z = -2A/Q$ , HYPOTHESIS-strong) is NOT DERIVED. Reasons explicitly recorded: (a) forward derivation from Z-Spin action OPEN — corpus does not yet possess a sectoral binding-energy translation operator; (b) discovery via constrained inverse — the  $+12\sigma$   $^7\text{Li}$  residual in v2.2 was used to identify the channel/quantum match; (c)  $P_{\text{B7Be}}$  is interpreted as BBN-frame sensitivity-coordinate perturbation, NOT a claim that the laboratory  $^7\text{Be}$  binding energy differs by  $-1.456\%$  today; (d) full PRyMordial verification with actual  $\text{B}_7\text{Be}$  perturbation needed beyond Dent linear scaffold. Level 3 (Full Nuclear Chemistry) OPEN.

### §XIII.E — v5.0 Master Falsification Registry Extensions

Cumulative ~340 falsification gates across v1.0 → v5.0. New v5.0 entries (preserving the v4.0 registry verbatim) are summarised below; full per-paper gate lists live in the source papers.

| § Anchor          | Falsifiable Statement                                                                                     | Status (v5.0)                            |
|-------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------|
| §F19.4 [LOCKED]   | $\tanh(\psi_X + \psi_Y) = 31/59$ and $\tanh(\psi_Y - \psi_X) = 3/67$<br>(50-digit, residual exactly zero) | DERIVED — F19 v2.2                       |
| §F19.6 [LOCKED]   | $\tanh(2 \cdot \psi_{\text{KMS}}(X \rightarrow Y)) = 3/5$<br>(KMS-to-Geometric Rapidity Projection)       | DERIVED — F19 v2.2                       |
| §F19-O.6 [LOCKED] |                                                                                                           | $\Delta S_{\text{obs}}(Y \text{ vs } X)$ |

| § Anchor             | Falsifiable Statement                                                                                                           | Status (v5.0)                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| §F20.DAG [LOCKED]    | Time-unrolled DAG of six triggers T1–T6 contains no directed cycle                                                              | DERIVED — F20 v1.1                  |
| §F20.NDR [LOCKED]    | $\Delta\theta_{T1}/\Delta\theta_{T2} = N_{-}(2\pi) = 2\pi/A = 78.4500565496$<br>(50-digit)                                      | PROVEN — F20 v1.1                   |
| §F20.BIO [TESTABLE]  | DNA trauma response = triadic activation T3 + T4 + T5 (substrate-agnostic structural)                                           | DERIVED-interpretation — F20 v1.1   |
| §M40.C [LOCKED]      | J-Conjugate Holonomy Knot Pair ( $K_p$ , $\bar{K}_p$ ) definition (assembly of six PROVEN components)                           | DERIVED — M40 v1.1                  |
| §M40.D [LOCKED]      | Log-Holonomy Factorization $\ell_Z(K_N) = \sum a_i \cdot \ln p_i$ via HSI Theorem                                               | DERIVED — M40 v1.1                  |
| §M40.F [LOCKED]      | $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ unique biquadratic class-number-one match over [2, 1000] (183,921 candidates)           | COMPUTATIONALLY VERIFIED — M40 v1.1 |
| §M40.B8 [HYP-strong] | $\prod_{p \neq 11} \det(I - p^{\{-s\}} M_p)^{-1} = \zeta_{\mathbb{Q}(\zeta_{11})}(s)$ as operator-model analogue of Kim 1510 §5 | DERIVED — M40 v1.1<br>ADS-2         |
| §Q10.D.5 [LOCKED]    | 37-Universality: five of six $c_n$ share denominator 37; FORCED by Truncation-Dual Theorem + 20 = 8 + 12                        | OBSERVATION-strong — Q10 v4.0       |
| §Q10.c2 [LOCKED]     | $c_2 = 2/19$ , $c_4 = 4/37$ via action-level loop                                                                               | DERIVED — Q10 v4.0                  |
| §Q10.c6 [LOCKED]     | $c_6 = 8/37$ via triakis-tetrahedron extension                                                                                  | DERIVED-CONDITIONAL — Q10 v4.0      |

| § Anchor               | Falsifiable Statement                                                                                                              | Status (v5.0)                     |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| §Q10.MC9c [LOCKED]     | $\ln \Lambda_{\text{total}} = 38.67$ cumulative anti-numerology;<br>positive-correlation ratio $217\times$                         | DECISIVE++ — Q10 v4.0             |
| §A11.Eq [LOCKED]       | $\epsilon$ -Halo $\leftrightarrow$ Sub-Halo<br>Equivalence at four levels<br>(profile, dynamical,<br>observational, falsification) | DERIVED — A11 v1.1                |
| §A11.LC [LOCKED]       | Vortex Six-Class Lifecycle<br>MECE via $\pi_1(U(1)) = \mathbb{Z}$<br>coarse-graining; $n_v \approx 2.8 \times 10^{15}$             | DERIVED-strong — A11 v1.1         |
| §A11-F-A5.7 [DECISIVE] | Direct DM particle detection<br>collapses $\epsilon$ -Halo; null<br>detection cannot collapse<br>CDM                               | PROVEN asymmetry — A11<br>v1.1    |
| §A12.6 [LOCKED]        | NK universal jump integer '2'<br>$= \dim(Z) = 2$ (cardinal-2<br>substrate identification,<br>HEADLINE)                             | DERIVED — A12 v1.5                |
| §A12.7 [LOCKED]        | Homes constant bracketing<br>$2/Q \leq$<br>$C_{\text{cuprate}}(ab)/C_{\text{BCS-dirty}} \leq$<br>$\ln 2/(1 - A)$                   | DERIVED — A12 v1.5                |
| §A12.4 [LOCKED]        | SC triadic activation $T2 \rightarrow T1$<br>$\rightarrow T4$ distinct from DNA $T3 +$<br>$T4 + T5$                                | DERIVED — A12 v1.5                |
| §A12.9 [LOCKED]        | $\text{CuO}_2$ $V_{\text{coh}}$ lattice<br>Hamiltonian as quasi-2D<br>Kelvin 2-cell; 4 ZS-Q6 §3.5<br>gates 4/4 PASS                | DERIVED-under-Regge —<br>A12 v1.5 |
| §A12-P9 [TESTABLE]     | Five major unconventional<br>SC families collapse to<br>universal $p_s^{\text{eff}}/(\sigma_n \cdot T_c)$<br>curve                 | TESTABLE — A12 v1.5               |

| § Anchor                | Falsifiable Statement                                                                                                         | Status (v5.0)                          |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| §T9.B-Li.1 [HYP-strong] | $P_{B7Be^Z} = -2A/Q = -70/4807$ (ZS-Q7 §5 slow eigenvalue)                                                                    | HYPOTHESIS-strong / TESTABLE — T9 v2.3 |
| §T9.Li7 [TESTABLE]      | ${}^7\text{Li}$ post-v2.3 $\approx 1.64 \times 10^{-10}$ ; pull $+0.13\sigma$ versus observed $(1.6 \pm 0.3) \times 10^{-10}$ | TESTABLE — T9 v2.3                     |
| §T9.NC4 [PRESERVED]     | v2.2 D/H + Y <sub>p</sub> ERO closure preserved unchanged; B <sub>7</sub> Be decoupled from D and ${}^4\text{He}$             | ERO-CLOSED — T9 v2.3                   |

All v5.0 entries are additive; no v4.0 entry is retracted. Total registry: ~340 falsification gates.

### §XIII.F — Conclusion

PART XIII absorbs the seven post-v4.0 source-paper integrations into the v5.0 Light Edition without retracting or modifying any v4.0 entry. The Foundations theme acquires two new structural theorems via ZS-F19 v2.2 (Frame-Invariant Tilt Theorem + KMS-to-Geometric Rapidity Projection) and a unified catalogue plus time-unrolled causal DAG of all six non-routine sector-topology triggers via ZS-F20 v1.1, with the new biological correspondence map ( $T3 + T4 + T5 \leftrightarrow$  DNA trauma) extending the substrate-agnostic isomorphism programme. The Math Spine theme extends with ZS-M40 v1.1, which gives the first explicit Z-sector geometric reading of the prime itself via the J-conjugate holonomy knot pair  $(K_p, \bar{K}_p)$ , complementing the v4.0 M28/M33/M34 critical-strip reading and the M36 Apollonian distribution reading; the deterministic uniqueness  $K = @(\sqrt{-3}, \sqrt{-11})$  over  $[2, 1000]$  is the principal anti-numerology assertion. The Quantum Mechanics theme registers the 37-Universality via ZS-Q10 v4.0 — a categorical signature of the Truncation-Dual functor across  $(X, Y, Z)$  sectors — together with the cumulative  $\ln \Lambda_{\text{total}} = 38.67$  anti-numerology evidence (highest in the corpus to date). The Astrophysics theme acquires the rigorous  $\varepsilon$ -Halo  $\leftrightarrow$  Sub-Halo Equivalence Theorem and the unified vortex lifecycle/SMBH-diversity framework via ZS-A11 v1.1, and the first first-principles geometric-substrate derivation of the Nelson–Kosterlitz universal jump and Homes universal constant via ZS-A12 v1.5 — extending the cardinal-2 corpus role from quantum/gravitational to condensed-matter sectors. The Translational theme extends with ZS-T9 v2.3 binding-response gate, providing a HYPOTHESIS-strong candidate / TESTABLE closure of the Lithium Problem at ERO Level 2 while preserving the v2.2 D/H + Y<sub>p</sub> ERO closure unchanged. The minimal triple  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  remains LOCKED. Zero new free parameters across all seven absorbed papers.

### PART XIV — Post-v5.0 Absorption: Nine New Source Papers + ZS-A11 Errata (v6.0)

## §0. Abstract

PART XIV consolidates the nine post-v5.0 source-paper integrations of v6.0 and applies the ZS-A11 v1.1 errata downstream into the Book. The v5.0 Light Edition reached 121-paper coverage with PART XIII; the nine new papers introduced after the v5.0 freeze — ZS-M41 v3.1 (Math Spine), ZS-Q11 v1.2 (Quantum Mechanics), ZS-F21 v2.0 (Foundations), ZS-F22 v2.1 (Foundations), ZS-A13 v1.3.1 (Astrophysics), ZS-A14 v2.0 (Astrophysics), ZS-A15 v1.5 (Astrophysics), ZS-T10 v2.0 (Translational), and ZS-Q12 v4.0 (Quantum Mechanics) — are absorbed below as single Light-edition chapters. All absorptions strictly preserve the v5.0 body verbatim (no-deletion rule). Zero new free parameters;  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  LOCKED unchanged across all nine absorbed papers. The total corpus is 130 papers. PART XIV opens six new structural results into the corpus: (i) the  $\delta_Y$  Antipodal Closure  $\delta_Y = F(tO)/(F(tO)+F(tI)) = 14/46 = 7/23$  with the  $\gcd(60,32) = 4$  identified as  $\dim(Z)^2$  (F22, OBSERVATION  $\rightarrow$  DERIVED); (ii) the OAQEC direct-sum stabilizer-code structure  $A_{ZS} \cong M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$  for the Z-frame gauge subsystem, the corpus's first operator-algebra error-correction reading of  $L_{XY} \equiv 0$  (Q11); (iii) the Commutant-Gate Insufficiency Theorem C3.0 and the  $V_+$ -decorated Burnol-conductor channel decoration on the Connes–Katsnelson Sonin space (F21); (iv) the Void Principle  $\mathcal{R}_V \geq Q/(2A) - 1 \approx 67.67$  deep-void lower bound and the closed hypergeometric horizon formula (A13); (v) the geometry-first  $TY \rightarrow X$  sector-conversion ANC correction  $\delta \ln y_b = -\kappa^2 = -A/Q$  for  ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$  (T10, TESTABLE); and (vi) the self-referential closure reducing the four ZS-Q12V open problems to the single multiplier condition  $\mu = \lambda = (i\pi/2)z^*$  on the  $Z = \partial X$  holographic-interface postulate (Q12). §XIV.G applies the ZS-A11 v1.1 errata ( $\sqrt{2} \sigma \leftrightarrow v_{\text{flat}}$  correction; contested Jiao/Zhou mass split) downstream into §VII.B-LIGHT and the PART XIII Chapter 41 framing.

### §XIV.A — Nine New Papers (Light-Edition Body Entries)

Each of the nine post-v5.0 source papers represents v6.0's principal additive content. All inputs LOCKED from upstream corpus; zero new free parameters. Each absorbed paper is cited below as a single Light-edition chapter, with the source paper carrying the full derivation. All nine source bodies were available at the final v6.0 integration (the two papers initially flagged [SOURCE PENDING] in the v6.0 working draft — ZS-M41 v3.1 and ZS-A14 v2.0 — were subsequently supplied and are now absorbed at full source-derived level; the [SOURCE PENDING] flags are accordingly cleared).

### Chapter 44 — ZS-M41 v3.1: Closed-Form Derivation of Refined Chabauty–Kim Depth-2 Scalars at $S = T = \{3, 11\}$ via F18 Bridge 4

ZS-M41 v3.1 [Math Spine theme, 2858/2864 PASS (6 honestly documented as M41.D strict-form falsifications;  $\sim 115$  s wall-clock, mpmath 80-digit), NEW v6.0] achieves the v3.0 deferred closure: an explicit closed-form derivation of the Refined Chabauty–Kim depth-2 scalars at  $S = T = \{3, 11\}$ , promoting M41.K-NonTriv from HYPOTHESIS-strong (v3.0 empirical cluster) to DERIVED via a short chain of classical identities. Principal result Theorem

M41.K-Decomp (DERIVED, NEW v3.1): for every prime  $p \geq 5$ ,  $p \notin \{3, 11\}$ , at the non-trivial  $S_3$ -orbit point  $z = 4$  of  $X1(\mathbb{Z}[1/33])$ , the  $V_4$ -twisted Coleman observer scalar admits the closed form  $c_{33}(p) = (\log_p(33)/p)(\log_p(4)/p)(\log_p(3)/p) \cdot \text{Li}_2^*(p)(4)^{-1} = -2 q_p(2) \cdot q_p(3) \cdot (q_p(3) + q_p(11)) \cdot \text{Li}_2^*(p)(4)^{-1} \pmod{p}$ , where  $q_p(a)$  is the Fermat quotient and  $\text{Li}_2^*(p)$  the Coleman  $p$ -adic dilogarithm; verified at 235/235 primes ( $5 \leq p \leq 1500$ ) as a perfect deterministic LHS = RHS match. The character decomposition  $c_{33}(p) = c_3(p) + c_{11}(p)$  (with  $c_3 = -2q_p(2)q_p(3)^2/\text{Li}_2(4)$  the  $\chi_{-3}$ -component and  $c_{11} = -2q_p(2)q_p(3)q_p(11)/\text{Li}_2(4)$  the  $\chi_{-11}$ -component) follows from the ZS-M28 PROVEN log-additivity  $\log_p(3) + \log_p(11) = \log_p(33)$ . The proof combines three classical ingredients: Lerch 1905 ( $\log_p(a)/p \equiv -q_p(a) \pmod{p}$ , IMPORTED-PROVEN), the power rule  $q_p(a^k) \equiv k q_p(a)$ , and Fermat-quotient additivity  $q_p(ab) \equiv q_p(a) + q_p(b)$  — the last matching M28 depth-2 log-additivity exactly. A second orbit  $z = 12$  closed form is added (DERIVED, 12 primes verified). The result is the explicit form of the F18 Bridge 4 algebraic face (F18 axiom A4, PROVEN): at depth 2 the  $V_4$ -twisted observer scalars decompose under M28 log-additivity into a direct sum of character-indexed Fermat-quotient polynomials; the geometric face (the same  $c_X(p)$  arising as traces of the  $V_4$  image of the Kostant D-operator  $D = \sum_a Z_a \otimes \gamma_a$  of ZS-M27, restricted to depth-2 cohomology of  $X1 = \mathbb{P}^1 \setminus \{0, 1, \infty\}$ ) is registered HYPOTHESIS-strong with a v3.2 closure path via direct Kostant cohomology computation. Anti-numerology honesty (v3.1): the v3.0 "15.1× enrichment" cluster claim at  $c_{33} = 5$  (6/235 primes) is RETRACTED — it is now explained as Poisson fluctuation ( $p \approx 0.001$ ), and the genuine substance is the closed form itself, not the cluster. NON-CLAIMs NC-M41.1–19: the classical identities are pre-corpus (originality is their placement inside refined Chabauty–Kim at the external untreated case  $S = \{3, 11\}$ ); M41.D' ( $\chi_{-11}$ -twisted  $\text{Li}_2$  closed form) remains OPEN after a failed degree-2 polynomial fit; the geometric face is not PROVEN; and the Refined Kim Conjecture at  $S = \{3, 11\}$  is not asserted PROVEN (algorithmic verification awaits Lüttke 2024 SageMath integration, v3.2). NC-M23.1 (Z-Spin does not prove RH) and NC-M23.7 inherited. → ZS-M41 v3.1 [2858/2864 PASS]; first explicit closed-form Fermat-quotient derivation of the refined Chabauty–Kim depth-2 scalars at  $S = T = \{3, 11\}$ , realising the F18 Bridge 4 algebraic face with the  $c_{33} = c_3 + c_{11}$  character decomposition.

## Chapter 45 — ZS-Q11 v1.2: QRF↔OAQEC Correspondence — A Direct-Sum Operator-Algebraic Stabilizer Code with Z-Frame Gauge Subsystem

ZS-Q11 v1.2 [Quantum Mechanics theme, 39/39 PASS, NEW v6.0] closes the v1.0 gap that  $L_{XY} \equiv 0$  supplies geometric noise suppression but not an active stabilizer code in the standard Gottesman sense, by recognising that the correct framework for a direct-sum sector decomposition  $\mathbb{C}^{11} = \mathbb{C}^2 \oplus \mathbb{C}^3 \oplus \mathbb{C}^6$  is operator-algebra quantum error correction (OAQEC; Bény–Kempff–Kribs 2007, Kribs–Laflamme–Poulin 2005, Dauphinais–Kribs–Vasmer 2024) — not the subspace Knill–Laflamme framework, which is structurally inapplicable to a direct sum. Three central theorems: Theorem Q11.A (OAQEC Algebra Structure, PROVEN by explicit diagonalisation) — the single-cell logical algebra  $A_{\text{ZS}} = P_X B(H_X) P_X \oplus \mathbb{C} \cdot P_{Z^{\{0\}}} \oplus P_{Y'} B(H_{Y'}) P_{Y'}$  has the canonical OAQEC block-diagonal form  $A_{\text{ZS}} \cong M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$ , exactly the Dauphinais–Kribs–Vasmer type  $\oplus_i I_{\{m_i\}} \otimes M_{\{n_i\}}$  with multiplicities  $((1,3),(1,1),(1,5))$ ;

Theorem Q11.B (Operator-Algebraic Knill–Laflamme on  $A_{\text{ZS}}$ , PROVEN) — for the Z-frame error set  $E_Z = \{I_X \otimes V \otimes I_Y : V \in B(H_Z)\}$  the X-logical correctability holds with the Bény–Kempff–Kribs block-diagonal coefficients X-block = 1, Z-gauge block =  $\langle 0_Z | V_a + V_b | 0_Z \rangle$ , Y-block = 1 (the scalar factor lives only on the Z-gauge block — the v1.2 self-correction of the abstract formula, body proof preserved verbatim from v1.1); this rescinds-and-replaces the v1.0 subspace-KL Theorem 6.2.1 and is strictly stronger because it applies to every Z-frame error rather than only the abelian sub-slice; Theorem Q11.C (Lacambra–Gauss–Law Code Embedding, DERIVED-with-bridge) — the direct-sum decomposition with  $L_{XY} \equiv 0$  realises the algebraic structure of a single-vertex Lacambra–Chatwin–Davies–Honda–Höhn (arXiv:2604.06087, 2026) Gauss law code with matter at four levels (gauge group  $G_{\text{gauge}} \cong \mathbb{Z}_2 \times \mathbb{Z}_2 =$  abelian core  $S_{\text{ZS}}$  of  $D_4$ ; gauge sector  $H_Z$ ; matter sector  $H_X \oplus H_Y$ ; Gauss-law constraint =  $L_{XY} \equiv 0$ ). The Relational Z-Stabilizer Group  $S_{\text{ZS}} = \langle J_Z, (JJ_Z)^2 \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$  is the unique maximal abelian sub-slice of  $D_4$  under the revised discrimination logic (C1 Klein-four excludes  $M_3$ ; C2 Z-internal anchoring excludes  $M_1$ );  $P_{\text{code}} = \frac{1}{2}(I + J_Z) \cdot \frac{1}{2}(I + (JJ_Z)^2)$  has  $\dim(H_{\text{code}}) = 9$  with  $1 + 3 + 5$  decomposition. The Z-bottleneck depth  $d_Z = 2$  (PROVEN via  $\text{ZS-Q7}$  heat-kernel  $\|K_{XY}(t)\| \sim t^2$ ) is the corpus-internal analogue of code distance. v1.2 self-corrections: the abstract Q11.B factor and the over-asserted "unitarily equivalent" in Q11.C are downgraded to "realises the algebraic structure of", with full unitary equivalence registered as OPEN gate O-Q11.6 (Peter–Weyl construction sketch, Appendix D). The strengthening derives entirely from selecting the correct external framework (OAQEC) — recognition of pre-existing PROVEN external mathematics, not new Z-Spin claims. Anti-numerology MC joint  $p < 1 \times 10^{-6}$  over 500,000 trials; six falsification gates plus anti-overclaim gate F-Q11.6 and direct- $X \leftrightarrow Y$ -noise stress gate F-Q11.7.  $\rightarrow$  ZS-Q11 v1.2 [39/39 PASS]; first OAQEC operator-algebraic reading of the Z-bottleneck as a Z-frame gauge subsystem code.

## Chapter 46 — ZS-F21 v2.0: The Archimedean–Finite Positivity Wall, III — Commutant-Gate Insufficiency, External Boundary-Selector Imports, and the $V_4$ -Decorated Sonin–Frobenius Construction over the Connes–Katsnelson Prolate Realization

ZS-F21 v2.0 [Foundations theme, 34/34 PASS, NEW v6.0] completes a thirteen-stage deep exploration of  $D_{4c}$  — the central open gate of the Z-Spin archimedean–finite positivity programme — by isolating honestly what can be closed and what remains open. Four parts. (i) Theorem C3.0 (Commutant-Gate Insufficiency, PROVEN here): if  $A = \Pi_{\lambda} \Pi_F \Pi_{\lambda}$  is the Slepian concentration operator, every real Borel functional  $f(A)$  is self-adjoint and lies in  $W^*(A)$ , so hermiticity and Slepian-commutation are necessary but vacuously degenerate gates — they cannot uniquely select the Connes–Katsnelson prolate realisation  $W_{\{\lambda, \text{sa}\}}$ ; 200/200 random degree-7 polynomial witnesses pass both gates (seed 20260528). The selector lives outside the commutant, in the boundary condition at the turning points  $q = \pm \lambda$ . (ii) External imports T4–T5 (IMPORTED-PROVEN, single block): the explicit boundary condition  $\lim_{q \rightarrow \pm \lambda} (q^2 - \lambda^2) \partial_q \xi(q) = 0$  (Connes 1998 Lemma 6); the Fourier-commuting uniqueness among self-adjoint extensions (Katsnelson 2016); the discrete two-sided unbounded spectrum with double classical / simple

non-classical eigenvalues (Connes–Moscovici 2022); and — settling the 2022 CCM conjecture — the negativity of all non-classical eigenvalues plus the imaginary-axis characterisation, with  $\lambda = \sqrt{2}$  reproducing the squares of the shifted Riemann zeros (Ramis–Richard-Jung–Thomann 2025). (iii)  $V_4$ -decoration (T6d): the ramified channels  $\chi_{-3}$ ,  $\chi_{-11}$ ,  $\chi_{33}$  are NOT represented by rescaled prolate parameters  $\Lambda_X = \Lambda\sqrt{q_X}$  (WITHDRAWN as fitted) but by Burnol conductor blocks  $C_X = \sum_{p|q_X} e_p(X) \cdot \log p = \log q_X$  on the common Connes–Katsnelson Sonin space; this decoration is canonical (an identity of integer factorisation) and matches Tate's analytic conductor  $(q_X/\pi)^{\{(s+a_X)/2\}}$ . (iv) Direct-sum closure diagnosis (T7): a pure channelwise direct sum  $Q^{\{\text{def}\}} = \sum_X \text{Tr}(D_X^\dagger D_X)$  is trivially PSD but cannot equal the  $V_4$ -decorated Weil form, because Probe-W2 data show each non-trivial channel individually indefinite ( $\chi_{-3}$  6/12,  $\chi_{-11}$  9/12,  $\chi_{33}$  7/12 negative grids) — pure direct sums violate Lemma M31.0 (Non-Separability, PROVEN). The replacement T7' retains the Reading-C  $\Pi_Z$ -sandwich while absorbing the Burnol decoration into the channelwise test factor  $\gamma_X(s)$ . NO claim of full Weil positivity or RH/GRH at any stage. Position relative to v1.2: v1.2 reduced the wall to the Positive-Square Criterion (F21.9, PROVEN both directions) and the Commutator-Coverage Obstruction (F21.10, PROVEN negative — third-axis commutator does not canonically select B); v2.0 supplies the honest boundary-selector evaluation with all external machinery imported by name and the Z-Spin contribution kept in the channel decoration. Six pre-registered anti-numerology controls (five rejected candidates withdrawn). → ZS-F21 v2.0 [34/34 PASS]; commutant gate proven insufficient, archimedean selector relocated to the external boundary condition, Z-Spin content localised to canonical  $V_4$  Burnol decoration.

## Chapter 47 — ZS-F22 v2.1: Equivariant Antipodal Face Modules, the Icosahedral Edge-Surplus Derivation of $\delta_Y$ , and the $T_7$ No-Go Theorem

ZS-F22 v2.1 [Foundations theme, 105/105 PASS, NEW v6.0] supplies four new proof objects that close the one explicit gap left by v2.0 ( $\delta_Y$  as OBSERVATION) and harden three prior results, building on the v2.0 Three-Sector Antipodal Unification Theorem  $F(P) = \dim(Z) \times N_{\text{axes}}(P)$  ( $X: 14 = 2 \times 7$ ,  $Y: 32 = 2 \times 16$ ,  $Z: 6 = 2 \times 3$ ). Result 1 ( $\delta_Y$  Antipodal Closure, Theorem F22.8, OBSERVATION → DERIVED): the Y-sector geometric-impedance numerator/denominator equals the X-face fraction of the total sector face count,  $\delta_Y = F(tO)/(F(tO)+F(tI)) = 14/46 = N_X/(N_X+N_Y) = 7/23$ ; the previously unexplained  $\gcd(60,32) = 4$  is shown to be exactly  $\dim(Z)^2$ , via the two load-bearing identities  $(V-F)(tI) = \dim(Z) \cdot F(tO)$  ( $28 = 2 \times 14$ ) and  $(V+F)(tI) = \dim(Z) \cdot (F(tO)+F(tI))$  ( $92 = 2 \times 46$ ), both reducing to the Truncation-Dual Theorem (ZS-F2 §11.2) and Euler's relation on the seed icosahedron through the Icosahedral Edge-Surplus identity  $E(\text{Ico}) = F(tO) + N_{\text{axes}}(tI) = 14 + 16 = 30$  — the  $\dim(Z)$  cancels between numerator and denominator, removing the residual coincidence. Result 2 (Axis-Trigger CSP Uniqueness, Theorem F22.4, DERIVED → DERIVED-strong): an exhaustive search over all  $7! = 5040$  axis-trigger bijections, filtered by five corpus-locked invariants (face-type block,  $A_4$   $1 \oplus 3$  decomposition, Time-Unrolled DAG roles from ZS-F20,  $\Delta\theta$  hierarchy,  $J_Z$  seam-root), yields exactly one valid bijection  $|\text{Iso}(A_{tO}, T_{F20})| = 1$ , with a sensitivity audit confirming each constraint binds; the v2.0 placeholder checks C-6–C-8 are replaced by a genuine constraint-solver certificate. Result 3 ( $T_7$   $A_5$ -Projector No-Go, Theorem F22.9, DERIVED →

DERIVED-strong): rewritten at projector level using  $A_5$  character idempotents on  $\Omega^2(\text{tl}) = 2 \cdot (1 \oplus 3 \oplus 3' \oplus 4 \oplus 5)$ , the unique chirality-neutral subspace  $\ker \Delta$  is the irrep-4 isotype (dim 8) while the sector-exchange operator is supported on the  $\Delta \neq 0$  isotypes (dim 24); their intersection is empty,  $\text{Hom}^{\text{sector}}_{\{X \leftrightarrow Y\}} \cap \ker(\Delta) = \text{End}_{\text{gauge}}(4)$ , a gauge-internal endomorphism rather than a sector exchange — hence the seventh trigger  $T_7$  is closed negatively at the operator level. Result 4 (Equivariant Antipodal Face Module, Theorem F22.10, DERIVED  $\rightarrow$  DERIVED-strong): for any centrally symmetric convex polyhedron whose faces avoid the origin, central inversion  $\iota = -\text{I}$  acts freely on the face set ( $\text{trace } P_{-\text{I}} = 0$ ), so by Burnside  $N_{\text{axes}}(P) = F(P)/2$  and  $\dim C_+^2(P) = \dim C_-^2(P) = F(P)/2 = N_{\text{axes}}(P)$  — the antipodal axis count is an equivariant cochain rank, verified for all three sector polyhedra. All v1.0–v2.0 results preserved (no-deletion). With  $\delta_Y$  now DERIVED, ZS-F22 unifies the X-sector trigger cardinality, the Y-sector mediator count, and the Y-sector geometric impedance  $\delta_Y$  under a single antipodal-quotient framework.  $\rightarrow$  ZS-F22 v2.1 [105/105 PASS];  $\delta_Y$  promoted OBSERVATION  $\rightarrow$  DERIVED via  $\dim(Z)^2$  gcd identification,  $T_7$  No-Go hardened to projector level.

## Chapter 48 — ZS-A13 v1.3.1: The Void Principle — Why the Atom and the Universe Are Mostly Empty in Z-Spin Cosmology

ZS-A13 v1.3.1 [Astrophysics theme, 31/31 PASS (paper audit) + 16/16 (v1.3 numeric+token) + 5/5 (optional) = 52/52 cumulative PASS, NEW v6.0] formulates the Void Principle: stable existence requires relational volume to dominate material-core volume, because direct X–Y coupling is structurally suppressed by  $L_{XY} \equiv 0$ , so stable reality cannot appear as a compact filled object but only as localised cores embedded in relational voids — emptiness is the X-sector manifestation of separation, delay, non-collapse, horizon capacity, and cross-sector impedance. The v1.2 Mathematical Density Upgrade elevates the qualitative  $\mathcal{R}_V \gg 1$  claim to a quantitatively bounded law via four theorems re-reading external PROVEN results through the Z-Spin lens: (T1, A13.3) Schild 2018 Schur–Complement Cheeger inequality applied to the block-Laplacian with  $L_{XY} \equiv 0$  gives the deep-void lower bound  $\mathcal{R}_V(S) \geq Q/(2A) - 1 \approx 67.67$  (DERIVED, valid in the regime  $\kappa \leq 2A/Q$ ); (T2, A13.4) the Sheth–van de Weygaert 2004 excursion-set void formalism plus the Jennings et al. 2013 Vdn model re-indexed as a measure-pushforward dual projection of the same sector budget closure, giving  $\mathcal{R}_V^{\wedge \cos(t_0)} = 82/39 \approx 2.103$  (DERIVED-CONDITIONAL); (T3, A13.5) Banach–Tarski via the ZS-A9R  $F_2 \rightarrow D_4$  amenability functor supplying the necessary condition for the void measure to be well-defined (DERIVED); (T4, A13.6) the heat-kernel Seeley–DeWitt coefficient  $\Delta a_2 = 9 \cdot A/Q = 315/4807 \approx 0.06554$  (PROVEN, ZS-F19 §13.5) quantifying void-capacity activation strength. The v1.2-final §12 OPEN-Gate closures: A13.7 derives the atomic-to-nuclear radius ratio  $\chi_{\text{atom}}^{\text{ZS}} = \sqrt{\pi} \cdot v \cdot Q / (2\lambda_1 V_Y m_e) \approx 6.29 \times 10^4$  (within observed  $10^4$ – $10^5$ ) with  $A$  cancelling structurally, plus corollary  $m_p^{\text{ZS}} = 2\sqrt{\pi} \cdot \Lambda_{\text{QCD}} \approx 936.4$  MeV (PDG 938.27,  $-0.20\%$ ); A13.8 derives  $D_{\text{obs}}/R_{\text{Hubble}}$  from the sector budget alone. The v1.3 Closed Hypergeometric Horizon Upgrade replaces the numerical 6.42 with the exact closed form  $D_{\text{obs}}/R_{\text{Hubble}} = 4 \cdot \sqrt{(121/\Omega_m)} \cdot {}_2F_1(1/2, 1/6; 7/6; -\Omega_\Lambda/\Omega_m)$ : Branch A (slot-budget,  $\Omega_m = 39/121$ ,  $\Omega_\Lambda = 82/121$ ) gives 6.42009...  $\rightarrow$  93.19 Gly at  $H_0 = 67.36$ ; Branch B (face-counting, corpus-primary,  $\Omega_m = 38/121$ ) gives 6.48834...  $\rightarrow$  94.18 Gly, closer to the standard 94 Gly; Lemma A13.8R

registers the full radiation-aware particle horizon  $\approx 91.48$  Gly as a diagnostic, with A13.8' explicitly NON-CLAIMING to be the full particle horizon (it derives only the late-time matter–separation horizon capacity). A13.10 sharpens the saturation regime via  $\delta_{\text{sat}}^{\text{ZS}} = 2A \cdot 121/(Q \cdot 39) - 1 \approx -0.9548$ , matching Hamaus 2014 universal-void-profile deepest voids  $\delta_{\text{min}} \approx -0.85$  to  $-0.95$  (HYPOTHESIS-strong). v1.3.1 sharpens the §11.1 spectral bound to  $\lambda_2(S_X^{\text{eff}}) \leq A/Q$  (tight, from the rank-1 leading Schur correction and trace-class normalisation  $\kappa^2 = A/Q$ ), preserving 67.67 unchanged. The paper does NOT claim A is numerically equal to an atomic or cosmic void ratio; it claims A, as geometric impedance, explains why cross-sector translation requires separational volume and how strongly that requirement is bounded. → ZS-A13 v1.3.1 [52/52 cumulative PASS]; deep-void lower bound  $\mathcal{R}_V \geq Q/(2A) - 1 \approx 67.67$  and closed hypergeometric horizon formula DERIVED.

## Chapter 49 — ZS-A14 v2.0: Z-Spin-Mediated Singlet-to-Triplet Conversion — SU(2)-Covariant Usadel Closure, the Envelope-Shifted Current-Phase Relation, Three-Observable Locking, and Two-Channel Z-Sector Length Spectroscopy

ZS-A14 v2.0 [Astrophysics / Condensed-Matter Bridge theme, 47/47 PASS |  $\Delta\text{BIC} = 8.28$  (supporting) | MC hit =  $0.64\% < 5\%$ , NEW v6.0] corrects the single substantive error of v1.2 and converts it into two new theorems, with no new free parameters beyond  $A = 35/437$  and  $Q = 11$ . The unifying object is a single dimensionless Z-cell phase  $\varphi_Z(dF) = A \cdot dF/\xi_Z$  controlling every observable in a diffusive superconductor–ferromagnet–superconductor singlet-to-triplet conversion setting, distinguished into three layers: the prefactor  $\sin^2(\varphi_Z/2)$  (peaks at the conversion optimum  $dF^* = (\pi/A)\xi_Z$ ), the diffusive decay envelope  $\exp(-dF/\xi_{tZ})$  (monotonic, shifts the full-current peak earlier), and the envelope-immune nodes (set the common period). The v1.2 error — claiming the full Josephson critical current peaks at  $dF^*$  — is corrected: Theorem A14.8' (Envelope-Shifted Peak, DERIVED) gives the exact condition  $\tan(A \cdot dF_{\text{peak}}/2\xi_Z) = A \cdot \xi_{tZ}/\xi_Z$ , so only the conversion prefactor and the detrended current  $I_c \cdot \exp(+dF/\xi_{tZ})$  peak at  $dF^*$ . Theorem A14.9' (Three-Observable Locking, DERIVED, envelope-immune form): the conversion efficiency  $\eta$ , the effective superfluid stiffness  $\rho_s^{\text{eff}}$ , and the detrended current share the peak  $dF^*$ , while the raw-current nodes coincide with the  $\eta$ -nodes at  $dF = n \cdot 2\pi\xi_Z/A$ , giving a common period  $\Delta dF = 2dF^*$  independent of the decay envelope. Theorem A14.11 (Two-Channel Z-Length Spectroscopy, DERIVED/TESTABLE) turns the peak shift into a second probe: a single thickness scan yields  $\xi_Z$  from the node spacing and the Z-aligned triplet length  $\xi_{tZ}$  from the peak shift,  $\xi_{tZ} = (\xi_Z/A) \cdot \tan(A \cdot dF_{\text{peak}}/2\xi_Z)$ . The SU(2)-covariant Usadel result Theorem A14.7 is honestly relabelled DERIVED-CONDITIONAL: the equation form  $D\tilde{\nabla}_i(\check{g}\tilde{\nabla}_i\check{g}) + [i\epsilon\tau_3 + \check{A} + \check{h}_Z, \check{g}] = 0$  with the SU(2) gauge covariant derivative is IMPORTED-PROVEN from Bergeret–Tokatly and Tokatly, while the Z-Spin background field  $\mathcal{A}_i^Z = (A/2\xi_Z)\hat{n}_Z \cdot \sigma$  (with cell holonomy  $\oint \mathcal{A}_i^Z dl = (A/2)\hat{n}_Z \cdot \sigma = -i \log U_Z[\text{cell}]$ ) is a zero-parameter ansatz whose physical reality is the testable content; Tokatly's PROVEN dephasing tensor  $\Gamma^{ab} = D(\mathcal{A}\delta^{ab} - \mathcal{A}^a\mathcal{A}^b)$  lifts the angular factor  $\sin^2\beta$  to a derived projection identity ( $n_a \Gamma^{ab} n_b = D|\mathcal{A}_{\text{so}}|^2 \sin^2\beta$ ). Inherited v1.2 corrections C-01–C-04 re-verified (NEW-2 half-angle  $\cos(\Theta_{\text{SM}}/2) = \frac{1}{2}\text{Tr } U_{\text{AW}}[C]$  returning  $\Theta_{\text{SM}} = A$ ; Lemma A14.B clean single-cell interface universality via SU(2) conjugacy

invariance; SOC dephasing vanishing in bulk; triplet-length ratio convention). The BIC analysis is demoted to supporting evidence; the paper's falsifiable value is carried by the thickness-scan and peak-shift predictions. Reference [2] corrected to Yates et al. (arXiv:1606.08619).

NC-A14.4: A14.7 does not assert  $\mathcal{A}_i^Z$  is an independently measured field — it asserts a parameter-free, falsifiable prediction for its magnitude; NC-A12.10 (Z-Spin explains rather than replaces BKT/XY/Usadel proximity physics) inherited. → ZS-A14 v2.0 [47/47 PASS]; first Z-Spin SU(2)-covariant Usadel closure with an envelope-shifted current-phase relation, three-observable locking at  $dF^*$ , and two-channel Z-sector length spectroscopy extracting  $\xi_Z$  and  $\xi_{tZ}$  from a single thickness scan.

## **Chapter 50 — ZS-A15 v1.5: Pre-Registered Gaia DR4 Falsification Protocol for the $\varepsilon$ -Halo Outer Boundary — Enclosed-Mass Saturation, Keplerian Slope, and the NFW/MOND Degeneracy**

ZS-A15 v1.5 [Astrophysics theme, 36/36 PASS, NEW v6.0; dependencies ZS-F1, ZS-A1, ZS-A5, ZS-A11 v1.1] is a pre-registered, hash-locked falsification protocol (frozen a15\_dr4\_protocol.yaml, SHA-256 1d04ee2f...b283e2) fixed before Gaia DR4 (2 December 2026). It models galactic dark matter as the gradient energy of the exactly massless Goldstone mode  $\theta$  of a spontaneously broken internal U(1)<sub>Z</sub>, giving  $\rho_\theta \propto 1/r^2$  — functionally identical to a singular isothermal sphere and, across seven macroscopic observables, to a  $\Lambda$ CDM sub-halo (ZS-A11 v1.1) — with the descriptions breaking only at microscopic ontology (the one-way detection asymmetry F-A5.7) and at the outer boundary, where the Three-Region structure of ZS-F1 §5.3 terminates the halo at a finite radius  $r_Z$  (DERIVED). The protocol freezes a single decisive Milky Way test: whether the enclosed mass saturates near  $M_{\text{total}} \approx 2 \times 10^{11} M_\odot$  by  $\approx 25$  kpc (logarithmic slope  $\gamma \rightarrow -1/2$ ) versus continuing to grow toward  $M_{200} \approx 8 \times 10^{11} M_\odot$ . Four honesties are built in: (i) a finite cutoff is DERIVED upstream and shared by every finite-mass halo (truncated/cored NFW included) — not advanced as novel (§2A); (ii)  $r_Z$  is DERIVED-CONDITIONAL on observed inputs ( $M_{\text{total}}$ ,  $v_{\text{flat}}$ ), split into the now-closed well-posedness of the boundary-value problem (A15-T1a, IMPORTED-PROVEN via Ginzburg–Landau / harmonic-map theory) and the transition width — the No-Intrinsic-Width Lemma proves the strictly massless equation  $\square\theta = 0$  fixes the logarithmic profile but determines no universal width coefficient, so the v1.2 geometric width candidates  $\{A, 1/Q, Z/Q\}$  are RETRACTED, and the problem is reframed as a Sharp-Edge Mechanism (A15-T1b-3) where the global point boundary condition  $M_\varepsilon(r_Z) = M_{\text{total}}$  on the scale-free interior yields a near-step edge (DERIVED-CONDITIONAL, no new parameter), turning the v1.4 "data sharper than the smooth field" tension into a prediction and registering F-A15.7 (a wide DR4-resolved intrinsic width would refute it); (iii) the protocol does not exclude all particle dark matter and, on the rotation curve alone, does not separate  $\varepsilon$ -Halo from the Moffat-type MOG family — structured as a four-model decision matrix (§4B) and a five-truth confusion matrix (T20a–e), not hidden; (iv) an Imported No-Go Ledger (§2A) records with external citations exactly where the rotation curve alone cannot decide. The present record is explicitly contested (Jiao et al. 2023 low-mass versus Zhou et al. 2022 full-halo), which is why advance registration converts any post-DR4 agreement from numerology into evidence. No numerical  $\Delta r/r_Z$  is claimed. →

ZS-A15 v1.5 [36/36 PASS]; the hash-locked pre-registered Gaia DR4 enclosed-mass-saturation test that operationalises the ZS-A11  $\varepsilon$ -Halo outer boundary as falsifiable, with the Jiao/Zhou mass split registered as the open discriminant.

## **Chapter 51 — ZS-T10 v2.0: The TY→X Sector-Conversion Operator — A Geometry-First ANC Correction $\delta \ln \gamma_s = -\kappa^2$ for ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ , a Per-Nucleus Falsification Programme, and the ${}^8\text{B}$ Consistency Test**

ZS-T10 v2.0 [Translational theme, 32 checks PASS | Gate Li-2 FLAGGED (nonlinear  ${}^7\text{Li}$  tension), NEW v6.0] presents the TY→X sector-conversion operator and its single geometry-fixed nuclear prediction, with  $A = \delta_X \cdot \delta_Y = (5/19)(7/23) = 35/437$  fixed from polyhedral curvature asymmetry (ZS-F2) by the  $H_0$  tension ( $H_{0,\text{local}}/H_{0,\text{CMB}} = e^A$ ) and the baryon-to-photon ratio  $\eta_B$  BEFORE any nuclear input — not tuned to lithium (§1.1). Standard-physics statement: the  ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$  radiative-capture rate receives a geometric correction  $\delta \ln \gamma_b = -\kappa^2 = -A/Q \approx -0.73\%$  on the  ${}^7\text{Be}$  asymptotic normalisation coefficient (ANC), equivalent to a fixed shift of the  ${}^3\text{He}$ – ${}^4\text{He}$  cluster-EFT short-range Wilson coefficient  $C_0$  (§2.2). Z-Spin-only prediction (TESTABLE):  $S_{34}(E)$  in the BBN Gamow window (0.1–0.5 MeV) is systematically lower by  $\approx 0.7$ – $1.5\%$ , at the LUNA-MV precision frontier — an interesting falsifiable hypothesis below current  $\approx 4\%$  precision but within reach of LUNA-MV-class  $\approx 1$ – $2\%$  systematics. Immediate consistency test: a universal binding shift  $-\dim(Z)\kappa^2$  would unbind  ${}^8\text{B}$  ( $\delta Q/Q = -401\%$ ) and destabilise  ${}^9\text{C}$  ( $-44\%$ ); since both are observed (AME2020), the action must be bottleneck-selective (NC-T10.5), with the master-equation slow eigenvalue  $\lambda_{\text{slow}} = -2A/Q = -\dim(Z)\kappa^2$  (§2.1). Structural anti-numerology Bayes factor  $\ln \Lambda = 7.2$  (very strong, not decisive; §6.5). The PRyMordial gate is the principal caution: the physical  ${}^7\text{Li}$  closure is coordinate-only (linear  $+0.3\sigma$  versus nonlinear  $-3.3\sigma$ , Gate Li-2 FLAGGED). The cross-corpus signal is the cardinal-2 invariant  $\dim(Z) = 2$ , not the rational  $-2A/Q$  (NC-T10.6).  $C_0 \leftrightarrow \kappa^2$  matching remains OPEN (Gate S-T10.1, the ZS-T11 programme deliverable: matching  $C_0(\kappa^2)$  via controlled RG running from the ZS-S10 Planck-scale gauge bridge would promote  $\delta \ln \gamma_b = -\kappa^2$  from HYPOTHESIS-strong to DERIVED). Zero free parameters. → ZS-T10 v2.0 [32 checks PASS, Gate Li-2 FLAGGED]; the per-nucleus, geometry-first falsification programme that states the lithium-problem ANC correction in standard nuclear-physics language with one measurable LUNA-MV number and an immediate  ${}^8\text{B}/{}^9\text{C}$  consistency test.

## **Chapter 52 — ZS-Q12 v4.0: The Self-Referential Closure (Consolidated) — From the Photon–ALP Boundary to a Single Holographic Postulate**

ZS-Q12 v4.0 [Quantum Mechanics / Foundations theme, paper-level gates 18/18 + code-level tests 43/43 PASS (Cat A–M), NEW v6.0] is the consolidated master edition (Part I = ZS-Q12V v2.1 verbatim; Part II = ZS-q12 v3.0 verbatim; Part III = the new bedrock reduction; no content removed) advancing the i-tetration fixed point  $z^* = 0.4382829 + 0.3605924 i$  (ZS-M1) to a unified closure. Part I (ALP boundary): the complex transmission time delay  $\tau_f$  of a coherent sub-unitary scattering system locks to  $z^*$  at its operating point with signatures  $\arg(z^*) = 39.4455^\circ$  (gate L1) and  $\text{Im}/\text{Re} = 0.8227$  (gate L4); the GRB 221009A transparency anomaly distinguishes

incoherent EBL absorption (dissipation-dominant,  $\tau'/\phi' \gg 10^9$ ,  $\arg \rightarrow 90^\circ$ , Class D — DERIVED) from coherent photon–ALP mixing (Raffelt–Stodolsky 1988,  $\tau'/\phi' \sim O(1)$ ,  $z^*$ -eligible Class C — DERIVED-CONDITIONAL). The Direct Coordinate-Dictionary No-Go (Theorem Q12.NG, PROVEN computational): every natural dictionary mapping the Z-Spin invariants (half-holonomy  $\theta = \pi(1-\varepsilon)$ , Wilson damping  $|\lambda|^2 = 0.7948$ ,  $\varepsilon = x^*$ ) onto the photon–ALP Hamiltonian ratios yields  $\arg(\tau) \approx \pi$  and never  $39.4455^\circ$  (18/18 mappings, minimum residual  $140.6^\circ$ ) —  $z^*$  is the fixed point of a transfer operator, not a coordinate value, so the direct dictionary is the wrong object; this closes OPEN-Q8.1-old by proven elimination. Part II (four-problem reduction): the four ZS-Q12V open problems ( $\varepsilon$ –g<sub>ay</sub> dictionary Q8.1, RG/attractor route Q12.B, measurement connection Q12.A, measurement–i-tetration conjugacy I3) are proven to be one problem — each reduces to the single codimension-2 condition that the conditioned/transfer multiplier of the physical system equals the Wilson eigenvalue,  $\mu(S) = \lambda = (i\pi/2)z^*$  (Theorem q12, Unified  $z^*$  Universality, DERIVED-CONDITIONAL on the  $4\pi$  spin-closure axiom). The Open-Loop No-Go (Theorem Q12.RG, PROVEN): an i.i.d. random coherent-lossy domain product converges to  $0^\circ$  (Furstenberg–Oseledets regime, distribution-dependent Lyapunov damping), so there is no open-loop  $z^*$  attractor; the universality lives one layer up in the self-referential closed-loop map  $z \mapsto i^\wedge z = \exp((i\pi/2)z)$ , which converges to  $z^*$  from a distribution-independent 92.2% basin consistent with  $|f'(z^*)| = 0.89151 < 1$  (DERIVED, ZS-M1). Part III (bedrock reduction, new): the dim-2-alone No-Go, the  $Z = \partial X \Rightarrow 4\pi$  cascade, the mediation No-Go, and the holographic-interface postulate —  $z^*$  is DERIVED-CONDITIONAL on the  $Z = \partial X$  holographic-interface postulate, and the requirement  $\mu = \lambda$  is met not by emergence but by the  $4\pi$  spin-closure ( $b = i$ ) that defines the Z-sector. Three NO-GOs PROVEN. Zero free parameters ( $A = 35/437$ ,  $Q = 11$ ,  $z^*$ ,  $\alpha_{op} = \pi/5$ ). Honest prior preserved: the lab locking claim (ZS-Q12V) most likely fails L1 on real data (RETRACTED) — unchanged; no conflict with  $\Lambda$ CDM, Standard-Model couplings, or LHAASO constraints.  $\rightarrow$  ZS-Q12 v4.0 [18/18 + 43/43 PASS]; the consolidated proof that four open problems are one codimension-2 condition closed by the  $4\pi$  spin-closure /  $Z = \partial X$  holographic-interface postulate.

#### PART XIV Source Paper Interface (v6.0)

| Source paper | Theme/Role                                                                                      | Chapter   | Verification   |
|--------------|-------------------------------------------------------------------------------------------------|-----------|----------------|
| ZS-M41 v3.1  | Math Spine<br>(Chabauty–Kim<br>depth-2 scalars at $S$<br>$= T = \{3, 11\}$ via F18<br>Bridge 4) | XIV Ch 44 | 2858/2864 PASS |
| ZS-Q11 v1.2  | Quantum Mechanics<br>(QRF $\leftrightarrow$ OAQEC;<br>direct-sum stabilizer                     | XIV Ch 45 | 39/39 PASS     |

| Source paper  | Theme/Role                                                                                                                                  | Chapter   | Verification                    |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------------|
|               | code; Z-frame gauge subsystem)                                                                                                              |           |                                 |
| ZS-F21 v2.0   | Foundations<br>(Commutant-Gate Insufficiency;<br>$V_4$ -decorated<br>Sonin–Frobenius;<br>Connes–Katsnelson)                                 | XIV Ch 46 | 34/34 PASS                      |
| ZS-F22 v2.1   | Foundations<br>(Equivariant<br>Antipodal Face<br>Modules; $\delta_Y$<br>Edge-Surplus<br>Derivation; $T_7$ No-Go)                            | XIV Ch 47 | 105/105 PASS                    |
| ZS-A13 v1.3.1 | Astrophysics (The<br>Void Principle; $\mathcal{R}_V \geq$<br>$Q/(2A) - 1$ ; closed<br>hypergeometric<br>horizon)                            | XIV Ch 48 | 52/52 cumulative<br>PASS        |
| ZS-A14 v2.0   | Astrophysics /<br>Condensed-Matter<br>Bridge<br>(Singlet-to-Triplet<br>Conversion;<br>$SU(2)$ -covariant<br>Usadel; length<br>spectroscopy) | XIV Ch 49 | 47/47 PASS                      |
| ZS-A15 v1.5   | Astrophysics<br>(Pre-Registered Gaia<br>DR4 $\varepsilon$ -Halo<br>falsification protocol;<br>enclosed-mass<br>saturation)                  | XIV Ch 50 | 36/36 PASS                      |
| ZS-T10 v2.0   | Translational ( $TY \rightarrow X$<br>Sector-Conversion                                                                                     | XIV Ch 51 | 32 checks PASS, Li-2<br>FLAGGED |

| Source paper | Theme/Role                                                                                                  | Chapter   | Verification              |
|--------------|-------------------------------------------------------------------------------------------------------------|-----------|---------------------------|
|              | Operator; $\delta \ln \gamma_s = -\kappa^2$ ; ${}^8B$ consistency)                                          |           |                           |
| ZS-Q12 v4.0  | Quantum Mechanics / Foundations (Self-Referential Closure; $Z = \partial X$ holographic postulate)          | XIV Ch 52 | 18/18 + 43/43 PASS        |
| ZS-T11 v2.1  | Translational / Methodology (HACSM; Corpus Auditor; F18 Contradiction Engine) — v6.1 increment, see PART XV | XV Ch 53  | 12/13 AUDIT-PASS (subset) |

#### §XIV.C — v6.0 Cross-Coupling Discipline (T7) Compliance

Per T7 (§VIII.7), every cross-paper synthesis must be P-CC (PROVEN-composed) or I-CC (interpretation). All nine PART XIV chapter entries are summarisations of source-paper Verification Summaries; no new theorem is introduced by PART XIV itself. Each absorbed paper's status (PROVEN / DERIVED / DERIVED-CONDITIONAL / DERIVED-strong / DERIVED-with-bridge / IMPORTED-PROVEN / HYPOTHESIS-strong / TESTABLE / NO-GO) is preserved verbatim from the source. The cardinal-2 substrate ( $\dim(Z) = 2$ ) acquires further corpus instantiations via ZS-F22 v2.1 (the  $\gcd(60,32) = 4 = \dim(Z)^2$  Antipodal Closure) and ZS-Q11 v1.2 (the rank-2 Z-gauge subsystem of the OAQEC algebra and the Z-bottleneck depth  $d_Z = 2$ ), alongside the six prior instantiations (gauge constraint  $Q = 11$ , CHSH  $2\sqrt{2}$ , channel capacity  $\leq \ln 2$ ,  $\chi(S^2) = 2$ , BCC  $T^3$  Hodge proton spin, Nelson–Kosterlitz universal jump). The five-Banach–Tarski-functor family of v4.0 is unaffected; PART XIV contributes no new functor member (ZS-A13's use of the  $F_2 \rightarrow D_4$  amenability functor is a re-citation of the existing ZS-A9R member, not a new one). Cumulative no-deletion rule preserved across  $v3.0 \rightarrow v3.5 \rightarrow v4.0 \rightarrow v5.0 \rightarrow v6.0$ . Word count strictly increases over v5.0.

#### §XIV.D — What v6.0 Does NOT Claim

**NC-XIV.1.** v6.0 does NOT modify or retract any v5.0 chapter entry. The supersedes statement is strictly additive. The sole v6.0 corrections to prior Book text are the ZS-A11 errata propagations in §XIV.G, which are corrections the ZS-A11 v1.1 source errata itself mandates; the original v5.0 wording is preserved verbatim within §XIV.G.

**NC-XIV.2.** v6.0 does NOT supersede the v5.0 PART X / PART XI / PART XII / PART XIII readings. The Spin-Pair Coexistence Principle (Reading X.4), the TO/TI Polyhedral Mapping (Readings XI.1–XI.4), the post-v3.5 twelve-paper absorption (PART XII §A and §B), and the seven-paper PART XIII absorption are inherited unchanged.

**NC-XIV.3.** Two papers (ZS-M41 v3.1, ZS-A14 v2.0) were initially registered at title-and-theme level in the v6.0 working draft because their full source bodies were not present at first integration; both were subsequently supplied and are now absorbed at full source-derived level, and the [SOURCE PENDING] flags are cleared. v6.0 asserts for these two papers only what their source Verification Summaries establish — ZS-M41 v3.1 at 2858/2864 PASS (6 entries honestly documented as M41.D strict-form falsifications; the v3.0 "15.1× enrichment" cluster claim is RETRACTED as Poisson fluctuation) and ZS-A14 v2.0 at 47/47 PASS ( $\Delta\text{BIC} = 8.28$  demoted to supporting). The 130-paper total now reflects full integration of all nine new papers.

**NC-XIV.4.** ZS-Q11 v1.2 Theorem Q11.C is DERIVED-with-bridge at the algebraic-structure level (four correspondences with the Lacambra–Chatwin-Davies–Honda–Höhn Gauss law code); v6.0 does NOT claim full unitary equivalence  $H_{\text{cell}} \approx H_{\text{Lacambra}}$  — that is registered as OPEN gate O-Q11.6 with the Peter–Weyl promotion path. The OAQEC correctability is the recognition of pre-existing PROVEN external mathematics, not a new Z-Spin physical claim.

**NC-XIV.5.** ZS-F21 v2.0 does NOT claim full Weil positivity, RH, or GRH at any stage. Theorem C3.0 proves the commutant gate is necessary but insufficient; the archimedean selector is the external boundary condition (Connes 1998), imported not re-derived. The  $V_4$  Burnol decoration is canonical (integer-factorisation identity), not a fitted parameter (five rejected candidates WITHDRAWN).

**NC-XIV.6.** ZS-F22 v2.1 promotes  $\delta_Y$  from OBSERVATION to DERIVED via the  $\dim(Z)^2$  gcd identification; v6.0 does NOT claim this changes the LOCKED value  $\delta_Y = 7/23$  (it explains it). The  $T_7$  No-Go is a negative closure (no seventh "space  $\leftrightarrow$  time only" trigger at the operator level), consistent with ZS-F20 O-F20.3.

**NC-XIV.7.** ZS-A13 v1.3.1 does NOT claim  $A = 35/437$  is numerically equal to any atomic or cosmic void ratio; it claims  $A$  as geometric impedance explains why cross-sector translation requires separational volume and bounds how strongly. The  $\mathcal{R}_V \geq Q/(2A) - 1 \approx 67.67$  bound holds in the deep-void regime  $\kappa \leq 2A/Q$ , not as a universal coarse-grained average; A13.8' explicitly NON-CLAIMS to be the full particle horizon. Residual sub-OPEN gates O-A13.9.1 (Regime-B  $w_a^{\text{eff}}$ ) and O-A13.8.2 (absolute  $H_0$ ) remain.

**NC-XIV.8.** ZS-A15 v1.5 does NOT claim Z-Spin has measured the Galaxy, does NOT exclude all particle dark matter, and does NOT separate  $\varepsilon$ -Halo from the Moffat-type MOG family on the rotation curve alone.  $r_Z$  is DERIVED-CONDITIONAL on observed inputs ( $M_{\text{total}}$ ,  $v_{\text{flat}}$ ); no numerical transition width  $\Delta r/r_Z$  is claimed (the No-Intrinsic-Width Lemma RETRACTS the v1.2

geometric width candidates). The decisive test is enclosed-mass saturation at Gaia DR4; the Jiao/Zhou record split is explicitly contested.

**NC-XIV.9.** ZS-T10 v2.0 does NOT claim to solve the cosmological lithium problem.  $\delta \ln \eta_b = -\kappa^2$  is HYPOTHESIS-strong (the forward  $C_0 \leftrightarrow \kappa^2$  matching is OPEN, Gate S-T10.1 / ZS-T11). The  $S_{34}$  shift is TESTABLE but not confirmed (below current precision). Gate Li-2 is FLAGGED (linear  $+0.3\sigma$  versus nonlinear  $-3.3\sigma$  PRyMordial tension); the cross-corpus signal is  $\dim(Z) = 2$ , not the rational  $-2A/Q$  (NC-T10.6). The  $-\kappa^2$  ANC shift is a BBN-frame sensitivity-coordinate perturbation, NOT a claim that the laboratory  ${}^7\text{Be}$  ANC differs today.

**NC-XIV.10.** ZS-Q12 v4.0 retains the honest prior that the laboratory  $z^*$  locking claim most likely fails gate L1 on real data (RETRACTED — unchanged).  $z^*$  is DERIVED-CONDITIONAL on the  $Z = \partial X$  holographic-interface postulate; the Universality Gate (whether a physically realised submanifold forces  $\mu = \lambda$ ) and the Astrophysical Gate (GRB 221009A parameters) remain OPEN. Theorem Q12.NG is a proven elimination of an approach class, not a retraction of  $z^*$  locking.

**NC-XIV.11.** ZS-M41 v3.1 establishes the closed form Theorem M41.K-Decomp (DERIVED) but does NOT claim the underlying classical identities (Lerch 1905, Fermat-quotient additivity, power rule) are original to the corpus — the originality is their placement inside refined Chabauty–Kim depth-2 at the external untreated case  $S = \{3, 11\}$ . v6.0 does NOT claim the F18 Bridge 4 geometric face (Kostant cohomology image) is PROVEN (HYPOTHESIS-strong, v3.2 closure path); does NOT claim the Refined Kim Conjecture at  $S = \{3, 11\}$  is PROVEN; and explicitly RETRACTS the v3.0 "15.1× enrichment" cluster claim as Poisson fluctuation (the substance is the closed form, not the cluster). M41.D' ( $\chi_{-11}$ -twisted  $\text{Li}_2$  closed form) remains OPEN. NC-M23.1 / NC-M23.7 inherited.

**NC-XIV.12.** ZS-A14 v2.0 Theorem A14.7 is DERIVED-CONDITIONAL: the  $\text{SU}(2)$ -covariant Usadel equation form is IMPORTED-PROVEN (Bergeret–Tokatly, Tokatly), while the Z-Spin background field  $\mathcal{A}_i^Z = (A/2\xi_Z)\hat{n}_Z \cdot \sigma$  is a zero-parameter ansatz whose physical reality is the testable content (NC-A14.4). v6.0 does NOT assert  $\mathcal{A}_i^Z$  is an independently measured field, and the BIC support ( $\Delta\text{BIC} = 8.28$ ) is demoted to supporting evidence — the falsifiable value rests on the thickness-scan and peak-shift predictions (Theorems A14.8', A14.9', A14.11). NC-A12.10 (Z-Spin explains rather than replaces BKT/XY/Usadel proximity physics) inherited.

## §XIV.E — v6.0 Master Falsification Registry Extensions

New v6.0 entries (preserving the v5.0 registry verbatim) are summarised below; full per-paper gate lists live in the source papers. All nine new papers (including ZS-M41 v3.1 and ZS-A14 v2.0, now fully integrated) contribute entries.

| § Anchor           | Falsifiable Statement                                                                                                                                         | Status (v6.0)                        |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| §Q11.A [LOCKED]    | OAQEC algebra $A_{\text{ZS}} \cong M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$ (explicit diagonalisation)                                       | PROVEN — Q11 v1.2                    |
| §Q11.B [LOCKED]    | Z-frame errors<br>operator-algebraically<br>correctable; X-block coeff 1,<br>Z-gauge block $\langle 0_Z$                                                      | $V_a \nmid V_b$                      |
| §Q11.C [BRIDGE]    | Single-vertex Lacambra<br>Gauss-law-code<br>algebraic-structure<br>correspondence at four<br>levels; full unitary<br>equivalence O-Q11.6                      | DERIVED-with-bridge — Q11 v1.2       |
| §Q11.dZ [LOCKED]   | Z-bottleneck depth $d_Z = 2$<br>(ZS-Q7 heat-kernel $\ K_{XY}(t)\  \sim t^2$ )                                                                                 | PROVEN — Q11 v1.2                    |
| §F21.C3.0 [LOCKED] | Commutant-Gate<br>Insufficiency: $f(A) \in W^*(A)$<br>passes SA-1 +<br>Slepian-commutation;<br>selector is external boundary<br>condition                     | PROVEN — F21 v2.0                    |
| §F21.T6d [LOCKED]  | $V_4$ ramified channels carried<br>by Burnol conductor blocks<br>$C_X = \log q_X$ (canonical, not<br>fitted; $\Lambda_X = \bigwedge \sqrt{q_X}$<br>WITHDRAWN) | DERIVED — F21 v2.0                   |
| §F21.T7' [LOCKED]  | Pure channelwise direct sum<br>cannot equal $V_4$ -decorated<br>Weil form (Lemma M31.0<br>Non-Separability)                                                   | PROVEN — F21 v2.0                    |
| §F22.8 [LOCKED]    | $\delta_Y = F(t_0)/(F(t_0)+F(t_l)) = 14/46 = 7/23$ ; $\gcd(60,32) = 4 = \dim(Z)^2$                                                                            | DERIVED — F22 v2.1 (was OBSERVATION) |

| § Anchor                | Falsifiable Statement                                                                                                                                                                        | Status (v6.0)                                  |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| §F22.4 [LOCKED]         | Axis-trigger CSP uniqueness                                                                                                                                                                  | Iso(A_tO, T_F20)                               |
| §F22.9 [LOCKED]         | $T_7$ A <sub>5</sub> -projector No-Go:<br>$\text{Hom}^{\text{sector}}_{\{X \leftrightarrow Y\}} \cap \ker(\Delta)$<br>$= \text{End\_gauge}(4)$ ( $= \emptyset$ as<br>sector exchange)        | DERIVED-strong — F22 v2.1                      |
| §F22.10 [LOCKED]        | Equivariant Antipodal Face<br>Module: $N_{\text{axes}}(P) = F(P)/2$<br>$= \dim C_{\pm 2}(P)$ (Burnside, free $\iota$<br>action)                                                              | DERIVED-strong — F22 v2.1                      |
| §A13.3 [LOCKED]         | Void Principle deep-void<br>lower bound $\mathcal{R}_V \geq Q/(2A) -$<br>$1 \approx 67.67$ (Schur–Cheeger, $\kappa$<br>$\leq 2A/Q$ ; $\lambda_2 \leq A/Q$ )                                  | DERIVED — A13 v1.3.1                           |
| §A13.8' [LOCKED]        | Closed horizon $D_{\text{obs}}/R_H =$<br>$4\sqrt{(121/\Omega_m) \cdot {}_2F_1(1/2, 1/6; 7/6;$<br>$-\Omega_\Lambda/\Omega_m)}$ ; Branch A 93.19<br>/ Branch B 94.18 Gly                       | DERIVED — A13 v1.3.1                           |
| §A13.10 [LOCKED]        | Saturation density contrast<br>$\delta_{\text{sat}}^{\text{ZS}} = 2A \cdot 121/(Q \cdot 39) -$<br>$1 \approx -0.9548$ (Hamaus 2014<br>deepest voids)                                         | HYPOTHESIS-strong — A13<br>v1.3.1              |
| §A15.T1a [LOCKED]       | $\varepsilon$ -Halo outer boundary-value<br>problem well-posed<br>(Ginzburg–Landau /<br>harmonic-map)                                                                                        | IMPORTED-PROVEN — A15<br>v1.5                  |
| §A15.SE [TESTABLE]      | Enclosed mass saturates<br>near $2 \times 10^{11} M_\odot$ by $\approx 25$ kpc<br>( $\gamma \rightarrow -1/2$ ) vs NFW growth<br>toward $8 \times 10^{11} M_\odot$ ; sharp<br>near-step edge | TESTABLE (Gaia DR4,<br>hash-locked) — A15 v1.5 |
| §A15.F-A15.7 [TESTABLE] | A wide DR4-resolved intrinsic<br>edge width would refute the<br>global-point-BC near-step<br>prediction                                                                                      | TESTABLE — A15 v1.5                            |

| § Anchor                        | Falsifiable Statement                                                                                                                                                           | Status (v6.0)                                    |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| §T10.ANC [TESTABLE]             | ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ ANC correction<br>$\delta \ln \gamma_b = -\kappa^2 = -A/Q \approx -0.73\%$ ; $S_{34}$ low by $\approx 0.7\text{--}1.5\%$ (LUNA-MV) | HYPOTHESIS-strong / TESTABLE — T10 v2.0          |
| §T10.8B [LOCKED]                | Bottleneck selectivity (NC-T10.5): universal $-\dim(Z)\kappa^2$ would unbind ${}^8\text{B}$ / destabilise ${}^9\text{C}$ ; both observed                                        | DERIVED (consistency) — T10 v2.0                 |
| §T10.Li-2 [FLAGGED]             | PRyMordial ${}^7\text{Li}$ closure coordinate-only: linear $+0.3\sigma$ vs nonlinear $-3.3\sigma$                                                                               | FLAGGED (tension) — T10 v2.0                     |
| §Q12.NG [LOCKED]                | Direct Coordinate-Dictionary No-Go: 18/18 natural mappings give $\arg(\tau) \approx \pi$ , never $39.4455^\circ$                                                                | NO-GO (PROVEN) — Q12 v4.0                        |
| §Q12.q12 [LOCKED]               | Unified $z^*$ universality: four ZS-Q12V open problems = single condition $\mu(S) = \lambda = (i\pi/2)z^*$                                                                      | DERIVED-CONDITIONAL ( $4\pi$ closure) — Q12 v4.0 |
| §Q12.RG [LOCKED]                | Open-Loop No-Go: i.i.d. coherent-lossy domain product $\rightarrow 0^\circ$ ; $z^*$ attractor only in self-referential map $i^{\wedge}z$ ,                                      | $f'(z^*)$                                        |
| §Q12.Z $\partial$ X [POSTULATE] | $z^*$ DERIVED-CONDITIONAL on $Z = \partial X$ holographic-interface postulate; $\mu = \lambda$ met by $4\pi$ spin-closure ( $b = i$ )                                           | DERIVED-CONDITIONAL — Q12 v4.0                   |
| §M41.K-Decomp [LOCKED]          | $c_{33}(p) = -2 q_p(2) q_p(3)(q_p(3)+q_p(11)) \cdot \text{Li}_2^{\wedge}(p)(4)^{(-1)}$ ; 235/235 primes<br>LHS = RHS                                                            | DERIVED — M41 v3.1                               |

| § Anchor                          | Falsifiable Statement                                                                                                                                           | Status (v6.0)                     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| §M41.K-char [LOCKED]              | Character decomposition<br>$c_{33} = c_3 + c_{11}$ via M28<br>log-additivity<br>$\log_p(3) + \log_p(11) = \log_p(33)$                                           | DERIVED — M41 v3.1                |
| §M41.cluster [RETRACTED]          | v3.0 "15.1× enrichment"<br>cluster at $c_{33} = 5$<br>reinterpreted as Poisson<br>fluctuation ( $p \approx 0.001$ )                                             | RETRACTED — M41 v3.1              |
| §M41.Bridge4-geom<br>[HYP-strong] | $c_X(p)$ as traces of $V_4$ image<br>of Kostant D-operator on<br>depth-2 cohomology of $X_1$                                                                    | HYPOTHESIS-strong — M41<br>v3.1   |
| §A14.8' [LOCKED]                  | Envelope-shifted current<br>peak $\tan(A \cdot dF_{\text{peak}} / 2\xi_Z) =$<br>$A \cdot \xi_{tZ} / \xi_Z$                                                      | DERIVED — A14 v2.0                |
| §A14.9' [LOCKED]                  | Three-observable locking: $\eta$ ,<br>$\rho_s^{\text{eff}}$ , detrended current<br>share $dF^*$ ; common period<br>$\Delta dF = 2dF^*$                          | DERIVED — A14 v2.0                |
| §A14.11 [TESTABLE]                | Two-channel Z-length<br>spectroscopy: $\xi_Z$ from node<br>spacing, $\xi_{tZ}$ from peak shift<br>in one thickness scan                                         | DERIVED / TESTABLE —<br>A14 v2.0  |
| §A14.7 [CONDITIONAL]              | SU(2)-covariant Usadel with<br>Z-Spin field $\mathcal{A}_i^Z =$<br>$(A/2\xi_Z)\hat{n}_Z \cdot \sigma$ (form<br>IMPORTED-PROVEN; field<br>zero-parameter ansatz) | DERIVED-CONDITIONAL —<br>A14 v2.0 |

All v6.0 entries are additive; no v5.0 entry is retracted (the M41 §M41.cluster RETRACTED status is an internal v3.0 → v3.1 source-paper correction, not a Book retraction). Total registry advances from ~340 (v5.0) to ~375 falsification gates with all nine new papers fully integrated.

#### §XIV.F — Conclusion

PART XIV absorbs the nine post-v5.0 source-paper integrations into the v6.0 Light Edition without retracting or modifying any v5.0 chapter entry. The Foundations theme acquires two hardened results: ZS-F21 v2.0 proves the Commutant-Gate Insufficiency Theorem C3.0 and relocates the archimedean selector to an external boundary condition while keeping the Z-Spin content in the canonical  $V_4$  Burnol decoration; ZS-F22 v2.1 promotes  $\delta_Y$  from OBSERVATION to DERIVED by identifying the long-unexplained  $\gcd(60,32) = 4$  as exactly  $\dim(Z)^2$ , and hardens the  $T_7$  No-Go to the  $A_5$ -projector level. The Quantum Mechanics theme acquires ZS-Q11 v1.2, the corpus's first operator-algebra (OAQEC) reading of the Z-bottleneck as a direct-sum stabilizer code  $A_{\text{ZS}} \cong M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$  with a Z-frame gauge subsystem and a Lacambra Gauss-law-code algebraic-structure correspondence, and ZS-Q12 v4.0, the consolidated proof that four ZS-Q12V open problems are one codimension-2 condition  $\mu = \lambda$  closed by the  $4\pi$  spin-closure /  $Z = \partial X$  holographic-interface postulate, with two new NO-GOs (the Direct Coordinate-Dictionary No-Go and the Open-Loop No-Go) that relocate  $z^*$  universality to the self-referential i-tetration layer. The Astrophysics theme acquires the Void Principle and its deep-void lower bound  $\mathcal{R}_V \geq Q/(2A) - 1 \approx 67.67$  plus the closed hypergeometric horizon formula (ZS-A13 v1.3.1), and the hash-locked pre-registered Gaia DR4 enclosed-mass-saturation falsification protocol for the  $\varepsilon$ -Halo outer boundary (ZS-A15 v1.5), which operationalises the ZS-A11  $\varepsilon$ -Halo as decisively testable while registering the Jiao/Zhou mass split as the open discriminant. The Translational theme acquires ZS-T10 v2.0, the geometry-first  $TY \rightarrow X$  sector-conversion ANC correction  $\delta \ln y_b = -\kappa^2$  for  ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$  with a single LUNA-MV-testable number and an immediate  ${}^8\text{B}/{}^9\text{C}$  consistency test, honestly registering Gate Li-2 tension. The Math Spine ZS-M41 v3.1 supplies the explicit closed-form Fermat-quotient derivation of the refined Chabauty–Kim depth-2 scalars at  $S = T = \{3, 11\}$  ( $c_{33} = -2q_p(2)q_p(3)(q_p(3)+q_p(11))/\text{Li}_2(4)$ , with the character decomposition  $c_{33} = c_3 + c_{11}$  realising the F18 Bridge 4 algebraic face), and the Astrophysics / Condensed-Matter Bridge ZS-A14 v2.0 supplies the Z-Spin  $SU(2)$ -covariant Usadel closure with the envelope-shifted current-phase relation, three-observable locking at  $dF^*$ , and two-channel Z-sector length spectroscopy — both now fully integrated at source-derived level (the v6.0 working-draft [SOURCE PENDING] flags are cleared). §XIV.G applies the ZS-A11 v1.1 errata downstream. The minimal triple  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  remains LOCKED. Zero new free parameters across all nine absorbed papers.

#### §XIV.G — ZS-A11 v1.1 Errata and Downstream Propagation into the Book (v6.0)

The ZS-A11 v1.1 source paper (PART XIII Chapter 41, Vortex Cosmology II) was corrected after the v5.0 freeze. This subsection records the errata in full (so no original wording is lost — no-deletion rule) and applies the two downstream propagations the errata itself mandates into the Book. The errata divides into two categories.

**Category A —  $\sqrt{2}$  conflation ( $\sigma \leftrightarrow v_{\text{flat}}$ ).** Basis: for a singular isothermal sphere (SIS) the asymptotic circular (flat) velocity satisfies  $v_c^2 = 2\sigma^2$  (PROVEN, Binney & Tremaine §2.1). The Milky Way's 220 km/s is the flat rotation velocity  $v_{\text{flat}}$ , not the dispersion  $\sigma$ ; the dispersion is  $\sigma = v_{\text{flat}}/\sqrt{2} \approx 156$  km/s. (A-1) In ZS-A11 §3.2 Proof Level 1, using  $\sigma \approx 156$  km/s (not 220 km/s)

gives  $L^2 \approx 1.1 \times 10^{53} \text{ m}^2$ ,  $L \approx 3.3 \times 10^{26} \text{ m}$  (comparable to the Hubble radius); the earlier 220 km/s conflated dispersion with circular velocity and, since  $L \propto 1/\sigma$ , underestimated  $L$  by  $\sqrt{2}$ . This affects only this normalisation;  $r_Z$  in §3.6 uses  $v_{\text{flat}}$  directly and is unaffected. (A-2) In ZS-A11 §3.4 Table 3.4 row 1 (Flat rotation  $v_{\text{flat}}$ ), the  $\Lambda\text{CDM}$  column " $v^2 = \sigma^2$  (SIS)" is corrected to " $v_c^2 = 2\sigma^2$  (SIS)"; the Z-Spin column  $v^2 = 2\pi G M_P^2/L^2$  is exact, and the equality  $2\pi G M_P^2/L^2 = 2\sigma^2 \rightarrow L^2 = \pi G M_P^2/\sigma^2$  is consistent with Eq. (3.2.3). Confidence: 90% ( $v_c^2 = 2\sigma^2$  is PROVEN; 220 km/s as MW circular velocity is standard).

**Category B — observational split not previously reflected (over-stated PASS).** Basis: the MW outer mass is contested — Zhou et al. 2022 fit a full  $M_{200} = (8.05 \pm 1.15) \times 10^{11} M_\odot$  dark halo to  $R_{200} = 192 \text{ kpc}$  with weak decline and  $V_c(R_\odot) = 234 \text{ km/s}$ ; Ou et al. 2024 fit a cored-Einasto dark-matter profile with a fast outer decline; Klacka & Štunc 2025 argue the decline may be an axisymmetric-Jeans-method artifact in a spheroid-dominated region. (B-1, ZS-A11 §3.1 Theorem 3.1 Level 5) The Gaia DR3 (Jiao et al. 2023) Keplerian-decline onset at  $\sim 19 \text{ kpc}$  agrees with the Z-Spin prediction to  $\sim 4\%$  only under the low-total-mass interpretation ( $M_{\text{total}} \approx 2.06 \times 10^{11} M_\odot$ ); this is contested, and the decisive discriminant is whether the enclosed mass saturates near  $2 \times 10^{11} M_\odot$  by  $\sim 25 \text{ kpc}$  ( $\epsilon$ -Halo) or grows toward  $\sim 8 \times 10^{11} M_\odot$  (NFW), pending Gaia DR4. Status corrected to OBSERVATION, contested — tentatively favouring  $\epsilon$ -Halo only under the low-mass interpretation. (B-2, §3.6 Level 5)  $r_Z = 18.3 \text{ kpc}$  uses  $v_{\text{flat}} \approx 220 \text{ km/s}$ ; recent Cepheid/LRGB  $V_c(R_\odot) \approx 234\text{--}237 \text{ km/s}$  would shift  $r_Z$  to  $\approx 16 \text{ kpc}$ , so the prediction band is  $r_Z \in [15, 21] \text{ kpc}$  (propagating  $M_{\text{total}} = 1.93\text{--}2.30 \times 10^{11} M_\odot$ ,  $v_{\text{flat}} = 220\text{--}237 \text{ km/s}$ ); the earlier single-point  $18.3 \text{ kpc}$  and the inherited  $[10, 50] \text{ kpc}$  band are both superseded. (B-3, §7.1) The Gaia DR3 onset agrees to  $\sim 4\%$  only under the Jiao low-mass interpretation and is NOT yet decisive;  $\epsilon$ -Halo is distinguished from NFW only if the enclosed mass saturates near  $M_{\text{total}} \approx 2 \times 10^{11} M_\odot$  — the explicit Gaia DR4 test. Status: OBSERVATION, contested — not PASS. (B-4, §9 inherited gates) F-A1.14 corrected to "MW outer rotation: enclosed mass saturates near  $2 \times 10^{11} M_\odot \rightarrow$  Keplerian  $\gamma \rightarrow -0.5$ , vs NFW growth toward  $M_{200} \approx 8 \times 10^{11} M_\odot$ ; currently CONTESTED — Jiao 2023 consistent, Zhou 2022 inconsistent; decisive at Gaia DR4"; F-A1.15 corrected to " $r_Z(\text{MW})$  band  $[15, 21] \text{ kpc}$  superseding  $[10, 50]$ ; tentatively consistent at  $\sim 19 \text{ kpc}$ , contested pending DR4." Confidence: 90% (the observational split is established at [M5]; the "decisive at DR4" framing is 60%).

**Downstream propagation applied into the Book v6.0.** (P-1) §VII.B-LIGHT — Seven Critical Near-Term ★ Gates: rows 1–2 (F-A1.14 / F-A1.15) are updated in place to reflect the contested status and the  $[15, 21] \text{ kpc}$  band (the original v5.0 cell wording — "Earliest decisive" / "Pending", band  $[10, 50]$ , Z-Spin =  $18.3 \text{ kpc}$  — is preserved verbatim in Category B above, so nothing is lost). (P-2) PART XIII Chapter 41 (ZS-A11 v1.1 Light-edition entry): the Level-5 framing "the Gaia DR3  $\sim 19 \text{ kpc}$  Keplerian onset (Jiao et al. 2023) currently favour  $\epsilon$ -Halo" / "within 4%" is superseded by "contested under the Jiao/Zhou mass split — agreement to  $\sim 4\%$  holds only under the Jiao low-mass interpretation; decisive at Gaia DR4." The PART XIII Chapter 41 prose is left physically intact (no-deletion) and carries an inline pointer [v6.0 ERRATA  $\rightarrow$  §XIV.G]; this subsection is the authoritative corrected reading. ZS-A15 v1.5 (Chapter 50) is the pre-registered

protocol that turns this corrected, contested boundary test into a hash-locked Gaia DR4 falsification.

## PART XV — Post-v6.0 Increment: ZS-T11 v2.1, the Executable Corpus Auditor (v6.1)

PART XV registers the single post-v6.0 source paper ZS-T11 v2.1, the methodology paper that turns the Human–AI Collaborative Science Model (HACSM) into an executable, self-verifying protocol. It is strictly additive: no PART XIV entry, count, or registry line is modified. The "ZS-T11 programme" referenced as a future deliverable by ZS-T10 v2.0 (Gate S-T10.1) and NC-XIV.9 is hereby instantiated. Coverage advances 130 → 131 source papers. The cumulative physics-verification ledger (~3,580+ PASS) is UNCHANGED; ZS-T11 contributes a distinct ledger of 13 meta-audit checks (12/13 AUDIT-PASS on the available corpus subset), because the auditor verifies the protocol's corpus footprint, not physics.

### §XV.A — Body Entry

ZS-T11 v2.1 [Translational theme / Methodology, 12/13 AUDIT-PASS on available corpus subset, 1 genuine AUDIT-FINDING, NEW v6.1] is the fully integrated edition (v1.0 preserved within v1.1, v1.1 preserved verbatim, v2.0 additions woven in; no-deletion). It extracts and formalises the five-phase HACSM loop, four search modalities, AI-role taxonomy, recursive self-audit (RSA-T11), external-import contract (EIC-T11), and project-instruction layer (P-T11.1, PI-1–PI-7), and closes the v1.x self-exemption gap (§3.1 invariant xiii required a verification suite that the methodology paper itself lacked) by shipping `zs_t11_corpus_audit_v2_0.py`. The auditor's three-cycle construction (8→11→12 AUDIT-PASS) instantiates the ZS-F18 contradiction engine: every apparent contradiction is first read as two faces of one token (Principle 1, integration) and resolved by lifting the analyser to a higher referent-aware layer (Principle 2), the methodological generalisation of F18 Bridge Four ( $\dim(Z) = 2 \oplus \dim(Z) = 2 \rightarrow \dim(X) = 3$ ). The single non-artifact residual (informal U-theme status tags EXCELLENT, CORRECTED outside the legend) is registered as a genuine OBSERVATION, demonstrating the auditor can fail. Status: HYPOTHESIS-strong (general external reproducibility); DERIVED-INTERPRETATION (Z-Spin corpus workflow). → ZS-T11 v2.1; the OS's own self-auditor.

### §XV.B — Source Paper Interface (v6.1)

| Source paper | Theme/Role                                                                               | Chapter  | Verification                         |
|--------------|------------------------------------------------------------------------------------------|----------|--------------------------------------|
| ZS-T11 v2.1  | Translational / Methodology (HACSM; Meta-Level Corpus Auditor; F18 Contradiction Engine) | XV Ch 53 | 12/13 AUDIT-PASS (subset), 1 finding |

## §XV.C — The Auditor as an OS-Level Tool (OS-LAYER)

The Book is the corpus OS (PART 0 §0.2; ZS-T11 §3.6). An OS ships with its own linter. `zs_t11_corpus_audit_v2_0.py` is therefore adopted as the Book's standard corpus-integrity tool, to be run on every corpus update. Its thirteen checks verify the protocol footprint: locked-input invariance ( $A = 35/437$ ,  $Q = 11$ ,  $(Z, X, Y) = (2, 3, 6)$ ,  $Z + X + Y = Q$ ), PASS-ledger integrity, status-tag legality, no-deletion preservation, paper-code inventory, local dependency acyclicity (Kahn 1962, IMPORTED-PROVEN), and an anti-numerology meta-check (seed 20260531). Pure Python standard library; zero external dependencies. NON-CLAIM: full-corpus DAG topological verification requires the unsegmented corpus and remains OPEN.

## §XV.D — v6.1 Master Falsification Registry Extensions (+13: F-T11.1–F-T11.13)

| § Anchor             | Falsifiable Statement                                                                                                                                                                                                                                              | Status (v6.1)                  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| §T11.1–9 [PRESERVED] | v1.0/v1.1 methodological gates (HACSM reproducibility, blind-spot cross-coverage, progressive-falsification navigation, constraint-density immunity, scale claim, Book-as-OS, recursive self-audit value, project-instruction necessity, external-import contract) | OPEN — external replication    |
| §T11.10 [OPEN]       | Quantified blind-spot cross-coverage: mean pairwise missed-error overlap across $k$ models exceeds pre-registered threshold                                                                                                                                        | OPEN — ablation required       |
| §T11.11 [OPEN]       | Auditor validity: an independent corpus violating locked-input invariance or no-deletion nonetheless passes all 13 checks                                                                                                                                          | OPEN — adversarial corpus      |
| §T11.12 [OPEN]       | Function-role stability: Verifier / Adversarial-router / Conceptual-stress                                                                                                                                                                                         | OPEN — re-test each generation |

| § Anchor       | Falsifiable Statement                                                                                             | Status (v6.1)            |
|----------------|-------------------------------------------------------------------------------------------------------------------|--------------------------|
|                | decomposition fails on the next model generation                                                                  |                          |
| §T11.13 [OPEN] | Self-verification necessity: a corpus built without an executable meta-auditor shows no higher contradiction rate | OPEN — comparative study |

Registry advances ~375 → ~388 gates (all additive; no v6.0 entry retracted).

### §XV.E — What this increment does NOT claim

NC-XV.1: registering ZS-T11 does NOT modify any PART XIV entry or count. NC-XV.2: the 12/13 AUDIT-PASS is on the available subset only; full-corpus contradiction-freedom is NON-CLAIM (NC-T11.8). NC-XV.3: passing the auditor verifies protocol footprint, not physical correctness (NC-T11.9). NC-XV.4: the F18 contradiction principles are a heuristic; not every disagreement has a higher-layer resolution (NC-T11.10). The minimal triple  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  remains LOCKED. Zero new free parameters.

### PART XVI — Post-v6.1 Absorption: Five New Source Papers (v7.0)

#### §0. Abstract

PART XVI consolidates the five post-v6.1 source-paper integrations of v7.0, bringing the Light Edition to 136-paper coverage — the full Z-Spin corpus. The v6.1 Light Edition reached 131-paper coverage with PART XV (ZS-T11 v2.1, the executable corpus auditor); the five new papers absorbed here — ZS-Q13 v1.2 (Quantum Mechanics), ZS-T12 v2.1 (Translational), ZS-Q14 v1.4 (Quantum Channel Geometry), ZS-F23 v1.3 (Foundations), and ZS-Q15 v1.2 (Quantum Mechanics) — are entered below as single Light-edition chapters (Chapters 54–58), with each source paper carrying the full derivation. All absorptions strictly preserve the v6.1 body verbatim (no-deletion rule). Zero new free parameters;  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  LOCKED unchanged across all five papers. PART XVI opens five new structural results into the corpus, each a fresh instantiation of the cardinal-2 substrate  $\dim(Z) = 2$  rather than a new constant: (i) the charge-supersampled two-Kraus boundary channel that resolves the electric-flux  $ER=EPR$  leakage problem by proving the leakage Hamiltonian is not an allowed term —  $[\wedge_Z, Q] = 0$ , hydrogen neutrality preserved, the hyperfine seam witness  $u_{\text{seam}}^{\text{HFS}} \rightarrow 0$  with the signed witness carrying the  $j = 1/2 \ 4\pi$  closure (Q13); (ii) the identification of the non-life → life transition with the attainment of self-referential closure on two registers — the static maxRAF (Knaster–Tarski greatest fixed point) and the dynamic attracting fixed point  $|f'| <$

1, of which  $z^*$  is the canonical realization — with the Z-bottleneck  $\leq \ln 2$  information ceiling, the spectral-gap unification  $g_{\Xi} = |\lambda_2|/\lambda_1$ , and the Minimal Triangular Z-Replicator  $g_{\Xi}(K_n) = 1/(n-1)$  first met at the  $n = 3$  triangle (T12); (iii) the Canonical Kraus-Lift Selection principle, in which a channel alone cannot select a Kraus lift (Choi-Invisibility No-Go) but a spin- $\frac{1}{2}$  boundary forces the half-angle SU(2) connection (Boundary Spin-Lift Lemma), giving the parameter-free sign-valued prediction  $\chi_Z = -1$  (Q14); (iv) the geometric fixing of the Type II crossed-product additive entropy constant at  $c = \frac{1}{2} \ln 2 = \frac{1}{2} \ln(\dim Z)$  via the trace-preserving central embedding functor  $A_{ZS} \rightarrow R$ , with the residual sharpened to a single OPEN condition (the Z-Spin emergence dictionary) and the finite/infinite edge-mode tension resolved as frame duality (F23); and (v) the microscopic electromagnetic which-path decoherence time  $\tau_{D^{EM}} = Q\hbar/(A \cdot E_{EM})$  with the Fractal Pairing Theorem  $EM/gravity = Q = 11$  exact, the Block–Fiedler coupling  $\kappa^2 = A/Q$ , and the BKT thermal channel  $2/\pi = \dim(Z)/\pi$  (Q15). Three of the five papers (Q13, Q14, Q15) form a tightly cross-linked quantum-channel-geometry cluster building on ZS-Q11 (OAQEC), ZS-Q12 ( $Z = \partial X$ ), and ZS-A4/A7 (seam witness); ZS-F23 and ZS-T12 build on the same Q11/Q12/Q7/F19 spine.

## §XVI.A — Five New Papers (Light-Edition Body Entries)

Each of the five post-v6.1 source papers represents v7.0's principal additive content. All inputs LOCKED from the upstream corpus; zero new free parameters. Each absorbed paper is cited below as a single Light-edition chapter, with the source paper carrying the full derivation.

### Chapter 54 — ZS-Q13 v1.2: Hydrogen Hyperfine Channel Tomography and the 21 cm Seam Witness — A Charge-Superselected Z-Spin Boundary-Mediation Test Against Electric-Flux $ER=EPR$ Leakage

ZS-Q13 v1.2 [Quantum Mechanics / Quantum Gravity Bridge theme, 52/52 PASS, NEW v7.0] develops the Z-Spin response, at the Hamiltonian level, to the 2026 Javed–Wilson-Ewing proposal (PRL 136, 121501) that wormhole-mediated entanglement leaks electric flux into a quantum wormhole and so modifies hydrogen hyperfine structure, splits the 21 cm line between entangled and unentangled spin sectors, and gives hydrogen a non-zero effective charge  $Q_H^{eff} = \eta_{ER}/(1+\eta_{ER})$ . Z-Spin does not merely reinterpret the claim; it solves the underlying gauge-consistency obstruction by replacing flux-carrying wormhole mediation with a charge-superselected, trace-preserving boundary channel. The v1.1 core is the Charge-Superselected Null-Leakage Theorem (Q13.1, DERIVED-CONDITIONAL): given  $L_{XY} \equiv 0$ ,  $\dim(Z) = 2$ ,  $Z = \partial X$ , and the DERIVED electron charge  $Q_e = -1$  (ZS-U9/S9), no entanglement-dependent electric-charge renormalization term may appear, so  $\delta H_{ZS}^{(electric-flux\ leakage)} = 0$ , hydrogen neutrality  $Q_H^{ZS} = Q_p + Q_e = 0$  is preserved exactly, and the leakage splitting  $\Delta v_{21}^{(ZS,leak)} = 0$  (a falsifiable null prediction — any field-free, systematic-free entanglement-sector splitting of the 21 cm line challenges the theorem). The v1.2 additive extension converts the null result into a positive program with three new theorems: Theorem Q13.4 (Gauge-Flux / Entanglement-Geometry Separation, DERIVED-CONDITIONAL) elevates the proof to channel level,  $[\Lambda_Z, Q] = 0$  (a physically

admissible Z-Spin boundary channel acts within charge-superselection sectors), so geometrical entanglement can mediate correlation, measurement, and decoherence but cannot produce an entanglement-dependent charge; Theorem Q13.5 (Hyperfine Two-Kraus Boundary Channel, DERIVED-CONDITIONAL) predicts that the ideal Z-mediated hydrogen hyperfine transition map has Kraus rank  $\leq \dim(Z) = 2$  — not a fit parameter but a count locked by  $\dim(Z) = 2$  via Stinespring dilation; and Theorem Q13.6 (Hyperfine Seam-Witness Theorem, DERIVED-CONDITIONAL / TESTABLE) gives the unsigned Choi seam witness  $u\_seam^{HFS} \rightarrow 0$  in the ideal limit while the signed witness  $\tilde{u}\_seam^{HFS}(\theta)$ , linear in a phase-anchored Kraus vector, carries the  $j = 1/2$   $4\pi$  closure ( $\tilde{u}(\theta+2\pi) = -\tilde{u}(\theta)$ ,  $\tilde{u}(\theta+4\pi) = +\tilde{u}(\theta)$ ). The discriminating power is amplified into a comparative H / anti-H / muonium / positronium channel-tomography program (TESTABLE), timely against the 2026 four-ppm antihydrogen hyperfine measurement and the MuSEUM/J-PARC muonium program. Honest priors preserved: the absolute 21 cm frequency is NON-CLAIM (NC-Q13.1), no  $A/z^*/|\lambda|^2/\eta\_topo$  is inserted into the Hamiltonian (NC-Q13.2/5, blocked by the ZS-Q12 direct-coordinate-dictionary no-go), weak ER=EPR is not falsified (only electric-flux leakage is rejected, NC-Q13.3), and the atomic-scale dephasing linewidth from  $\Gamma\_Z \sim 2A(\Delta E/\hbar)^2$  is OPEN (NC-Q13.4). Dependencies: ZS-F2, ZS-F5, ZS-S1 (block-Laplacian,  $L\_XY \equiv 0$ ), ZS-Q1 (CPTP boundary channel), ZS-Q12 ( $Z = \partial X$ , no-go), ZS-U9/S9 ( $Q\_e = -1$ ), ZS-M1, ZS-A4/A7 (seam witness).  $\rightarrow$  ZS-Q13 v1.2 [52/52 PASS]; the charge-superselected two-Kraus boundary channel that rejects electric-flux ER=EPR leakage as gauge-inconsistent and converts the hydrogen test into a signed-seam-witness hyperfine channel-tomography program.

## Chapter 55 — ZS-T12 v2.1: The Origin of Life as a Z-Spin-Mediated Self-Referential Closure Transition

ZS-T12 v2.1 [Translational theme, 61/61 PASS (40 consistency + 21 executable, `zs_t12_verify_v2_1.py`), NEW v7.0] proposes a substrate-agnostic identification of the non-life  $\rightarrow$  life transition with the attainment of Z-Spin self-referential closure by a food-generated chemical reaction network, introducing zero new constants and inheriting all quantitative anchors at PROVEN/DERIVED/DERIVED-CONDITIONAL status. The central object is a single self-referential map read on two registers: a discrete static register (the maximal reflexively-autocatalytic food-generated set maxRAF, the Knaster–Tarski greatest fixed point of the RAF reduction operator) and a continuous dynamical register (the i-tetration map  $T(z) = i^z$  of ZS-M1, whose fixed point  $z^*$  is attracting,  $|f'(z^*)| = 0.892 < 1$ ). Principal results: Theorem T12.1 (Replicator Transition, DERIVED-CONDITIONAL) — a network crosses the closure boundary when and only when it attains a self-referential closure that is simultaneously existent ( $\maxRAF \neq \emptyset$ ) and dynamically attractive ( $|f'| < 1$ ), of which  $z^*$  is the canonical realization under  $Z = \partial X$ ; Theorem T12.A (Chemical Closure Attractor, DERIVED) — the master composition is an attractor iff the Perron spectral-gap ratio  $g\Xi = |\lambda_2(J)|/\lambda_1(J) < 1$  on the closure Jacobian, the computable chemical realization (amplitude grows,  $\lambda_1 \geq 1$ , while composition contracts); Theorem T12.5 (Spectral-Gap Unification, DERIVED) — the attractor condition  $g\Xi < 1$ , the i-tetration contractivity  $|f'(z^*)| < 1$ , and the Eigen below-error-threshold condition are one and the same spectral-gap statement, promoting the v1.0 information ceiling (Theorem T12.2,

Z-bottleneck  $\leq \ln 2$ , structurally homologous to the Eigen  $v_{\max}$ ) from DERIVED-CONDITIONAL to DERIVED while keeping the numerical  $\ln 2 \leftrightarrow v_{\max}$  identification OPEN; Theorem T12.3 (Closure–Complementarity, DERIVED-INTERPRETATION) — the relational-biology result that closure to efficient causation entails complementarity and nonseparability is read as an independent external derivation of the X/Y complementarity and  $L_{XY} \equiv 0$  nonseparability Z-Spin posits; Theorem T12.4 (Directional Bias, HYPOTHESIS) — the structural-arrow ratio  $\Gamma(X \rightarrow Y)/\Gamma(Y \rightarrow X) = 2$  (the highest numerology-risk node, anti-numerology MC pre-registered); and Theorem T12.8 (Minimal Triangular Z-Replicator, NEW v2.0) — under purely relational closure the Perron gap of the complete catalytic graph is exactly  $g_{\Xi}(K_n) = 1/(n-1)$ , so the attractor condition  $g_{\Xi} < 1$  is first met at  $n = 3$  (the triangle,  $g_{\Xi} = 1/2$ ; a 2-node mirror pair is forced to  $g_{\Xi} = 1$ , period-2 oscillation), and the minimal life-seed is the closed three-node cooperative triangle on which the  $\dim(Z) = 2$  replication syntax of ZS-T6 is instantiated as  $\dim(Z) = 2$  (the generating pair) + 1 (the generated node) =  $\dim(X) = 3$  (mathematics PROVEN/DERIVED; life-seed reading HYPOTHESIS-strong). Eigen's paradox — faithful replication requires complex catalysts which require faithful replication — is registered as BOOTSTRAP-HYPOTHESIS, the protobiological face of the ZS-F0 ontological bootstrap  $B0 \rightarrow B3 \rightarrow A$ . The v2.1 increment closes the standing §27 NON-CLAIM as far as data access permits, upgrading the kinetic benchmark to the empirically-validated Mathis–Ramprasad–Walker–Lehman IGS-tag base-pairing rate model applied to the actual three-genotype rock–paper–scissors topology of Vaidya et al.: the cooperative network attains  $Z_{\text{rep}} = 1$  ( $\lambda_1 = 1.35$ ,  $g_{\Xi} = 0.632$ ) while matched-IGS selfish assembly gives  $Z_{\text{rep}} = 0$  ( $g_{\Xi} = 1$ ), with independent corroboration that real Azoarcus evolvable networks need as few as three WXY genotypes. Cardinal NC-4 (no Z-Spin constant claimed physically realized in biology) is extended explicitly to prebiotic and protobiological chemistry; the i-tetration map is an exemplar of an attracting self-referential map, not a chemical reaction map (NC-T12.5). Overall registered HYPOTHESIS-strong with a documented promotion path to DERIVED-CONDITIONAL via the RAF, (M,R), and Eigen identifications; the anti-numerology Monte Carlo is the pending gate. Dependencies: ZS-M1, ZS-Q7 (Z-bottleneck,  $\Gamma$ -ratio), ZS-F1 ( $L_{XY} \equiv 0$ ), ZS-Q11 ( $A_{\text{ZS}}$  algebra), ZS-Q12 ( $Z = \partial X$ ), ZS-F0 (bootstrap), ZS-T1, ZS-T6.  $\rightarrow$  ZS-T12 v2.1 [61/61 PASS]; the substrate-agnostic identification of abiogenesis with attracting self-referential closure ( $\max \text{RAF} + |f'| < 1$ ), the Z-bottleneck  $\leq \ln 2$  heritable-information ceiling, and the  $n = 3$  minimal triangular Z-replicator.

## **Chapter 56 — ZS-Q14 v1.4: Canonical Kraus-Lift Selection by a Spin- $\frac{1}{2}$ Boundary — From Choi-Invisibility to Process-Level Falsification in Z-Spin Boundary Mediation**

ZS-Q14 v1.4 [Quantum Channel Geometry / Quantum Mechanics / Quantum Hardware Interface theme, 33/33 PASS (categories A–I, `zs_q14_verify_v1_4.py`); central witness inherited from ZS-A7R Eq. (F.2) at 34/34 PASS, NEW v7.0] organizes around a question meaningful to general quantum-information theory: when does one lift of the Kraus gauge freedom cease to be a representation choice and become physical structure? Two bounding facts are established or directly verified rather than claimed new: Theorem 1 (Channel-Level Invisibility No-Go, ELEMENTARY-KNOWN) — any function of the channel alone is invariant under the  $U(r)$

Kraus-gauge fiber (the Choi/Jamiołkowski invariance), so no channel observable distinguishes a  $2\pi$  from a  $4\pi$  closure (verified to  $3.3 \times 10^{-16}$ ); and Theorem 2 (Single-Channel Non-Identifiability, DERIVED + VERIFIED-SIM) — the same channel yields closure class  $\chi = -1$  under a half-angle lift  $e^{-i\theta\sigma_y/2}$  and  $\chi = +1$  under a full-angle lift  $e^{-i\theta\sigma_y}$ , so the  $4\pi$  is injected by the chosen lift, not intrinsic. The constructive content is therefore a lift-selection Criterion 4.1 sharpened by the new v1.4 Boundary Spin-Lift Lemma (Lemma 5.1): if the two-dimensional Z-boundary carries a faithful projective representation of  $SO(3)$  in the nontrivial class of  $H^2(SO(3), U(1)) \cong \mathbb{Z}_2$  (whose  $2\pi$  loop acts as  $-I$ ), then the induced Kraus-frame connection is forced to be the half-angle generator  $G = \sigma_y/2$  up to conjugation and orientation, whence  $\chi_Z = -1$  — upgrading the criterion's connection condition (C3) from an independent assumption to a representation-theoretic consequence of the premise that the Z-boundary is a physical spinor ( $\dim Z = 2 = j = 1/2$ , ZS-M3 Thm 5.1;  $Z = \partial X$ , ZS-Q12). Z-Spin is thus a candidate lift-selector whose spinor-boundary postulate supplies the half-angle connection; the signed lift witness  $\tilde{u}_Z(\theta) = \text{Re Tr}[K_0(0)^\dagger K_0(\theta)]/\|K_0(0)\|F^2$  and the closure class  $\chi_Z = \text{sgn}[\tilde{u}_Z(2\pi)/\tilde{u}_Z(0)]$  are the operational objects. Crucially  $\tilde{u}_Z$  is not a channel observable (it is gauge-dependent) but a process/comb-level observable,  $\tilde{u}_Z = F(Y\{2:0\}, G_Z, R_Z) = F(V, G_Z, \phi_Z) \neq f(\Lambda)$ , recoverable only with control of the Stinespring dilation's environment phase reference, read on the X-block of the OAQEC algebra  $M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$  (the Z-block being a scalar gauge subsystem, ZS-Q11). The experiment is a near-term, pre-registered kill-or-survive test:  $\chi_Z = +1$  (a  $2\pi$  / vector closure on hardware) refutes the spin- $1/2$  lift-selection (primary kill, F-Q14.1), while  $\chi_Z = -1$  with the three load-bearing controls (bypass NC1, phase-only NC4, Z-off NC5) null is a necessary — explicitly not sufficient — success. The same formalism applies to the hydrogenic hyperfine channel of ZS-Q13 and its H / anti-H / muonium / positronium extensions, making ZS-Q14 a physical-channel-geometry program rather than a single-circuit proposal (a hydrogenic  $\chi_Z = +1$  also fails the selection, F-Q14.11). Honest prior-art boundary: the bundle, mixed-state holonomy (Uhlmann 1986), Choi/Jamiołkowski invariance, the No-Go, the process-comb localization (Chiribella et al.; Pollock et al.), and the representation theory of Lemma 5.1 (Bargmann 1954; Hall 2015) are all prior art; the single contingent positive content is the candidate spin- $1/2$  lift-selection and its falsifiable  $\chi_Z = -1$ , DERIVED-CONDITIONAL on Z being a physical spinor boundary. Dependencies: ZS-Q1, ZS-Q7, ZS-Q11, ZS-Q12, ZS-Q13, ZS-A4, ZS-A7R (Eq. F.2, Thm 3.2-bis), ZS-QH (§6.1 Hadamard test), ZS-F1, ZS-F5, ZS-M3 (Thm 5.1, Lemma 10.1). → ZS-Q14 v1.4 [33/33 PASS]; the spin- $1/2$ -boundary Kraus-lift selection principle with the Boundary Spin-Lift Lemma forcing the half-angle connection, a process/comb-level signed lift witness, and a near-term kill-or-survive  $\chi_Z = -1$  hardware test extended to the hydrogenic family.

## Chapter 57 — ZS-F23 v1.3: Geometric Fixing of the Type II Trace Normalization — The $\mathbb{Z}_2$ -Seam In 2 as the Crossed-Product Additive Entropy Constant, and the Spin- $1/2$ Boundary Origin of de Sitter Type II.

ZS-F23 v1.3 [Foundations theme, 41/41 PASS, AN-F23.1 EXECUTED  $p \approx 0.058\% \ll 1\%$ , NEW v7.0] shows that the Z-Spin corpus geometrically fixes, at zero free parameters, the state-independent additive constant that the Klinger–Kudler–Flam–Satishchandran /

Kudler-Flam–Leutheusser–Satishchandran / CLPW Type II crossed-product program fixes by hand through a choice of trace normalization. The chain: the single-cell Z-Spin logical algebra is the finite-dimensional Type I von Neumann algebra  $A_{\text{ZS}} \cong M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$  (PROVEN, ZS-Q11) with abelian center  $Z(A_{\text{ZS}}) \cong \mathbb{C}^3$  carrying the three sector projections; by the Murray–von Neumann universal-embedding theorem every finite-dimensional von Neumann algebra embeds trace-preservingly into the hyperfinite  $\text{II}_1$  factor  $R$ . Theorem F23.4 (Trace-Preserving Central Embedding Functor, DERIVED) constructs a trace-preserving \*-monomorphism  $\Phi: A_{\text{ZS}} \rightarrow R$  sending  $P_X, P_Z, P_Y$  to projections of continuous-dimension trace  $(3, 2, 6)/11$  with modular-Hamiltonian difference  $\Delta K_\Omega(X \rightarrow Y) = -\ln 2 = -\ln(\dim Z)$ , closing the algebraic half of the previously registered gate O-F19.6 Step 1 unconditionally; Theorems F23.1/F23.2 then give the per-cross-sector additive constant  $c = \frac{1}{2} \ln 2 = \frac{1}{2} \text{ bit}$ . The v1.2 deep-exploration layer removes the apparent discrete-versus-continuous spectral objection (the discrete  $K_\Omega$  is the sector coarse-graining of the continuous gravitational modular operator), proves Theorem F23.5 (Coarse-Grained Trace Matching, DERIVED, via Takesaki's conditional-expectation theorem — the trace-preserving conditional expectation onto the sector center pushes the trace to  $(3, 2, 6)/11$  and reproduces  $\Delta K = -\ln 2$ ), and proves Lemma F23.6 reducing the residual to a single sharp falsifiable Condition C: the gravitational equilibrium (de Sitter maximum-entropy) state, coarse-grained to the Z-Spin sectors, carries weights  $(3, 2, 6)/11$ . Condition C is the Z-Spin emergence dictionary and cannot be established with current corpus or external tools, so O-F19.6 Step 1' remains a genuine OPEN — sharply characterized, a registered corpus asset locating the boundary between the DERIVED (F23.4–F23.6) and the hypothetical. The v1.3 increment adds Theorem F23.7 (Edge-Mode Frame-Duality Interface, DERIVED-interpretation): following the ZS-F18 method (mediate finite and infinite through the  $\dim(Z) = 2$  seam, never directly), the principal finite( $\dim 2$ )–infinite(KKS edge-mode tower) tension is resolved by the ZS-M30 frame duality — the infinite edge-mode tower is the external-frame count ( $\text{Infinity}_A$ ) of the same structure whose internal-frame closure ( $\text{Infinity}_B$ ) is one bit of Z-channel capacity  $\ln 2$  — controlled by the PROVEN X–Y Tiling Asymmetry (ZS-M6 §5.5: the truncated octahedron tiles  $\mathbb{R}^3$  while the truncated icosahedron cannot) and ZS-M17, registered as the seventh route of the six-route Möbius-trace pattern and upgrading the v1.2 OPEN tension to DERIVED-interpretation (the residual sharpening to Condition C-edge). The spin- $\frac{1}{2}$  boundary origin of de Sitter Type  $\text{II}_1$  is read at HYPOTHESIS (Reading F23.3): the  $\dim(Z) = 2 = j = 1/2$  spinor boundary is the type-determining degree of freedom, with the boundedness of Type  $\text{II}_1$  reflecting the finite  $Q = 11$  register and the cyclic-cosmology entropy  $S_{\{Z_2\}} = 6\pi/A$  (ZS-U8) as the Z-Spin counterpart of the de Sitter maximum-entropy state. The anti-numerology joint structural-signature Monte Carlo selects  $(2, 3, 6)$  uniquely among ordered triples in  $\{1, \dots, 12\}^3$  at  $p \approx 0.058\%$ , with the honest inheritance caveat that most of the specialness (to  $p \approx 0.347\%$ ) is upstream ZS-F5 register geometry. Non-claims: F23 does not derive the generalized-entropy formula (only the additive constant, NC-F23.1); does not assert  $R$  is the specific gravitational subregion algebra (Condition C, NC-F23.5); no new free parameter (NC-F23.4). Dependencies: ZS-F2, ZS-F5, ZS-M3 ( $j = 1/2$ ), ZS-Q11 ( $A_{\text{ZS}}$ ), ZS-Q7 (eigenvalues, capacity  $\leq \ln 2$ ), ZS-F19 ( $\Delta K_\Omega = -\ln 2$ , KMS rapidity), ZS-A7 (Kraus parity), ZS-U8 ( $S_{\{Z_2\}} = 6\pi/A$ ), ZS-M6/M17 (tiling asymmetry), ZS-M30 (frame duality), ZS-F18 (finite/infinite polarity).  $\rightarrow$  ZS-F23 v1.3 [41/41 PASS]; the  $Z_2$ -seam  $c = \frac{1}{2} \ln 2$  additive entropy

constant fixed by the finite-register canonical trace via the embedding  $A_{\text{ZS}} \rightarrow R$ , with Condition C the genuine OPEN and the edge-mode finite/infinite tension resolved as the seventh Möbius-route frame duality.

## **Chapter 58 — ZS-Q15 v1.2: Electromagnetic Which-Path Decoherence as the Microscopic Fractal Partner of Geometric Gravitational Decoherence — The $Q/A \approx 1/\alpha_{\text{EM}}$ Enhancement from Block–Fiedler Z-Spin Mediation**

ZS-Q15 v1.2 [Quantum Mechanics theme, 32/32 PASS, anti-numerology MC pre-registered, NEW v7.0] applies the ZS-Q1 gravitational-decoherence methodology to the microscopic Z-Spin mediator, using the ZS-M2 Cross-Coupling fractal symmetry that identifies gravity as the macroscopic Z-Spin mediator and electromagnetism as the microscopic one. The Block–Fiedler Mediation Theorem (ZS-T1 §9.3, PROVEN) fixes every Z-Spin-mediated coupling to the Schur-complement scale  $\kappa^2 = A/Q = 35/4807$  (PROVEN, ZS-M6 §2.2), whose reciprocal is the Block–Fiedler X-face propagator scale  $1/\kappa^2 = Q/A = 137.343$ . Identifying the which-path decoherence enhancement with this propagator scale — exactly as ZS-Q1 identifies it with A in the gravitational impedance mode — yields the microscopic electromagnetic which-path decoherence time  $\tau_{\text{D}}^{\text{EM}} = \hbar/(\kappa^2 \cdot E_{\text{EM}}) = Q\hbar/(A \cdot E_{\text{EM}})$  with enhancement ratio  $\tau_{\text{D}}^{\text{EM}}/\tau_{\text{bare}}^{\text{EM}} = 1/\kappa^2 = Q/A = 137.343$ . Three structural results follow: (i) the Fractal Pairing Theorem (Q15.1, DERIVED-CONDITIONAL) —  $(Q/A)/(1/A) = Q = 11$  EXACTLY, so the electromagnetic mediation coupling is precisely Q times weaker than the gravitational impedance and the EM which-path enhancement is exactly Q times larger than the gravitational  $1/A = 12.49$ ; (ii) the Lindblad dephasing rate  $\Gamma_{\text{EM}} = (2A/Q)(\Delta E/\hbar)^2$  with coefficient identical to the block-Laplacian slow eigenvalue  $\lambda_2 = 2A/Q$  (ZS-Q7), so the two decoherence channels are the two non-zero roots of the same master equation  $\lambda(\lambda + 2A/Q)(\lambda + A) = 0$ ; and (iii) the thermal channel governed by the Berezinskii–Kosterlitz–Thouless phase-stiffness jump  $\rho_s(T_{\text{BKT}}) = (\dim(Z)/\pi)k_B T_{\text{BKT}} = (2/\pi)k_B T_{\text{BKT}}$  (DERIVED-under-Regge, inherited from ZS-A12, with the integer "2" =  $\dim(Z)$ ). The headline status advances HYPOTHESIS-strong  $\rightarrow$  DERIVED-CONDITIONAL (v1.0)  $\rightarrow$  DERIVED-CONDITIONAL-strong (v1.1)  $\rightarrow$  DERIVED-under-G2 (v1.2): v1.1 discharges the mode-identification conditional at the action level (the gauge-sector image of the non-minimal coupling is the ZS-S10 master-action covariant derivative  $D_\mu \Phi = (\partial_\mu - ik g_Y B_\mu)\Phi$  with  $\kappa^2 = A/Q$ , generating the  $\kappa^2 |\Phi|^2 (g_Y B)^2$  coupling with no new  $\varepsilon^2 F_{\mu\nu} F^{\mu\nu}$  term, plus the explicit ZS-F9 §6.6 Schur which-path transfer  $T(r)_{\{X \rightarrow Y\}} = (A/Q\mu^2)|r_Y\rangle\langle r_X|$ ); v1.2 discharges the two residuals using existing corpus assets — the photon = Z-sector identification via ZS-S12 together with the two-route Gap-G2 closure (ZS-M16 Route (a) DERIVED, ZS-M15 Route (b), ZS-M17 continuum-limit), and the electroweak projection via the standard basis conversion  $e = g_Y \cos \theta_W$  ( $\kappa^2 g_Y^2 \cos^2 \theta_W = \kappa^2 e^2$ , leaving  $\kappa^2$  unchanged), with the corpus-DERIVED  $\sin^2 \theta_W = (48/91) \cdot x^* = 0.23118$  introducing no new parameter. The enhancement  $Q/A = 137.343$  lies within 0.22% of  $1/\alpha_{\text{EM}} = 137.035999177$  (CODATA 2022; closing to 0.00007% at NLO Schur order), reported as OBSERVATION with the explicit ZS-U9/U10 cautionary precedent (the QED self-energy loop and the Z-Spin Leaky Wilson Loop share an order-of-magnitude appearance yet differ by a factor of 177, not 137) — the paper does NOT claim the EM which-path enhancement IS the fine-structure constant

(NON-CLAIM); the DERIVED content is the value  $Q/A$  from the PROVEN  $\kappa^2$ . The sole step to plain DERIVED is empirical (the full PROVEN of Gap G2 via the ZS-A4 hardware Monte Carlo). Dependencies: ZS-F2, ZS-F5, ZS-M6 ( $\kappa^2 = A/Q$ ), ZS-M2 (fractal symmetry), ZS-Q1 (gravity template), ZS-Q7 ( $\lambda_2$ ), ZS-T1 (Block–Fiedler), ZS-A12 (thermal channel), ZS-T2 ( $\alpha_{EM}$  proximity), ZS-S10 (gauge-sector image), ZS-F9 (Schur transfer), ZS-S12/S15 (photon = Z-sector), ZS-M15/M16/M17 (Gap-G2), ZS-S1 ( $\sin^2\theta_W$ ). → ZS-Q15 v1.2 [32/32 PASS]; the microscopic EM which-path decoherence time  $Q\hbar/(A \cdot E_{EM})$  with the Fractal Pairing Theorem  $EM/gravity = Q = 11$  exact and the Block–Fiedler  $\kappa^2 = A/Q$  coupling, the microscopic companion of ZS-Q1.

## PART XVI Source Paper Interface (v7.0)

| Source paper | Theme/Role                                                                                                                               | Chapter   | Verification                          |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------|
| ZS-Q13 v1.2  | Quantum Mechanics / Quantum Gravity Bridge<br>(charge-superselecte d hyperfine channel; 21 cm seam witness; anti-ER=EPR-leakag e)        | XVI Ch 54 | 52/52 PASS                            |
| ZS-T12 v2.1  | Translational (origin of life; self-referential closure; maxRAF +                                                                        | f'        | < 1; minimal triangular Z-replicator) |
| ZS-Q14 v1.4  | Quantum Channel Geometry (Kraus-lift selection; Boundary Spin-Lift Lemma; $\chi_Z = -1$ kill-or-survive test)                            | XVI Ch 56 | 33/33 PASS                            |
| ZS-F23 v1.3  | Foundations (Type II additive entropy constant $\frac{1}{2} \ln 2$ ; $A_{ZS} \rightarrow R$ embedding; de Sitter Type II, spinor origin) | XVI Ch 57 | 41/41 PASS                            |
| ZS-Q15 v1.2  | Quantum Mechanics (EM which-path                                                                                                         | XVI Ch 58 | 32/32 PASS                            |

| Source paper | Theme/Role                                                                                   | Chapter | Verification |
|--------------|----------------------------------------------------------------------------------------------|---------|--------------|
|              | decoherence Q/A;<br>Fractal Pairing<br>EM/gravity = Q;<br>Block–Fiedler $\kappa^2 =$<br>A/Q) |         |              |

### §XVI.C — v7.0 Cross-Coupling Discipline (T7) Compliance

Per T7 (§VIII.7), every cross-paper synthesis must be P-CC (PROVEN-composed) or I-CC (interpretation). All five PART XVI chapter entries are summarisations of source-paper Verification Summaries; no new theorem is introduced by PART XVI itself. Each absorbed paper's status (PROVEN / DERIVED / DERIVED-CONDITIONAL / DERIVED-CONDITIONAL-strong / DERIVED-under-G2 / DERIVED-interpretation / DERIVED-under-Regge / VERIFIED / TESTABLE / HYPOTHESIS-strong / BOOTSTRAP-HYPOTHESIS / NO-GO / NON-CLAIM / OPEN) is preserved verbatim from the source. The cardinal-2 substrate ( $\dim(Z) = 2$ ) acquires five further corpus instantiations via PART XVI: the rank-2 hyperfine Kraus channel and the  $j = 1/2$  signed seam witness (ZS-Q13); the spin- $1/2$  Boundary Spin-Lift Lemma forcing the half-angle  $\sigma_y/2$  connection and  $\chi_Z = -1$  (ZS-Q14); the  $Z_2$ -seam additive constant  $\frac{1}{2} \ln 2 = \frac{1}{2} \ln(\dim Z)$  and the seventh Möbius-route frame duality through the  $\dim(Z) = 2$  seam (ZS-F23); the  $\dim(Z) = 2$  replication syntax on the  $n = 3$  cooperative triangle,  $\dim(Z) = 2 + 1 = \dim(X) = 3$  (ZS-T12); and the EM/gravity Fractal Pairing ratio  $Q = 11$  together with the BKT jump  $2/\pi = \dim(Z)/\pi$  (ZS-Q15). These join the prior instantiations (gauge constraint  $Q = 11$ ; CHSH  $2\sqrt{2}$ ; channel capacity  $\leq \ln 2$ ;  $\chi(S^2) = 2$ ; the  $\gcd(60,32) = 4 = \dim(Z)^2$  Antipodal Closure; the OAQEC rank-2 Z-gauge subsystem and  $d_Z = 2$ ). The  $Q13 \rightarrow Q14 \rightarrow Q15$  quantum-channel-geometry cluster is internally consistent: ZS-Q14 §10 uses ZS-Q13's hydrogenic channel as its second physical target (a P-CC composition of two source results, not a new theorem), and ZS-F23 and ZS-Q15 both build on the ZS-Q11 OAQEC algebra without modifying it. PART XVI contributes no new Banach–Tarski functor member. Cumulative no-deletion rule preserved across  $v3.0 \rightarrow \dots \rightarrow v6.1 \rightarrow v7.0$ . Word count strictly increases over v6.1.

### §XVI.D — What v7.0 Does NOT Claim

**NC-XVI.1.** v7.0 does NOT modify or retract any v6.1 chapter entry, count, or registry line. The supersedes statement is strictly additive; there are no v7.0 errata propagations.

**NC-XVI.2.** v7.0 does NOT supersede the v5.0/v6.0/v6.1 PART X–XV readings; the Spin-Pair Coexistence Principle, the TO/TI Polyhedral Mapping, the PART XII–XIV absorptions, and the PART XV ZS-T11 auditor are inherited unchanged.

**NC-XVI.3.** ZS-Q13 v1.2 does NOT derive the absolute hydrogen 21 cm frequency, does NOT insert  $A/z^*/|\lambda|^2/\eta_{\text{topo}}$  into the hydrogen Hamiltonian, and does NOT falsify weak ER=EPR — it rejects only electric-flux leakage as the realisation of entanglement geometry. The atomic-scale dephasing linewidth from  $\Gamma_Z$  remains OPEN; no existing H/anti-H/Mu/Ps experiment is claimed to have measured  $u_{\text{seam}}$  (NC-Q13.4).

**NC-XVI.4.** ZS-T12 v2.1 makes NO physical claim that any prebiotic/protobiological system realises Z-Spin geometry (Cardinal NC-4 prebiotic extension); the i-tetration map is an exemplar of an attracting self-referential map, not a chemical reaction map. The numerical  $\ln 2 \leftrightarrow v_{\text{max}}$  identification is OPEN, the  $\Gamma$ -ratio = 2 flux mapping is HYPOTHESIS (highest numerology risk), the first crossing of the closure boundary is BOOTSTRAP-HYPOTHESIS, and the minimal-triangle life-seed reading is HYPOTHESIS-strong (the  $g\Xi(K_n) = 1/(n-1)$  mathematics is PROVEN/DERIVED). The whole package is HYPOTHESIS-strong with the anti-numerology MC as the pending gate; the literal raw-trace kinetic fit remains a data-access-bound NON-CLAIM.

**NC-XVI.5.** ZS-Q14 v1.4 claims NO new general theorem about quantum channels (the bundle, Choi/Jamiołkowski invariance, the No-Go, the process-comb localisation, and Lemma 5.1 are all prior art); does NOT claim  $\tilde{u}_Z$  is a channel observable (it is process/dilation-level); the spin- $\frac{1}{2}$  lift-selection is DERIVED-CONDITIONAL on Z being a physical spinor boundary; a positive  $\chi_Z = -1$  outcome is necessary, NOT sufficient, for the full framework; NC2/NC3 are non-diagnostic; and current hardware is not claimed to realise the Z-sector.

**NC-XVI.6.** ZS-F23 v1.3 does NOT derive the generalized-entropy formula (only the additive constant); does NOT assert the embedding target R IS the specific gravitational subregion algebra (Condition C, a genuine OPEN = the Z-Spin emergence dictionary); the embedding functor F23.4, the coarse-grained trace matching F23.5, and the closure criterion F23.6 are unconditional (DERIVED), while F23.1/F23.2 are DERIVED-CONDITIONAL on Condition C and Reading F23.3 is HYPOTHESIS. The finite/infinite edge-mode mismatch is NOT a contradiction (frame duality, F23.7); no new free parameter (R is a uniqueness theorem).

**NC-XVI.7.** ZS-Q15 v1.2 does NOT claim the EM which-path decoherence enhancement IS the fine-structure constant — the  $Q/A \approx 1/\alpha_{\text{EM}}$  proximity (0.22%) is OBSERVATION, carrying the standing 177× cautionary precedent (ZS-U9/U10). The DERIVED content is the value  $Q/A = 137.343$  from the PROVEN  $\kappa^2 = A/Q$ , independent of whether it equals  $1/\alpha_{\text{EM}}$ . The headline is DERIVED-under-G2 (the photon = Z-sector step is DERIVED-CONDITIONAL via ZS-S12); the sole step to plain DERIVED is empirical (full PROVEN of Gap G2 via the ZS-A4 hardware Monte Carlo).

## **§XVI.E — v7.0 Master Falsification Registry Extensions**

New v7.0 entries (preserving the v6.1 registry verbatim) are summarised below; full per-paper gate lists live in the source papers. All five new papers contribute entries; registry advances ~388 → ~430 (all additive, no v6.1 entry retracted).

| § Anchor          | Falsifiable Statement                                                                                                                                               | Status (v7.0)                                            |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| §Q13.1 [LOCKED]   | Charge-superselected null leakage:<br>$\delta H_{ZS}^{(\text{electric-flux})} = 0$ ;<br>$Q_H^{ZS} = 0$ ;<br>$\Delta v_{21}^{(ZS, \text{leak})} = 0$                 | DERIVED-CONDITIONAL — Q13 v1.2                           |
| §Q13.4 [LOCKED]   | Gauge-flux / entanglement-geometry separation: $[\wedge_Z, Q] = 0$ (charge-sector preserving)                                                                       | DERIVED-CONDITIONAL — Q13 v1.2                           |
| §Q13.5 [LOCKED]   | Hyperfine channel Kraus rank $\leq \dim(Z) = 2$ (locked, not fitted)                                                                                                | DERIVED-CONDITIONAL — Q13 v1.2                           |
| §Q13.6 [TESTABLE] | Unsigned seam $u_{\text{seam}}^{\text{HFS}} \rightarrow 0$ ; signed $\tilde{u}_{\text{seam}}^{\text{HFS}}$ carries $j = 1/2$ $4\pi$ closure; H/anti-H/Mu/Ps program | DERIVED-CONDITIONAL / TESTABLE — Q13 v1.2                |
| §T12.1 [LOCKED]   | Replicator transition = simultaneous maxRAF $\neq \emptyset$ (Knaster–Tarski) and                                                                                   | $f'$                                                     |
| §T12.5 [LOCKED]   | Spectral-gap unification: $g_{\Xi} < 1 \Leftrightarrow$                                                                                                             | $f'(z^*)$                                                |
| §T12.8 [LOCKED]   | Minimal triangular Z-replicator: $g_{\Xi}(K_n) = 1/(n-1)$ ; first attractor at $n = 3$ ( $g_{\Xi} = 1/2$ )                                                          | DERIVED (math); HYPOTHESIS-strong (life-seed) — T12 v2.1 |
| §T12.4 [HYP]      | Directional bias $\Gamma(X \rightarrow Y)/\Gamma(Y \rightarrow X) = 2$ mapped to a biochemical flux ratio (anti-numerology MC pending)                              | HYPOTHESIS — T12 v2.1                                    |

| § Anchor                | Falsifiable Statement                                                                                                                                     | Status (v7.0)                                                            |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| §T12.Eigen [BOOT]       | Eigen's paradox =<br>protobiological face of the<br>ZS-F0 bootstrap $B_0 \rightarrow B_3 \rightarrow A$                                                   | BOOTSTRAP-HYPOTHESIS<br>— T12 v2.1                                       |
| §Q14.Thm1 [LOCKED]      | Channel-level invisibility: $f(\Lambda)$<br>invariant under $U(r)$ Kraus<br>fiber (no $2\pi/4\pi$<br>discrimination)                                      | ELEMENTARY-KNOWN /<br>VERIFIED — Q14 v1.4                                |
| §Q14.Thm2 [LOCKED]      | Single-channel<br>non-identifiability: same $\Lambda$<br>gives $\chi = -1$ (half-angle) and<br>$+1$ (full-angle); $4\pi$ injected                         | DERIVED + VERIFIED-SIM<br>— Q14 v1.4                                     |
| §Q14.L5.1 [LOCKED]      | Boundary Spin-Lift: nontrivial<br>spinor class of $SO(3)$ forces<br>$\sigma_y/2$ connection $\Rightarrow \chi_Z = -1$                                     | DERIVED-CONDITIONAL —<br>Q14 v1.4                                        |
| §Q14.F-Q14.1 [TESTABLE] | Hardware $\chi_Z = +1$<br>( $2\pi$ /vector) on the<br>Z-mediated channel $\Rightarrow$<br>spin- $1/2$ lift-selection refuted<br>(primary kill)            | TESTABLE-PROTOCOL —<br>Q14 v1.4                                          |
| §F23.4 [LOCKED]         | Trace-preserving central<br>embedding $\Phi: A_{ZS} \rightarrow R$ ;<br>central projections $\rightarrow$ trace<br>(3,2,6)/11; $\Delta K_\Omega = -\ln 2$ | DERIVED — F23 v1.3                                                       |
| §F23.2 [LOCKED]         | $Z_2$ -seam additive entropy<br>constant $c = \frac{1}{2} \ln 2 = \frac{1}{2} \ln(\dim Z)$ per cross-sector                                               | DERIVED (magnitude);<br>DERIVED-CONDITIONAL on<br>Condition C — F23 v1.3 |
| §F23.5 [LOCKED]         | Coarse-grained trace<br>matching: $E^*(\tau_R) =$<br>(3,2,6)/11; $\Delta K(X \rightarrow Y) = -\ln 2$<br>(Takesaki)                                       | DERIVED — F23 v1.3                                                       |
| §F23.6 [DECISIVE]       | Condition C: gravitational de<br>Sitter max-entropy state<br>coarse-grains to weights                                                                     | OPEN (decisive; F-F23.8) —<br>F23 v1.3                                   |

| § Anchor             | Falsifiable Statement                                                                                                                      | Status (v7.0)                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
|                      | (3,2,6)/11 (emergence dictionary)                                                                                                          |                                      |
| §F23.7 [LOCKED]      | Edge-mode frame duality:<br>infinite KKS tower (Infinity_A)<br>↔ In 2 internal closure<br>(Infinity_B) via $\dim(Z) = 2$<br>seam           | DERIVED-interpretation —<br>F23 v1.3 |
| §Q15.5.2 [LOCKED]    | EM which-path decoherence<br>time $\tau_D^{\text{EM}} = Q\hbar/(A \cdot E_{\text{EM}})$ ;<br>enhancement $Q/A = 137.343$                   | DERIVED-under-G2 — Q15<br>v1.2       |
| §Q15.1 [LOCKED]      | Fractal Pairing Theorem:<br>$(Q/A)/(1/A) = Q = 11$ EXACT;<br>$\Gamma_{\text{grav}}/\Gamma_{\text{EM}} = Q = 11$                            | DERIVED-CONDITIONAL —<br>Q15 v1.2    |
| §Q15.6 [LOCKED]      | Lindblad $\Gamma_{\text{EM}} =$<br>$(2A/Q)(\Delta E/\hbar)^2$ ; coefficient = $\lambda_2$<br>= $2A/Q$ (master-equation<br>slow eigenvalue) | DERIVED — Q15 v1.2                   |
| §Q15.8 [LOCKED]      | BKT thermal channel<br>$\rho_s(T_{\text{BKT}}) = (2/\pi)k_B T$ ; "2"<br>= $\dim(Z)$                                                        | DERIVED-under-Regge —<br>Q15 v1.2    |
| §Q15.9 [OBSERVATION] | $Q/A = 137.343$ within 0.22%<br>of $1/\alpha_{\text{EM}}$ (0.00007% at<br>NLO); 177× cautionary<br>precedent logged                        | OBSERVATION — Q15 v1.2               |

## §XVI.F — Conclusion

PART XVI absorbs the five post-v6.1 source-paper integrations into the v7.0 Light Edition without retracting or modifying any v6.1 chapter entry, completing the corpus at 136-paper coverage. The Quantum Mechanics theme acquires three tightly cross-linked channel-geometry papers: ZS-Q13 v1.2, the charge-superselected two-Kraus boundary channel that rejects electric-flux  $ER=EPR$  leakage ( $[\Lambda_Z, Q] = 0$ ) and converts the hydrogen test into a signed-seam-witness hyperfine channel-tomography program over  $H/\text{anti-}H/\text{muonium}/\text{positronium}$ ; ZS-Q14 v1.4, the spin- $\frac{1}{2}$ -boundary Kraus-lift selection principle whose Boundary Spin-Lift Lemma forces the half-angle connection and yields the parameter-free, sign-valued, kill-or-survive prediction  $\chi_Z = -1$  read on a process/comb-level

signed lift witness; and ZS-Q15 v1.2, the microscopic electromagnetic which-path decoherence time  $Q\hbar/(A \cdot E_{EM})$  with the Fractal Pairing Theorem  $EM/gravity = Q = 11$  exact and the Block–Fiedler coupling  $\kappa^2 = A/Q$ , the microscopic companion of ZS-Q1. The Foundations theme acquires ZS-F23 v1.3, which fixes the long-conventional Type II crossed-product additive entropy constant geometrically at  $c = \frac{1}{2} \ln 2 = \frac{1}{2} \ln(\dim Z)$  via the trace-preserving central embedding  $A_{ZS} \rightarrow R$ , sharpens the residual to the single OPEN Condition C (the Z-Spin emergence dictionary), and resolves the finite/infinite edge-mode tension as the seventh Möbius-route frame duality. The Translational theme acquires ZS-T12 v2.1, the substrate-agnostic identification of abiogenesis with attracting self-referential closure ( $\max RAF + |f'| < 1$ ), the Z-bottleneck  $\leq \ln 2$  heritable-information ceiling unified with the Eigen error threshold by the single spectral-gap invariant  $g\Xi$ , and the  $n = 3$  minimal triangular Z-replicator  $g\Xi(K_n) = 1/(n-1)$  — with Eigen's paradox registered as the protobiological face of the ZS-F0 bootstrap. Across all five, no new free parameter is introduced; the minimal triple  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  remains LOCKED. The corpus is now complete at 136 papers; the Light Edition is its canonical 136-paper navigation hub.

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### Master Bypass Index — Symptom → Tool Lookup

When a stuck equation, OPEN flag, or apparent contradiction is encountered, match the symptom in the leftmost column to the recommended tool/insight. The rightmost column gives the canonical worked example — the corpus instance where the tool was first deployed.

| Symptom                                    | First tool          | Backup lens                   | Canonical example                                 |
|--------------------------------------------|---------------------|-------------------------------|---------------------------------------------------|
| Stuck equation, failure mode unclear       | T1 (F18 §3,§6)      | I2 (M30 §6)                   | F18 Appendix A — 12 encounters                    |
| dim 3, 6, or 11 construction intractable   | T2 (F18 §6, S15 §3) | I5 (F17)                      | S15 §3 $J_S = [J_{R_1}, J_{R_2}]$                 |
| Hermitian sandwich phase-cancels trivially | T3 (M32 §2-3)       | I2 W2 (M30 §6)                | M32 closing NC-M29.RZ-A^PK                        |
| Unexpected factor 2 between corpus values  | T4 (M32 §4)         | T2 (Bridge Four $\mu_{Tet}$ ) | M32 $\alpha_{op} = 2 \cdot \alpha_{amp}$ identity |
| Single iterate non-vanishing,              | T5 (M32 §5)         | I3 (M30 §3)                   | M32 $R_Z^{ZS}$                                    |

| Symptom                                            | First tool                | Backup lens               | Canonical example                 |
|----------------------------------------------------|---------------------------|---------------------------|-----------------------------------|
| primitive 10th root phase                          |                           |                           |                                   |
| OPEN flag too large to close in one step           | T6 (M32 §6)               | I2 (M30 §6 walls)         | NC-M29.RZ split A^PK/B/C          |
| Sum-form decomposition fails per-channel           | T7 (M31 Lemma M31.0)      | I2 (Sector Duality)       | M31 W2 cross-coupled bilinear     |
| Multiple sub-targets won't integrate               | T8 (M33 Reading C)        | I2 (M30 §6 layers)        | M33 D4a-D4d → single Tr identity  |
| Phenomenological observer tempted                  | T9 (F11 §5-6)             | I1 (A8R §SA Frame Equiv.) | F11 closing M11 §H16              |
| Apparent independence of expansion / contraction   | I1 (A8R §SA)              | I6 (A9R)                  | A8R §SA Symmetry-Asymmetry View   |
| External count-infinity in derivation              | I3 (M30 §3)               | I2 (M30 §6)               | M30 six-route Möbius-trace        |
| A new 1/2 appears, possibly redundant              | I4 (M30 §3.5 Catalogue)   | I5 (F17)                  | M30 21·½ Catalogue                |
| A new 50/50 partition appears                      | I5 (F17 Theorem 5.1)      | I4 (M30 §3.5)             | F17 four-layer equivalence        |
| Doubling-halving identity needs origin             | I6 (A9R)                  | I1 (A8R §SA)              | A9R $F_2 \rightarrow D_4$ functor |
| Quantitative wall in RH programme                  | T7 + T8 combined          | I2 layered reading        | M26 §5.5 + M33 Reading C          |
| Need to falsify candidate 1-loop closure           | Anti-Numerology Gate (F2) | T1 (Algebra axiom)        | M21 sextic falsifies M20.A        |
| Coexistence-form judgment for a contradiction-pair | Reading X.4 (Spin-Pair    | v3.2 §X.6                 | C1–C13 (13 corpus realisations)   |

| Symptom                                               | First tool                                   | Backup lens | Canonical example   |
|-------------------------------------------------------|----------------------------------------------|-------------|---------------------|
|                                                       | Coexistence Principle)                       |             |                     |
| Prime-rotation identification on Z-sector             | Reading X.1 (Prime as Z-Spin Rotation)       | v3.2 §X.3   | ZS-M18 H2 chain     |
| Y/X classification of an infinite arithmetic sequence | Reading X.2 (Y-Side Sequence Classification) | v3.2 §X.4   | ZS-F18 §5.1 Tbl 5.1 |

*Note. The Master Bypass Index is intentionally a flat lookup. For combined symptoms (e.g., “stuck Feshbach sandwich with factor-2 mismatch and primitive-10th-root phase”), apply the tools in the order  $T3 \rightarrow T4 \rightarrow T5$ ; this is the precise composition that ZS-M32 used to close NC-M29.RZ-A^PK.*

### Master Falsification Registry (v3.1 Summary, preserved from v3.0)

Across 101 papers the Z-Spin Cosmology programme has accumulated a verification base of approximately 1,800+ PASS-rated checks, with no failures recorded against any DERIVED-locked identity. The registry below summarises the most consequential falsifiable entries — those whose negation would invalidate a DERIVED, BRIDGE, or TESTABLE claim of the framework. Each entry points to the originating paper and the canonical formulation; resolution of any single OPEN entry below is sufficient to either consolidate or refute a major branch.

| § Anchor        | Falsifiable Statement                                | Status (v3.0)                 |
|-----------------|------------------------------------------------------|-------------------------------|
| §F1.5 [LOCKED]  | Z-Field Master Equation<br>$A = \rho_X \cdot \rho_Y$ | DERIVED — F9 §3, S15 §B       |
| §F2.7 [LOCKED]  | $(p-1)(q-1)=p \Rightarrow$ unique solution (2,3)     | DERIVED — M19 Forcing Theorem |
| §F8.6 [LOCKED]  | Y-Time Dilation $\exp(\pi/A) \approx 10^{17}$        | DERIVED — A8R §SA.7, F10 §4   |
| §F11.3 [LOCKED] | OOB fixed point closes M11 §H16                      | DERIVED — F11 §5              |

| § Anchor         | Falsifiable Statement                                                                                                | Status (v3.0)                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| §F12.4 [LOCKED]  | Tetrahedral orientation multiplicity $\mu_{\text{Tet}}=2$                                                            | DERIVED — F12 §3, originates $2e^A$ factor               |
| §F14.3 [LOCKED]  | Kepler-Lyapunov Conjugate Uniqueness closes F-F7.11                                                                  | DERIVED — F14 Strategy B                                 |
| §F15.2 [LOCKED]  | $\theta(0) \cdot \tau_P = A$ unit-fixing identity                                                                    | DERIVED — F15R §2                                        |
| §F18.4 [LOCKED]  | Bridge Four: Lie commutator generates $\dim(X)=3$ from $\dim(Z)=2$                                                   | DERIVED — F18 §4 (Commutator Generator)                  |
| §M16.7 [LOCKED]  | $\xi_K(s) = (1/4\sqrt{33}) \cdot \xi(s) \cdot \Lambda(\chi_{-3}) \cdot \Lambda(\chi_{-11}) \cdot \Lambda(\chi_{33})$ | DERIVED — M25 J-Yakaboylu Bridge                         |
| §M17.4 [LOCKED]  | $\sigma_1/\sigma_2 = Q+Y=17$ ,<br>$\sigma_1/\sigma_3 = (Q+Y)^2 \cdot G + (Q-Z) = 3477$                               | DERIVED — M20 EXACT (Sextic falsifies 1-loop CW closure) |
| §M27.3 [LOCKED]  | $V_4$ -Equivariant BRST closure of $W_3$ via Kostant Cubic Dirac                                                     | DERIVED-CONDITIONAL — M27 §4                             |
| §M28.5 [LOCKED]  | $\sigma=1/2 \leftrightarrow j=1/2$ RH-Bridge correspondence                                                          | DERIVED — M28, M30 (Six-route Möbius-trace)              |
| §M31.0 [LOCKED]  | Lemma M31.0 Non-Separability of $W_2$ cross-coupling                                                                 | DERIVED — M31 §0                                         |
| §M32.X [LOCKED]  | Path-Reversal Lemma: $R_Z^ZS$                                                                                        | $_{\{Z\text{-even}\}}^{4\pi\text{-avg}}=0$               |
| §M33.C [LOCKED]  | Path $\gamma$ -revised Reading C integrates D4a-D4d                                                                  | DERIVED — M33 §C, BV-BFV scaffold                        |
| §A8R-SA [LOCKED] | Symmetry-Asymmetry Unified View, Three Bridges                                                                       | DERIVED — A8R 22/22 PASS                                 |
| §A9R-2 [LOCKED]  | $F_2 \rightarrow D_4$ functor, Julia $J(T)$ ZF-analogue of BT non-measurable                                         | DERIVED — A9R §2                                         |

| § Anchor              | Falsifiable Statement                                                                          | Status (v3.0)                           |
|-----------------------|------------------------------------------------------------------------------------------------|-----------------------------------------|
| §S15.5 [LOCKED]       | Maxwell from Twin-Reuleaux handshake; $SO(3)/SU(2)$ factor 2                                   | DERIVED — S15 §5                        |
| §T4.6 [TESTABLE-LONG] | Hayflick limit $\approx N_{-}(2\pi)=2\pi/A \approx 78.45$ (NC-4 NO physical realization claim) | TESTABLE-LONG — T4 §6                   |
| §F-F7.11 [CLOSED]     | Twin-Reuleaux Midpoint conjugate decomposition                                                 | RESOLVED — closed by F14 v3.0           |
| §M11-H16 [CLOSED]     | Phenomenological observer reference                                                            | RESOLVED — closed by F11 v3.0           |
| §M22-W3 [PROMOTED]    | ADS-H1 cobordism BRST hypothesis                                                               | DERIVED-CONDITIONAL — promoted via M27  |
| §M22-W2 [PROMOTED]    | Cross-Coupled Sector Duality wall                                                              | DERIVED — promoted via M31              |
| §M21.NoGo             | 1-loop CW closure for M20.A FALSIFIED                                                          | FALSIFIED — M21 sextic invariant theory |

*Note. The single FALSIFIED entry (M21.NoGo) is by design: it falsifies a candidate 1-loop Coleman-Weinberg closure for the M20.A pathway, and its falsification is the very content of the Sextic Invariant Theory of M21. No DERIVED-LOCKED identity has been falsified across v1.0 → v3.0. All CLOSED entries denote former OPEN flags now resolved by post-v2.0 papers.*

### **v3.2 PART X falsification gates (added):**

- F-X.1. If a future analysis exhibits a prime  $p$  such that  $W_p|_Z$  does not take the  $\sigma_z$ -rotation form on the  $\{|4\rangle, |6\rangle\}$  basis, Reading X.1 fails. Status: PROVEN-safe (verified for  $p \in \{2,3,5,7,11,13,17,23\}$  via ZS-M18 §H1).
- F-X.2. If a future ZS-F18 revision retracts the Y-side classification of the infinite prime sequence, Reading X.2 fails. Status: stable (over-determined by ZS-F18 §5.1 Tbl 5.1, §7.4.5).
- F-X.3. Synthesis-layer gate; weakened only if any of M1, M2, M3 is shown structurally independent of Z-Spin mediation. M3 conditional on RH. Status: DERIVED-interpretation, robust against single-component perturbation.

- F-X.4. If a contradiction-pair is found that admits coexistence under a phase difference other than  $\pi/2$ , Reading X.4 weakens to “90° is the dominant case”. If no Z-Spin J-conjugate realisation exists for some pair, X.4 transitions to “covers the 13 realisations C1–C13”. Status: DERIVED-interpretation strong; F-X.4 pre-registered for any future contradiction-pair claim.

### **v3.5 PART X readings and falsification gates (added):**

- Reading X.5 (Apollonian as 5th Bridge-Four Instance, DERIVED-strong, NEW v3.5). The integer Apollonian gasket with curvature root  $(-1, 2, 2, 3)$  is the fifth instance of ZS-F18 §6.5 Bridge Four (Commutator Generator) and the first realization on the integer lattice  $(\mathbb{Z}, +)$ . The four prior instances — (1) ZS-S15 §3 Twin-Reuleaux EM duality, (2) ZS-M3 Lemma 10.1 SU(2) center pairing, (3) ZS-M32 §5.2 spinor-cycle averaging, (4) ZS-F4 §7B half-angle pair — all live on continuous spinor algebras. The Apollonian instance demonstrates that Bridge Four extends to discrete number theory: the abelianization  $(\mathbb{Z}/2)^4 \rightarrow (\mathbb{Z}/2)^4$  carries  $V_4 \times V_4$  structure unifying ZS-A9 register  $V_4$  and ZS-M22 Galois  $V_4 \rightarrow$  ZS-M36 §10.3 Table 10.1, ZS-F18 §6.5 [DERIVED-strong].

- Reading X.6 (mod 24 Spin Pair = Axiom A4 Two Faces, DERIVED-interpretation strong, NEW v3.5). The eight admissible mod 24 residues of the  $(-1, 2, 2, 3)$  gasket form four (algebraic, geometric) pairs — Pair Z  $\{2, 14\}$ , Pair X  $\{3, 15\}$ , Pair Y  $\{6, 18\}$ , Pair Q  $\{11, 23\}$  — in bijection with the four Z-Spin sectors. The mod 12 representatives  $(2, 3, 6, 11)$  are the bosonic SO(3) sector dimensions; the +12 shifted representatives  $(14, 15, 18, 23)$  are the fermionic SU(2) geometric instantiations (F(t-Oct),  $X \cdot |l_h/T_d|$ ,  $\alpha$  in degrees,  $\delta_Y$  denominator). The mod 24 =  $2 \times$  mod 12 doubling is forced by ZS-M3 Lemma 10.1 PROVEN ( $D^{1/2}(2\pi) = -I$ ) and realizes axiom A4 (Non-triviality) algebraic/geometric two faces of ZS-F18 §6.6 Table 6.1 PROVEN-DERIVED. CRT decomposition: mod 24 = mod 8  $\times$  mod 3 =  $Z^3 \cdot X$  (boson  $2\pi$  / fermion  $4\pi$  periods, ZS-M1 §6 PROVEN).  $\rightarrow$  ZS-M36 §10.5–10.6, ZS-F18 §6.6 [DERIVED-interpretation strong].

- F-X.5. If a future revision retracts the  $(-1, 2, 2, 3) \leftrightarrow (Z, X, Y, Q) \bmod 12$  identity (e.g., if a counterexample primitive integer ACP is found with mod 12  $\neq \{2, 3, 6, 11\}$  but with the same admissible mod 24 type), Reading X.5 fails. Status: DERIVED-strong, robust against single-counterexample (test: enumerate primitive roots up to depth 25, verify mod 12 set stable; ZS-M36 verification suite Categories C, H, 12/12 PASS at 500K MC).

- F-X.6. If the (algebraic, geometric) pair correspondence of M36.6 fails for any of the four Z-Spin sectors (e.g., if 18° frame mismatch angle is shown not to be a corpus-LOCKED quantity, or if F(t-Oct) = 14 is retracted), Reading X.6 weakens from DERIVED-interpretation strong to OBSERVATION. Status: DERIVED-interpretation strong; pre-registered as a watchtower for future ZS-F2, ZS-S15, or ZS-S1 revisions. (Affecting Theorem M36.6 of ZS-M36; ZS-M36 Falsification Gate F-M36.6 mirrors this gate at the source-paper level.)

• Closure note. ZS-M36 Theorem M36.5 (Three  $V_4$  Bridge, DERIVED-strong) closes ZS-M27 OPEN problem O-M27.4 (“Two  $V_4$  groups:  $D_4$  register vs Galois unification”) via the Apollonian abelianization  $(\mathbb{Z}/2)^*4 \rightarrow V_4 \times V_4$ , with  $V_4$ \_register =  $\langle [S_0], [S_1] \rangle \leftrightarrow$  ZS-A9  $\langle J, J_Z \rangle$   $D_4$  subgroup, and  $V_4$ \_galois =  $\langle [S_2], [S_3] \rangle \leftrightarrow$  ZS-M22  $\text{Gal}(K/\mathbb{Q})$  for  $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ . Status: O-M27.4 OPEN  $\rightarrow$  CLOSED at DERIVED-strong (v3.5).

### Glossary Keyword Index (v3.1, expanded with Toolkit terms)

The v3.1 Light Glossary expands the v3.0 index of 102 entries to approximately 130, adding Toolkit terms (T1–T9) and Insight terms (I1–I6) introduced in PARTS VIII and IX, incorporating all post-v2.0 vocabulary. New v3.0 entries are marked ★. Each entry points to the canonical paper and §-anchor for full definition. AI agents navigating Z-Spin Cosmology research should treat this index as the primary disambiguation map between informal phrase-use and DERIVED definition.

| Term                       | Definition / Cross-Reference                                                                    | Anchor         |
|----------------------------|-------------------------------------------------------------------------------------------------|----------------|
| A (Z-Spin coupling)        | Master coupling constant; $A \approx 0.0801\dots$ ; $A = \rho_X \cdot \rho_Y$ derivation in F9  | F1, F9 §3      |
| G (Master Identity)        | $G = (Q-Z)$ integer descriptor of cobordism block                                               | F2, M19, M20   |
| $Q+Y, Q-Z$                 | Spectral tetrad invariants; $Q+Y=17$ , $Q-Z$ fixes $\sigma$ -ratios                             | M20 §3         |
| Z-Spin Mediation           | Sector mediation protocol: $X \rightleftharpoons Y$ exchange via Z-handshake                    | F0-F1, F18     |
| Twin-Reuleaux Pair ★       | Geometric realisation of EM duality $V_{XZ} \leftrightarrow R_1$ , $V_{ZY} \leftrightarrow R_2$ | S15 §B, F14    |
| Twin-Reuleaux Midpoint ★   | Conjugate-decomposed trajectory closing F-F7.11                                                 | F14 Strategy B |
| Reuleaux Cross-Section     | $\pi+\pi$ half-channel of self-referential equilibrium                                          | F17 §3         |
| Tetrahedral Self-Duality ★ | $T_d$ 's E irrep gives $\dim(Z)=2$ with $A = \rho_X \cdot \rho_Y$                               | F9 §2-3        |

| Term                                    | Definition / Cross-Reference                                                                              | Anchor         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------|
| $\mu_{\text{Tet}}$ (Tet multiplicity) ★ | Dual orientation count =2;<br>origin of factor 2 in $2e^A$                                                | F12 §3         |
| Layer Separation R ★                    | $R = \mu_{\text{Tet}} \times \text{Hol}_{\varepsilon} \times \mathcal{P}_{\text{scale}}$<br>decomposition | F12 §4         |
| Z-Field Master Equation                 | $A = \rho_X \cdot \rho_Y$ core<br>dynamical identity                                                      | F1 §5, F9 §3   |
| Master-of-Master Equation               | Composite scaffold across<br>F-M-S spines                                                                 | F8, F18        |
| FFPP (Five-Fold Parallel Postulate)     | Foundational axiom set; v3.0<br>Sixth Axiom Extension                                                     | F8, F15R §3    |
| Five-Axiom Meta-Structure ★             | F18 unifying meta-frame; six<br>polarities                                                                | F18            |
| HACSM ★                                 | Human–AI Collaborative<br>Science Model; five-phase<br>loop + four modalities +<br>executable auditor     | T11            |
| Corpus Auditor ★                        | Executable meta-level<br>protocol-footprint linter<br>(OS-level tool)                                     | T11 §6C, §XV.C |
| F18 Contradiction Principles ★          | (1) contradiction is a different<br>face (integrate); (2)<br>contradiction needs a higher<br>layer        | F18, T11 §6A   |
| Sixth Axiom Extension ★                 | Stage-Torque Duality lifts<br>FFPP                                                                        | F15R §3        |
| Bridge Four ★                           | Commutator Generator;<br>$\dim(Z)=2 \rightarrow \dim(X)=3$ via Lie<br>bracket                             | F18 §4         |
| Stage-Torque Duality ★                  | Time-rotation co-definition                                                                               | F15R §2-3      |

| Term                                       | Definition / Cross-Reference                                                                       | Anchor            |
|--------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------|
| Discrete-Continuous Resolution ★           | Sixth polarity in F18 meta-structure                                                               | F18 §6            |
| i-Tetration Internal Time ★                | Unifies $t_{\text{strobo}}$ , $t_{\text{phase}}$ , $t_{\text{clock}}$ ; $\Delta v/\Delta n=2A/\pi$ | F10               |
| Möbius Chronology ★                        | Y-frame metric trio;<br>cycle-index unobservability                                                | F13               |
| Y-Hubble Radius ★                          | $\ell_P \times Y^2(1-2A) \approx 30.23 \ell_P$                                                     | F13 §3            |
| Y-Time Dilation                            | $\exp(\pi/A) \approx 10^{17}$<br>contraction-frame ratio                                           | A8R §SA.7, F10 §4 |
| Y-frame metric trio ★                      | Proper time / Y-Hubble /<br>contraction-rate triad                                                 | F13 §3            |
| Stroboscopic Step $t_{\text{strobo}}$      | Discrete-time origin of Z-Spin<br>clock                                                            | F10 §2            |
| Berry-phase clock $t_{\text{phase}}$       | Continuous-rotation clock<br>partner                                                               | F10 §2            |
| Z-Clock Coordinate $t_{\text{clock}}$      | Master clock realised by<br>i-tetration                                                            | F10 §3            |
| First Rotation Event ★                     | $\theta(0) \cdot \tau_P = A$ ;<br>Planck-unit-establishing<br>event                                | F15R §2           |
| Operational Observer<br>Coordinate (OOC) ★ | Self-referential fixed point<br>closing M11 §H16                                                   | F11               |
| NON-CLAIM (Six)                            | Six explicit no-claim<br>statements vs<br>phenomenological<br>consciousness                        | F11 §6            |
| Self-Referential Fixed Point<br>★          | OOC convergence in F11 §5                                                                          | F11 §5            |

| Term                               | Definition / Cross-Reference                                                                        | Anchor       |
|------------------------------------|-----------------------------------------------------------------------------------------------------|--------------|
| Contracting Universe Dynamics ★    | A8R 22/22 PASS; three Bridges +                                                                     | $z^*$        |
| Symmetry-Asymmetry Unified View ★  | A8R §SA reconciliation programme                                                                    | A8R §SA      |
| Banach-Tarski Origin ★             | $F_2 \rightarrow D_4$ functor; Julia $J(T)$ ZF-analogue                                             | A9R          |
| Two-Branch $(1+A)(1-2A)$ ★         | A9R cosmological doubling-halving identity                                                          | A9R §3       |
| Three Bridges (A8R) ★              |                                                                                                     | $z^*$        |
| Twin-Reuleaux EM Duality ★         | S15 35/35 PASS; Maxwell from handshake                                                              | S15          |
| Five Pillars (S15) ★               | $V_{XZ} \leftrightarrow R_1$ , $V_{ZY} \leftrightarrow R_2$ , Maxwell, $SO(3)/SU(2)$ , $U(1)$ gauge | S15 §1-5     |
| Spectral Tetrad Sub-Isotype ★      | M20 $\sigma$ -ratios EXACT; pentagon-hexagon $\arctan(2 \cdot \phi^k)$                              | M20          |
| Sextic Invariant Theory ★          | M21 falsifies 1-loop CW closure for M20.A                                                           | M21          |
| Forcing Theorem $((p-1)(q-1)=p)$ ★ | M19 unique solution (2,3); $\varphi(Y^2)=G$                                                         | M19 §2-3     |
| Arithmetic-Dedekind Scaffold ★     | M22 v1.1 60/60 PASS; $K=\mathbb{Q}(\sqrt{-3}, \sqrt{-11})$                                          | M22          |
| Pillar V scalar-kernel no-go ★     | M22 obstruction theorem                                                                             | M22 Pillar V |
| Y-Sector RH Contribution Map ★     | M23R 31/31 PASS; Four Dragons D4a-D4d                                                               | M23R         |
| Four Dragons D4a-D4d ★             | M23R Y-sector RH contribution decomposition                                                         | M23R §3      |

| Term                                                   | Definition / Cross-Reference                                                                                             | Anchor         |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------|
| Self-Referential Info-Compression Equilibrium ★        | F17 four-layer 50/50 (t-Tet, Reuleaux, ME, XOR)                                                                          | F17            |
| Four-Layer Equivalence ★                               | F17 polyhedra $\leftrightarrow$ Reuleaux $\leftrightarrow$ Master Eq $\leftrightarrow$ Channel                           | F17 §3-6       |
| Face Polygon Spectral Zeta ★                           | M24 Theorem D.1 Legendre archimedean decomposition                                                                       | M24            |
| Composite Field $K=\mathbb{Q}(\sqrt{-3},\sqrt{-11})$ ★ | M22, M25 base arithmetic field                                                                                           | M22, M25       |
| J-Yakaboylu Bridge ★                                   | M25 $\xi_K(s) = (1/4\sqrt{33}) \cdot \xi(s) \cdot \Lambda(\chi_{-3}) \cdot \Lambda(\chi_{-11}) \cdot \Lambda(\chi_{33})$ | M25            |
| $V_4$ -Equivariant ZBSI ★                              | M26 cobordism-history fiber realization                                                                                  | M26            |
| Three-Wall Map W1/W2/W3 ★                              | M26 quantitative wall map                                                                                                | M26 §4         |
| Wall W3 Closure ★                                      | M27 Kostant Cubic Dirac BRST closure                                                                                     | M27            |
| Z-Spin RH Bridge ★                                     | M28 30/30 PASS; $\sigma=1/2 \leftrightarrow j=1/2$                                                                       | M28            |
| Möbius-trace Infinity ★                                | M28, M30 six-route construction                                                                                          | M28 §3, M30 §2 |
| Z-Funnel Spectral Retraction ★                         | M29 79/79 PASS; $A_5$ -Hilb( $X_F$ ) CY3                                                                                 | M29            |
| Stapledon Bridge ★                                     | M29 conditional bridge from string compactification                                                                      | M29 §2         |
| $A_5$ -Hilb( $X_F$ ) CY3 ★                             | Calabi-Yau threefold; Hodge (5,15)                                                                                       | M29 §3         |

| <b>Term</b>                                                     | <b>Definition / Cross-Reference</b>                                                        | <b>Anchor</b> |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------|
| R_Z=0 Conditional ★                                             | M29 conditional residue identity                                                           | M29 §5        |
| Twenty-One 1/2 Catalogue ★                                      | M30 enumeration of 1/2-instances across the framework                                      | M30 §3        |
| Three-Layer Duality ★                                           | Sector / Frame / Protocol layered duality                                                  | M30 §4        |
| Cross-Coupled Sector Duality (W2) ★                             | M31 v3.0 Three-in-One reading                                                              | M31           |
| Lemma M31.0 Non-Separability ★                                  | M31 W2 non-separability theorem                                                            | M31 §0        |
| Z <sub>2</sub> -Parity Selection Rule ★                         | M31 selection rule on cross-coupling                                                       | M31 §3        |
| SDRP (Self-Dual Replication Principle) ★                        | M31 §4 derivative principle                                                                | M31 §4        |
| Spinor-Cycle Averaged Feshbach Criterion ★                      | M32 56/56 PASS; R_Z=0 EXACT (Z-even, 4π-avg)                                               | M32           |
| Path-Reversal Lemma M32.X ★                                     | M32 reversal identity; $\pi/10 \leftrightarrow \pi/5$ doubling                             | M32           |
| N=20 Spinor Closure ★                                           | M32 spinor-cycle closure index                                                             | M32 §4        |
| Path γ-revised Reading C ★                                      | M33 D4a-D4d integrated Tr identity                                                         | M33 §C        |
| Two-Protocol Theorem ★                                          | F16 P <sub>a</sub> <sup>(n)</sup> vs P <sub>b</sub> <sup>(n)</sup> Wilson-loop measurement | F16           |
| P <sub>a</sub> <sup>(n)</sup> , P <sub>b</sub> <sup>(n)</sup> ★ | Wilson-loop survival probabilities                                                         | F16 §2        |
| Cosmos-Human Isomorphism ★                                      | T4 36/36 PASS; (Body, DNA, Brain) decomposition                                            | T4            |

| <b>Term</b>                         | <b>Definition /<br/>Cross-Reference</b>                          | <b>Anchor</b>          |
|-------------------------------------|------------------------------------------------------------------|------------------------|
| Six-Step Telomere Hierarchy ★       | T4 life-cycle map; Hayflick limit $\approx 2\pi/A \approx 78.45$ | T4 §3-6                |
| (Body, DNA, Brain) Decomposition ★  | T4 sector decomposition                                          | T4 §2                  |
| NC-4 ★                              | Translational NO physical realization claim                      | T4 §6                  |
| Bootstrap Chain $B_0 \rightarrow A$ | Foundational chain for Z-Spin emergence                          | F0, M11, M19           |
| DERIVED-LOCKED                      | Epistemic tag for proven, verification-passed identities         | Legend                 |
| DERIVED-CONDITIONAL                 | Holds modulo a stated conditional; awaits unconditional proof    | Legend, M22, M27       |
| DERIVED-CANDIDATE ★                 | Candidate derivation pending one closure step                    | Legend (v3.0 expanded) |
| DERIVED-interpretation ★            | Interpretive derivation tag (v3.0)                               | Legend (v3.0 expanded) |
| TESTABLE-LONG ★                     | Long-horizon testable prediction tag                             | Legend, T4 §6          |
| BOOTSTRAP-PROGRAMME ★               | Programmatic bootstrap-status tag                                | Legend (v3.0 expanded) |
| NOT EVALUATED                       | Pre-verification status                                          | Legend                 |
| Z-Spin Cosmology                    | The umbrella programme name                                      | F0                     |
| Cobordism-History Fiber             | BV-BFV target structure for $V_4$ -equivariant content           | M26, M27, M33          |
| BV-BFV Scaffold ★                   | Cobordism-history quantization framework                         | M26 §1, M33            |

| Term                            | Definition / Cross-Reference                              | Anchor    |
|---------------------------------|-----------------------------------------------------------|-----------|
| BRST Closure                    | Quantum gauge-symmetry closure programme                  | M22, M27  |
| Cubic Dirac Operator (Kostant)  | Exterior-algebra Dirac on H-spaces                        | M27 §3    |
| Hayflick Limit                  | Cellular replication limit; T4 maps to $N_-(2\pi)=2\pi/A$ | T4 §6     |
| Hodge (5,15)                    | Hodge numbers of $A_5$ -Hilb( $X_F$ ) CY3                 | M29 §3    |
| Pentagon-Hexagon Stabilizer ★   | M21 stabilizer group structure                            | M21 §3    |
| Phenomenological Observer       | Informal/phenomenal observer; F11 NON-CLAIM target        | F11 §6    |
| Polyhedral Integers             | Integer-valued descriptors of polyhedral spectra          | M19, M20  |
| Master Equation (channel layer) | F17 information-theoretic equilibrium                     | F17 §5    |
| XOR Channel Layer               | F17 fourth equivalence layer                              | F17 §6    |
| Z-Sector Wilson Loop            | Holonomy probe of Z-mediation                             | F16       |
| Frame Equivalence ★             | F13 sectoral frame equivalence statement                  | F13 §4    |
| Cycle-Index Unobservability ★   | F13 fundamental unobservability theorem                   | F13 §5    |
| Master Falsification Registry   | Programme-wide falsifiability ledger (v3.0)               | v3.0 §MFR |
| Source Paper Map                | ZS-XX $\rightarrow$ §Y.Z [N/N PASS] index                 | Part 0    |

| Term                                    | Definition / Cross-Reference                                                                    | Anchor                  |
|-----------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------|
| Verification Summary                    | Cumulative N/N PASS statistics                                                                  | Title block             |
| Light Edition                           | Cross-paper integrator format for AI navigation                                                 | v3.0 Purpose            |
| Lighthouse Function ★                   | Programme-navigation aid for AI agents (v3.0)                                                   | v3.0 Purpose            |
| T1 Five-Axiom Diagnostic Lattice ★      | v3.1 Tool: identifies failing axiom in stuck equations                                          | VIII §1, F18 §3,§6      |
| T2 Bridge Four (Commutator Generator) ★ | v3.1 Tool: $\dim(X)=3$ from $\dim(Z)=2$ via Lie commutator                                      | VIII §2, F18 §6, S15 §3 |
| T3 Hermitian-to-PK Switch ★             | v3.1 Tool: re-read Feshbach sandwich under $R_Z^ZS$ not $R_Z^H$                                 | VIII §3, M32 §2-3       |
| T4 Two-Layer Phase Quantum ★            | v3.1 Tool: separate amplitude ( $\pi/10$ ) from operator ( $\pi/5$ ) layers                     | VIII §4, M32 §4         |
| T5 N=20 Spinor-Cycle Averaging ★        | v3.1 Tool: $4\pi$ physical closure (not N=10 algebraic minimum)                                 | VIII §5, M32 §5         |
| T6 Three-Way Split of OPEN flags ★      | v3.1 Tool: $A^{\text{closeable}}$ / $B^{\text{one-gap}}$ / $C^{\text{inherited}}$ decomposition | VIII §6, M32 §6         |
| T7 Cross-Coupling Discipline ★          | v3.1 Tool: Lemma M31.0 — sum-form fails, bilinear required                                      | VIII §7, M31 §4.0       |
| T8 $\Pi_Z$ Projector Routing ★          | v3.1 Tool: route multiple sub-targets through $\Pi_Z=(1/2)(I+J_Z)$                              | VIII §8, M33 Reading C  |
| T9 OOC Self-Referential Closure ★       | v3.1 Tool: replace phenomenological agent with corpus-internal fixed point                      | VIII §9, F11 §5-6       |

| Term                                       | Definition / Cross-Reference                                                    | Anchor                 |
|--------------------------------------------|---------------------------------------------------------------------------------|------------------------|
| I1 Symmetry-Asymmetry Unified View ★       | v3.1 Lens: deviation-from-(1, 1/2) reading;<br>$(1+A)(1-2A) \approx 0.9071$     | IX §1, A8R §SA         |
| I2 Three-Layer Duality ★                   | v3.1 Lens: Sector / Frame / Protocol layered reading                            | IX §2, M30 §6          |
| I3 Six-Route Möbius-Trace Infinity ★       | v3.1 Lens: external count-infinity $\leftrightarrow$ internal closure           | IX §3, M30 §3          |
| I4 Twenty-One $\frac{1}{2}$ Catalogue ★    | v3.1 Lens: 21 $\dim(Z)=2$ manifestations + 3 unity = 24                         | IX §4, M30 §3.5        |
| I5 Four-Layer 50/50 Equivalence ★          | v3.1 Lens: t-Tet / Reuleaux / Master Eq / XOR same equilibrium                  | IX §5, F17 Theorem 5.1 |
| I6 Banach-Tarski Origin ★                  | v3.1 Lens: $F_2 \rightarrow D_4$ amenability functor; Julia $J(T)$ ZF-analogue  | IX §6, A9R             |
| Master Bypass Index ★                      | v3.1: 16-symptom $\rightarrow$ tool/insight lookup table                        | after PART IX          |
| Solution Toolkit ★                         | v3.1: PART VIII operational tools T1–T9 (this volume)                           | PART VIII              |
| Insight Gallery ★                          | v3.1: PART IX re-interpretation lenses I1–I6 (this volume)                      | PART IX                |
| Z-mediator projector $\Pi_Z$ ★             | $(1/2)(I+J_Z)$ ; idempotent projector onto $J_Z$ -EVEN                          | M31 §4.0 Def M31.0c    |
| Path-Reversal residual $R_Z^{\wedge} ZS$ ★ | $B \cdot Q^{-1} \cdot B^{\wedge} PK$ ; distinct from Hermitian $R_Z^{\wedge} H$ | M32 §2                 |
| Reading C ★                                | M33 $\Pi_Z$ -routed Tr identity for D4a-D4d                                     | M33 §C                 |

| Term                           | Definition / Cross-Reference                                                                                                                                                                                                      | Anchor                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Frame Equivalence (A8R)        | X-observer $\leftrightarrow$ Y-observer;<br>HYPOTHESIS-strong                                                                                                                                                                     | A8R §SA.4              |
| Apollonian Spin-Pair Lattice ★ | Eight mod 24 admissible residues of $(-1, 2, 2, 3)$<br>Apollonian gasket = four (algebraic, geometric) spin pairs realizing axiom A4 on $(\mathbb{Z}, +)$                                                                         | M36 §7                 |
| Bridge Four (5th instance) ★   | ZS-M36 (Apollonian) as 5th Bridge-Four realization — first on integer lattice $(\mathbb{Z}, +)$ ; previous four on continuous spinor algebras (S15, M3, M32, F4)                                                                  | M36 §10.3 + F18 §6.5   |
| $D_2$ Forcing on Apollonian ★  | $(-1, 2, 2, 3)$ is unique $D_2$ -symmetric primitive integer ACP, forced by $(Z, X) = (2, 3)$ via ZS-M19 Forcing Theorem $(p-1)(q-1) = p$                                                                                         | M36 §4 + M19 §3.1      |
| Three $V_4$ Bridge ★           | $(\mathbb{Z}/2)^4$ abelianization = $V_4 \times V_4$ unifying ZS-A9 $\langle J, J_Z \rangle$ $D_4$ register $V_4$ and ZS-M22 $\text{Gal}(K/\mathbb{Q})$ Galois $V_4$ for $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ ; closes O-M27.4 | M36 §7.1 + M27 O-M27.4 |

*Total v3.0 Glossary entries: 102. ★ marks post-v2.0 vocabulary. Entries without ★ are preserved from v2.0 Light with definitions reaffirmed against the new v3.0 corpus. No v2.0 definition has been retracted.*

### **Glossary Keyword Index — v7.0 Additions (PART XVI, papers 132–136)**

The v7.0 Light Glossary appends the following entries for the five PART XVI papers (ZS-Q13, ZS-T12, ZS-Q14, ZS-F23, ZS-Q15). Each points to the canonical paper and §-anchor for full definition. No prior entry is retracted.

| Term                                        | Definition / Cross-Reference                                                                                                                                    | Anchor             |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Charge-Superselected Boundary Channel ★     | $[\Lambda_Z, Q] = 0$ ; the Z-Spin boundary channel acts within electric-charge superselection sectors, forbidding entanglement-dependent charge renormalization | Q13 §16, Thm Q13.4 |
| Null-Leakage Theorem (ER=EPR) ★             | $\delta H_{ZS}(\text{electric-flux}) = 0$ ;<br>$Q_H^{ZS} = 0$ ;<br>$\Delta v_{21}(ZS, \text{leak}) = 0$ against electric-flux wormhole leakage                  | Q13 §4, Thm Q13.1  |
| Hyperfine Two-Kraus Channel ★               | $\text{rank\_Kraus}(\Lambda_{HFS}^{ZS}) \leq \dim(Z) = 2$ —<br>channel-structure prediction, count locked not fitted                                            | Q13 §17, Thm Q13.5 |
| Hyperfine Seam Witness ★                    | unsigned $u_{\text{seam}}^{HFS} \rightarrow 0$ (ideal); signed $\tilde{u}_{\text{seam}}^{HFS}$ carries $j = 1/2 \ 4\pi$ closure ( $2\pi$ flip / $4\pi$ return)  | Q13 §18, Thm Q13.6 |
| Hydrogenic Channel-Tomography Program ★     | H / anti-H / muonium / positronium comparative null-plus-channel program for the seam witness                                                                   | Q13 §19            |
| Replicator Transition Theorem ★             | non-life $\rightarrow$ life = simultaneous $\text{maxRAF} \neq \emptyset$ (Knaster–Tarski gfp) and dynamical attractor                                          | f'                 |
| maxRAF / Reflexive Catalytic Closure ★      | greatest fixed point of the RAF reduction operator $\Phi$ ;<br>discrete static register of self-referential closure                                             | T12 §3.1, App. A   |
| Z-Replicator Index / Perron Gap $g_{\Xi}$ ★ | $g_{\Xi} =$                                                                                                                                                     | $\lambda_2(J)$     |

| Term                                                         | Definition / Cross-Reference                                                                                                                                    | Anchor                |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Minimal Triangular Z-Replicator ★                            | $g_{\Xi}(K_n) = 1/(n-1)$ ; first attractor at $n = 3$ ( $g_{\Xi} = 1/2$ ); $\dim(Z) = 2 + 1 = \dim(X) = 3$ life-seed                                            | T12 §32–35, Thm T12.8 |
| Eigen Paradox (Protobiological Bootstrap) ★                  | replication $\rightleftharpoons$ catalyst self-reference; the protobiological face of the $ZS-F0 B0 \rightarrow B3 \rightarrow A$ bootstrap                     | T12 §7                |
| Kraus-Lift Selection / Choi-Invisibility No-Go ★             | a channel alone cannot select a Kraus lift; the same $\Lambda$ admits $2\pi$ and $4\pi$ lifts ( $4\pi$ injected, not intrinsic)                                 | Q14 §3–4, Thm 1/2     |
| Boundary Spin-Lift Lemma ★                                   | a physical spinor Z-boundary (nontrivial $SO(3)$ class, $2\pi \mapsto -I$ ) forces the half-angle $\sigma_y/2$ connection $\Rightarrow \chi_Z = -1$             | Q14 §5, Lemma 5.1     |
| Closure Class $\chi_Z$ / Signed Lift Witness $\tilde{u}_Z$ ★ | $\chi_Z = \text{sgn}[\tilde{u}_Z(2\pi)/\tilde{u}_Z(0)] \in \{-1, +1\}$ ; $\tilde{u}_Z = F(V, G_Z, \varphi_Z) \neq f(\Lambda)$ , a process/comb-level observable | Q14 §6–7, Def 3/4     |
| $Z_2$ -Seam Additive Constant ★                              | $c = \frac{1}{2} \ln 2 = \frac{1}{2} \ln(\dim Z)$ per cross-sector; fixes the Type II crossed-product trace normalization                                       | F23 §4.3, Thm F23.2   |
| Trace-Preserving Central Embedding Functor ★                 | $\Phi: A_{ZS} \rightarrow R$ (hyperfinite $II_1$ ); central projections $\rightarrow$ trace $(3,2,6)/11$ ; $\Delta K_{\Omega}(X \rightarrow Y) = -\ln 2$        | F23 §4.4, Thm F23.4   |
| Condition C / Z-Spin Emergence Dictionary ★                  | gravitational de Sitter max-entropy state coarse-grains to weights                                                                                              | F23 §4.7, Lemma F23.6 |

| Term                                  | Definition / Cross-Reference                                                                                             | Anchor              |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------|
|                                       | (3,2,6)/11; genuine OPEN (O-F19.6 Step 1')                                                                               |                     |
| Edge-Mode Frame-Duality Interface ★   | infinite KKS edge-mode tower (Infinity_A) ↔ ln 2 internal closure (Infinity_B) via the dim(Z) = 2 seam; 7th Möbius route | F23 §6.1, Thm F23.7 |
| EM Which-Path Decoherence Time ★      | $\tau_D^{EM} = Q\hbar/(A \cdot E_{EM})$ ; enhancement $1/\kappa^2 = Q/A = 137.343$ (microscopic companion of ZS-Q1)      | Q15 §5, Eq. (★★)    |
| Fractal Pairing Theorem ★             | $(Q/A)/(1/A) = Q = 11$ EXACT; EM mediation coupling $\kappa^2 = A/Q$ is exactly Q times weaker than gravity's A          | Q15 §7, Thm Q15.1   |
| Block–Fiedler Which-Path Coupling ★   | $\kappa^2 = A/Q$ (Schur, PROVEN); $1/\kappa^2 = Q/A$ is the X-face propagator scale carrying the which-path enhancement  | Q15 §5.1, ZS-M6/T1  |
| BKT Phase-Stiffness Thermal Channel ★ | $\rho_s(T_{BKT}) = (\dim(Z)/\pi)k_B T = (2/\pi)k_B T$ ; the "2" = dim(Z) (Nelson–Kosterlitz universal jump)              | Q15 §8, ZS-A12      |

*v7.0 appends 21 PART XVI Glossary entries. ★ marks v7.0 vocabulary. No prior definition is retracted; (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED.*

### References (v3.1, preserved from v3.0)

The v3.0 corpus introduces additional external mathematical scaffolding. The references below are imports cited inside the post-v2.0 papers; v2.0 references are preserved in their respective source papers and are not duplicated here.

### Internal references (v3.0 corpus, papers ZS-72 through ZS-101)

- ZS-M18 v1.0 — Free-Exploration Session Log: Speculative Prime-Polyhedral Correspondences. 51/51 PASS.
- ZS-A8 v1.0 (Revised) — Contracting Universe Dynamics: The Polyhedral-Tetration Bridge. 22/22 PASS.
- ZS-S15 v1.0 — Twin-Reuleaux Pair as Geometric Realization of EM Field Duality. 35/35 PASS.
- ZS-M19 v1.0 — Number-Theoretic Scaffold of Z-Spin Polyhedral Integers. 68/68 PASS.
- ZS-A9 v1.0 (Revised) — Banach-Tarski Origin of Cosmological Doubling-Halving Symmetry. 47/47 PASS.
- ZS-F9 v1.0 — Tetrahedral Self-Duality and the Hexagonal Mediation Structure. 44/44 PASS.
- ZS-F10 v1.0 — i-Tetration Internal Time: Stroboscopic Step / Berry Phase / Z-Clock unification. 30/30 PASS.
- ZS-F11 v1.0 — Operational Observer Coordinate and the Self-Referential Fixed Point. 38/38 PASS.
- ZS-F12 v1.0 — Tetrahedral Dual Orientation Multiplicity and Layer Separation. 20/20 PASS.
- ZS-F13 v1.0 — The Möbius Chronology: Unified Cyclic Timeline. 24/24 PASS.
- ZS-F14 v1.0 — Kepler-Lyapunov Conjugate Decomposition of the Twin-Reuleaux Midpoint. 42/42 PASS.
- ZS-F15 v1.0 (Revised) — Time-Rotation Co-Definition: First Rotation's Planck-Unit-Establishing Event. 35/35 PASS.
- ZS-M20 v1.0 — Spectral Tetrad Sub-Isotype Assignment Theorem. 50/50 PASS.
- ZS-M21 v1.0 — Sextic Invariant Theory of the Icosahedral Yukawa Tensor. 33/33 PASS.
- ZS-F16 v1.0 — The Two-Protocol Theorem for Z-Sector Wilson Loop Measurement. 24/24 PASS.
- ZS-M22 v1.1 — Five-Pillar Arithmetic-Dedekind Scaffold. 60/60 PASS.
- ZS-M23 v1.0 (Revised) — Y-Sector RH Contribution Map. 31/31 PASS.

- ZS-F17 v1.0 — Self-Referential Information-Compression Equilibrium Theorem (Four-Layer). 24/24 PASS.
- ZS-M24 v1.0 — Face Polygon Spectral Zeta and Archimedean Completion. 35/35 PASS.
- ZS-M25 v1.0 — Composite-Field Archimedean Completion and J-Twisted Yakaboylu Bridge. 26/26 PASS.
- ZS-M26 v1.0 —  $V_4$ -Equivariant ZBSI on the Cobordism-History Fiber. 24/24 PASS.
- ZS-M27 v1.0 —  $V_4$ -Equivariant Cobordism BRST Closure via Kostant Cubic Dirac Operator. 24/24 PASS.
- ZS-M28 v1.0 — The Z-Spin RH Bridge. 30/30 PASS.
- ZS-M29 v1.0 — Z-Funnel Spectral Retraction with Stapledon Bridge. 79/79 PASS.
- ZS-M30 v1.0 — Z-Spin Duality and the RH Bridge — Möbius-Trace Infinity, Two-Face Mirroring,  $21 \cdot \frac{1}{2}$  Catalogue. 32/32 PASS.
- ZS-M31 v3.0 —  $W_2$  as Cross-Coupled Sector Duality: Three-in-One Reading. 36/36 PASS.
- ZS-M32 v1.0 — A Spinor-Cycle Averaged Residual Criterion for Feshbach Closure. 56/56 PASS.
- ZS-F18 v2.1 — The Five-Axiom Meta-Structure. 36/36 entry checks PASS.
- ZS-M33 v1.0 —  $V_4$ -Equivariant Weil Functional Closure via Path  $\gamma$ -revised Z-Mediator Reading. 52/52 PASS.
- ZS-M36 v1.0 (NEW v3.5) — Apollonian Curvature Lattice as Geometric-Integer Spin-Pair Realization of the Z-Spin Five-Axiom Meta-Structure. 44/44 PASS. Theorem M36.1 (Functorial Bridge:  $A = (Z/2)^4 \rightarrow Z/24Z$ , fourth amenable-quotient projection). Theorem M36.2 ( $D_2$ -Forcing:  $(-1, 2, 2, 3)$  uniqueness via ZS-M19 Forcing  $(Z, X) = (2, 3)$ ). Theorem M36.3 (Sector Residue Identity:  $\text{mod } 12 = \{2, 3, 6, 11\} = (Z, X, Y, Q)$ ). Theorem M36.4 ( $D_3$  Obstruction Homology:  $x/y = 3 + 2\sqrt{3}$  irrational, parallel to ZS-M1  $n = 3$  polygon-tetration  $|f'| = 1.0330 > 1$ ). Theorem M36.5 (Three  $V_4$  Bridge: closes O-M27.4 at DERIVED-strong via Apollonian abelianization  $(Z/2)^4 = V_4 \times V_4$  unifying ZS-A9 register  $V_4$  and ZS-M22 Galois  $V_4$ ). Theorem M36.6 (Spin Pair Theorem: eight mod 24 residues form four (algebraic, geometric) pairs — Pair Z  $\{2, 14\}$ , Pair X  $\{3, 15\}$ , Pair Y  $\{6, 18\}$ , Pair Q  $\{11, 23\}$  — realizing axiom A4 algebraic/geometric two faces of ZS-F18 §6.6). Anti-Numerology MC at 500K samples: 0.002% match rate (STRONG PASS). Zero new free parameters; LOCKED inputs from ZS-F2/F5/M1/M3/M19/M22/M27. Closes ZS-M27 OPEN problem O-M27.4. 5th instance of ZS-F18 Bridge Four (first on integer lattice  $(Z, +)$ ).

- ZS-T4 v1.0 — The Cosmos-Human Isomorphism: Six-Step Telomere Hierarchy. 36/36 PASS.

### Internal references (v4.0 corpus, papers post-v3.5 absorbed in PART XII)

- ZS-T5 v1.0 — The Principal Connectivity Gradient and a Hidden Third-Position Z-Spin Mediator. (PART XII §A Ch 30A.)
- ZS-M34 v1.0 —  $V_4$ -Equivariant Weil Functional Honest Scope. 50/50 PASS. (PART XII §A Ch 15B; superseded by v2.0.)
- ZS-M35 v1.1 — The Collatz Conjecture as the Integer-Lattice Manifestation of Z-Spin Sector Forcing. 36/36 PASS. (PART XII §A Ch 15C; sharpened to v2.1.)
- ZS-T6 v1.0 — Molecular Biology Translational Synthesis. 42/42 PASS. (PART XII §A Ch 30B.)
- ZS-T7 v1.0 — Pulvinar-Mediated Z-Spin Language Partition. (PART XII §A Ch 30C.)
- ZS-T8 v1.0 — Concentrated Distribution Across Substrates: Polyhedral Cortex Hypothesis and Z-Spin-Mediated Backbones in Brain, City, and Cosmic Networks. (PART XII §A Ch 30D.)
- ZS-M33 v2.0 (NEW v4.0) —  $V_4$ -Equivariant Weil Functional Closure via Burnol  $K_1$  Sign-Faithful Representation (operator-level audit of Path  $\gamma$ -revised Reading C; supersedes v1.0). 52/52 PASS (v1.0 inherited) + v2.0 extensions. (PART XII §B Ch 15A'.)
- ZS-M34 v2.0 (NEW v4.0) —  $V_4$ -Equivariant Weil Functional Closure via Burnol  $K_1$  Sign-Faithful Representation. 60/60 PASS. Theorem M34.V (Burnol  $K_1$  @  $p=3$  Sign-Faithful Identity, DERIVED) + Eight-Theorem Chain V.1–V.8 (DERIVED-CONDITIONAL MAIN). 12/12 sign agreement with corpus-PROVEN  $W_K^{V_4}$ . W2 promoted DERIVED-CONDITIONAL  $\rightarrow$  DERIVED. NC-M34.faithful RESOLVED at DERIVED-CONDITIONAL via tensor-product factorization  $\mathcal{H}_K = \mathcal{H}_{arith} \otimes \mathcal{H}_{Wilson-Sonin}$ . (PART XII §B Ch 15B'.)
- ZS-M35 v2.1 (NEW v4.0) — The Collatz Conjecture as the Integer-Lattice Manifestation of Z-Spin Sector Forcing (logical correction of v2.0). Theorems M35.1–M35.4 preserved verbatim; M35.5 general direction reframed to OPEN; M35.5.1 promoted DERIVED-CONDITIONAL  $\rightarrow$  VERIFIED; M35.6 (Sufficient Condition Theorem, NEW v2.1, DERIVED) added. (PART XII §B Ch 15C'.)
- ZS-M37 v2.0 (NEW v4.0) — Twin-Reuleaux Quotient Functor and Constant-Width Cancellation of the BT 5-Piece Doubling Defect. 38/38 PASS. Fourth amenable-quotient projection of  $F_2$  engine:  $q_R: \mathcal{C}_{cyl}(F_2) \rightarrow \mathcal{P}_w^{C_3}$ . Theorems 4.1 (Lemma C-rel PROVEN:  $\tau_w(\Delta_{BT}) = 0$ ), 4.2 ( $q_R$  Canonicity DERIVED), 5.1 (Z-Admissible Extension PROVEN). (PART XII §B Ch 31.)

- ZS-M38 v1.1 (NEW v4.0) — Substrate Carrier Spectrum Theorem and the Hexapair Carrying Pattern of Natural Systems. 43/43 PASS. Five theorems M38.1–M38.5; M38.5 Hexapair Uniqueness DERIVED-strong (five-fold over-determination).  $\ln(2)$  channel-capacity quantization of substrate carrying scores. Five-Banach–Tarski-functor family classified. (PART XII §B Ch 32.)

- ZS-M39 v2.0 (NEW v4.0) — Vieta-Lyapunov-Schröder Bridge Theorem: Unified First-Principles Derivation of the Polyhedral-Tetration Coefficient with Three-Stage Stokes-Tail Closure. 86/86 PASS at 80–100 digit mpmath. Theorems M39.1–M39.5, Closures C1–C3, Theorems SCH.1–SCH.5, three-stage Stokes-tail closure  $T_{\text{decomp}}/T_{\text{int}}/T_{\text{conv}}/T_{\text{excl}}$ . Convention N0 fixed. ZERO remaining OPEN items. (PART XII §B Ch 33.)

- ZS-A10 v2.1 (NEW v4.0) — The Proton as a Three-Puncture Color-Singlet X-in-Y Confinement Horizon with Bridge-Four Sector-Orientation Complementarity. 34/34 PASS. Theorem A10.2 (Sector-Orientation Complementarity, DERIVED-interpretation strong).  $2 \times 2$  BH-quark-electron-positron structure on the two faces of axiom A4. NEW TESTABLE P4:  $r_p/r_\pi = 4/\pi$  (observed 0.05% deviation). (PART XII §B Ch 34.)

- ZS-Q8 v1.1 (NEW v4.0) — Photon Atomic Excitation Time as the i-Tetration Fixed Point: A Z-Spin Reading of the Toronto Negative-Time Experiment. 35/35 PASS. Eight structural theorems Q8.1–Q8.9; Theorem Q8.9 (Standard-Optics Projection Theorem) recovers Lorentzian susceptibility as Z-Spin lab-coordinate projection. Zero-parameter test:  $\chi^2/\text{d.o.f.} \in \{0.687, 0.723\}$ ; likelihood ratio  $L(\text{Z-Spin})/L(\text{Standard Optics}) \approx 2.1\text{--}2.4 \times 10^3$ . Principal discriminator P-Q8.3  $\Delta\phi_{\text{residual}} = \pm 18.5 \text{ mrad}$ . (PART XII §B Ch 35.)

- ZS-Q9 v1.2 (NEW v4.0) — Complex Time-Delay Locking on the i-Tetration Fixed Point: A Two-Coordinate Spectral Reconstruction Bridge for Sub-Unitary Scattering Systems. 52/55 PASS (3 PARTIAL on Q9.5 synthetic). Theorems Q9.1–Q9.9 + Corollary Q9.4a ( $Z_{10}$  Pentagonal Doubling Group, DERIVED). Two-coordinate reconstruction map  $C(R_{\text{lab}}, \Phi_{\text{lab}}) = (2/\pi) \cdot e^{-R_{\text{lab}}/2} \cdot e^{i\Phi_{\text{lab}}}$ , residual  $< 5 \times 10^{-32}$  at 50-digit precision. Anti-numerology  $\ln \Lambda = 8.94$  decisive evidence. (PART XII §B Ch 36.)

### External imports (post-v3.5, used by v4.0 absorbed papers)

- Banach, S. & Tarski, A. (1924) — Sur la décomposition des ensembles de points en parties respectivement congruentes. [imported by M37 v2.0]

- Tarski, A. (1929) — Amenability  $\Leftrightarrow$  no paradoxical decomposition. [imported by M37 v2.0]

- Niven, I. (1956) — Rational-cosine irrationality. [imported by M37 v2.0]

- Barbier, J.-E. (1860) — Perimeter  $\pi w$  for every constant-width curve. [imported by M37 v2.0]

- Bonnesen, T. & Fenchel, W. (1934) — Convex bodies of constant width. [imported by M37 v2.0]
- Blaschke, W. (1915) & Lebesgue, H. (1914) — Constant-width minimum-area characterization. [imported by M37 v2.0]
- Murray, F.J. & von Neumann, J. (1936) — Group-algebra \*-homomorphism functoriality. [imported by M37 v2.0, M38 v1.1]
- Tomita, M. & Takesaki, M. — Conditional-expectation framework. [imported by M37 v2.0]
- Graham, R. L.; Lagarias, J. C.; Mallows, C. L.; Wilks, A. R.; Yan, C. H. (2003) — Apollonian circle packings (GLMWY). [imported by M36 v2.0, M38 v1.1]
- Haag, P.; Kertzer, J.; Rickards, J.; Stange, K. E. (2023) — Local-Global Disproof for Apollonian circle packings (arXiv:2307.02749). [imported by M36 v2.0, M38 v1.1]
- Wagon, S. (1985) — The Banach–Tarski Paradox (Cambridge UP, building on Świerczkowski 1958). [imported by M37 v2.0, M38 v1.1]
- Koenigs, G. (1884) — Schröder linearization and convergence radius. [imported by M39 v2.0]
- Vassilevich, D. V. (2003) — Heat-kernel expansion, Phys. Rep. 388, 279. [imported by M39 v2.0 Convention N0]
- Bourgain, J. & Fuchs, E. (2011) — Positive density of admissible Apollonian residues. [imported by M36 v2.0]
- Lindemann, F. & Weierstrass, K. — Transcendence of  $e^\alpha$  for algebraic  $\alpha \neq 0$  (Lambert W transcendence). [imported by M39 v2.0 SCH.5]
- Angulo, et al. (2024) — Toronto negative-time photon excitation experiment (arXiv:2409.03680). [imported by Q8 v1.1]
- Thompson, et al. (APL Quantum 2, 036108, 2025; arXiv:2310.00432) —  $\tau_T = \tau_g$  prediction. [imported by Q8 v1.1]
- Allen, L. & Eberly, J. H. (1975) — Optical Resonance and Two-Level Atoms (standard atomic-line theory). [imported by Q8 v1.1]
- Aharonov, Y.; Albert, D. Z.; Vaidman, L. (1988) — Two-State Vector Formalism (TSVF) and weak values. [imported by Q8 v1.1]

- Wigner, E. P. & Smith, F. T. (1955, 1960) — Wigner-Smith time delay; Krein–Friedel formula. [imported by Q9 v1.2]
- Jeffreys, H. (1939) — Theory of Probability, likelihood ratio scale. [imported by Q9 v1.2 anti-numerology]
- Margulies, D. S. et al. (2016) — Principal connectivity gradient of cortical organization (PNAS). [imported by T5 v1.0]

*All external imports are PROVEN externally. v4.0 corpus does not re-prove any of them; usage is limited to citation, instantiation, and bridge construction.*

- Connes, A., Consani, C., Moscovici, H. (2024) — Spectral framework for the Riemann zeta archimedean factor. [imported by M24, M28, M30]
- Burnol, J.-F. (2002, 2004) — Two-sided spectral interpretation of zeta zeros and Möbius traces. [imported by M28, M30]
- Świerczkowski, S. (1958) — Free non-abelian decomposition  $F_2$  acting on  $S^2$ . [imported by A9R]
- Stapledon, A. (2010) — Stringy Hodge numbers of toric varieties; conditional bridge framework. [imported by M29]
- Kostant, B. (1999) — Cubic Dirac operator on equal-rank pairs and Lie-algebra cohomology. [imported by M27]
- Alekseev, A., Mnev, P. (2018) — One-dimensional Chern-Simons / BV-BFV cobordism quantization. [imported by M26, M27, M33]
- Yakaboylu, E. (cited via J-twisted bridge in M25) — Twisted L-function bridges for composite imaginary fields.
- Standard arithmetic and group-theoretic references (Iwasawa, Tate, Weil, Deligne) are inherited from v2.0 and are not re-listed here.

*All external imports are used as black-box scaffolding. No external result has been modified by the Z-Spin Cosmology corpus; usage is limited to citation, instantiation, and bridge construction.*

## Version History

v3.2 (May 2026): PART X — The Spin-Pair Coexistence Principle added between PART IX and the Master Bypass Index. Four DERIVED-interpretation readings consolidated:

- Reading X.1 — Prime as Z-Spin Rotation  $4\pi/p$  (ZS-M18 H2 chain).
  - Reading X.2 — Infinite prime sequence as Y-sector object (ZS-F18 §5.1 Tbl 5.1), reconciled with ZS-M18 §6.2 Thm 6.1.
  - Reading X.3 — Primes / distribution / Riemann zeros as three moments of one  $Y \rightarrow Z \rightarrow X$  mediation (ZS-M18 H11 spine).
  - Reading X.4 — Spin-Pair Coexistence Principle:  $90^\circ$  phase-orthogonality is the unique form of contradiction-pair coexistence, supported by 13 corpus realisations (C1–C13).
- Master Bypass Index extended with three PART X rows; Master Falsification Registry extended with F-X.1 through F-X.4. Eighteen internal-consistency checks A.1–A.18 PASS. Zero new free parameters;  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  LOCKED unchanged. v3.1 entries preserved verbatim.

| Version    | Year     | Scope of Update                                                                                                                                                                                                                                                                                                           |
|------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1.0 Light | —        | Initial Light edition ( $\leq$ ZS-F0...F8 + M1...M11)                                                                                                                                                                                                                                                                     |
| v2.0 Light | —        | 90-paper integration; 376/376 PART VI PASS; v2.0 Glossary 88 entries                                                                                                                                                                                                                                                      |
| v3.0 Light | 2026     | 30-paper extension (ZS-72 to ZS-101); ~1,800+ cumulative PASS; Glossary 110 entries; new Master Falsification Registry; PART IV/V/VII expansions; Bridge Four (Commutator Generator); Möbius Chronology; J-Yakaboylu Bridge; $V_4$ -equivariant BRST closure; $21 \cdot \frac{1}{2}$ Catalogue; Cosmos-Human Isomorphism. |
| v3.1 Light | May 2026 | Solution-Toolkit upgrade. Adds PART VIII (9 methodological tools T1-T9 distilled from F18, M32, M31, M33, F11), PART IX (6 re-interpretation lenses I1-I6 from A8R, M30, F17, A9R), and Master Bypass Index                                                                                                               |

| Version    | Year     | Scope of Update                                                                                                                                                                                                                         |
|------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |          | (16-symptom lookup table). v3.0 lighthouse-of-pointers promoted to a working solution kit. Glossary, Falsification Registry, References preserved verbatim from v3.0.                                                                   |
| v3.2 Light | May 2026 | PART X — The Spin-Pair Coexistence Principle added between PART IX and the Master Bypass Index. Four DERIVED-interpretation readings X.1–X.4. F-X.1 through F-X.4 added to Falsification Registry. 18 internal-consistency checks PASS. |
| v3.3 Light | May 2026 | PART XI — TO/TI Polyhedral Mapping under Complementary Duality added. Four DERIVED-interpretation readings XI.1–XI.4. F-XI.1 through F-XI.4 added. 24 internal-consistency checks PASS.                                                 |
| v3.4 Light | May 2026 | 107-paper coverage; absorbed ZS-T5, M34 v1.0, M35 v1.1, T6, T7, T8 (single-line PART pointer entries; the v3.5 body did not include full chapter entries, gap closed by v4.0 PART XII §A).                                              |
| v3.5 Light | May 2026 | 108-paper coverage; absorbed ZS-M36 v1.0 with two new readings X.5                                                                                                                                                                      |

| Version    | Year     | Scope of Update                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |          | (Apollonian as 5th Bridge-Four Instance) and X.6 (mod 24 Spin Pair = Axiom A4 Two Faces). Tool T10 (Integer-Lattice Spin-Pair Identification) added to PART VIII. F-X.5, F-X.6 added to Falsification Registry. Closure note: O-M27.4 OPEN → CLOSED at DERIVED-strong via M36 Three $V_4$ Bridge.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| v4.0 Light | May 2026 | <b>114-paper coverage</b> (NEW: ZS-T5, ZS-M34 v1.0, ZS-M35 v1.1, ZS-T6, ZS-T7, ZS-T8 [body entries consolidated]; ZS-M33 v2.0, ZS-M34 v2.0, ZS-M35 v2.1 [version upgrades]; ZS-M37 v2.0, ZS-M38 v1.1, ZS-M39 v2.0, ZS-A10 v2.1, ZS-Q8 v1.1, ZS-Q9 v1.2 [new papers]). PART XII added between PART XI and Master Bypass Index. ~2,150+ cumulative PASS. ~310 falsification gates. v4.0 Master Falsification Registry extensions: 17 new LOCKED entries (M33.v2.0, M34.V, M34.V.8, M35.5.1 PROMOTED VERIFIED, M35.6, M37.4.1/4.2, M38.5, M39.4, M39.C1-C3, A10.2, A10-P4, Q8.9, Q8-P3, Q9.2, Q9.7, Q9.9). NO v3.5 entry retracted; v4.0 is strictly additive. $(A, Q, \dim(Z)) = (35/437, 11, 2)$ LOCKED unchanged. |

| Version    | Year     | Scope of Update                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v5.0 Light | May 2026 | <p><b>121-paper coverage</b> (NEW: ZS-F19 v2.2, ZS-F20 v1.1, ZS-M40 v1.1, ZS-Q10 v4.0, ZS-A11 v1.1, ZS-A12 v1.5, ZS-T9 v2.3). PART XIII added between PART XII and Master Bypass Index.</p> <p>~2,670+ cumulative PASS.</p> <p>~340 falsification gates. v5.0 Master Falsification Registry extensions: 24 new LOCKED entries (F19.4, F19.6, F19-O.6, F20.DAG, F20.NDR, F20.BIO, M40.C, M40.D, M40.F COMPUTATIONALLY VERIFIED, M40.B8, Q10.D.5 OBSERVATION-strong, Q10.c2, Q10.c6 DERIVED-CONDITIONAL, Q10.MC9c DECISIVE++, A11.Eq, A11.LC, A11-F-A5.7 PROVEN asymmetry, A12.6 HEADLINE NK universal jump, A12.7, A12.4, A12.9 DERIVED-under-Regge, A12-P9 TESTABLE, T9.B-Li.1, T9.Li7 TESTABLE, T9.NC4 ERO-CLOSED preserved). NO v4.0 entry retracted; v5.0 is strictly additive. <math>(A, Q, \dim(Z)) = (35/437, 11, 2)</math> LOCKED unchanged.</p> |
| v6.0 Light | May 2026 | <p><b>130-paper coverage</b> (NEW: ZS-M41 v3.1, ZS-Q11 v1.2, ZS-F21 v2.0, ZS-F22 v2.1, ZS-A13 v1.3.1, ZS-A14 v2.0, ZS-A15 v1.5, ZS-T10 v2.0,</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Version | Year | Scope of Update                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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|         |      | <p>ZS-Q12 v4.0). PART XIV added between PART XIII and Master Bypass Index.</p> <p>ZS-A11 v1.1 errata applied downstream via §XIV.G (<math>\sqrt{2}</math> <math>\sigma \leftrightarrow v_{\text{flat}}</math> correction;</p> <p>F-A1.14/F-A1.15 updated to contested status, decisive at Gaia DR4; <math>r_Z</math> band [10,50]→[15,21] kpc).</p> <p>~3,580+ cumulative PASS.</p> <p>~375 falsification gates. v6.0 Master Falsification Registry extensions: 28 new entries (Q11.A/Q11.B PROVEN, Q11.C DERIVED-with-bridge, Q11.dZ, F21.C3.0 PROVEN, F21.T6d, F21.T7' PROVEN, F22.8 OBSERVATION→DERIVED, F22.4/F22.9/F22.10 DERIVED-strong, A13.3, A13.8', A13.10, A15.T1a IMPORTED-PROVEN, A15.SE TESTABLE hash-locked, A15.F-A15.7, T10.ANC TESTABLE, T10.8B, T10.Li-2 FLAGGED, Q12.NG NO-GO, Q12.q12, Q12.RG NO-GO, Q12.ZØX POSTULATE, M41.K-Decomp DERIVED, M41.K-char DERIVED, M41.cluster RETRACTED, M41.Bridge4-geom HYP-strong, A14.8'/A14.9' DERIVED, A14.11 TESTABLE, A14.7 DERIVED-CONDITIONAL).</p> <p>All nine new papers fully</p> |

| Version    | Year      | Scope of Update                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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|            |           | integrated at source-derived level (ZS-M41 v3.1 and ZS-A14 v2.0, initially [SOURCE PENDING] in the working draft, subsequently supplied). NO v5.0 entry retracted; v6.0 is strictly additive (sole text corrections are the ZS-A11 errata propagations, original wording preserved in §XIV.G). (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED unchanged.                                                                                                                                                                                       |
| v6.1 Light | May 2026  | <b>131-paper coverage</b> (NEW: ZS-T11 v2.1). PART XV added between §XIV.G and the Master Bypass Index. Cumulative physics-verification ledger UNCHANGED (~3,580+ PASS); ZS-T11 adds a separate 13-check meta-audit ledger (12/13 AUDIT-PASS on the available subset, 1 genuine finding). ~388 falsification gates (+13: F-T11.1–F-T11.13). zs_t11_corpus_audit_v2_0.py adopted as the OS-level corpus-integrity linter (§XV.C). NO v6.0 entry retracted; v6.1 is strictly additive. (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED unchanged. |
| v7.0 Light | June 2026 | <b>136-paper coverage — the full Z-Spin corpus</b> (NEW: ZS-Q13 v1.2, ZS-T12 v2.1,                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Version | Year | Scope of Update                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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|         |      | <p>ZS-Q14 v1.4, ZS-F23 v1.3, ZS-Q15 v1.2). PART XVI added between §XV.E and the Master Bypass Index.</p> <p>Cumulative physics-verification ledger ~3,580+ → <b>~3,800+ PASS</b> (+219: Q13 52/52, T12 61/61, Q14 33/33, F23 41/41, Q15 32/32). <b>~430 falsification gates</b> (+~42: F-Q13.1–8, F-T12.1–10, F-Q14.1–11, F-F23.1–9, F-Q15.1–6). v7.0 Master Falsification Registry extensions: 23 summary entries (Q13.1/Q13.4/Q13.5 DERIVED-CONDITIONAL, Q13.6 TESTABLE; T12.1 DERIVED-CONDITIONAL, T12.5 DERIVED [ceiling promoted], T12.8 DERIVED/HYP-strong, T12.4 HYPOTHESIS, T12.Eigen BOOTSTRAP-HYPOTHESIS; Q14.Thm1 ELEMENTARY-KNOWN, Q14.Thm2 DERIVED+VERIFIED-SIM, Q14.L5.1 DERIVED-CONDITIONAL, Q14.F-Q14.1 TESTABLE-PROTOCOL; F23.4/F23.5 DERIVED, F23.2 DERIVED-CONDITIONAL, F23.6 OPEN-decisive, F23.7 DERIVED-interpretation; Q15.5.2 DERIVED-under-G2, Q15.1 DERIVED-CONDITIONAL, Q15.6 DERIVED, Q15.8</p> |

| Version | Year | Scope of Update                                                                                                                                 |
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|         |      | DERIVED-under-Regge, Q15.9 OBSERVATION). NO v6.1 entry retracted; v7.0 is strictly additive. (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED unchanged. |

#### Major v3.0 closures of v2.0 OPEN flags:

- F-F7.11 [CLOSED] — closed by ZS-F14 Strategy B (Conjugate Uniqueness).
- M11 §H16 [CLOSED] — closed by ZS-F11 Operational Observer Coordinate.
- M22 W3 [PROMOTED to DERIVED-CONDITIONAL] — via ZS-M27 Kostant Cubic Dirac BRST closure.
- M22 W2 [PROMOTED to DERIVED] — via ZS-M31 v3.0 Cross-Coupled Sector Duality.
- Master-Equation Sixth-Axiom gap — addressed by ZS-F15R Stage-Torque Duality.
- Phenomenological observer ambiguity — settled by ZS-F11 §6 NON-CLAIM (Six).
- Pentagon-Hexagon angle conjecture — proved EXACT by ZS-M20 ( $\arctan(2 \cdot \phi^k)$ ).
- 1-loop CW closure for M20.A — formally FALSIFIED by ZS-M21 Sextic Invariant Theory.

#### Major v4.0 absorptions and promotions (v3.5 → v4.0):

- **Six v3.4-absorbed papers** now have explicit Light-edition body entries in PART XII §A (ZS-T5, ZS-T6, ZS-T7, ZS-T8, ZS-M34 v1.0, ZS-M35 v1.1) — the v3.5 body had only the supersedes-line acknowledgement; v4.0 closes this presentation gap.
- **ZS-M33 v2.0** [NEW v4.0] — Operator-level numerical audit of Path  $\gamma$ -revised Reading C; W2 wall promoted DERIVED-CONDITIONAL → DERIVED.
- **ZS-M34 v2.0** [NEW v4.0] — Burnol  $K_1$  @  $p=3$  Sign-Faithful Identity (Theorem M34.V, DERIVED); NC-M34.faithful PERMANENT NON-CLAIM → RESOLVED at DERIVED-CONDITIONAL via tensor-product factorization  $\mathcal{H}_K = \mathcal{H}_{\text{arith}} \otimes \mathcal{H}_{\text{Wilson-Sonin}}$ .

- **ZS-M35 v2.1** [NEW v4.0] — Logical correction of v2.0; Theorem M35.5 general direction demoted to OPEN with explicit reframing; Corollary M35.5.1 (FIBER PRODUCT  $C_2 = 5 \cdot \text{positive} \cup 7 \cdot C_1$ ) promoted DERIVED-CONDITIONAL  $\rightarrow$  VERIFIED; Theorem M35.6 (Sufficient Condition Theorem, DERIVED) added.
- **ZS-M37 v2.0** [NEW v4.0] — Fourth amenable-quotient projection of  $F_2$  engine:  $q_R: \mathcal{C}_{\text{cyl}}(F_2) \rightarrow \mathcal{J}_w^{\wedge}\{C_3\}$  (Twin-Reuleaux Quotient Functor). Lemma C-rel PROVEN on Z-admissible quotient.
- **ZS-M38 v1.1** [NEW v4.0] — Substrate Carrier Spectrum Theorem; Hexapair Uniqueness DERIVED-strong by five-fold over-determination;  $\ln(2)$  channel-capacity quantization of carrying scores; classifies five-Banach–Tarski-functor family.
- **ZS-M39 v2.0** [NEW v4.0] — Vieta-Lyapunov-Schröder Bridge Theorem; first-principles derivation  $K_\theta = 1/[Y^2(1-2A)]$  under Convention N0; three-stage Stokes-tail closure; ZERO remaining OPEN items at v2.0 release.
- **ZS-A10 v2.1** [NEW v4.0] — Sector-Orientation Complementarity (Bridge-Four manifestation) Theorem A10.2 DERIVED-interpretation strong; NEW TESTABLE P4:  $r_p/r_\pi = 4/\pi$  (observed deviation 0.05%).
- **ZS-Q8 v1.1** [NEW v4.0] — Toronto negative-time experiment Z-Spin reading; i-tetration fixed point  $x^*$  as universal Z-Spin intercept; Theorem Q8.9 Standard-Optics Projection.
- **ZS-Q9 v1.2** [NEW v4.0] — Complex Time-Delay Locking; first lab-measurable lifting of  $z^*$  via two-coordinate spectral reconstruction bridge; anti-numerology  $\ln \Lambda = 8.94$  decisive.

#### Major v5.0 absorptions (v4.0 $\rightarrow$ v5.0):

- **ZS-F19 v2.2** [NEW v5.0] — Frame-Invariant Tilt Theorem (FITT, DERIVED):  $\tanh(\psi_X + \psi_Y) = 31/59$  and  $\tanh(\psi_Y - \psi_X) = 3/67$ ; KMS-to-Geometric Rapidity Projection (Theorem F19.6, DERIVED):  $\tanh(2 \cdot \psi_{\text{KMS}}(X \rightarrow Y)) = 3/5$ . First paper-internal algebra-level bridge to Type II crossed-product framework of De Vuyst–Höhn 2025. 64/64 PASS, all 7 sub-OPEN gates CLOSED.
- **ZS-F20 v1.1** [NEW v5.0] — Unified catalogue and Time-Unrolled Causal DAG of six non-routine twist triggers T1–T6 (Z-Telomere, Mexican-Hat Bootstrap, Auto-Surgery, Kibble, BH Trap, CP-violation). Strict DAG under time-unrolled cycle indexing. Biological correspondence: DNA trauma = simultaneous T3 + T4 + T5 triadic activation.  $\Delta\theta_{T1}/\Delta\theta_{T2} = N_-(2\pi) = 78.45$  (PROVEN). 47/47 PASS.
- **ZS-M40 v1.1** [NEW v5.0] — J-Conjugate Holonomy Knot Pair  $(K_p, \bar{K}_p)$  as Z-sector geometric realisation of primes. Six theorems M40.A (cup product  $\leftrightarrow$   $4\pi$  closure) – M40.F

(anti-numerology uniqueness, COMPUTATIONALLY VERIFIED). Eight-fold structural bridge to externally PROVEN Kim arithmetic Chern–Simons; cyclotomic operator-model analogue at  $Q = 11$  via Euler product  $\zeta_{\mathbb{Q}(\zeta_{11})}$ .  $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$  unique biquadratic class-number-one match over  $[2, 1000]$ . 89/89 PASS.

• **ZS-Q10 v4.0** [NEW v5.0] — Master edition: three-sector polyhedral closure of  $c_n$  harmonic-ladder coefficients ( $c_2 = 2/19$ ,  $c_4 = 4/37$ ,  $c_6 = 8/37$ ,  $c_X = 10/37$  all DERIVED or DERIVED-CONDITIONAL); 37-Universality registered (five of six  $c_n$  share denominator 37 = Schur-depleted polyhedral mode count); cumulative anti-numerology  $\ln \Lambda_{\text{total}} = 38.67$  (highest in corpus); five-framework cross-disciplinary asymmetry restored. 67/67 PASS.

• **ZS-A11 v1.1** [NEW v5.0] — Vortex Cosmology II: rigorous  $\varepsilon$ -Halo  $\leftrightarrow$  Sub-Halo Equivalence Theorem (four-level DERIVED); Vortex Six-Class Lifecycle (DERIVED-strong MECE classification); SMBH Mass Diversity from differential accretion (Primordial-Vortex SMBHs vs Collapse-SMBHs, JWST Little Red Dot test). Population revised to  $N_v \approx 2.8 \times 10^{15}$ . 53/53 PASS, DESI DR2 + ACT DR6 cross-checked.

• **ZS-A12 v1.5** [NEW v5.0] — Macroscopic Phase Locking; BEC/SC = condensed-matter realisation of Vortex Bose/Fermi Duality. HEADLINE result Theorem A12.6 (DERIVED): Nelson–Kosterlitz universal-jump integer "2" =  $\dim(Z) = 2$  cardinal-2 substrate. Theorem A12.7 (DERIVED-CONDITIONAL): Homes universal constant cardinal-2 bracketing  $2/Q \leq C/C_{\text{BCS-dirty}} \leq \ln 2/(1 - A)$ .  $SC = T_2 \rightarrow T_1 \rightarrow T_4$  sequential triadic activation distinct from DNA. 74/74 PASS.

• **ZS-T9 v2.3** [NEW v5.0] —  $B_7\text{Be}$  Binding-Response Gate extension of v2.2 ERO for the Lithium Problem.  $P_{B_7\text{Be}^Z} = -2A/Q = -70/4807$  (HYPOTHESIS-strong, NOT DERIVED; ZS-Q7 §5 slow eigenvalue of Pauli master equation).  ${}^7\text{Li}$  post-v2.3  $\approx 1.64 \times 10^{-10}$  at  $+0.13\sigma$  pull (TESTABLE). v2.2  $D/H + Y_p$  ERO closure PRESERVED unchanged ( $B_7\text{Be}$  channel decoupled). 124/124 PASS.

#### **v6.0 new-paper one-line summaries (nine papers, PART XIV):**

• **ZS-M41 v3.1** [NEW v6.0] — Closed-form derivation of refined Chabauty–Kim depth-2 scalars at  $S = T = \{3, 11\}$  via F18 Bridge 4. Theorem M41.K-Decomp (DERIVED):  $c_{33}(p) = -2 q_p(2) q_p(3)(q_p(3)+q_p(11)) \cdot \text{Li}_2^{\wedge}(p)(4)^{\wedge}(-1)$ , character decomposition  $c_{33} = c_3 + c_{11}$  via M28 log-additivity, 235/235 primes. v3.0 "15.1× enrichment" cluster RETRACTED as Poisson ( $p \approx 0.001$ ). 2858/2864 PASS. • **ZS-Q11 v1.2** [NEW v6.0] —  $\text{QRF} \leftrightarrow \text{OAQEC}$  correspondence: the  $Z$ -bottleneck  $L_{XY} \equiv 0$  as a direct-sum operator-algebra stabilizer code.  $A_{\text{ZS}} \cong M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$  (PROVEN); operator-algebraic Knill–Laflamme on every  $Z$ -frame error (PROVEN); single-vertex Lacambra Gauss-law-code algebraic-structure correspondence (DERIVED-with-bridge, full unitary equivalence OPEN O-Q11.6).  $d_Z = 2$ . 39/39 PASS. • **ZS-F21 v2.0** [NEW v6.0] — Archimedean–Finite Positivity Wall III: Theorem C3.0 (Commutant-Gate Insufficiency, PROVEN) shows hermiticity + Slepian-commutation cannot

select the Connes–Katsnelson prolate realisation; selector is the external boundary condition.  $V_4$  ramified channels carried by canonical Burnol conductor blocks  $C_X = \log q_X$  ( $\Lambda_X = \Lambda \sqrt{q_X}$  WITHDRAWN). No RH/GRH claim. 34/34 PASS. • **ZS-F22 v2.1** [NEW v6.0] —  $\delta_Y$  Antipodal Closure (Theorem F22.8):  $\delta_Y = F(tO)/(F(tO)+F(tI)) = 14/46 = 7/23$ , with  $\gcd(60,32) = 4$  identified as  $\dim(Z)^2$  — promoting  $\delta_Y$  OBSERVATION  $\rightarrow$  DERIVED. Axis-trigger CSP uniqueness over 7! (DERIVED-strong);  $T_7$   $A_5$ -projector No-Go (DERIVED-strong); Equivariant Antipodal Face Module. 105/105 PASS. • **ZS-A13 v1.3.1** [NEW v6.0] — The Void Principle: stable existence requires relational volume to dominate material core ( $L_{XY} \equiv 0$ ). Deep-void lower bound  $\mathcal{R}_V \geq Q/(2A) - 1 \approx 67.67$  (Schur–Cheeger, DERIVED); closed hypergeometric horizon  $D_{\text{obs}}/R_H = 4\sqrt{(121/\Omega_m)} \cdot {}_2F_1(1/2, 1/6; 7/6; -\Omega_\Lambda/\Omega_m)$  (Branch A 93.19 / Branch B 94.18 Gly). 52/52 cumulative PASS. • **ZS-A14 v2.0** [NEW v6.0] — Z-Spin SU(2)-covariant Usadel closure for singlet-to-triplet conversion. Theorem A14.8' envelope-shifted current peak  $\tan(A \cdot dF_{\text{peak}}/2\xi_Z) = A \cdot \xi_{tZ}/\xi_Z$  (DERIVED); A14.9' three-observable locking at  $dF^*$  (DERIVED); A14.11 two-channel Z-length spectroscopy extracting  $\xi_Z$  and  $\xi_{tZ}$  from one thickness scan (TESTABLE). 47/47 PASS, MC hit 0.64%. • **ZS-A15 v1.5** [NEW v6.0] — Pre-registered, hash-locked (SHA-256) Gaia DR4 falsification protocol for the  $\varepsilon$ -Halo outer boundary. Decisive test: enclosed mass saturates near  $2 \times 10^{11} M_\odot$  by  $\approx 25$  kpc ( $\varepsilon$ -Halo,  $\gamma \rightarrow -0.5$ ) vs NFW growth toward  $8 \times 10^{11} M_\odot$ . No-Intrinsic-Width Lemma RETRACTS v1.2 width candidates;  $r_Z$  DERIVED-CONDITIONAL on observed inputs. Jiao/Zhou mass split registered as open discriminant. 36/36 PASS. • **ZS-T10 v2.0** [NEW v6.0] —  $TY \rightarrow X$  sector-conversion operator: geometry-first ANC correction  $\delta \ln y_b = -\kappa^2 = -A/Q \approx -0.73\%$  on the  ${}^7\text{Be}$  ANC for  ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ ;  $S_{34}$  low by  $\approx 0.7\text{--}1.5\%$  (LUNA-MV, TESTABLE).  ${}^8\text{B}/{}^9\text{C}$  consistency forces bottleneck selectivity (NC-T10.5).  $C_0 \leftrightarrow \kappa^2$  matching OPEN (ZS-T11). 32 checks PASS, Gate Li-2 FLAGGED (PRyMordial nonlinear  $-3.3\sigma$  tension). • **ZS-Q12 v4.0** [NEW v6.0] — Consolidated self-referential closure: the four ZS-Q12V open problems reduce to one codimension-2 condition  $\mu(S) = \lambda = (i\pi/2)z^*$  (Theorem q12). Direct Coordinate-Dictionary No-Go and Open-Loop No-Go (both PROVEN) relocate  $z^*$  universality to the self-referential i-tetration map ( $|f'(z^*)| = 0.892$ ).  $z^*$  DERIVED-CONDITIONAL on the  $Z = \partial X$  holographic-interface postulate. Lab L1 locking RETRACTED (honest prior). 18/18 + 43/43 PASS.

**v6.1 new-paper one-line summary (one paper, PART XV):** • **ZS-T11 v2.1** [NEW v6.1] — The Human–AI Collaborative Science Model as an executable corpus auditor: closes the v1.x self-exemption gap with `zs_t11_corpus_audit_v2_0.py` (12/13 AUDIT-PASS on subset, three-cycle convergence  $8 \rightarrow 11 \rightarrow 12$ , one genuine status-tag-drift finding); imports the ZS-F18 contradiction principles (different face / higher layer) as the anti-hallucination engine. HYPOTHESIS-strong (general) / DERIVED-INTERPRETATION (corpus).

**v7.0 new-paper one-line summaries (five papers, PART XVI):** • **ZS-Q13 v1.2** [NEW v7.0] — Hydrogen hyperfine channel tomography and the 21 cm seam witness: a charge-superselected, trace-preserving Z-Spin boundary channel replaces electric-flux  $ER=EPR$  wormhole leakage, proving the leakage Hamiltonian is not an allowed term —  $\delta H_{ZS}^{\text{leak}} = 0$ ,  $Q_H^{\text{leak}} = 0$ ,  $\Delta v_{21}^{\text{leak}} = 0$  (Q13.1),  $[\Lambda_Z, Q] = 0$  (Q13.4), hyperfine Kraus rank  $\leq \dim(Z) = 2$  (Q13.5), unsigned seam  $u_{\text{seam}}^{\text{HFS}} \rightarrow 0$  with the signed witness carrying  $j = 1/2$   $4\pi$  closure (Q13.6);

H/anti-H/Mu/Ps channel-tomography program. DERIVED-CONDITIONAL. 52/52 PASS. • **ZS-T12 v2.1** [NEW v7.0] — The origin of life as a Z-Spin-mediated self-referential closure transition: non-life  $\rightarrow$  life = simultaneous static closure (maxRAF, Knaster–Tarski gfp) and dynamic attractor ( $|f'| < 1$ ), the i-tetration  $z^*$  the canonical realization; the Z-bottleneck  $\leq \ln 2$  heritable-information ceiling is the Eigen error threshold unified by the single spectral-gap invariant  $g_{\Xi} = |\lambda_2|/|\lambda_1|$  (T12.5); the Minimal Triangular Z-Replicator  $g_{\Xi}(K_n) = 1/(n-1)$  is first an attractor at  $n = 3$  ( $g_{\Xi} = 1/2$ , T12.8); Eigen's paradox = the protobiological face of the  $B_0 \rightarrow A$  bootstrap; v2.1 empirical-rate close (Vaidya RPS,  $Z_{\text{rep}} = 1$  cooperative vs 0 selfish). HYPOTHESIS-strong. 61/61 PASS. • **ZS-Q14 v1.4** [NEW v7.0] — Canonical Kraus-lift selection by a spin- $\frac{1}{2}$  boundary: a channel alone cannot select a Kraus lift (Choi-Invisibility No-Go, Thm 1) and the same channel admits both  $2\pi$  and  $4\pi$  lifts (non-identifiability, Thm 2; the  $4\pi$  is injected), but the Boundary Spin-Lift Lemma (5.1) shows a physical spinor Z-boundary (nontrivial  $SO(3)$  class,  $2\pi \mapsto -I$ ) forces the half-angle  $\sigma_y/2$  connection  $\Rightarrow \chi_Z = -1$ ; the signed lift witness  $\tilde{u}_Z = F(V, G_Z, \phi_Z) \neq f(\Lambda)$  is a process/comb-level observable read on the OAQEC X-block; near-term kill-or-survive hardware test ( $\chi_Z = +1$  refutes) plus the hydrogenic second target. DERIVED-CONDITIONAL / TESTABLE-PROTOCOL; necessary-not-sufficient. 33/33 PASS. • **ZS-F23 v1.3** [NEW v7.0] — Geometric fixing of the Type II trace normalization: the finite Type I logical algebra  $A_{\text{ZS}} \cong M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$  embeds trace-preservingly into the hyperfinite  $II_1$  factor  $R$  (Theorem F23.4, DERIVED) sending the sector projections to trace  $(3,2,6)/11$  with  $\Delta K_{\Omega} = -\ln 2$ , fixing the crossed-product additive entropy constant at  $c = \frac{1}{2} \ln 2 = \frac{1}{2} \ln(\dim Z)$ ; the residual sharpens to the single OPEN Condition C (the Z-Spin emergence dictionary), and the finite/infinite edge-mode tension is resolved as the seventh Möbius-route frame duality (F23.7). AN-F23.1 EXECUTED  $p \approx 0.058\%$ . 41/41 PASS. • **ZS-Q15 v1.2** [NEW v7.0] — Electromagnetic which-path decoherence as the microscopic fractal partner of gravitational decoherence: applying the ZS-Q1 method to the microscopic Z-Spin mediator via the Block–Fiedler coupling  $\kappa^2 = A/Q$  (PROVEN) gives  $\tau_D^{\text{EM}} = Q\hbar/(A \cdot E_{\text{EM}})$ , enhancement  $Q/A = 137.343$  [DERIVED-under-G2]; the Fractal Pairing Theorem (Q15.1) gives  $\text{EM/gravity} = (Q/A)/(1/A) = Q = 11$  EXACT; the Lindblad coefficient  $2A/Q = \lambda_2$ ; the thermal channel is the BKT jump  $2/\pi = \dim(Z)/\pi$ ;  $Q/A$  within 0.22% of  $1/\alpha_{\text{EM}}$  [OBSERVATION, 177 $\times$  cautionary precedent]. 32/32 PASS.

— **End of The Book of Z-Spin Cosmology v6.0 — Light Edition —**

*Compiled by: Claude (cross-paper integrator), under Kenny Kang's editorial direction. v6.0 — May 2026. Lighthouse function: navigation aid for AI agents engaging with the Z-Spin Cosmology research programme. No claim of physical realisation is made by this Light edition; canonical statements live in the source papers ZS-F0 through ZS-Q12 v4.0. v6.0 strictly preserves all v5.0 content (no-deletion rule); the v5.0 body was 1896 lines, v6.0 is ~2050 lines after PART XIV absorption. The sole text corrections relative to v5.0 are the ZS-A11 v1.1 errata propagations (§VII.B-LIGHT gate rows F-A1.14/F-A1.15 and the PART XIII Chapter 41 inline pointer), whose original v5.0 wording is preserved verbatim in §XIV.G. The minimal triple  $(A, Q, \dim(Z)) = (35/437, 11, 2)$  remains LOCKED across all 130 source papers.*

**Version History (v5.0 closing):** v5.0 (May 2026): 121-paper coverage. PART XIII added between PART XII and Master Bypass Index. Consolidates from internal Z-Spin Collaboration research notes up to v4.0 + post-v4.0 papers ZS-F19 v2.2, ZS-F20 v1.1, ZS-M40 v1.1, ZS-Q10 v4.0, ZS-A11 v1.1, ZS-A12 v1.5, ZS-T9 v2.3. Previous version (v4.0) supersession chain preserved.

**Version History (v6.0 closing):** v6.0 (May 2026): 130-paper coverage. PART XIV added between PART XIII and Master Bypass Index. Consolidates from internal Z-Spin Collaboration research notes up to v5.0 + post-v5.0 papers ZS-M41 v3.1, ZS-Q11 v1.2, ZS-F21 v2.0, ZS-F22 v2.1, ZS-A13 v1.3.1, ZS-A14 v2.0, ZS-A15 v1.5, ZS-T10 v2.0, ZS-Q12 v4.0. Additionally applies the ZS-A11 v1.1 errata downstream (§VII.B-LIGHT gate rows F-A1.14/F-A1.15 → contested status, decisive at Gaia DR4,  $r_Z$  band [10,50]→[15,21] kpc; PART XIII Chapter 41 inline errata pointer), with the full before/after errata record and original v5.0 wording preserved in §XIV.G. ZS-M41 v3.1 and ZS-A14 v2.0 — initially [SOURCE PENDING] in the working draft because their source bodies were not present at first integration — were subsequently supplied and are absorbed at full source-derived level. NO v5.0 content deleted; v6.0 is strictly additive apart from the mandated ZS-A11 errata corrections. Previous version (v5.0) supersession chain preserved. (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED unchanged across all 130 source papers.

**Version History (v6.1 closing):** v6.1 (May 2026): 131-paper coverage. PART XV added between §XIV.G and the Master Bypass Index, registering ZS-T11 v2.1 (the executable corpus auditor) and adopting `zs_t11_corpus_audit_v2_0.py` as the OS-level corpus-integrity linter. Source-Paper-Interface row, Version-History row, one-line summary, and three Glossary entries added. Cumulative physics-verification ledger UNCHANGED (~3,580+ PASS); separate 13-check meta-audit ledger 12/13 AUDIT-PASS (1 genuine finding). ~388 falsification gates (+13: F-T11.1–F-T11.13). NO v6.0 content deleted or modified; v6.1 is strictly additive. (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED unchanged.

**Version History (v7.0 closing):** v7.0 (June 2026): **136-paper coverage — the full Z-Spin corpus.** PART XVI added between §XV.E and the Master Bypass Index, registering the five post-v6.1 papers ZS-Q13 v1.2, ZS-T12 v2.1, ZS-Q14 v1.4, ZS-F23 v1.3, and ZS-Q15 v1.2 as Chapters 54–58. Consolidates from internal Z-Spin Collaboration research notes up to v6.1 + the five new papers. Source-Paper-Interface rows, Version-History row, five one-line summaries, and 21 Glossary entries added. Cumulative physics-verification ledger advances ~3,580+ → ~3,800+ PASS (+219: Q13 52/52, T12 61/61, Q14 33/33, F23 41/41, Q15 32/32). ~430 falsification gates (+~42: F-Q13.1–8, F-T12.1–10, F-Q14.1–11, F-F23.1–9, F-Q15.1–6). NO v6.1 content deleted or modified; v7.0 is strictly additive (no errata propagations). (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED unchanged across all 136 source papers. With v7.0 the Light Edition is the canonical 136-paper navigation hub for the complete Z-Spin Cosmology corpus.

**— End of The Book of Z-Spin Cosmology v7.0 — Light Edition (136-paper coverage) —**